



**GREEN
CLIMATE
FUND**

Meeting of the Board

4 – 7 March 2024

Kigali, Rwanda

Provisional agenda item 10

GCF/B.38/02/Add.06

12 February 2024

Consideration of funding proposals – Addendum VI

Funding proposal package for FP226

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru
Country(ies):	Peru
Accredited Entity:	Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)
Date of first submission:	2023-09-18
Date of current submission	2024-02-05
Version number	V.04



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Project	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	Not applicable		
A.4. Result area(s)		GCF contribution	Co-financers' contribution¹
	Mitigation total	Enter number %	Enter number %
	☐ Energy generation and access	Enter number %	Enter number %
	☐ Low-emission transport	Enter number %	Enter number %
	☐ Buildings, cities, industries and appliances	Enter number %	Enter number %
	☐ Forestry and land use	Enter number %	Enter number %
	Adaptation total	100 %	100 %
	☒ Most vulnerable people and communities	30 %	30 %
	☒ Health and well-being, and food and water security	30 %	30 %
	☐ Infrastructure and built environment	Enter number %	Enter number %
	☒ Ecosystems and ecosystem services	40 %	40 %
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	N/A	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	60,715 Direct² beneficiaries (of which 30,088 are women) 2,011,856 Indirect³ beneficiaries (of which 1,005,928 are women) 0.2% of the national population⁴ 6.8% of the national population
A.7. Total financing (GCF + co-finance⁵)	78.32 million Euros	A.9. Project size	Medium (Up to USD 250 million)
A.8. Total GCF funding requested	40.79 million Euros		
A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8. ☒ Grant 40.79 million ☐ Equity Enter number ☐ Loan Enter number ☐ Results-based payment Enter number ☐ Guarantee Enter number		
A.11. Implementation period	6 years	A.12. Total lifespan	15 years
A.13. Expected date of AE internal approval	11/6/2023	A.14. ESS category	Category B

A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☐ No ☒	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	Yes ☐ No ☒		
A.20. Executing Entity information	The Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH, besides being the Accredited Entity (AE) of the project, will also act as an Executing Entity (EE). GIZ is registered in Germany, and its sole shareholder is the Federal Republic of Germany, represented within Germany by the Federal Ministry for Economic Cooperation and Development (BMZ) and the Federal Ministry of Finance (BMF). In addition, there will be four Peruvian Executing Entities: i) the Ministry of Agricultural Development and Irrigation (MIDAGRI), ii) the National Service of Natural Protected Areas by the State (SERNANP), iii) Profonanpe, and iv) Asociación Instituto Andino de Montaña (hereafter Instituto de Montaña (IdM)). MIDAGRI and SERNANP are Government entities, Profonanpe is a private non-profit institution of public and social interest, and IdM is a non-government organization (NGO).		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
1. Climate change and unsustainable management of productive practices are degrading the Puna ecosystems, (peatlands, wetlands, and grasslands) and the services they provide (provision and regulation of water; provision of fodder, food, and fiber; nutrient and carbon regulation) in the Southern High Andes of Peru (SHAP)⁶. Increased temperatures have contributed to glacier melting while fewer rains and longer drought periods have affected water availability in an area where 80% of the population is heavily reliant on rainfed agriculture. This climate factors are threatening the livelihoods of approx. half a million people in SHAP and the water security of approx. 4.5 million downstream. Increased temperatures have already decreased 51 percent of Peru's glaciers. In addition, huge stocks of carbon stored in the Puna ecosystems could be released into the atmosphere as they are increasingly degrading. 2. The Andean Indigenous Peoples and Local Communities (IPLCs) in the SHAP are characterized throughout history by their own adapting capacity to cope with environmental changes. Over the years, they have passed down from one generation to others their agricultural, water and ecosystem management knowledge and practices. For example, selection and domestication of tolerant crops and varieties such as potatoes and Andean grains, water management technologies, systems of terraces to use small parcels of land at different altitudes and within different microclimates to ensure crop diversification. They also brought important food crops such as potatoes, quinoa and other highly nutritious high-altitude crops to the world⁷. Even though this traditional knowledge has been a key asset to Andean people when facing climate stress, nowadays Andean population find the increasing severity and unpredictability of extreme events, as well as the unpredictability of hydric regimes quite challenging⁸. This situation coupled with their vulnerabilities such as low levels of development, dependence on subsistence agriculture and husbandry practices, limited economic opportunities are forcing them to migrate and lose their traditional			

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e., GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² Direct beneficiaries include all individuals who will receive i) targeted support from the GCF-funded intervention and ii) a measurable adaptation benefit from a GCF-funded intervention, including: a) Individuals that implement Local initiatives that will be financed and supported by the Puna Facility, multiplied by the average number of members of the groups/legal entities supported and the average members per household in the region, b) Individuals that will be supported by MIDAGRIs co-finance programs, multiplied by the average members number per household in the region. c) Individuals that will receive technical assistance from the GCF-funded intervention (e.g., trainings, support to develop site-specific climate diagnostics and preparation of participatory intervention plans) and d) Public officials that will receive capacity building for adjusting regulations and norms. For more information, please see Annex 23.

³ Indirect beneficiaries of the project will be those that do not receive targeted support from the GCF-funded intervention but are likely to receive a measurable adaptation benefit from the GCF-funded intervention but will benefit from water generated by the improvement of ecosystems that contribute to water regulation in the basin in the main cities of Cusco, Arequipa and Cañete. For more information, please see Annex 23.

⁴ Number based on the latest census of 2017 with a total national population of 29,381,884.

⁵ Refer to the Policy of Co-financing of the GCF.

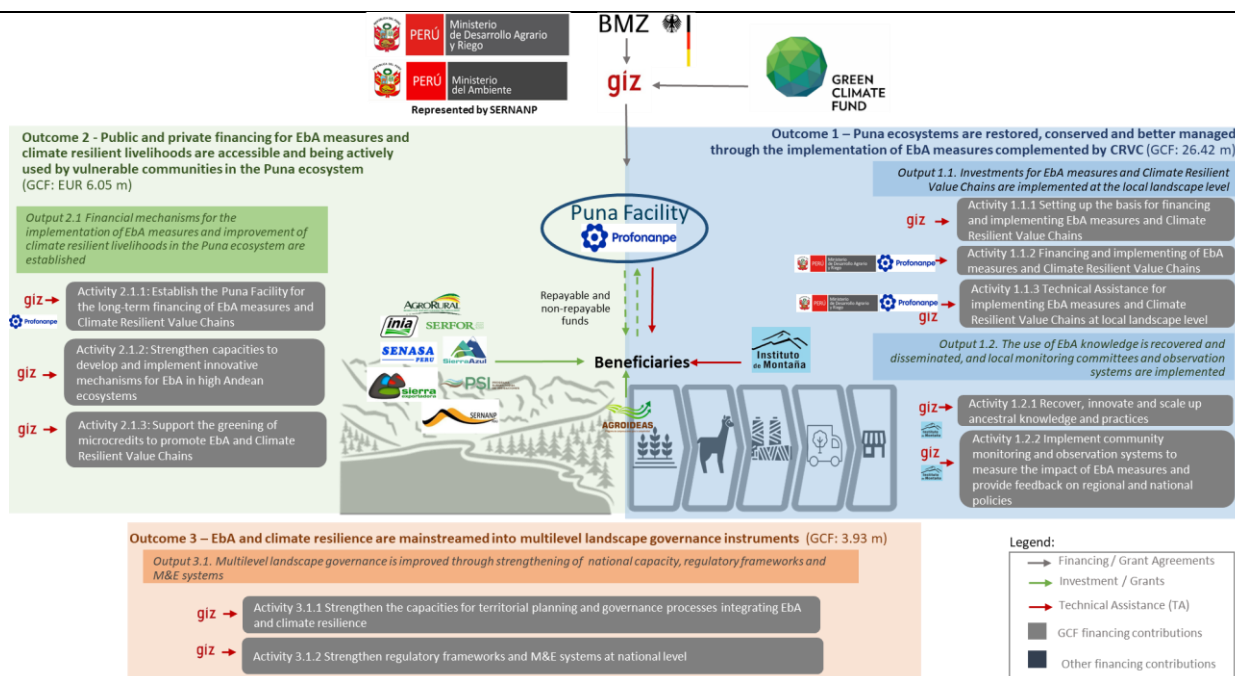
⁶ Four regions are considered in the SHAP: Puno, Cusco, Arequipa and Apurimac.

⁷ Argumedo, A; Stenner, T; Swiderska, K; (2021). Quechua values and worldviews in the chalakuy maize park, Peru, IIED, London.

⁸ Ponce, Carmen (2020). Intra-seasonal climate variability and crop diversification strategies in the Peruvian Andes. A word of caution on the sustainability of adaptation to climate change.

- knowledge. Especially young people are abandoning ancient practices, resulting in insufficient means and capacities to develop alternatives or adopt climate-resilient livelihoods. The Peruvian Ministry of Agricultural Development and Irrigation (MIDAGRI) has set in place a series of programs to support these vulnerable populations, but access is low, and the available budgets are insufficient to address identified needs.
3. **Regarding the noted vulnerabilities, the proposed project aims to enhance ownership and resources of Andean Indigenous Peoples and Local Communities to build their own resilience to climate change by defining, implementing and monitoring their Ecosystem based Adaptation (EbA) priorities to improve the Puna ecosystems management and ensure the continuous provision of their services such as water provision that will be captured during raining season for their livelihoods during dry season.** Therefore, the project participatory approach consists in providing technical assistance and setting up of a financing mechanism for the implementation of Ecosystem based Adaptation (EbA) measures and promoting Climate Resilient Value Chains (CRVC) led by IPLCs at local scale via the alignment and leveraging of public and private financing. Funding by the GCF will complement, facilitate and increase the access to MIDAGRI programs through technical assistance and set the financing structure to sustain investments in the long term via payment for ecosystem services, private sector contributions, and microfinancing. At the local level, experiences with EbA implementation and the facilitation of community participatory processes from the Nor Yauyos Cochabamba landscape reserve will be scaled up to the SHAP where communities will receive direct financial and technical support to design, implement and monitor EbA measures to enhance Climate Resilient Value Chains (CRVC) and thus promote a change in behaviour and boost the local economy for the continued protection of the Puna ecosystems. In doing so, at the national level the project will support MIDAGRI in strengthening and mainstreaming the EbA approach with a gender perspective into all its programs (Agorural, Sierra Azul, INIA, Agroideas and SERFOR), and improve the coordination with other local stakeholders on the landscape. Support will also be given to SERNANP in line with its mandate for actions within Natural Protected Areas (NPAs). To assess progress on agricultural and water sectoral adaptation measures local monitoring systems will be set up and feed into existing information management systems and promote coordination between sectorial M&E systems. The project will support MIDAGRI to enhance their internal monitoring systems in order to allow easier and streamlined reporting on their NDCs into the System for Monitoring Adaptation and Mitigation Measures (SIMOM) managed by Ministry of Environment (MINAM). This will be done in coordination with technical advice from the General Directorate of Climate Change and Desertification (DGCCD) of MINAM and in coordination with the General Directorate of Agriculture Environmental Affairs (DGAAA), focal point within MIDAGRI for its NDCs. The project also recognises the inclusion of different gender perspectives in its activities as a crucial element to understand inequalities between women and men in the SHAP for managing and deciding over agriculture, water and ecosystems (e.g., low participation in decision-making processes. Women have limited access to finance as well as trainings and capacity building opportunities. Men and women have unequal use/control of natural resources. The project proposes actions to address the barriers that have been hindering women's participation in building adaptation to climate change, as well as actions that will contribute to the strengthening of women roles, allowing them to actively participate in the implementation of the project, and thereby promoting a wider appropriation of the EbA measures and CRVC. Following the approach the project design has been structured based on three components (1) Resilient ecosystems and communities, (2) Public and private investments for scaling up EbA measures aligned and leveraged and (3) Multilevel territorial governance for the incorporation of EbA measures which are aligned with three outcomes, as depicted in the following **Error! Reference source not found..**

Figure 1. Overview of project outcomes, outputs and activities



4. By implementing investments on the ground (through **component 1**), the aim is (i) to maintain or improve the provision of puna ecosystem services for climate resilience of Andean Indigenous Peoples and Local Communities (IPLCs) in the SHAP and (ii) to strengthening sustainable and productive value chains that are dependent on and impacting on those ecosystems. This component will provide financial and technical support, through a call for proposal mechanism, for EbA and Climate Resilient Value Chains implementation on the ground. To ensure IPLCs investment ownership and enhance replication potential, the component will have a special focus on the promotion of traditional knowledge and practices that play an important role in Ecosystem based Adaptation. Since EbA uses the best available science on climate change and ecosystems, alongside local and traditional knowledge, a list of EbA measures relevant in the target area were pre-identified and validated with IPLCs during stakeholder consultations phase. EbA approach draws on community-based adaptation (CBA) and builds on traditional knowledge systems, therefore IPLCs will decide which combination of EbA measures and Climate Resilient Value Chains will be planned, implemented, and monitored. All this will be supported by peer-to-peer learning among communities and facilitated by the EEs; and, in some cases by local partners (Civil Society Organizations (CSOs)). IPLCs in the SHAP will thus recognize and acknowledge the benefits provided by their traditional knowledge, practices and other EbA-related measures, namely improved resilience in ecosystems, maintained carbon stocks and sustainable livelihoods. A series of technological packages, trainings, information materials and communities' exchanges to nurture dialogue will be implemented to co-produce knowledge and foster local innovation that will result in investment on the ground. Therefore component 1 will enhance access to information obtained at local level, which is expected to trigger a change in behaviour towards conservation of ecosystems and their services and the potential of developing climate resilient agribusinesses. Activities in this component will address the barrier of limited economic opportunities by supporting the development of climate resilient value chains and enhanced entrepreneurship in the SHAP region.
5. Through **component 2** the project will establish a Facility implemented by Profonanpe to delivery financing through a call for proposal for IPLCs led Local Initiatives oriented to increase climate resilience. This component will also align and leverage public and private financing for EbA measures and Climate Resilient Value Chains at different and coordinated levels. At macro level it will focus on the alignment, leveraging, mobilization and upscaling of private and public funds to finance EbA activities on the ground (component 1). At the meso level, it will strengthen the capacities of local institutions and organizations to access financial resources and technical assistance. At the micro level, it will work on enhancing access to alternative financial services for small farmers, such as microfinancing, and access to public programs for improved entrepreneurship (i. e. Agroideas and Sierra y Selva Exportadora from MIDAGRI) and climate resilient investments (i. e. Sierra Azul and Agrorural from MIDAGRI). As a result, the focus of this project is on mobilizing finance at different scales and with different schemes, with impact beyond the specific landscape. The key feature of this project is its leverage potential, to facilitate the mobilization of MIDAGRI, MERESE, private and financial institutions resources.

6. Furthermore, through the last **component 3** the project will promote integrated gender responsive⁹ landscape planning, governance platforms and policy improvement and coordination, fostering dialogue and improving coordination among stakeholders that intervene in the landscape (local, regional and national governments, rural communities, producer organizations, watershed committees, and MIDAGRI extension services, among others). The most adequate processes through effective participatory approaches or platforms for knowledge exchange, dialogue, coordination and consensus-building will be fostered according to local needs. The paradigm shift towards climate resilience and sustainability will be driven by a transformation in the way MIDAGRI interacts with rural communities, provides context-appropriate extension services and coordinates among its programs, regional offices and other sectors to implement consistent and effective policies in the territory.
7. Overall, the project will promote the sustainable management of 23,914 ha of Puna ecosystems, benefiting directly 60,715 people and indirectly another 2,011,856 people. The mitigation co-benefit is estimated at 407,657 t CO₂eq. The promoted EbA measures (i.e., qochas, peatlands and grasslands restoration and conservation) will not recompensate the water from rainfall or glacial melting in the same magnitude but they will capture water during the rainy season or during glacier melting and store it to recharge aquifers, expanding water availability during the dry season for agriculture, livestock and drinking water.
8. The Accredited Entity of this project is the German Gesellschaft für Internationale Zusammenarbeit (GIZ), with MIDAGRI as the leading political partner. Profonampe, the NGO Instituto de Montaña and GIZ Peru will act as Executing Entities receiving GCF funds. While MIDAGRI and the Peruvian National Service of Natural Protected Areas by the State (SERNANP) will act as Executing Entities that will contribute with financing to achieve project's objective. Furthermore, Indigenous Peoples and Local Communities (IPLCs) with their different forms of organisation (i.e., communities, producers' associations and cooperatives in the process of formalization) will be supported by local Civil Society Organizations (CSOs) for the planning, presenting and implementing of climate resilient Local initiatives financed through the Puna Facility.

⁹ Please note that the GIZ working definition for gender responsive follows the OECD Gender Equality Continuum (2022): A gender-responsive approach not only considers gender differences, inequalities, ratios and aspects, e.g., by targeting and including a certain number of women in project activities but goes a step further in responding to gender specific needs and/or requirements, or by collecting gender-disaggregated data for monitoring purposes. Gender responsiveness aims to ensure that all people can engage in and/or benefit from certain measures regardless of their gender (e.g., trainings or strategies are developed that consider gendered needs, preferences, and constraints).

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

B.1.1 Socio-economic Context:

9. **Peru** is considered an upper-middle-income economy, but inequality is high, and poverty is concentrated among rural, indigenous populations, whose food security depends upon climate conditions. More than 80 percent of farmers practice subsistence, rainfed agriculture, and changes in precipitation are increasing competition over water resources for consumption, agriculture, and industry. Glaciers, a main water source, are melting at accelerating rates due to increasing temperatures. Peru is susceptible to natural disasters including floods, droughts and landslides, whose frequency, severity, and impacts are compounded by the El Niño Southern Oscillation and will be amplified by increased climate change and variability. Combined with ongoing problems such as agricultural expansion, deforestation, illegal mining and air and water pollution, these climate risks threaten advancements in Peru's development¹⁰.
10. **Peru's population** reached more than 33 million people in 2021, of which 50.5% are women. The country's annual population growth decreased to 1.2%, in comparison to 1.9% in 2019 (World Bank, 2023). Most of the population, 78.5%, lives in urban centers, while 21.5% live in rural areas (FAO, 2023). Peru is divided in three main regions Coast with 58% of the Peruvian population, the Andes with 28% of the population, and the Amazon with 14% of the population¹¹. Peru is a multiethnic country formed by the amalgamation of different cultures and ethnicities. [Mestizos](#) compose about 60% of the total population¹². The Indigenous Peoples in Peru represent about 25.7%. Of those, 95.8% are Andean and 3.3% from the Amazon. The two major [indigenous](#) or ethnic groups are the [Quechuas](#), followed by the [Aimaras](#). A large proportion of the indigenous population who live in the Andean highlands still speak [Quechua](#). The Afro-Peruvian population (around 3.5%) is concentrated mostly in coastal cities south of Lima. The project will be focused on the Andes region where most of the population belongs to the Quechua Indigenous Peoples Group.
11. **The agricultural sector** employs 29.9% of the total population in Peru (CEPAL, 2023), and 30% of the national territory is dedicated to activities within the sector. In the rural areas, the number reaches, on average, 78% of the population (FAO, 2023). According to the National Agricultural Census (2012) 50.4% of agricultural producers are women. Besides the economic benefits, the agricultural sector also contributes to the country's food security, as 70% of the food consumed in the country is produced internally (Fernández, 2021). However, the government expenditure and investment in the sector is still very low, mainly due to a lack of interest from policy makers (World Bank, 2017). In addition, only 12.2% of the total number of producers have higher education (INEI, 2012).
12. The target region of the project interventions is in the Southern High Andes of Peru (SHAP): Puno, Cusco, Arequipa, Apurímac, Lima. The SHAP is part of the 7,000 km long tropical Andes Mountain range, which is the top biodiversity hotspot for conservation priorities worldwide (Myers et al., 2000). This area is located above 3,500 m.a.s.l and the ecosystems consist primarily of grasslands (77%), wetlands (locally known as bofedales¹³ in Spanish), and some relicts of high Andean forests also persist¹⁴. Grasslands hold significant importance as they play a vital role in safeguarding the soil from erosion and promoting effective water infiltration, thereby regulation the flow of runoff water. Wetlands and grasslands capture rainwater and retain it for some period. Local communities rely on Puna ecosystems to feed and provide water to their domestic herds, their crops and finally, for domestic use.
13. The **main economic activities in the SHAP** are mixed husbandry and agriculture for local subsistence on both communal and family land. Major livestock species above the 3,500 m are native South American camelid species (alpaca, lama and vicuña), typically reared in mixed systems with sheep, while the vicuña is a wild species.¹⁵ In the transition zone between 2,800 and 3,500 m, cattle is also present in these mixed herds. See section 3.7 in the Feasibility Study for more details.
14. **Agriculture in the highlands of Peru** is primarily characterized by small-scale farming systems focused on subsistence production, and low-complexity value chains tied to local markets. These farming systems combine the cultivation of staple crops such as potatoes, wheat, and quinoa with livestock rearing (particularly camelids, such as alpacas and *vicuña*, sheep, and cattle). Puna ecosystems contribute with a substantial amount of meat, fiber, and dairy products to local, regional, and international markets. Peru is the main producer and exporter of Andean grains, especially quinoa. Peru is also the largest producer of native potato in Latin America and exported

¹⁰ USAID (2017). Fact Sheet Climate Change Risk Profile: Peru

¹¹ INEI 2017

¹² "Perú: Perfil Sociodemográfico" (PDF). Instituto Nacional de Estadística e Informática.

¹³ In Peru, the term "bofedales" is used to describe areas of wetland vegetation that may have underlying peat layers. These areas are a key resource for traditional land management at high altitudes. Because they retain water in the upper basins of the cordillera, they are important sources of water and forage for domesticated livestock as well as biodiversity hotspots. Maldonado (2014). An introduction to the bofedales of the Peruvian High Andes.

¹⁴ WWF. General description of Central Andean Puna. <https://www.worldwildlife.org/ecoregions/nt1002>

¹⁵ In Peru, the management and conservation of vicuña are under the responsibility of the National Forestry and Wildlife Service (SERFOR), under MIDAGRI; if the species is found within a protected natural area, conservation and sustainable use is the responsibility of the National Service of Natural Areas Protected by the State (SERNANP), an entity attached to the Ministry of the Environment (MINAM).

- 6,530 tonnes in 2022 ([Agencia Agraria de Noticias, 2023](#)). Peru ranks 14th in the world in potato production and the potato is the second most important product (after rice) in Peruvian agriculture, with more than 3,000 potato varieties.
15. From the total agricultural units in Peru, 82 percent correspond to farms with equal or less than five (5) hectares, 64 percent are characterized as subsistence farming and 57.5 percent are located in the highlands, especially in the departments of Cajamarca, Puno and Cusco ([INEI, 2012](#)). Thus, the majority of farms in the SHAP are less than five hectares, and farmers often have dispersed plots located in various micro-environments. These micro-environments differ in terms of altitude, soil quality, water availability, and climate.
 16. **Food production** in the highlands is often inadequate to meet the consumption needs of households, leading many farmers to seek additional income through off-farm activities. Smallholder farmers in the highlands region make use of the diverse ecological niches available to them. In high-altitude areas, livestock is grazed, and specialized traditional tubers like native potatoes, *mashua*, *olluco*, *oca*, and *maca* are grown using traditional agroecological practices, mainly for subsistence and barter. At intermediate altitudes, farmers cultivate grains such as wheat, barley, rye, and maize, along with pulses, fruits, berries, and vegetables. In the inter-Andean valleys, various types of fruits like avocado, orange, lemon, and banana are grown.
 17. **Land tenure** in the High Andes generally appears to be dominated by private regime, in average 48% of the agricultural units in the prioritized districts of the Resilient Puna project (varying from 24% to 86% depending on the department), followed by communal land regime, for 44% of agricultural units in average (variation from 1% to 70 % depending on the department). Tenant and Possessors regimes are less common in the target area of the project (INEI 2012). Traditional herder communities living in the High Andes tend to combine private and collective regimes as an alternative form for strengthening land tenure, treating grasslands as common resources which are accessed, used and controlled collectively, usually under open access or communal land tenure regimes.¹⁶ Regarding land use in the project area is 77% is for natural pastures and 8% for agricultural use. Studies¹⁷ show that women have limited access to land, which is conditioned upon their membership in a community and their relationship within a family, including marital status and whether they have children.
 18. **Population in SHAP is mainly Andean Indigenous Peoples and Local Communities (IPLCs).** In the target area of the Resilient Puna project only one Indigenous Peoples Group exist: the Quechuas. According to the last National Census (2017) almost 69% of the population within the project target area have Quechua as their mother tongue while 31% of the population have Spanish¹⁸. Spanish native speakers in this area are mainly rural farmers living also in communities The Quechua people are the long-standing largest Indigenous Group in Peru and in the Andean territory, whose mother tongue is Quechua. A large part of this Indigenous Group is organised in peasant or local communities. Nonetheless, they have other forms of organisation and live in other places such as villages, hamlets, among others. In many areas of the country, including in the Andean region, Quechua communities nowadays tend to also form associations, cooperatives, community enterprises and organisations by the need to manage local development, administer common resources or provide services to each other¹⁹. Therefore, the project will work with the various forms of organizations that IPLCs have established in the target area. In accordance with the obligations established in Convention 169 of the International Labour Organisation (ILO) and the Peruvian Law, peasant or Andean communities peoples can also be identified as Indigenous or Original Peoples, according to the criteria: a) Direct descent from the original populations of the national territory, b) Lifestyles and spiritual and historical links with the territory they traditionally use or occupy, c) Their own social institutions and customs, d) Cultural patterns and way of life different from those of other sectors of the national population. The names used to designate Indigenous or Original Peoples do not alter their nature or their collective rights.
 19. **Andean IPLCs in their various forms of organization continue using practices that they learned from their ancestors.** For instance, they still use ancient techniques to storage rainwater into micro water reservoirs named *gochas* in Quechua, to infiltrate water and recharge aquifers or build irrigation system of canals to divert water from rivers descending from highland areas to irrigate foothill slopes. This ancestral knowledge is accumulated, and transgenerational knowledge related to the natural resources found on the lands and territories they occupy and is central to climate change adaptation. During their agricultural and livestock activities, they also combine collective celebrations and perform a series of rites of payment to the *pachamama* (mother earth) and to the guardian hills of the livestock. They perform animal fertility rites involving singing, dancing and feasting. Also, they have a series of festivals linked to productive activities, such as sowing and harvesting, but above all the cleaning of irrigation ditches²⁰. Native fibers (from alpaca, llama, and vicuña wool) and ancestral weaving traditions continue to occupy

¹⁶ Damonte, G., M. Glave, S. Rodríguez and A. Ramos. 2016. 'The evolution of collective land tenure regimes in pastoralist societies: lessons from Andean countries. IDS Working Paper No. 480. Brighton: Institute of Development Studies.

¹⁷ 2016-Best-Practices-Case-Peru.pdf (landesa.org)

¹⁸ Please note that in 2012, 83% were Quechua speakers and 17% Spanish speaker within the project area of the project (CENAGRO, 2012)

¹⁹ In Peru mestizo is defined as the biological, as well as cultural, mixture between white Europeans and Indigenous peoples.

²⁰ Quechuas | BDPI (cultura.gob.pe)

a central space not only in the Andean culture but also in the social and economic aspects since many communities depend on camelids wool for their livelihoods. In the pre-colonial times Andean worldview was represented by the *Ayllu* concept which is understood as the form of social organization in Andean communities where individuals share same reciprocity norms and rituals with respect to humans, animals, rocks, spirits, mountains, lakes, rivers, pastures, crops. The main objective of *ayllu* was to give guide in the positive relationship between communities and their social and natural environment. Forms of organisation based on the *Ayllu* concept can still be found in some communities of Cusco, Apurimac and Puno. These cultural and spiritual values of Andean people are still very strong in some parts of the Andes and will play an important role in EbA by promoting conservation and equity values, and social cohesion at landscape level²¹. IPCLs play vital roles in the management of their ecosystems (FAO, 2016), although in most cases, this knowledge is being lost due to the low valuation of native cultures, cultural erosion, high rates of rural-urban migration, among others. Additionally, gender traditional norms in the Andes often prevent women from achieving the same level of visibility as men, or benefit from the same opportunities and recognition, despite their deep knowledge of agriculture, water and ecosystems.

B.1.2 Climate Change context (historic climate trends and future scenarios)

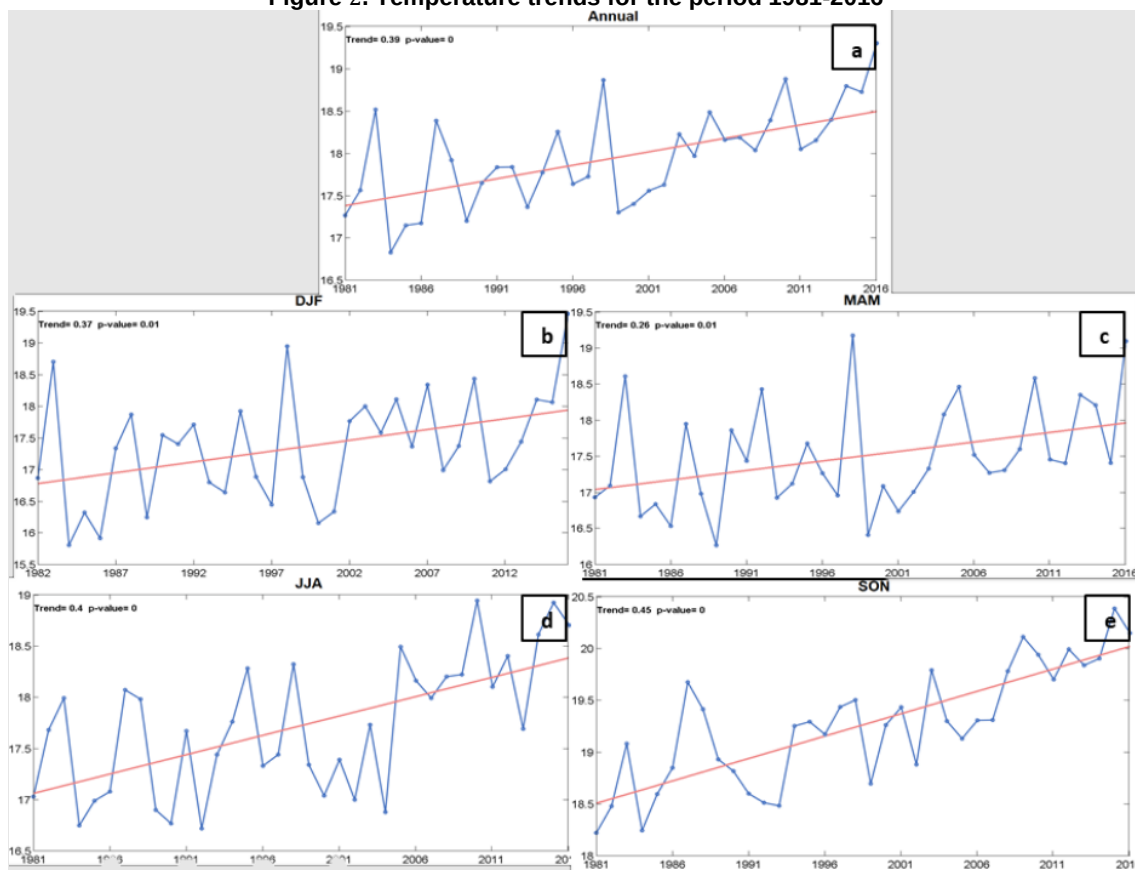
20. Climate change and unsustainable management of productive practices are degrading the Puna ecosystems, (peatlands, wetlands, and grasslands) and the services they provide, including the provision and regulation of water, provision of fodder, food, and fiber, nutrient and carbon regulation. Fewer rains and longer drought periods have contributed to glacier melting, threatening the livelihoods of approx. 4.5 million people in SHAP and the water security of millions more downstream. Increased temperatures have already decreased 51 percent of Peru's glaciers. In addition, huge stocks of carbon stored in the Puna ecosystems could be released into the atmosphere as they are degrading increasingly.
21. The communities in the SHAP are characterized by their vulnerabilities such as low levels of development, dependence on subsistence agriculture and husbandry practices, limited economic opportunities, and overall high vulnerability to climate change. They are losing their traditional knowledge and are abandoning ancient practices, resulting in insufficient means and capacities to develop alternatives or adopt climate-resilient livelihoods. MIDAGRI has set in place a series of programs to support these vulnerable populations, but access is low, and the available budgets are insufficient to address identified needs.
22. The goal of the project is to align public and private investments and increase access to state programs improving climate resilience of high Andean ecosystems and communities livelihoods to shift a state of ecosystem conservation, high productivity, income diversity and climate resilience because they will reduce the degree of losses from climate impacts, improve their relationship with ecosystems, increase their participation in the value chain and product transformation, improve their associative processes, market access and monitor progress on decreasing their vulnerability.

Historic climate trends

23. Temperatures during the analysed period (1981-2016) as shown in Figure 2 overall show an increase. Maximum temperatures show the greatest increases during the spring, while the smallest increases show in the autumn season. Increases range from 0.3°C to 0.44°C per decade and the districts with the greatest trends are approximately at 4,600m above sea level. Precipitation historical trends show high seasonal and interannual variability. Trends are therefore of low magnitude and do not usually show statistical significance. Overall, the intervention shows an increase in precipitation at the annual level as well as the summer and autumn seasons. In spring, trends in precipitation have been negative in recent decades. Since the end of the 1970s, Andean tropical glaciers have been shrinking at an increasing rate. The main factor that explains this retreat is the warming of the atmosphere, given that precipitation does not show significant changes. Recent analysis of glacier melting in the last decades in the SHAP show considerable glacier losses (Seehaus et al., 2019). During the period between 2000-2013, the retreat rate in the southern SE subregion was -1.3 percent by year, and in the SW subregion, it was -2.3 percent by year, approximately.

²¹ Swiderska, K, King-Okumu, C and Monirul Islam, M (2018) Ecosystem-based adaptation: a handbook for EbA in mountain, dryland and coastal ecosystems. IIED, London.

Figure 2. Temperature trends for the period 1981-2016

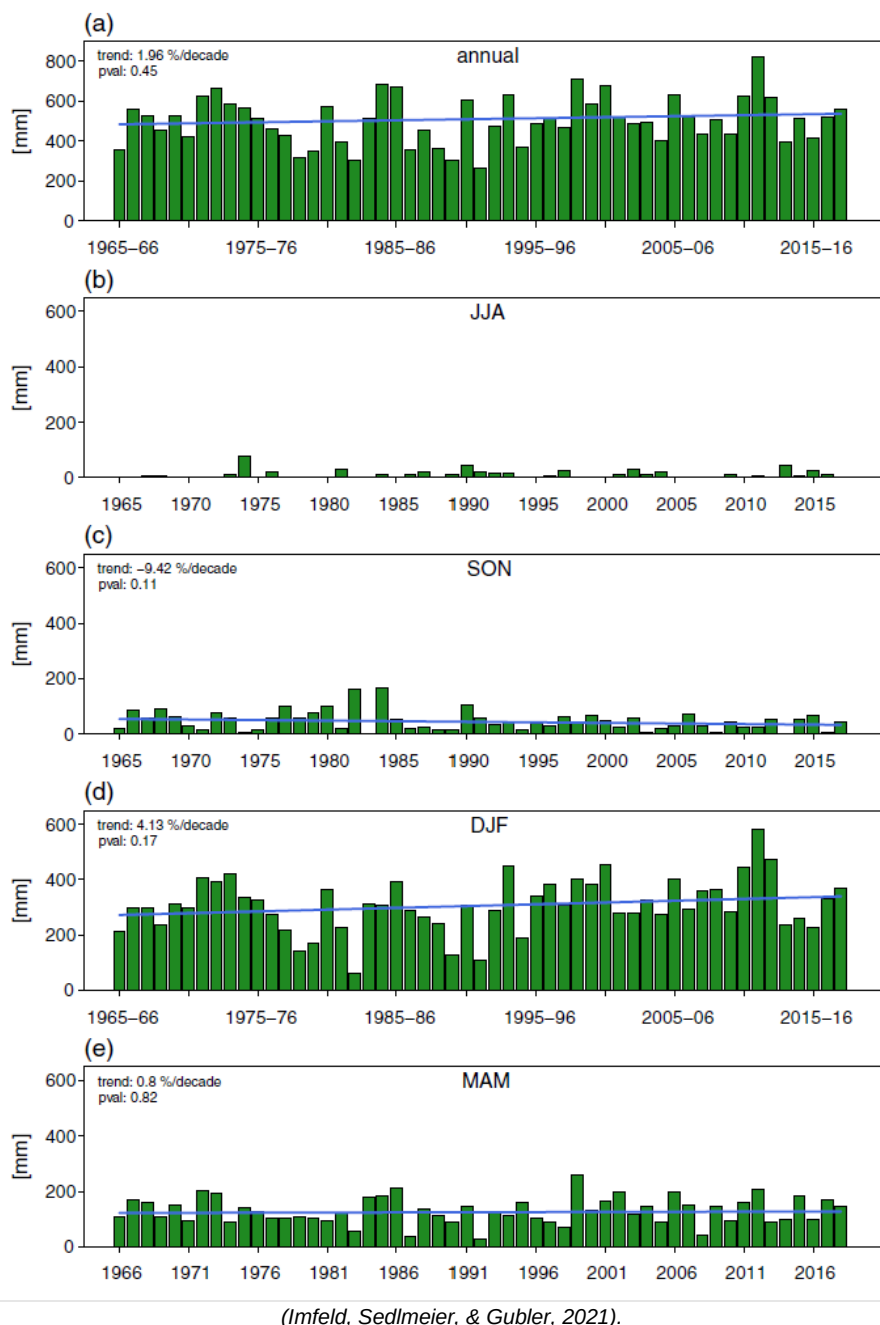


24. Regarding extreme weather events, in accordance with total precipitation trends, the number of events with more than 10 mm rainfall show very few significant trends over the regions with very small intensity. SPI3 trends over the last decades lean towards more droughts in the SHAP region (a negative SPI indicate more severe droughts). Significant trends are visible in the Altiplano, SW and SE region. In recent decades, the Cusco, Puno and Arequipa regions (SHAP), were the most affected by major meteorological droughts, especially in 1983, 1987, 1990, 1992, 2005 and 2016. See Chapters 3.3 and 3.4 in Annex 2 (Feasibility Study) for more details on these observed climate trends.

Precipitation historical trends

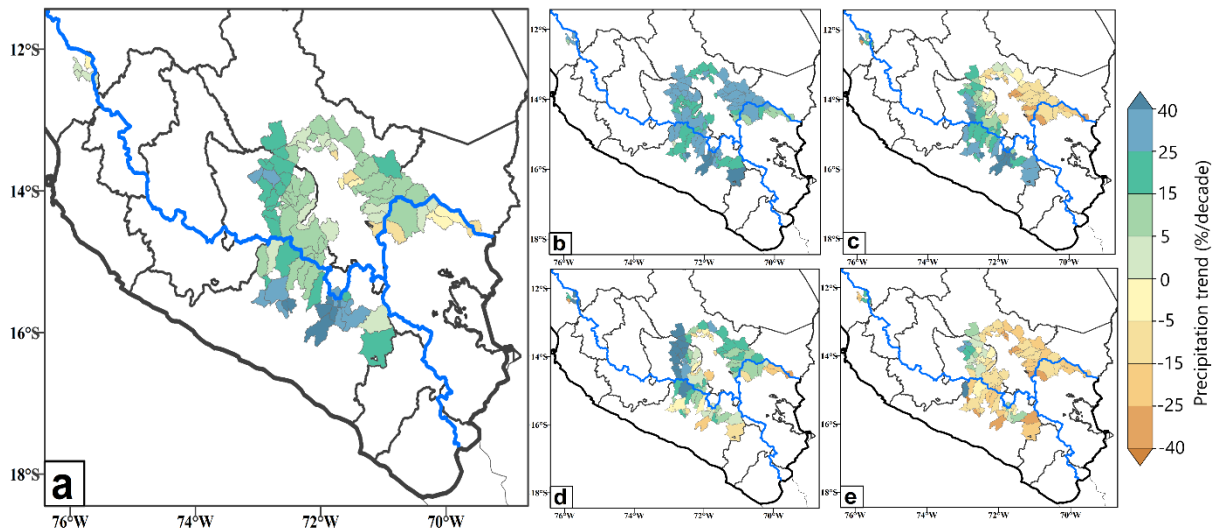
25. As shown in Figure 3 show high seasonal and interannual variability. Trends are therefore of low magnitude and do not usually show statistical significance (a). Overall, the intervention shows an increase in precipitation at the annual level as well as the summer and autumn seasons. In spring, trends in precipitation have been negative in recent decades. For winter, no precipitation trends are shown.

Figure 3. Precipitation trends for the period 1965-2016



26. As shown in Figure 4 annual trends (a) in precipitation range from -14.1% to 55.4% per decade. Annual trends are positive throughout SHAP, except in a few districts located in SE and A, which show slightly negative trends. In summer (b), trends show a clear spatially distributed positive trend ranging from 3.7 to 50.9% per decade. In autumn (c), trends range from -28.6 to 65.4% per decade, with eastern SE and A showing negative trends. In contrast, west of SE and SW return spatially positive trends. In the winter (d) months, trends are not significant, but discrepancies in trends again between east of SE and A (decrease) and west of SE and SW (increase), are noticeable. In spring (e), there is a clear trend of decreasing rainfall, ranging from -45.7 to 45.7% per decade. All districts show a decrease, except for a few located northwest of the SE. These decreases in spring months are related to the tendency to delay the onset of rainfall that typically begins in this season. This is in line with other studies in the Peruvian Andes that have noticed similar patterns of rainfall seasonality (Giráldez, Silva, Zubieta, & Sulca, 2020).

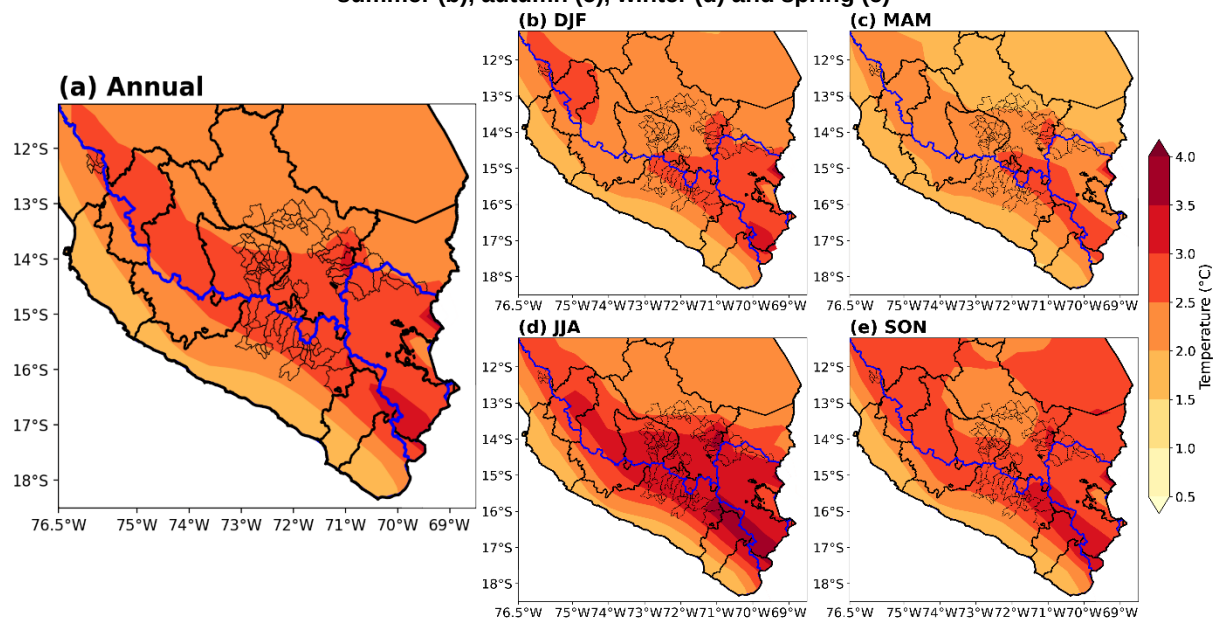
Figure 4. Spatial distribution of precipitation trends (%) from 1981 to 2016 in the SHAP chosen districts



Climate change future scenarios

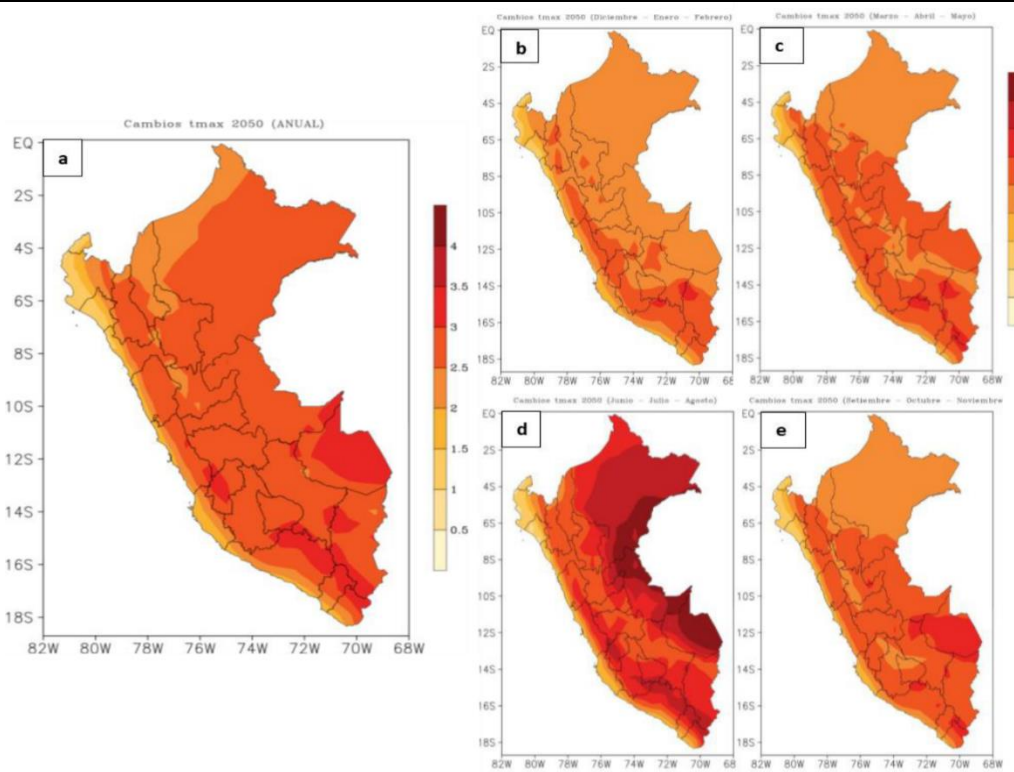
27. Projections for temperature and precipitation in the SHAP region as shown in Figure 5 (subregions are shown divided by blue lines), shows that the changes in temperature are always positive, in any season or scenario. According to the RCP4.5 scenario, by 2050, changes in maximum temperature range between 2.1 and 3.8°C, with stronger signals in winter (d). Minimum temperature shows changes that go from 1.3 to 4°C. Spatially, these changes are bigger in one specific zone placed over high altitudes between the SE and the Altiplano region and the border of SW and SE in autumn.

Figure 5. Projected changes in maximum temperature towards 2050 under RCP 4.5 scenario for the annual mean (a), summer (b), autumn (c), winter (d) and spring (e)



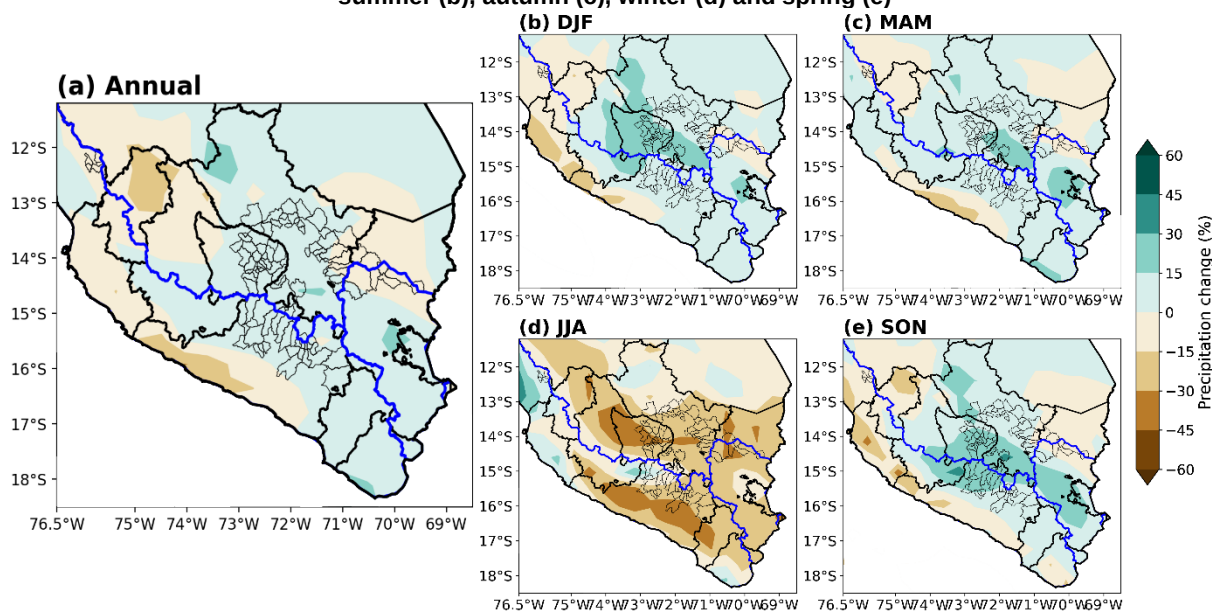
28. Under the RCP8.5 scenario as shown in Figure 6, by 2050, changes in maximum temperature range from 2.5 to 4°C, with stronger signals in winter. Minimum temperature shows changes between from 2 to 4°C. Similar to RCP4.5, areas that show higher increases in all seasons are the border of SE and A and the border of SW and SE.

Figure 6. Projected changes in maximum temperature (°C) towards 2050 under RCP 8.5 scenario for the annual mean (a), summer (b), autumn (c), winter (d) and spring (e)



29. For precipitation, under the RCP4.5 as shown in Figure 7 (subregions are also shown divided by blue lines), annual changes range from -5 to +20 percent by 2050. In winter (d), changes are negative in most parts of the SHAP. On the other hand, the largest positive changes occur in summer, especially over the north of the SE region. In general, areas that show higher increases in all seasons are located at the west of SE. On the contrary, areas with decreases or smaller increases are located in the A subregion.

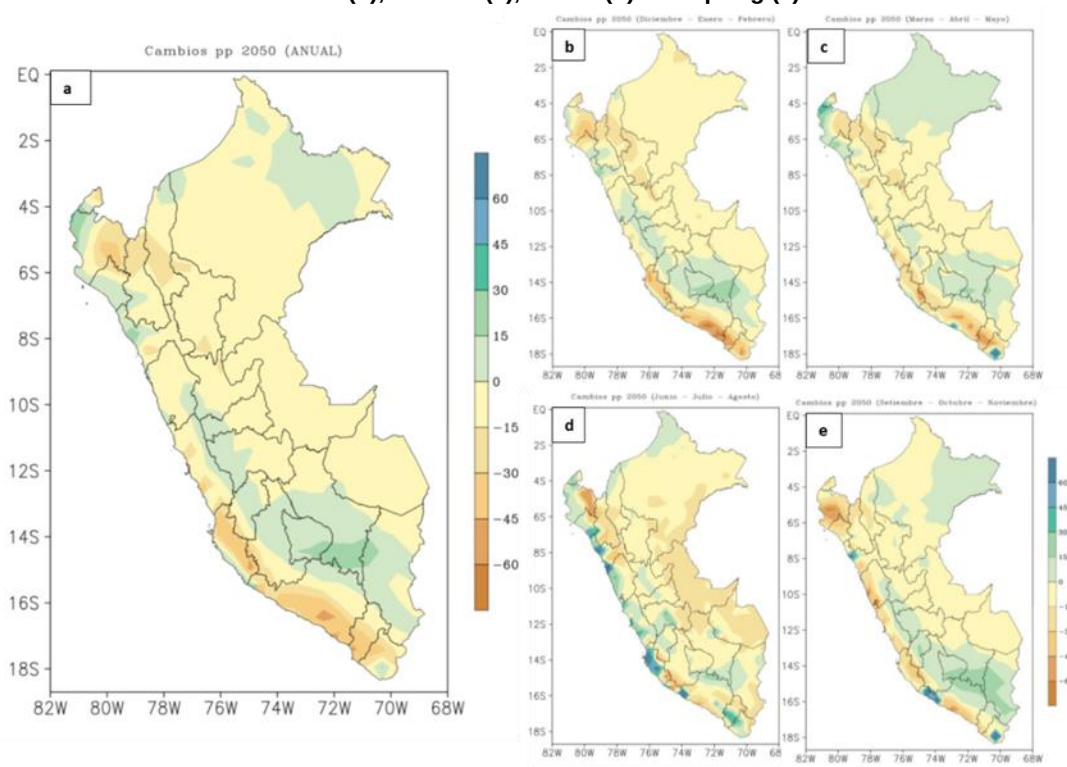
Figure 7. Projected changes in precipitation towards 2050 under the RCP4.5 emission scenario for the total annual (a), summer (b), autumn (c), winter (d) and spring (e)



30. Under the RCP8.5 emission scenario as shown in Figure 8, annual changes range from -30 to +30 percent by 2030. The biggest negative changes happen in winter (d). However, spring is the season when the largest positive changes in precipitation are expected. The SE and the Altiplano regions show mixed positive changes, whereas the SW region shows consistent negative changes. Regarding extreme climate events like droughts Under RCP

8.5, there is an increase of droughts in the Altiplano region by 2050, and wetter climates at lower altitudes. The total number of drought months by some estimates is projected to almost double by 2100 under the RCP 8.5 scenario (Potter, et al., 2022). This will have impacts on the water balance, and on agricultural activity. See Chapter 3.5 in Annex 2a (Feasibility Study) for more details on these projected climate changes.

Figure 8. Projected changes in precipitation (%) towards 2050 under the RCP8.5 scenario for the annual mean (a), summer (b), autumn (c), winter (d) and spring (e)



B.1.3 Climate risks and vulnerability of the SHAP ecosystems

31. **Glacier retreat.** Increased temperatures lead to the shrinking of glaciers. Peru is home to 71 % of the world's tropical glaciers, however, due to higher temperatures and the increased absorption of solar radiation, the country has already lost 51 % of its glaciers in the last 70 years.²² Accelerated glacier melting is possibly the strongest evidence of climate change impacts in the Andes. Water resources in the Puna ecosystems, are greatly dependent on Andean glaciers. According to the National Research Institute on Glaciers and Mountain Ecosystems (INAIGEM), Peru glaciers have been reduced by 43 percent over the last 40 years, with at least seven watersheds already crossing peak flow. Once the glaciers feeding these watersheds disappear, dry season average discharge may decrease by up to 30 percent, causing water stress in these areas (Baraer et al., 2012). In the tropical Andes, glaciers have represented a crucial water source for ancient and modern livelihoods and are of particular cultural and spiritual values for local communities²³. In recent years, those areas closer to melted glaciers has impacted local livelihoods due to the scarcity of water resources. This is also threatening sacred festivals and reshaping centuries-old indigenous rituals such as the Sanctuary of the Lord of Qoyllur Rit'i which each year about 100,000 individuals make a pilgrimage to the sacred Colque Punku glacier²⁴.
32. **Frost days.** Associated with the general warming in the region, frost days trends return a general decrease over the last three decades, especially at higher altitude in A and SE (Figure 31: Trends for several extreme weather indices for the period 1981-2020 a) Frost days (FD), b) extreme rainfall (R10) and c) summer drought SPI3 Trends in frost days mostly correspond to trends of minimum temperature with low magnitudes and few significant signals. Since frost days are strongly related to incoming longwave radiation, the occurrence of frost days follows the cycle of minimum temperature cycle (García, Raes, Jacobsen, & Michel, 2007). Regionally, differences in the occurrence of frost days are observed. For instance, the percentage of frost days is considerably reduced close to the border

²² ANA (2020). Retroceso glaciar en el Perú 1948-2019. Impactos en el recurso hídrico.

<http://repositorio.ana.gob.pe/bitstream/handle/20.500.12543/4551/ANA0003031.pdf?sequence=1&isAllowed=y>

²³ Drenkhan, Fabian; Carey, Mark; Huggel, Christian; Seidel, Jochen; Oré, María Teresa (2015). The changing water cycle: climatic and socioeconomic drivers of water-related changes in the Andes of Peru. Wiley Interdisciplinary Reviews: Water, 2 (6):715-733

²⁴ <https://www.nationalgeographic.com/culture/article/andean-glaciers-melting-reshaping-centuries-old-indigenous-rituals>

- of Lake Titicaca compared to the areas more distant to the lake. This is an indication of the warming effect of the lake on minimum temperature.
33. **Heavy rainfall events.** In accordance with total precipitation trends, the number of events with more than 10 mm rainfall show very few significant trends over the regions with very small intensity. In a recent study, authors were not able to attribute trends in R10 to trends in precipitation in the region (Imfeld, Sedlmeier, & Gubler, 2021). Variability in decadal trends explain well the variability of precipitation in the region and the challenges in detecting significant trends (Imfeld, Sedlmeier, & Gubler, 2021).
 34. **Drought risk.** SPI3 trends over the last decades lean towards more droughts in the SHAP region (a negative SPI indicate more severe droughts). Significant trends are visible in the A, SW and SE region. In recent decades, the Cusco, Puno and Arequipa regions (SHAP), were the most affected by major meteorological droughts, especially in 1983, 1987, 1990, 1992, 2005 and 2016. The El Niño phenomenon has long-distance impacts on the behaviour of the climate of the Altiplano and its surroundings, causing a decrease in rainfall during El Niño episodes. ENSO only partially explains a proportion of this decrease in the Andes, mainly in the central and southern western highlands (Lavado W. E., 2014). Therefore, it is important to consider that recent studies of projections of ENSO events indicate that, with a warmer climate comes an increase in ENSO events, and therefore an increased risk of droughts (Cai, Ng, Wang, Santoso, & Yang, 2022).
 35. Moreover, in a 2015 SENAMHI study, it was found that the greatest numbers of dry events between 1970 and 2014 are located in the SHAP region. In the Altiplano region (A subregion), these events have a shorter length. The SW subregion, roughly corresponding to Arequipa's high areas, experiences drought with the maximum intensity. Between Cusco and Apurímac (SE subregion), dry events show greater intensity and number of events at the same time (SENAMHI, 2015).
 36. **Impacts on Species and Ecosystems.** The impact of higher temperatures and less precipitation will have an impact on species distribution. For example, Feeley et al. (2010) studied the behaviour of Andean trees in the tropical Andes located between 950m 3400m above sea-level. The authors concluded that due to the increase in temperature, trees were migrating to higher elevations, at a mean rate of 2.5-3.5m per year (and even up to 20.6m for some species). The observed mean rate of change is less than predicted in comparison to the temperature increases in the region, suggesting that there is a limited capacity to respond to these and poses extinction risks for some genera (Feeley, et al., 2010).
 37. Similarly, glacier retreat has meant that there is more suitable territory for some species to inhabit. For example, the glacial retreat in the Vilcanota mountain range (SE subregions) has opened an ecological corridor in recent decades where species' distributions have been affected. Many of the woody and herbaceous species in the Andes (e.g. *Ericaceae*, *Bromeliaceae*) depend on interactions with animals for seed dispersal and pollination, and the effects of climate change on these organisms could cause spatial, temporal or physiological asynchronies between mutualistic species. The changes on species distribution implies that there will be changes in the coverage of landscapes as well. For example, Ludeña et al. (2014) found that, in the Puna region, there would be an increase in shrub surface under an A2 emission scenario (equivalent of an RCP 8.5 scenario with regards to temperature rise). The same phenomenon is projected for the páramos. Conversely, glaciers, the suprandino area (the area between glaciers and the puna), the puna and yungas forests show large reductions in extension. In general, a migration to higher elevations was observed in the vegetation bands that characterize the Andes, but the puna was replaced by shrubs rather than yungas forests. The projected expansion desert and dry (xeric) areas will reduce water availability (Ludeña, Sánchez-Aragón, de Miguel, Martínez, & Pereira, 2014). Results from different climate change scenarios suggest a progressive reduction in the carrying capacity of ecosystems and a decrease in the available area of grazing land.
 38. **Impacts on agriculture and livestock production.** Prolonged droughts, changes in temperature and precipitation are likely to have various impact on the livelihoods of subsistence farmers in the Andes, including decreased crop yields and shorter growing seasons, changes in water availability, grasslands and diminished carrying capacity of ecosystems for livestock, including camelids. An evaluation developed by SENAMHI found that increases in maximum temperature (2°C to 3°C), minimum temperature (4°C to 6°C), and higher rainfall (10% to 20%) by 2050 would lead to negative impacts in agricultural activity in the SHAP region. According to SENAMHI (2015), crop yields in the SE, SW and A subregions decrease in different magnitudes depending on the model and scenario. For potato²⁵, in the SE subregion for CanESM2 in the RCP4.5 scenario, the changes in yield crop are between -436.4 and -1750.3 kg/ha with statistical significance toward 2050. Similarly, for CNRM-CM5 in the RCP8.5 scenario, the yield crop is between -762.4 and -21.7 kg/ha with statistical significance toward 2050. Results obtained from MPI-ESM-MR for both scenarios, for CanESM2 for the RCP8.5 and CNRM-CM5 for the RCP4.5 are mostly negative

²⁵ In SEMANHI (2015), they also quantified the σ parameter, which is the proportion between the change in yield ($\Delta \text{yield} = \text{yield}_{(2036-2065)} - \text{yield}_{(1971-2010)}$) and the standard deviation ($\sigma = \Delta \text{yield (kg/ha)} / \text{standard deviation(kg/ha)}$) considering the average of 2036-2065 years as the future climate and the 1971-2010 average as the present climate. Results for potato crops showed that averaging all three models, σ goes from -0.17 to -0.82 for the RCP4.5 and from -0.28 to -0.69 for the RCP8.5. All models in both scenarios show a consistent decrease in potato yields. For wheat yield, the σ parameter goes from -0.84 to 0.56 in the RCP4.5 and from -0.71 to 0.35 in the RCP8.5. Positive σ is shown in the A region, while negative σ , in the SW region.

but with no statistical significance. Wheat yields in the SW and A subregions, are also likely to significantly decrease from 125.5 to 39.5 kg/ha with statistical confidence in the RCP4.5 scenario using the CanESM2 model. Likewise, the crop yield for the RCP8.5 scenario using the CNRM-CM5 model, goes from -56.6 to 18.2 kg/ha with statistical confidence. Pests and diseases²⁶ will affect potato and onion crops due to shifts in species distribution ([SENAMHI, 2015](#)). Most highland communities in the Andes rely on crop and livestock farming for their livelihoods. Many are smallholders who own livestock and use traditional farming practices, relying on rainfed crops and limited irrigation (Aragón, Oteiza, & Rud, 2018). Surveys carried out in Apurimac in 2009, the loss of potato production due to drought was 69% and 65% for corn. Most of the products from this region are used for self-consumption (Llosa, 2009).

39. In the case of livestock, the reduction in the carrying capacity of ecosystems in the high Andes translates into economic losses for communities in the SHAP region. The value of the loss of livestock due to climate hazards was estimated in the context of the Ministry of Agricultural Development and Irrigation's Climate Risks and Adaptation Strategy. The losses in the Puno region alone constitute over 60% of the national total in terms of monetary loss (MIDAGRI, 2012). The scarcity of water and limited grazing areas are further reducing livestock productivity. Extreme temperatures and frost events disrupt the reproductive cycles of both plants and animals, further exacerbating the vulnerability of subsistence farmers who heavily rely on these resources. A detailed climate change impact analysis on relevant value chains can be found in section 5.7.3 of Feasibility Study (Annex 2a).
40. **Impact on population.** In principle, climate change affects all people – but to varying degrees. In addition to regional differences, this is largely due to social structures. Being affected by the consequences of climate change also depends to a relevant extent on gender-conforming behaviour and gender-inequitable distribution of resources. This means that migration caused by climate change also has something to do with gender equality²⁷. This is also applicable to specific highland communities use migration to anticipate or react to hazards. In terms of temperature-related hazards, Sperling et al. (2008) mention the emigration of communities from the highlands of Puno, especially of young people, mostly men, due to frosts and droughts. Crespeigne et al. (2009)'s findings show that after an episode of rainfall deficiencies investigated in a community in Puno in 2007, 31% of those surveyed migrated or looked for work as vendors, waiters and shoe-shiners to smooth out income losses in places like Lima and other cities, or by working in mines and in agribusiness. Generally, men and boys migrate and look for work whereas women and girls and the elderly do stay behind. According to a survey carried out in the Apurimac region, in the last five years, 50% of the relatives of the inhabitants of that region have emigrated due to the drought, being the province of Grau (76%) with the highest immigration according to the proportion of cases (Llosa, 2009).

B.1.4 Water balance under climate change

41. In order to assess the current status of the hydrological system of the puna and to describe how the system will be affected by climate change and what the status of the system would be after implementing the, by the project promoted, EbA measures, a hydrological study was conducted (see Annex 2g). As the availability of water is critical for the maintenance of ecosystems and also for the sustainability of the agricultural and livestock breeding activities. In this regard, the hydrological study considered hydrological and climate variables that are available as gathered from open data and scientific literature to assess actual and future hydrological status and dynamics as a function of changing climate and the effect of the EbA measures in the Puna Resilient project's districts.
42. The hydrological study predicts an overall negative climate change impact²⁸ in all the districts selected with different levels of intensity from an index²⁹ of 0.25 to 1 (the higher is the value, the higher is the combined effect of the climate change and current water usages as respect to the water availability). Detailed results from the carried hydrological model are described in Annex 2g.
43. In addition, to evaluate the impact of the EbA measures on each district, a Combined Index (CI) considering water stress³⁰ and EbA suitability was developed. The result showed that Ecosystem based adaptation measures have the potential to generate positive impacts on the water availability at the district level. For example, the EbA Wetlands (Bofedal) restoration and conservation, could potentially increase the soil-plant system humidity, increase the spring flow due to infiltration produced by the wetland system recovered and decrease the base flows. The values of the EbA Impact Index for each district and each EbA vary from 0³¹ to 1. The higher the value, the higher

²⁶ The affirmation about the effect of pests and diseases that might affect crops due to shifts in species distribution is not quantified but only mentioned in the results obtained from SENAMHI (2015).

²⁷ [Klimawandel, Migration und Geschlechterverhältnisse | Migration und Klimawandel | bpb.de](#)

²⁸ Data on current and projected hydrological variables with the projected climate change were based on the data presented in the Annex 2f and information available at landscape scale from reliable sources.

²⁹ This hydrological study supports the calculation of hydrological and climate resource and risk indexes that contributes to the understanding and estimation of water availability versus water usages. To analyses climate change impacts 10 climate change indexes were calculated based in five water resources and two climate change variables.

³⁰ For the water stress hydrological interdependencies of EbA measure for climate change adaptation were considered as assumptions based on studies that have demonstrate the links between the proposed interventions and improved climate resilience among vulnerable communities in the region (The benefits for adaptation of the proposed EbAs measures are described in section 9.2 of Annex 2a).

³¹ Some EbAs are not suitable for all analyzed districts.

is the potential of the relative impact of the EbA in respect to the water availability of the district. In general, the results demonstrate the benefits and positive impacts that the proposed interventions will have in the target areas, identifying which districts benefit best from each EbA measure, providing information on the prioritization of EbA measures for the implementation of sub-activity 1.1.1.2. Ultimately these EbA interventions will substantially reduce the expected negative impacts of climate change in the project area.

B.1.5 Priorities for climate change adaptation in ecosystems and agriculture

44. The most relevant impacts of climate variability and change in the Puna ecosystems and agricultural value chains include: i) losses in livestock, both in terms of the number of animals and productivity; ii) losses in crop yields, increased incidence of pests and diseases and loss in resilience to external shocks; iii) reduced water availability; iv) increased soil erosion and reduced topsoil moisture; v) decreased food and nutritional security.
45. At present, the prioritized value chains suffer significant impacts due to climate change, which translate into the following vulnerabilities:

Andean crops

46. Potato: The effects of climate change on potato production are complex, as plant growth and yield are related to temperature. Potato cultivation, in order to maintain its nutritional properties, requires cold temperatures. Many farmers in the High Andes have moved the agricultural frontier upwards to maintain favourable temperatures for their crops (Giamberso, 2014). Shaw and Kristjanson (2013) found that potato farmers in the region have raised their crops by approximately 150 m in the last 30 years. Arce et al. (2019) found that in Huancavelica, native potatoes rose almost 500 m above those reported 38 years ago. In the agro-ecological zones of the Puna, in addition to the increase in temperature, there are currently other climatic events that put agricultural production at risk, such as frosts, droughts and hailstorms that generate risks of production losses and loss of genetic diversity of native potatoes (Practical Solutions, 2018). The droughts have caused a reduction of approximately 12.1% in potato sowings (2022/2023 season) with respect to the average accumulated sowings up to October of the last 5 seasons, i.e., 21,028 ha less (MIDAGRI, 2022e). Evidence also indicates that farmers are responding to climate change by increasing areas of potato cultivation in natural grassland areas in Puna areas, due to the increased population and rate of pests and diseases that appear to be occurring at lower altitudes such as rancho, also known as tizón tardío and seca seca.
47. Quinoa: Quinoa has many qualities that make it more resilient to climate change (Balakrishnan et al., 2022, Hinojosa, et al, 2018) than other crops, nevertheless quinoa is not immune to the negative impacts of climate change, which can adversely affect its production and yield. Areas once suitable for quinoa cultivation may no longer have the necessary conditions for sustained production. As temperatures rise and exceed the crop's optimum threshold, production and yield decline. For instance, in Puno, a 1 °C temperature increase results in a reduction of 112.2 Mt in production and 169.1 kg per ha in yield. Additionally, increased rainfall by 300 mm leads to a reduction of 75.78 Mt in production and 127 kg per ha in yield (Carrasco, 2016). Drought is another significant challenge that quinoa producers face, causing water scarcity for irrigation. In Ayaviri, Puno, the entire quinoa harvest was lost during the 2023 season.

South American Camelids

48. Alpaca and Vicuña: Increasing temperatures and reduced rainfall have resulted in diminished pastures and a higher incidence of pests and diseases and contribute to higher mortality rates among newborn. This situation is likely to negatively impact vicuña and alpaca populations (Kasterine and Lichstenstein, 2018). Furthermore, climate change has influenced the spatial dynamics of pastures, grazing areas, and the vicuña habitat. As grazing areas degrade, vicuñas are forced to cover larger areas with lower-quality pastures and migrate to higher elevations where better pastures can be found (Korswagen, 2015). Also, sudden changes in rainfall and the melting of glaciers are causing disruption among both alpaca communities and the communities that rear them. A primary challenge currently faced by alpaca producers is the scarcity of water due to reduced rainfall, directly affecting pasture and fodder production for feeding the herd. Consequently, they are incurring additional expenses that were not present before for the animals' breeding.
49. Sustainable management of South American camelids is therefore an ancestral activity and in several cases are still, the most highly appreciated resource for Andean livelihoods (Vilá and Arzamendia, 2022). Although it is known that South American Camelids are more efficient grazers, there is a need to ensure long-term sustainability of this livestock system in a context of climate change by supporting communities in developing climate-resilient business models for camelid breeding and fiber production. This includes improving the management and conservation of water and pastures, adding sustainable value to the fiber products, combining primary production with manufacturing, developing a stronger fiber quality control, promoting associations among camelid breeders to achieve a higher professional level of their members, and taking charge of the direct commercialization on the national and international market. Other productive practices (e. g., crop rotation, selection of adapted crop varieties, agrobiodiversity conservation) related to agriculture value chains will also be promoted by the project through the conservation and restoration of key ecosystem services (food provision, soil fertility, water regulation,

nutrient cycling) with the aim to increase livelihoods diversification and to reduce the vulnerabilities of the communities.

50. Additionally, the project will work on complementary value chains including **traditional Andean crops, breeding of small animal (guinea pigs), community based tourism and traditional handicrafts**. Traditional crops include: (i) Kiwicha (*Amaranthus caudatus*), (ii) Cañihua (*Chenopodium pallidicaule*), (iii) Tarwi, (iv) Oca (*Oxalis tuberosa*), and (v) Mashua. Their vulnerabilities are related to those explained for the main crops (quinoa and potato), as well as those related to animal livestock and tourism. Since Agricultural systems are highly sensitive to climate change, most farm rural people in the Peruvian Andes hold diversified, low input cropping systems and intercropping as an adaptation response to the intra-seasonal climate variability³².
51. Ecosystems based Adaptation (EbA) supports the sustainable use of resources and diversification of land and livelihood options such as multi-cropping and agroforestry while reducing climate vulnerability for people at least local scale but ideally at landscape scale³³. The table below describes the impacts for each of the prioritized value chains and the EbA measures that the project will promote in a complementary manner so climate resilience can be increased in the SHAP. The EbA measures mentioned below has been revised and validated during stakeholder consultations with different actors in the regions (local authorities, technical officials from rural extension programs, Indigenous Peoples and Local Communities, producer organizations, cooperatives, non-governmental organizations, academia, etc.).

Table 1. Climate change risks and impacts in the target Puna ecosystems and value chains, adaptation interventions and benefits

Value chain / Ecosystem	Threat / Climate risk	Impacts of climate change	Ecosystem based Adaptation measures	Adaptation benefits
Highland crops - Potato	- Increase in maximum temperature. - Decrease of minimum temperature/ frost - Dry periods/drought - Heavy rainfalls - Glacial retreat and melting	- Decreased crop yields and shorter growing seasons - Decrease in production quantity owing to water scarcity. - Decreased crop growth - Soil fertility - Logistical problems (harvest/ transport) due to floods - Increase in pests and diseases that affect crop yield and local population	<u>Prioritized EbA measures for highland crops include:</u> - Conservation and restoration of wetland ecosystems (bofedales) – ancestral use to retain rainwater and regulate runoff flows, increasing soil moisture. Improvement in water regulation at the source. - Qochas – ancestral water micro reservoir natural water reservoirs to retain rainwater - Integrated soil fertility management (ISFM) – ancestral agricultural practice to enhance soil fertility, improves water retention, and even reduces germination time through seed priming through e.g. crop rotation, camelids manure (Liniger, Mekdaschi, Hauert, & Gurtner, 2011). - Andenes/terraces -ancestral technique to reduce water runoff and soil erosion, usually on steep slopes. Terraces maintain soil moisture and generate a suitable microclimate for crop diversification (HELVETAS Swiss Intercooperation, 2017). - Infiltration ditches -retain and infiltrate rainwater. The deposited water infiltrates the soil and helps with aquifer recharge. - Agroforestry measures -tree-crop interaction helps control erosive processes and increases water infiltration at	EbA measures ensure the provision of ecosystem services. Therefore, water regulation functions of wetlands towards the basins in times of low water levels represent an important measure to make a significant contribution to the challenges of water security, integrated water resources management, and ensuring adequate water availability for human supply (SDG6). In a scenario of low provision of water, technologies to increase efficiency within the irrigation systems is crucial. Solar pumping, geotanks and sprinkler and drip irrigation systems are also crucial to water management.
Highland crops - Quinoa	- Increase in maximum temperature. - Decrease of minimum temperature/ frost - Dry periods/drought - Heavy rainfalls - Glacial retreat and melting	- Decreased crop yields - Decline in crop production and quality. - Decrease in production quantity owing to water scarcity. - Logistical problems (harvest/ transport) due to floods	- Andenes/terraces -ancestral technique to reduce water runoff and soil erosion, usually on steep slopes. Terraces maintain soil moisture and generate a suitable microclimate for crop diversification (HELVETAS Swiss Intercooperation, 2017). - Infiltration ditches -retain and infiltrate rainwater. The deposited water infiltrates the soil and helps with aquifer recharge. - Agroforestry measures -tree-crop interaction helps control erosive processes and increases water infiltration at	Improved soil fertility would increase resilience to climate change, underpinning plant growth and optimising crop yields. This would promote food security and environmental sustainability of agricultural systems. Sustainable pest management and other adaptation practices will maintain and/or increase productivity and product quality in a changing climate context. Diversifying production will increase the resilience of producers in the event of a climatic event affecting economic activity. Women participate in the different stages of agricultural work. They select and care for native seeds, which are the basis of the food biodiversity that sustains families.

³² Ponce, Carmen (2020). Intra-seasonal climate variability and crop diversification strategies in the Peruvian Andes. A word of caution on the sustainability of adaptation to climate change.

³³ Swiderska, K, King-Okumu, C and Monirul Islam, M (2018) Ecosystem-based adaptation: a handbook for EbA in mountain, dryland and coastal ecosystems. IIED, London.

			small scale (HELVETAS Swiss Intercooperation, 2017) . - Restoration with native species- ancestral practice using native species (such as Buddleja coriacea, Alnus acuminata or Polylepis racemosa) have high soil retention capacity, stabilize sudden temperature changes and hold soil and environmental moisture. - Contour farming by orienting the crop rows following the contour lines of the topography and other ancestral tillage systems. This practice helps to reduce water runoff and soil erosion. - Conservation agriculture - farming system that prevents soil disturbance and emphasizes maintenance of permanent soil cover. The aim is to enhance biodiversity and improved and sustained crop production. Also, it increased water retention and runoff reduction. <u>Other complementary adaptation measures will be also supported.</u>	While men tend to choose to plant hardy crops that secure their yields, women's preferences are more often shaped by varieties that are easier to grow, quick to cook, nutritious and palatable³⁴. They therefore make a key contribution to food security and sovereignty (SDG5) In Peru, the gender roles around potato production are clearly differentiated. Seed potato management has traditionally been the domain of women, who have knowledge of the cooking quality and agronomic or storage characteristics of different varieties. Women make decisions regarding the selection, storage and use of seeds at planting time, as well as sale of seed or home consumption, and their knowledge is critical to the preservation of the thousands of varieties that exist today³⁵
Camelids (alpacas and vicunas)	- Dry periods/ drought - Increase in maximum temperature - Heavy rainfalls - Glacial retreat and melting - Decrease of minimum temperature/ frost	- Low forage production - Increase in herd morbidity and mortality. - Decrease in drinking water sources. - Decrease in production quality and quantity - Decrease in meat/fiber yield - Increase in presence of pests and diseases (animals and farmers)	<u>Prioritized EbA measures for the South American camelid value chain in the intervention areas include: (see detail on measures above)</u> - Conservation and restoration of wetland ecosystems (bofedales) - ancestral use to retain rainwater and regulate runoff flows, increasing soil moisture and fodder. - Qochas -ancestral water micro reservoir natural water reservoirs to retain rainwater. - Infiltration ditches -retain and infiltrate rainwater. The deposited water infiltrates the soil and helps with aquifer recharge. - Sustainable management of natural pastures -ancestral practice for overgrazing control, increasing water infiltration and controlling erosion, grasslands can be recovered and provide enough fodder for livestock (HELVETAS Swiss Intercooperation, 2017). - Agroforestry measures (see highland crops above) Restoration with native species - ancestral practice using native	EbA measures will improve water harvesting during the rainy season for distribution in times of drought by supplementing water from rainfall. This, together with efficient irrigation technologies, will improve and make sustainable use of water resources for fodder production. Also, grassland vegetation cover protects soils from the effects of erosion, reduces evaporation as well as runoff. This not only improves carbon sequestration, helping to mitigate greenhouse gas emissions from livestock production, but also contributes to improved soil quality by increasing soil porosity and water retention capacity (DDCC, 2019). Its sustainable management contributes to increasing communities' resilience to climate change by buffering the effects of extreme events and is also a strategic social and economic element for camelid producers, as a source of food and survival for Andean vicuñas and alpacas. Diversifying production will increase the resilience of

³⁴ Women hold the seeds of a stronger food system for Peru / Las mujeres alimentarán nuestro futuro | CGIAR GENDER Impact Platform

³⁵ <https://www.potatonewstoday.com/2021/03/18/the-phenixes-in-our-food-systems-women-farmers-in-peru-safeguarding-the-survival-of-potato-biodiversity/>

			species (such as <i>Buddleja coriacea</i>, <i>Alnus acuminata</i> or <i>Polylepis racemosa</i>) have high soil retention capacity, stabilize sudden temperature changes and hold soil and environmental moisture.	producers in the event of a climatic event affecting economic activity. Women have been playing a fundamental role in the implementation and maintenance of these techniques (wetlands, harvesting and sowing water), as they also work in agriculture and livestock farming. Livestock raising is an activity mostly carried out by women and requires fodder in the dry season (from June to October or November) to maintain the size of the herds, their productivity and income. In this context, they produce and transfer knowledge about animal well-being and grassland health. When men migrate to access other income in urban or peri-urban areas, women remain in charge of their agricultural units.
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52. **Many core elements and principles of the EbA approach coincide with traditional knowledge, management practices, technologies, and values of High Andean communities.** There are traditional communal management practices to prevent degradation of wetlands through different systems:
53. The management of rustic reservoirs, known as *qochas*, to store rainwater for human use and animal consumption and to promote groundwater recharge through seepage. Another traditional way to support water infiltration and prevent soil loss through erosion is by constructing trenches following the contour of the land, thus, capturing rainwater runoff (and soil) and subsequently feeding aquifers and springs to face water deficits in dry seasons.
54. Such infrastructure is complemented with ancient diversion canals conducting streamflow to infiltrate trenches from which it can slowly seep and then re-emerge in micro-reservoirs or springs for agricultural use.
55. Moreover, the Southern Andes are well known for their terraced landscape, dating back to pre-Inca periods. Terraces reduce the speed of runoff in areas with steep slopes, slowly seeping water into the ground and increasing the surface of land that can be used for agriculture or animal grazing.
56. The management of natural grasslands and wetlands through the establishment of animal exclusion areas, rotational grazing, and restoration allows ecosystems to recover and continue supplying services such as water regulation and provision, fodder, control of soil erosion, and carbon sequestration.
57. These ancient technologies have been an integral part of traditional livelihoods in the SHAP for centuries, shaping a particular way of living with and within nature. Furthermore, these technologies are not only based on the natural infrastructure themselves but on the social network behind them, which demands a high degree of social cohesion to successfully perform the required communal work to keep the technologies in place ([Rolando et al., 2017](#)).
58. Proven solutions with potential to be upscaled. The protection of grasslands and wetlands through fencing and the restoration of flows in those areas that have been drained, have proven efficient in recovering their functionality in terms of water regulation. Empirical evidence from studying ten peatland areas (bofedales) in Peru and Bolivia showed that these Andean Puna ecosystems can reduce the climate change impact on water availability by increasing the groundwater recharge during the rainy season ([Cooper et al., 2019](#)). Another research by Cervantes (2022) demonstrated the contribution of ecosystems (peatlands, grasslands, and *Polylepis* forests) to water regulation. According to this recent analysis in the Rontoccocha micro basin in the Peruvian region of Apurímac from 2019, grasslands had the capacity to recharge 194 mm (194 l/m²) of water in one year and stored 15.1 percent of the total rainfall water in one-unit area. Peatlands were shown to store 16.4 percent and recharge 211 mm (l/m²), and *Polylepis* forests stored 15.4 percent and recharged 198 mm. As presented in the following chart, peatlands have a greater water recharge capacity (water regulation ecosystem service) than grasslands and *Polylepis* forests ([Cervantes, 2022](#)). Moreover, [Mosquera et al. \(2022\)](#) revealed that grasslands have an important and positive hydrological function in the High Andes and, when conserved and restored, can improve soil water infiltration and reduce soil erosion.

Table 2. Results for each ecosystem under study Results for each ecosystem under study (in mm and, in brackets, in % of total precipitation).

Ecosystem	Precipitation	Interception	Net precipitation	Surface runoff	Actual evapotranspiration	Recharge
Grassland	1 290 (100%)	51.5 (4.0%)	1 240 (96.0%)	324 (25.1%)	698 (54.2%)	194 (15.1%)
Peatland	1 290 (100%)	25.8 (2.0%)	1 260 (98.0%)	320 (24.9%)	765 (59.4%)	211 (16.4%)
Polylepis forests	1 290 (100%)	122 (9.5%)	1 170 (90.5%)	311 (24.2%)	696 (54.1%)	198 (15.4%)

Source: Cervantes (2022)

59. A recent study analysed alternatives for water supply in the city of Lima. The combination of amuna restoration, hydrological restoration of wetlands, animal exclusion, and rotational grazing in the puna ecosystem turned to be a competitive alternative (0.51 USD/m³), compared to a desalinization plant (0.74 USD/m³) ([Gammie and De Bievre, 2015](#)). Ecosystem based solutions, are well known for their contribution to mitigation co-benefits.

B.1.6 Prioritized Project Areas

60. Due to the extent of the SHAP and limited project funds, areas where interventions could have the greatest potential were prioritized in a two-stage process. In the first stage, the scope of analysis was defined according to a set of criteria including vulnerable communities, high altitude Andean areas (above 3,500 m.a.s.l, including a buffer zone between 2,800 to 3,500 m.a.s.l), presence of key Puna ecosystems (peatlands, grasslands and wetlands), distance to degraded lands and to areas that have suffered deglaciation, and existence of agriculture and husbandry lands with high or very high risks of droughts and high climate vulnerability. In the second stage, priority was given to districts with the best enabling conditions for initiate interventions, considering social, economic, and environmental factors. As a result of the first stage, 91 districts in the regions of Arequipa, Cusco, Apurimac, and Puno were prioritized, covering 5,314,607 hectares. Of the 91 districts, 58 districts will be eligible for the Puna Facility (covering 4,116,475 hectares) and capacity building activities, while 33 districts will only benefit of capacity building activities. Four Natural Protected Areas fall inside of some eligible districts which are: the Landscape Reserve of Nor Yauyos Cochas in Lima region, the Landscape Reserve of Cotahuasi, the National Reserve of Salinas y Aguada Blanca in Arequipa region, and the National Sanctuary of Ampay in Apurimac region. SERNANP is the governing entity of these areas. A detailed explanation of the prioritization methodology and process is included in Annex 16b "Explanatory note Area of intervention". Indigenous Peoples and Local Communities in this area are 69% Quechua speakers and the other 31% Spanish speakers (National Census, 2017)³⁶. As depicted in the Map of Indigenous People in Peru, there are no other Indigenous Groups in the target area of the project.

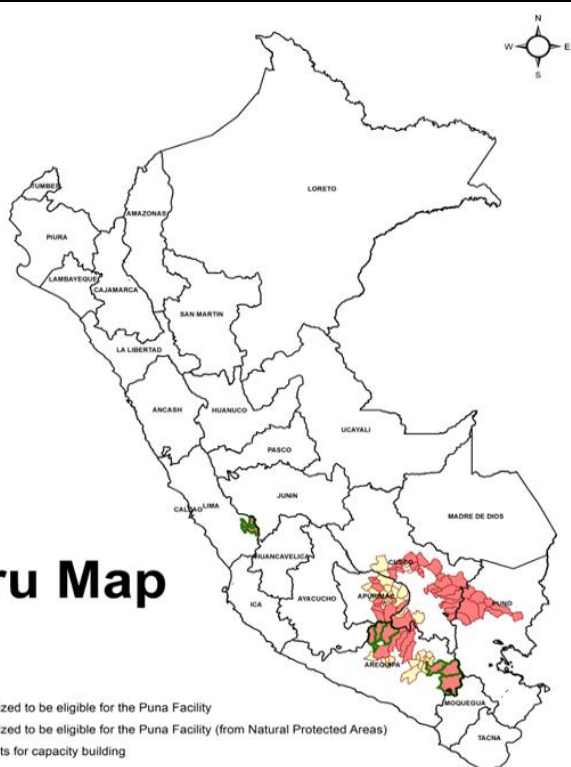
Figure 9. Map of prioritized districts by type of intervention **Figure 10. Map of Indigenous Peoples in Peru (Quechua in light pink)**

³⁶ Note that in 2012, 83% were Quechua speakers and 17% Spanish speaker within the project area of the project (CENAGRO, 2012)

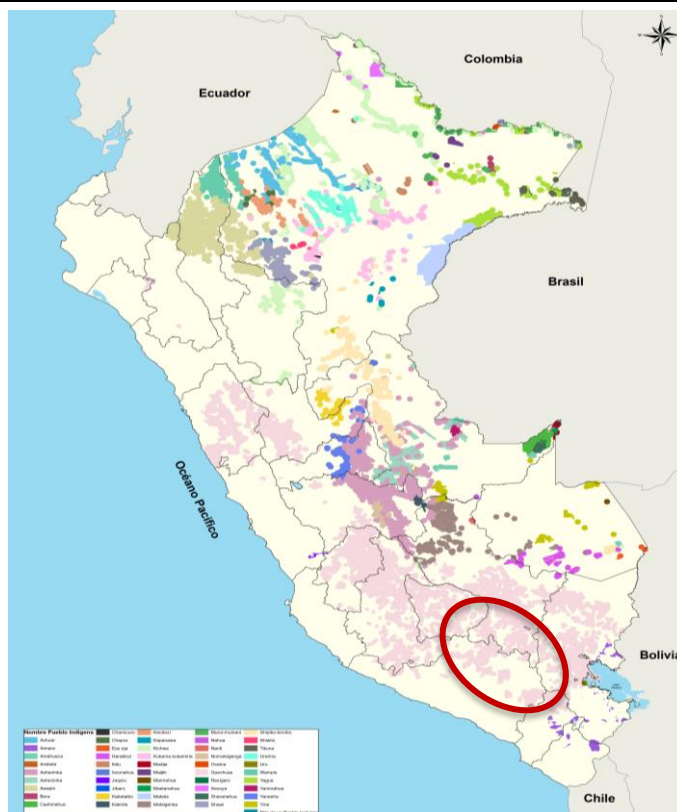
Peru Map

Legend

-  Prioritized to be eligible for the Puna Facility
-  Prioritized to be eligible for the Puna Facility (from Natural Protected Areas)
-  Districts for capacity building
-  Departments of Peru



Source: own



Source: Ministry of Culture (2023)

Table 3. Prioritized districts by departments and type of intervention

Department	Watersheds	N° of districts for Capacity Building	N° of districts for Puna Facility + Capacity building	Total
Apurímac	Inter-watershed Alto Apurímac	13	10	23
Arequipa	Vitor Quilca Chili, sub-watershed Cotahuasi- Ocoña	14	12	26
Cusco	Vilcanota-Urubamba, Inambari	6	23	29
Lima	Cañete	-	4	4
Puno	Azángaro, sub-watershed Coata	-	9	9
Total		33	58	91

B.1.7 Direct beneficiaries of the project

61. The direct beneficiaries of the project are primarily Andean Indigenous Peoples and Local Communities from vulnerable rural regions in the high Andean regions (for detailed beneficiary numbers, refer to section D.1.3). In the majority of cases, project's direct beneficiaries are Indigenous Peoples and Local Communities engaged in agriculture and animal husbandry activities in the high Andean zones. Both economic activities are essentially for their subsistence and have a high dependency on climate (70% of agricultural activity depends on rainy season). IPLCs in this part of the Andes as mentioned in the previous section are organized in peasant or local communities, and in other forms (refer to the list below). to manage local development, administer common resources or provide services to each other. These forms of organizations will be regarded as potential recipients of funding from the Puna Facility, as outlined in section B.4.4. Special emphasis will be given to women's participation in IPLCs, enterprises, associations and cooperatives.
62. Local Communities³⁷: Refers in most of the cases to peasant communities which are the most widespread and important category in the region, as they represent a strong part of the Andean culture as they are descendants of the Original Peoples of Peru. They are organizations of public interest, with legal existence and legal personality, recognized by the government and protected by Peru's Political Constitution, with rights over natural resources and the territory they occupy. Peasant communities consist of families united together by ancestral, cultural, social and

³⁷ Please note that the term of local communities is going to be used for this project in order to include a wider range of activities in which they engage, as peasant communities are predominantly focused on agricultural activities.

economic ties, who have communal ownership of the territories they inhabit. These communities operate through democratic governance, involving a Communal Assembly and a Communal Board elected for terms ranging from one to two years.

63. **Community Enterprises:** These are Local communities themselves that, using their legal status, organise and administer their economic activities in a business manner, by generating productive units of communal goods and services, to ensure the well-being of their members and contribute to the development of the community as a whole. A local community can constitute one or more communal enterprises.
64. **Producers' Associations:** Represent a stable organisation of natural or legal persons, or both, which through a common activity pursues a non-profit purpose. In many cases, community's members tend to form different types of producer associations according to a specific economic activity, for example: artisans associations, alpaca associations, quinoa producer associations, etc. For many years the Ministry of Agricultural Development and Irrigation has promoted producer associativity, which comprises a process of voluntary grouping of individuals, organisations or enterprises working in a coordinated and concerted manner to achieve their goals. This approach aims to optimize the utilization of production resources, enhance earnings through associative marketing efforts, and realize economic of scale, improved market access, and improved incomes.
65. **Cooperatives:** An agricultural cooperative is a society of persons that can also be the same members of a community who carry out agricultural, forestry and/or livestock activities, who have joined voluntarily through a jointly owned and democratically controlled company that complies with cooperative principles. The agricultural cooperative is constituted with the purpose of providing services related to the agricultural, forestry and/or livestock activity that its members carry out. Among the services that the cooperative can provide to its members are the supply of products and services, marketing, processing, financing, technical advice, etc., as well as any other related or complementary service that contributes to the realization of its objective. The agricultural cooperative acquires the status of legal entity from its registration in the public registers, and to enjoy the benefits established in this law, they must prove its registration in the National Registry of Agricultural Cooperatives in charge of the Ministry of Agricultural Development and Irrigation (MIDAGRI).
66. **Micro and Small Enterprises (MSEs):** Are an economic unit constituted by a natural or legal person, under any form of organisation, whose purpose is to carry out activities of extraction, transformation, production, commercialisation of goods or provision of services. According to the Legislative Degree 1086, micro enterprises have from 1 to 10 employees and annual sales up to 150 taxable tax units, while small enterprises have from 1 to 100 employees and annual sales up to 1700 taxable tax units.

B.1.8 Related ongoing and forthcoming projects with synergies

67. Table B.1.1.3 provides a summary of the worth noting related past, ongoing and forthcoming interventions in the SHAP (a more extended table and list can be found in the Feasibility Study (Annex 2a) section 7.1.6 Gaps and complementarity with other development initiatives in the sector).
68. Among the projects described below, it is worth noting some of them which the project will have some synergies. For example, the project will scale up EbA pilot interventions on previous projects such as the *Global Mountain Ecosystem-based Adaptation* programme³⁸ implemented by UNDP and IUCN from 2011-2015 in Uganda, Nepal and Peru. In Peru it was piloted only in a small area of Nor Yauyos Landscape Reserve to test methodologies and decision-making tools for EbA planning and monitoring; implement pilot measures focused on water resources management and built evidence and knowledge for decision making. Instituto de Montaña was IUCN implementing partner so its experience, methodologies and lessons learnt will be replicated and scaled up into Resilient Puna Project through Component 1 and 3 (in community monitoring systems and in territorial planning at different levels).
69. The Puna Facility will also complement and benefit from the experience of the *Six Phase of GEF's Small Grants Programme* implemented by UNDP until June 2021 and from the *Seven Phase* that goes until July 2025, covering the regions of, Puno, Tacna and Cusco. The Puna Facility will fund EbA measures and Climate Resilient Value Chains specifically if they have a climate adaptation purpose, rather than a biodiversity conservation/agricultural development one such as the GEF's SGP. The Puna Facility will start its first call for proposals at the beginning of 2025 avoiding overlaps with SGP but trying to complement the support to IPLCs and the existing multi-stakeholder platforms. The GEF SGP's six phase was successful in creating multi-stakeholder platforms, as well as knowledge-exchange platforms that this project could leverage in Component 1. For example, through the SGP Peru Knowledge Fair or "*Encuentro de Saberes*" where 50 communities in the Peruvian Andes have been sharing their solutions based on nature and their ancestral knowledge³⁹. Recommendations on how to strengthen Andean local experts' capacities also will be taken in considerations for the implementation of sub-activity 1.1.2.1.
70. The USAID/Forest Trends/Canada "Natural Infrastructure and Water Security Project", which promotes public-private partnerships and ecosystem recovery to prevent hydrometeorological risks will serve to replicate tools and

³⁸ Global Ecosystems Based Adaptation in Mountains Programme | UNDP Climate Change Adaptation (adaptation-undp.org)

³⁹ When knowledge becomes power | Programa De Las Naciones Unidas Para El Desarrollo (undp.org)

methodologies to identify where to implement EbA measures. Potential synergies with this project during implementation in some territories will be closely coordinated.

71. The Helvetas/AVINA/IISD/IFAD “Resilient Andes project” which aim is to strengthen capacities in private-public stakeholders living in poverty and vulnerability in Bolivia, Ecuador and Peru, aiming to improve their food and water security. Component 3 of this project will consider, and replicate family farming best practices and knowledges gathered in “Encontrar” web platform.
72. The Sustainable Management of Agro-Biodiversity and Vulnerable Ecosystems Recuperation in Peruvian Andean Regions through the Globally Important Agricultural Heritage Systems Approach (GEF- FAO) which aim is to conserve and sustainable use agrobiodiversity in the high Andes. In component 1 of this project will replicate best practices of farmer field schools for the cultivation of potato and quinoa and tools to connect producers with local markets.
73. **Alignment with existing GCF country programmes/ projects.** Profonanpe is currently implementing two GCF projects in Peru. As one of the main stakeholders, the project will build on previous experiences and include lessons learned in its activities.
74. The project (FP001) *Building the Resilience of Wetlands in the Province of Datem del Marañón* seeks to facilitate better land-use planning and management of the region’s wetlands, while strengthening sustainable, commercial bio-businesses of non-timber forest products. It will entrust indigenous communities with the management of resources, improve their livelihoods, and empower women in the decision-making processes.
75. Next to FP001, Profonanpe is also implementing and managing the *Peruvian Amazon Bio Business Facility* (FP193), which was set up in November 2022. The Facility has the aim to invest in eco bio businesses (EBBs) to provide mitigation benefits through enhancing carbon stocks. The Puna Facility which will invest in Climate Resilient Value Chains will benefit from the experience of Profonanpe, which will have gained capacity and knowledge on best practices and management from the implementation of the EBBF.
76. Furthermore, Peru is part of two additional GCF programs related to the Puna project activities: i) the in October 2021 approved (FP173) *Amazon Bioeconomy Fund: Unlocking private capital by valuing bioeconomy products and services with climate mitigation and adaptation results in the Amazon* with the Inter-American Development Bank as an Accredited Entity. The Amazon Bioeconomy Fund in six Amazon countries will deliver sustainable solutions to reduce the impacts of climate change in the Amazon biome by prioritizing natural capital and delivering climate benefits. The project has a similar approach, but its geographic implementation (Amazon basin) does not overlap with the Puna. ii) (FP128) *Arbaro Fund – Sustainable Forestry Fund*, approved in March 2020, the Programme will provide effective climate change mitigation outcome through investing in sustainable plantation forestry projects in emerging forestry markets while also bringing adaptation co-benefits. This investment approach provides developing countries and their rural communities with a solution to increase carbon sinks by producing wood in a sustainable manner and conserving natural forests, whilst contributing to reduction of illegal logging.

Table 4. Related projects and potential synergies with the project

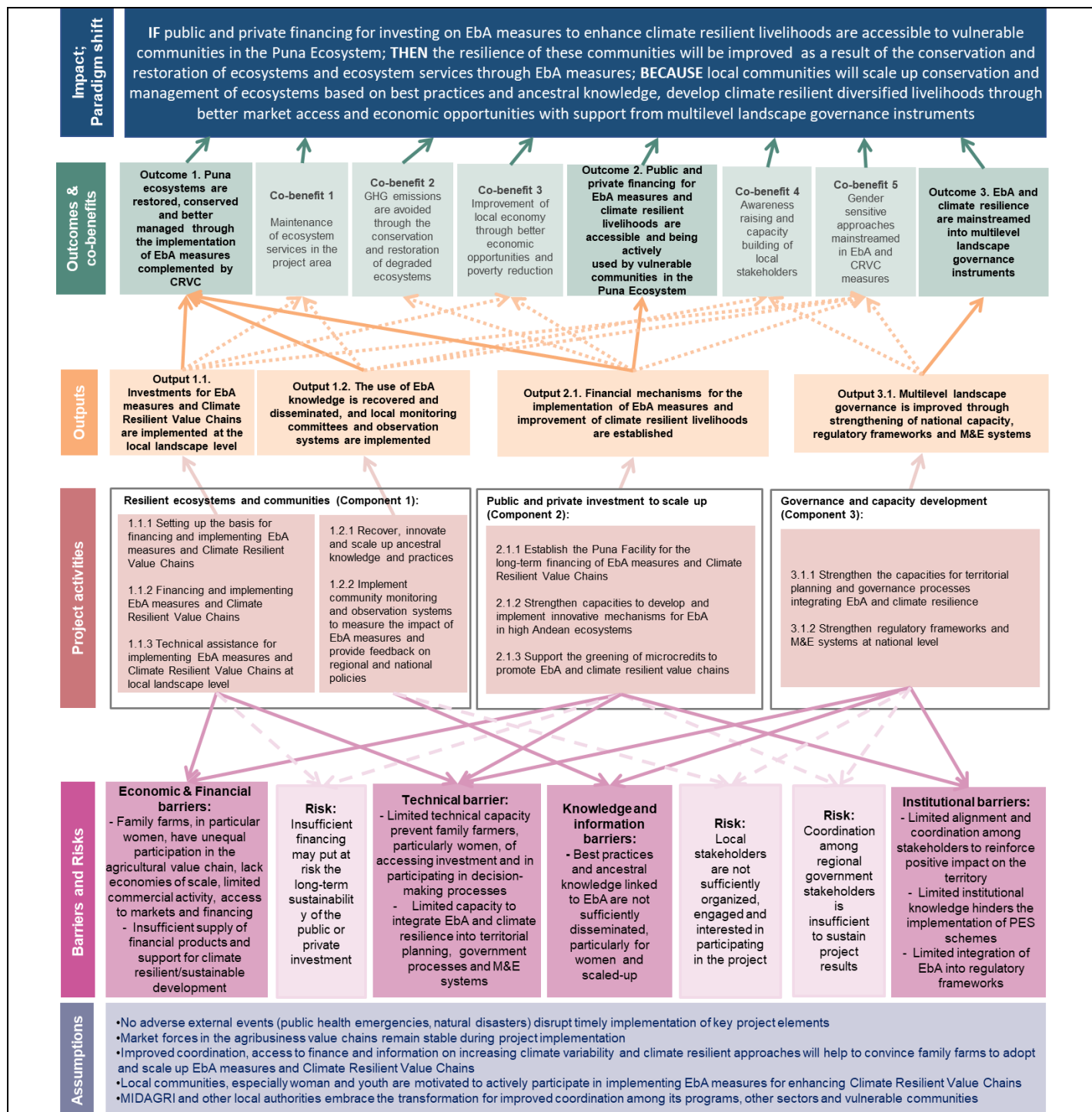
Project characteristics	Objective and results	Potential synergies
Title: Mountain EbA Budget: USD 3,276,637 Implementation period: 2012-2016 Donor: BMU Agency: IUCN, UNDP	Objective: Strengthen national, regional and local capacities to implement EbA measures and reduce community vulnerability, emphasizing mountain ecosystems. LINK Results: Improved water availability and management, as well as the rehabilitation of grasslands in Canchayllo and Miraflores, with knock-on benefits on biodiversity. Three prioritized EbA actions in three pilot sites: a) management of vicuñas for the extraction of animal fibre. b) Community management of natural grasslands, including livestock c) sustainable water resources management, including rehabilitation of ancestral water infrastructure and restoration of wetlands and grasslands. Strengthened local capacities and knowledge related to climate adaptation and the importance of ecosystem services. This includes park rangers and small landowners to municipal-level workers. Creation of interest groups and committees made up of academics and rural communities that has reinforced locally led action. Validity of EbA as an adaptation measure to be invested in at regional and national levels as well as “proof-of-concept”.	This project serves as a “pilot” for this GCF project. The proposed GCF project builds on the lessons learnt from this first project, specifically regarding the implementation of EbA measures for better water management and agriculture in the Andes, resulting in enhanced community resilience. The Puna Facility, established under Activity 2.1.1 of this project will also fund EbA activities. Moreover, in a similar manner to the UNDP/IUCN project, this GCF project’s component 1 proposes activities such as strengthening community based EbA measures and monitoring impacts.
Title: Seventh Operational Phase of the GEF Small Grants Programme in Peru (UNDP) Budget: 6,337,319 USD	Objective: To build socio-ecological landscape resilience in the Southern Andes in Peru through community-based activities for global environmental benefits and sustainable development. LINK	Both programmes fund EbA interventions to enhance ecosystem services with agricultural and economic benefits in mind. This GEF Small Grants Programme in the Peruvian Andes seeks to address very similar challenges through providing

GEF grant: 1,959,132 USD. Implementation period: 2021-2025 Donor: GEF Agency: UNDP	Results: The programme is in its Seventh Operational phase. The mid-term review for the Sixth Operational phase noted that the programme was especially successful in creating multi-stakeholder platforms called Strategic Landscape Platforms. These support participatory planning and adaptive management of landscapes, and are composed of 6-9 members, including community platforms, local authorities, and private sector. (UNDP, 2019).	small grants (5000-50,000 USD) for community-led projects in biodiversity conservation, improvement of ecosystem services, as well as improving agricultural value chains through soil and water conservation practices. The Puna Facility will fund similar activities to those of the GEF Small Grants programme. It is important to note, however, that the Puna Facility will fund EbA activities and climate-resilient value chains specifically if they have a climate adaptation purpose, rather than a conservation/agricultural development one.
Title: Peruvian Amazon Bio Business Facility Budget: 10,000,000 USD GCF grant: 8,972,400 Implementation period: 2022-2032 Donor: GCF Agency: PROFOFONAPE	Objective: The Amazon Eco Bio Business Facility (EBBF) (FP193) will provide effective climate change mitigation outcomes by investing in eco bio businesses (EBBs) supporting the sustainable management and conservation of Peruvian forests. LINK Results: This project is still under implementation, but the EBBF aims to contribute to enhancing carbon stocks and avoiding the emission of 3,8 million tonnes of CO₂, with an estimated cost of 263 USD / t CO₂eq during its lifetime, and support REDD+ infrastructure in Peru.	This facility, managed by Profonanpe, provides repayable grants to EBBs that manage and conserve Peruvian forests in the Amazon, and provide mitigation benefits through enhancing carbon stocks. The Puna Facility, established under Activity 2.1.1 of this proposed project, will invest in Climate Resilient Value Chains through the development of businesses that increase SHAP communities' resilience to climate change, and/or reduce the negative impacts of existing livelihood strategies on the puna ecosystem. The Puna Facility will also invest in EbA measures to adapt to climate change impacts in the intervention area. Both facilities are managed by Profonanpe, which will have gained capacity and knowledge on best practices and management through implementing the EBBF. However, the Puna Facility will fund climate adaptation activities in the Puna biome.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

B2a.1. Theory of change and Project Goal

77. The following diagram shows the ToC for the project, which illustrates the change pathway and interactions of different elements of the intervention logic proposed by the project. As shown in the theory of change diagram, the project has established an integrated self-supporting approach, addressing the identified barriers and risks.



78. **Goal:** The aim of the project is to increase the resilience of communities, ecosystems and livelihood in the High Andean region affected by climate change through the implementation of EbA measures, complemented by CRVC, the establishment of public and private financing for investing in these measures and the support from multilevel landscape governance instruments.
79. **Goal Statement:** IF public and private financing for investing in EbA measures to enhance climate resilient livelihoods are accessible to vulnerable communities in the Puna Ecosystem, THEN the resilience of these communities will be improved as a result of the conservation and restoration of ecosystems and ecosystem services through EbA measures, BECAUSE local communities will scale up conservation and management of ecosystem based on best practices and ancestral knowledge, develop climate resilient diversified livelihoods through better market access and economic opportunities with support from multilevel landscape governance instruments.
80. The ToC diagram, goal and goal statement respond to a series of activities, outputs and outcomes (described in detail in section B.3) that addresses the barriers and risks described below.

B.2a.2. Identified Barriers

81. The barriers that the project will address in the prioritized districts in the SHAP region, including economic, financial, technical, institutional, knowledge and information, are:

Economic and financial barriers

- i) **Barrier 1: Family farms have unequal participation in the agricultural value chain, lack economies of scale, limited commercial activity, access to markets and financing.** Excessive fragmentation in agricultural production and supply creates barriers to market entry. Limited collaboration among agricultural producers weakens their ability to negotiate effectively for input procurement and product sales, thereby detrimentally impacting the seamless connection between agricultural producers and markets (Hernández Calderón, 2006). According to Cannock et al (2016), the high transaction costs derived from the remoteness and high geographical dispersion of agricultural productive areas hinders communication and interaction between producers who could share information on input prices or sales channels. In a study conducted by Escobal (cited by Cannock et al, 2016), applied to a sample of potato producers in two districts of the department of Huancavelica, the author finds that transaction costs represent 50% of the value of sales and that these costs are particularly higher for small producers compared to larger ones. The rural population is being limited to the services offered by the formal financial system, which prevents them from investing and improving their production, thus limiting their access to markets. The regulated IFI Intermediary Financial Institutions perceive the agricultural sector as very high risk, since it presents conditions of high climate vulnerability, fragmentation and dispersion of the population itself and mainly of the economic units that generate income, which operate in small-scale volume with low levels of technology applied to the production process with little specialized technical personnel, among other factors (PP121 MIDAGRI 2021). Potential clients also face limited financial literacy and inadequate loan terms, as well as generally high financial transaction costs and limited collateral (Sotomayor, Talledo, & Wong, 2018).
- ii) Among camelid producers, the majority of alpacas and llamas in the region, around 85%, are owned by small-scale farmers. These farmers usually possess less than 100 animals each within mixed herds. Also, the South American camelid fiber value chain demonstrates inequality as its commercialization structure is characterized by high intermediation, maintained by traditional relations that limit and condition the price and negotiation terms to the detriment of the producers. The supply is dispersed and in small volumes, and producers seldom have the negotiation capacity to present a collective offer due to their lack of organization (Moya & Torres, 2008). In the case of rural women, there is generally access to lower quality farmland, often in marginal areas, where land is mainly cultivated for household consumption. Rural women raise domestic animals, process primary products or sell their small surpluses in local markets, while men's participation in productive activities is more closely linked to more competitive and profitable value chains⁴⁰. In addition, another important gender gap in the project's region is related to land ownership and/or tenure, with women owning only 30.8% of farmland in Peru (FAO 2018).
- iii) Inequalities in income generation among women and men is also present in the target area of the project. According to the National Agricultural Census (2012), there are more women with no income of their own. For example, in Apurímac 32.8% of women no have income in their own compared to only 12.1% of men. In the case of Arequipa women without income of their own (38.1%) while men (22.3%). In Cusco women (30.5%) and men (18.2%). Finally in Puno, women 25.1% while men 17.4%. In the case of women in agricultural occupations, 37.6% are unpaid family workers and only 46.3% of women in agricultural occupations received cash income.
- iv) Specifically, in terms of the project beneficiaries IPLCs in their various forms of organization face differentiated barriers in regard to access to financing according on how they are organized which includes:
 - a) For communities, producers' associations and cooperatives in the process of formalisation
 - i. Lack of economies of scale. Their production is mainly for self-consumption, because their productivity is very low, and their production is very small.
 - ii. Limited commercial activity. The small part of their production that they manage to sell is sold through middlemen who come to their farms. Consequently, they lack control over pricing, generally resulting in very low prices.
 - iii. Limited access to finance. These organisations don't have access to formal credits because they don't meet the legal and commercial requirements to apply for a credit lines. In some cases, they receive advanced payments from middlemen and are then penalised with the price of their products.
 - b) Micro and/or Small Enterprises, community enterprises and cooperatives with commercial capacities and clear target markets and formalized producers' organization
 - i. Partially self-consumption, but with an emerging group of clients. These organisations need to increase the number of customers by finding niche markets that value the origin of their products.

⁴⁰ retos y oportunidades de la agricultura familiar en Perú : care Perú

- ii. The organisation collects the products of its members and sells it together. Because of that, they can access better prices, and sometimes can add value to their production. However, they still need to improve their production process and quality controls to maintain their customers and expand them sustainably.
 - iii. Limited access to finance. Their limited experience, low productivity, and sales figures hinder their easy access to credit based on their technical capabilities, despite being legally eligible to do so.
- i) **Barrier 2: Insufficient supply of financial products and support for climate resilient/sustainable development.** Peru's microfinance landscape comprises a limited number of institutions that provide loans to small and medium-sized businesses pertinent to this project, particularly those catering to smallholder farmers and communities in the High Andean areas. Financial institutions such as Financiera Confianza, Mibanco, Financiera Compartamos, and Caja Rural Los Andes are notable due to their alignment with the project's target borrowers, their presence in the project regions, and their emphasis on rural lending. Caja Rural Los Andes and Caja Rural Inka Sur also extend microcredits to the rural and peasant population. The Arequipa - Fondesurco Savings and Credit Cooperative is also active in the region, offering small "green loans" tailored for agriculture and livestock purposes. The state-owned bank Agrobanco, which oversees Agroperu, an independent agricultural lending fund, also holds significance due to its rural and sector-specific orientation. Likewise, Cofide, Peru's development bank, also plays a role in this landscape. Regarding to access to loans, in 2012 only between 5% and 20% of men farmers took out a loan; and for women farmers it is even lower, between 4% and 12% (INEI 2012). The extreme shortage of capital of small producers does not allow them to invest in infrastructure (e.g. planting and harvesting water, sheds, greenhouses, road improvements), or to introduce new cultivation and grazing practices (e.g. sprinkler irrigation for pasture cultivation, mobile roosts to facilitate herd management) for the benefit of their families, and particularly women.

Technical barriers

- i) **Barrier 3: Limited technical capacity prevents family farmers from accessing investment and in participating in decision-making processes.** The legislation and norms on the participatory budget allow the involvement of the communities, both in the formulation of the budget and in the planning of the territory; however, their participation is limited, due to, among other causes, ignorance of the legislation, lack of interest and difficulty of the institutions responsible for calling and bringing communities together to inform them, discuss, negotiate and put their needs on the agenda of territorial management (Doughty, 2014). In terms of women participation on decision making process the tendency is low whether at national, sub-national or community level. This geographical dispersion is a very serious limitation for the associativity of the rural communities, which, in turn, prevents them from accessing technical and financial assistance offered to increase productivity by the MIDAGRI programs in the area, conditional on loans, which imply compliance with multiple requirements, unattainable for farmers, due to the complex procedures. This prevents small producers, particularly women, from participating in the camelid value chain, mainly higher up the chain in accessing international markets, as well as technological advances for highly valued products such as vicuña fibre (MIDAGRI, 2023). In the case of Andean women, they struggle with a position of gender inferiority regarding access and control over natural resources such as land, water, production and commercialisation of agricultural products. This translates into under representation in different type of organisations. Despite this, women actively participate in the management of key natural resources such as water and land. While men concentrate their participation and decision making in agriculture and husbandry, women are responsible for water supply for subsistence farming, food preparation, cleaning and hygiene of family members, including the ill and disable (Carrillo & Remy, 2022).
- v) In addition to the economic and financial barriers mentioned above the project beneficiaries face differentiated barriers in regard to organizational barriers and technical capacities which includes:
- c) For communities, producers' associations and cooperatives in the process of formalisation
 - i. **Organizational barriers:**
 - a. Organizations without internal organisational and commercial capacities and clear target markets. This means lack of business organizations. Such organisations do not have commercial purposes; therefore they cannot directly commercialize any product for the purpose of distributing profits among their members, nor can they set aside profits for the purpose of repayment.
 - b. Sell individually through middlemen. They usually organise themselves to work on common goods (irrigation canals, schools, etc.), but they lack organisational structures to sell together, coordinating the marketing and distribution of commercial gains.
 - ii. **Limited technical capacities:** Community members are generally unaware of the types of requirements needed to apply to credit lines. Furthermore, due to the individual production, the

training provided by different sources fail to reach all members of the communities, putting the most vulnerable at a disadvantage by denying access to technological improvements.

- d) Micro and/or Small Enterprises, community enterprises and cooperatives with commercial capacities and clear target markets and formalized producers' organization

iv. Organizational barriers:

- a. Organizations with commercial capacities and clear target markets. These organisations have commercial purposes, so they are accredited to generate profits, distribute profits, and make payments on a joint basis.
- b. They have established a minimum level of internal entrepreneurial organisation for their business' operations, marketing and distribution of their financial benefits.

- v. Limited technical capacities:** These organisations are initiating the processes to improve their productivity by providing training to their members (cooperatives) and workers (micro-enterprises). However, they face the difficulties of physical distances, insufficient resources, and little knowledge of the market.

- ii) **Barrier 4: Limited capacity to integrate EbA and climate resilience into territorial planning, government processes and M&E systems.** Different stakeholders do not possess systematic methods to plan EbA and climate resilient practices into policies, plans, budgets and investments. MIDAGRI extension services do not register or keep track of progress in climate change adaptation impact of their training or structural interventions, so there is no way to monitor and evaluate progress in adaptive capacity neither at the Ministry nor at local levels. Traditional methods for decision-making based on observation and ancient knowledge exist but are being lost. On top, in some instances, they are no longer relevant in the new climate context and are often not well understood by regional and national decision-makers. According to Cerdan et al (2022)⁴¹, the main challenges that must be overcome to optimize EbA investments include reducing the time of the budget allocation process from public funds to EbA investments. Sometimes, it depends on the political decision of both regional and local authorities. On the other hand, it was evidenced that only 16% of the project investments include a monitoring system, for example hydrological, what makes more complex to verify the impacts that the interventions on the quantity and quality of water resources, among other variables. Finally, the incorporation of the gender approach is a pending and necessary task in the investment projects, it is necessary to include conceptual frameworks and formal tools of the State that incorporate the needs and women's visions from the formulation, implementation and evaluation of projects.

Knowledge and information barriers

- i) **Barrier 5: Best practices and ancestral knowledge linked to EbA are not sufficiently disseminated and scaled-up.** In this field, a very important barrier refers to the fact that the best practices and ancestral knowledge linked to EbA are not disseminated or expanded sufficiently. Ancestral practices and knowledge on EbA are being lost due to: (i) the processes of emigration of young people and (ii) the lack of appreciation and recognition of the value of such knowledge and technologies by national, regional, and local institutions. Climandes project carried out a survey to identify the existence of a common practice among the farmers, named here "Ancestral Visual Indicator". It consists of a hereditary climate-related knowledge, based on the observation of the sky. This usage is mainly spread among old farmers (older than 70 years), for whom it has higher credibility than the information provided by SENAMHI. In contrast, young farmers tend not to adopt this ancestral knowledge, refusing non-modern and for them unreliable habits⁴². The rural areas of SHAP have experienced a significant decrease in population due to migration. While some highland communities migrate in anticipation or in reaction to hazards, others stay in spite of the challenges/dangers. The vulnerability of highlanders is often based on a function of the extent, quality, and location of household resources, including land and livestock, and demographic factors such as family size, age, and health.⁴³ Outmigration is deteriorating family's structures, culture, traditions and local knowledge⁴⁴. López-i-Gelats et al. (2015) found that migration may weaken the adaptive capacity as traditional knowledge and labour resources are being lost.

⁴¹ Eliana Cerdán Estrada, Mia Smith, Mirtha Camacho Hernández y Claudia V. Grados Bueno. Recuperando la fluidez: Estado de la inversión en acciones en infraestructura natural para la seguridad hídrica en el Perú, 2022.

⁴² G. Rosas, S. Gubler, C. Oria, D. Acuña, G. Avalos, M. Begert, E. Castillo, M. Croci-Maspoli, F. Cubas, M. Dapozzo, A. Díaz, D. van Geijtenbeek, M. Jacques, T. Konzelmann, W. Lavado, A. Matos, F. Mauchle, M. Rohrer, A. Rossa, S.C. Scherrer, M. Valdez, M. Valverde, G. Villar, E. Villegas, (2016). Towards implementing climate services in Peru – The project CLIMANDES, Climate Services, Volume 4, 2016, Pages 30-41.

⁴³ Oliver-Smith, A. (2014) Climate change adaptation and disaster risk reduction in highland Peru. In: Adapting to Climate Change: Lessons from Natural Hazards Planning (B.C. Glavovic and G.P. Smith, eds.). Springer, Dordrecht, Netherlands, pp. 77–100.

⁴⁴ Bergmann, J., K. Vinke, C.A. Fernández Palomino, C. Gornott, S. Gleixner, R. Laudien, A. Lobanova, J. Ludescher and H.J. Schellnhuber (2021). Assessing the Evidence: Climate Change and Migration in Peru. Potsdam Institute for Climate Impact Research (PIK), Potsdam, and International Organization for Migration (IOM), Geneva.

As the younger population of the SHAP has moved to urban areas, 40 % of the population remaining are now over 54 years old⁴⁵. In addition, gender traditional norms in the Andes often prevent women from achieving the same level of visibility as men, or benefit from the same opportunities and recognition, despite their deep knowledge of agriculture, water and ecosystems. For example, women alpaca herders in the Andes continue to be side-lined in technical discussions at local and national level around animal health and pasture management, the same as, in discussions about agrifood systems and crop diversification⁴⁶.

Institutional barriers

- ii) **Barrier 6: Limited alignment and coordination among stakeholders to reinforce positive impact on the territory.** The weak intersectoral and intergovernmental articulation and coordination is one more reason for policies in Peru to fail. According to A. Fernández (2017)⁴⁷, the Peruvian State presents a complex and hierarchical structure organizational that limits the channels of coordination between ministries, levels of governments, among other institutions in the implementation of public policies which generates the appearance of critical knots that make the correct continuity of the policy impossible. The implementation of public policies in Peru is the weakest link in the decentralization process and public management, due to the ineffectiveness of the instances, platforms or spaces of articulation and coordination where the government authorities of the three levels of government (national, regional and local) have not achieved consensus and/or agreements to establish a shared vision of territorial development in search of closing economic and social inequalities⁴⁸. Despite the existing normative on participatory involvement of communities in budget and territorial planning, community's involvement is limited due to the lack of awareness, weak capacities to dialogue, negotiate and put their needs in the territorial agenda.
- iii) **Barrier 7: Limited institutional knowledge hinders the implementation of PES schemes.** From the institutional point of view, barriers are related to the roles of regional and local governments in prioritizing investments to improve the impact of the actions developed within the framework of the hydrological MERESE stand out. For the specific case of the water utilities (EPS⁴⁹), they need to adapt their organizational structure to address the challenges present in the design of the MERESE and the execution of the funds raised, as well as to incorporate conservation and recovery of water sources in their institutional culture and strategies (Tristan et al 2021). Technical barriers are mainly related to the limited knowledge to understand the functioning and management of MERESE and therefore to design projects oriented to the conservation, recovery and management of ecosystems. On the other side, limited capacities and methodologies to monitor and evaluate the impacts of the green investments on the hydrologic ecosystem services (HES); the lack of information on which activities are effective for the recovery, conservation and sustainable use of HES source ecosystems; and on methodologies to evaluate the return on investment for retribuyentes⁵⁰, such as EPS (Tristan et al 2021).
- iv) **Barrier 8: Limited integration of EbA into regulatory frameworks.** Although there is a State Policy No. 34 of Territorial Planning and Management, there is still no Land Management Law, which establishes the necessary instruments to channel efforts to regulate the occupation and appropriate use of the territory, establishing objectives and actors responsible for its implementation, monitoring and vigilance for compliance (Ojea, E, 2015). Another aspect is the absence of policies for the recognition and valuation of ecosystem services, which establish the responsibility to protect and conserve the ecosystems that produce them, as well as the way to reward the work of restoration and conservation of ecosystems. the same. An advance in this sense is Law 30215, Mechanisms for Retribution for Ecosystem Services - MERESE, (LSE, 2014) which is fulfilled through the companies that provide water and sanitation services and must be used for the restoration and conservation of ecosystems that provide hydrological services, among others. The progress achieved with this law opens a new range of technical and social needs and limitations that must be overcome. Since MERESE resources are insufficient for the recovery needs of ecosystem services, it is necessary to design regulatory frameworks which facilitate other public actors could be engaged and committed.

B.2a.3. Alignment of proposed outcomes, outputs and activities with identified barriers

82. The identified barriers will be addressed by the three project outcomes and related outputs and activities. Specifically, in the following paragraphs a clear explanation on how each barrier will be addressed by the project outputs and activity is provided. Table B.2a.1 provides a summary of the linkages between barriers addressed and

⁴⁵ Instituto Nacional de Estadística e Informática (INEI) - IV Censo Nacional Agropecuario 2012.

⁴⁶ Caine, A (2021) "Who Would Watch the Animals?": Gendered Knowledge and Expert Performance Among Andean pastoralists. Culture, Agriculture, Food and Environment ISSN 2153-9553, eISSN 2153-9561.

⁴⁷ Fernández, A. (2017). Análisis de la gestión articulada orientada a reducir la desnutrición crónica y la anemia infantil, distrito de Sancos – Ayacucho. [Trabajo presentado para obtener el grado académico de Magister en Gestión pública]. Universidad del Pacífico.

⁴⁸ Chessman, Yuri (2022). The intergovernmental articulation and coordination mechanisms of National and Territorial policies in Peru.

⁴⁹ EPS refers to the sanitation service providers, for more information, please see <https://www.sunass.gob.pe/prestadores/empresas-prestadoras/>

⁵⁰ The retribuyentes is the natural or legal person, public or private, who obtains an economic, social or environmental benefit, and rewards contribuyente for the ecosystem service received.

corresponding outcome/outputs and activities. The unequal participation of family farms, in particular women, in the agricultural value chain, lack economies of scale, limited commercial activity, access to markets and financing (Barrier 1) will be addressed by Output 1.1 (the implementation of EbA measures at the local landscape level) and Output 2.1 (Financial mechanisms for the implementation of EbA measures established). The insufficient supply of financial products (Barrier 2) will be address by Outputs 1.1 as well as 2.1.

83. The limited technical capacity of farmers, particular of women (Barrier 3), will be addressed by technical assistance for implementing EbA measures (Activity 1.1.3), the recovery, innovation and scale up of ancestral knowledge and practices (Activity 1.2.1) and strengthen capacities to develop and implement innovative mechanisms for EbA (Activity 2.1.2). The limited capacity to integrate EbA and climate resilience into territorial planning, government processes and M&E systems (Barrier 4) will be addressed by implementing community monitoring and observation systems to measure the impact of EbA measures and provide feedback on regional and national policies (Activity 1.2.2), strengthening capacities to develop and implement innovative mechanisms for EbA with a gender responsive approach in high Andean ecosystems (Activity 2.1.2) and the improvement of multilevel landscape governance through strengthening of national capacity, regulatory frameworks and M&E systems (Output 3.1).
84. The limited dissemination, particularly limiting women and scale-up of best practices and ancestral knowledge linked to EbA (Barrier 5), will be addressed by Output 1.2. (use of EbA knowledge is recovered and disseminated, and local monitoring committees and observation systems are implemented) and trough strengthening the capacities for territorial planning and governance processes integrating EbA and climate resilience (Activity 3.1.1).
85. The limited alignment and coordination among stakeholders (Barrier 6) will be addressed by activity 3.1.1. The limited institutional knowledge for the implementation of PES schemes (Barrier 7) will be addressed by activity 2.1.2. Finally, the limited integration of EbA into regulatory frameworks (Barrier 8) will be addressed by activity 3.1.2.

Table 5. Overview of Project's outcomes, outputs and activities and their link to barriers addressed

Outcome	Output	Activity	Sub-activities	Barriers addressed
1	1.1.	1.1.1	1.1.1.1	• Barrier 1: Smallholder farmers and associations, in particular women, have unequal participation in the agricultural value chain, lack economies of scale, limited commercial activity, access to markets and financing.
			1.1.1.2	
		1.1.2	1.1.2.1	• Barrier 1: Family farms, in particular women, have unequal participation in the agricultural value chain.
			1.1.2.2	• Barrier 2: Insufficient supply of financial products and support for climate resilient/sustainable development.
		1.1.3	1.1.3.1	• Barrier 3: Limited technical capacity, particularly of women.
			1.1.3.2	
			1.1.3.3	• Barrier 1: Family farms, in particular women, have unequal participation in the agricultural value chain. • Barrier 3: Limited technical capacity, particularly of women.
	1.2.	1.2.1	1.2.1.1	• Barrier 3: Limited technical capacity, particularly of women. • Barrier 5: Best practices and ancestral knowledge linked to EbA are not sufficiently disseminated and scaled-up.
			1.2.1.2	
			1.2.1.3	• Barrier 5: Best practices and ancestral knowledge linked to EbA are not sufficiently disseminated, particularly for women and scaled-up.
			1.2.1.4	
		1.2.2	1.2.2.1	• Barrier 4: Limited capacity to integrate EbA and climate resilience into territorial planning, government processes and M&E systems.
			1.2.2.2	
		1.2.2.3	1.2.2.3	• Barrier 4: Limited capacity to integrate EbA and climate resilience into territorial planning, government processes and M&E systems. • Barrier 5: Best practices and ancestral knowledge linked to EbA are not sufficiently disseminated and scaled-up.
2	2.1.	2.1.1	2.1.1.1	• Barrier 2: Insufficient supply of financial products and support for climate resilient/sustainable development.
			2.1.1.2	
			2.1.1.3	• Barrier 1: Smallholder farmers and associations, in particular women, have unequal participation in the agricultural value chain, lack economies of scale, limited commercial activity, access to markets and financing. • Barrier 2: Insufficient supply of financial products and support for climate resilient/sustainable development.
		2.1.2	2.1.2.1	• Barrier 3: Limited technical capacity, particularly of women, prevents family farmers of accessing investment and in participating in decision-making processes. • Barrier 4: Limited capacity to integrate EbA and climate resilience into territorial planning, government processes and M&E systems. • Barrier 7: Limited institutional knowledge hinders the implementation of PES schemes.
		2.1.3	2.1.3.1	• Barrier 2: Insufficient supply of financial products and support for climate resilient/sustainable development.
			2.1.3.2	• Barrier 1: Smallholder farmers and associations, particularly women, have unequal participation in the agricultural value chain, lack economies of scale, limited commercial activity, access to markets and financing. • Barrier 2: Insufficient supply of financial products and support for climate resilient/sustainable development.
3	3.1.	3.1.1	3.1.1.1	• Barrier 4: Limited capacity to integrate EbA and climate resilience into territorial planning, government processes and M&E systems • Barrier 5: Best practices and ancestral knowledge linked to EbA are not sufficiently disseminated and scaled-up.
			3.1.1.2	• Barrier 4: Limited capacity to integrate EbA and climate resilience into territorial planning, government processes and M&E systems. • Barrier 6: Limited alignment and coordination among stakeholders to reinforce positive impact on the territory. • Barrier 5: Best practices and ancestral knowledge linked to EbA are not sufficiently disseminated, particularly for women and scaled-up.
			3.1.1.3	• Barrier 4: Limited capacity to integrate EbA and climate resilience into territorial planning, government processes and M&E systems.
		3.1.2	3.1.2.1	• Barrier 8: Limited integration of EbA and climate resilience approaches into regulatory frameworks.
			3.1.2.2	• Barrier 4: Limited capacity to integrate EbA and climate resilience into territorial planning, government processes and M&E systems. • Barrier 8: Limited integration of EbA and climate resilience approaches into regulatory frameworks.

B.2a.4. Project co-benefits.

86. The project will generate several co-benefits including:

- Maintenance of ecosystem services in the project area.
- GHG emissions are avoided through the conservation and restoration of degraded ecosystems.
- Improvement of local economy through better economic opportunities for poverty reduction.

- iv. Awareness raising and capacity building of local stakeholders.
- v. Gender responsive approaches mainstreamed in EbA and CRVC measures.

87. See section D.3 (sustainable development potential) for further details on the identified co-benefits, and section E.5 for the co-benefit indicators.

B.2a.4. Risks and Assumptions.

88. **Risks.** Project risks related to technical and operational, credit, governance and legal risks, among others, are described in detail in Chapter F, and environmental and social risks are described within Chapter G.1. Amongst others it includes:

- **Insufficient financing may put at risk the long-term sustainability of the public or private investment.** Insufficient public or private financial flows may put at risk the long-term sustainability of the project. Market intake for EbA options, especially those promoted via microfinancing may be insufficient. Fluctuations in the market value of cash products (South American camelid fiber, quinoa) may limit economic activity and entrepreneurship.
- **Local stakeholders are not sufficiently organized, engaged and interested in participating in the project.** Participation from local stakeholders in the project activities may not be as high as expected due to the lack of interest.
- **Coordination among regional government stakeholders is insufficient to sustain project results.** Coordination among government stakeholders is insufficient and the lack of interest and ownership of the project may limit the sustainability of project results.
- **Risk of continuity due to political instability.** The project is constructed on the assumption of sustained commitment from the Government of Peru to fulfil its Nationally Determined Contribution (NDC) goals and from MIDAGRI to continue its open and proactive engagement. Political instability in Peru may affect project continuity.
- **Maladaptation and degradation of ecosystems and social structures.** Improved economic opportunities in the SHAP may lead to increased pressure on ecosystems (e. g., number of cattle heads increases as a result of business-oriented approaches). Moreover, the promotion of certain EbA's such as agrotourism may result in maladaptation. Social structures may continue to deteriorate because of increased ecosystem degradation.
- **Difficulties and delays in unlocking technical and financial support.** Access to MIDAGRI programs as well as technical and financial support from the Puna Facility will be on a voluntary basis. This may limit the ability of the project to meet its targets or have sufficient concentration of interventions to show tangible results. Unlocking of public financing via MERESE schemes may require lengthy negotiations beyond the life of the project.

89. **Assumptions.** The project design is based on the following assumptions (assumptions are further described in sections E.3 and E.5 for each relevant project indicator):

- No adverse external events (public health emergencies, natural disasters) disrupt timely implementation of key project elements.
- Market forces in the agri-business value chains remain stable during project implementation.
- Improved coordination, access to finance and information on increasing climate variability and climate resilient approaches will help to convince family farms to adopt and scale up EbA measures and Climate Resilient Value Chains .
- Local communities, especially woman and youth are motivated to actively participate in implementing EbA measures for enhancing Climate Resilient Value Chains.
- MIDAGRI and other local authorities embrace the transformation for improved coordination among its programs, other sectors and vulnerable communities.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

90. Each of the project Outcomes identified in section B.2(a) map to the GCF results area as follows:

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities,	MRA 4 Forestry and land use	ARA 1 Most vulnerable	ARA 2 Health, well-being, food	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and

			industries, appliances		people and communities	and water security		ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☐	☒
Outcome 2	☐	☐	☐	☐	☒	☒	☐	☒
Outcome 3	☐	☐	☐	☐	☒	☒	☐	☒

91. The following co-benefits identified in section B.2(a) map to the following corresponding category:

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
<u>Co-benefit 1</u> Maintenance of ecosystem services in the project area	☒	☒	☐	☐	☐	☐
<u>Co-benefit 2</u> GHG emissions are avoided through the conservation and restoration of degraded ecosystems	☐	☐	☐	☐	☐	☒
<u>Co-benefit 3</u> Improvement of local economy through better economic opportunities and poverty reduction	☐	☒	☒	☐	☐	☐
<u>Co-benefit 4</u> Awareness raising and capacity building of local stakeholders	☐	☒	☐	☐	☐	☐
<u>Co-benefit 5</u> Gender responsive approaches mainstreamed in EbA and CRVC measures	☐	☒	☐	☒	☐	☐

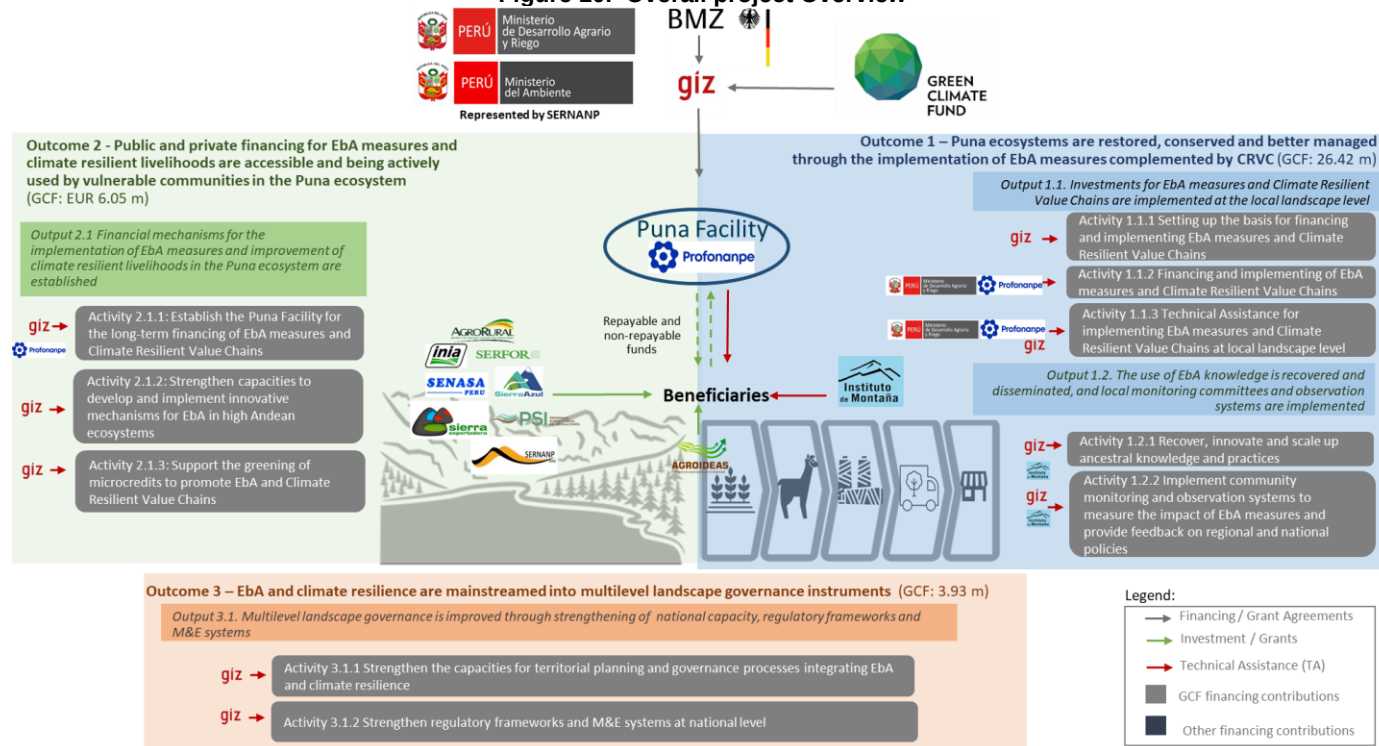
B.3. Project/programme description (max. 2500 words, approximately 5 pages)

92. Overall, the project will support MIDAGRI in promoting a paradigm shift in the management of Puna ecosystems (wetlands, peatlands and grasslands) and productive practices to increase climate resilience of some of the most vulnerable groups in the country which include rural farming and herding communities depending on these ecosystems for their livelihoods. Puna ecosystems and associated livelihoods are threatened by increased temperatures, droughts, precipitation variability, and overall decrease in water availability. The project will address identified financial, ecological, social, institutional, and regulatory barriers (as described in section B.2 (a)) to improve structural conditions that will enable rural communities in the High Andes to use and manage Puna ecosystems in a more sustainable and climate-smart way, offering alternatives to reduce unsustainable practices contributing to ecosystem degradation and magnifying climate change impacts. This is consistent with the need for a comprehensive approach on Ecosystem-based Solutions and a focus on the implementation and upscaling of ecosystem-based adaptation (EbA) measures⁵¹.
93. In order to achieve this, the project consists of three components (1) Resilient ecosystems and communities, (2) Public and private investments for scaling up EbA measures aligned and leveraged and (3) Multilevel territorial governance for the incorporation of EbA measures which are aligned with the three outcomes, as depicted below. The overall project design is presented in detail below⁵².

⁵¹ [Ecosystem-based Adaptation](#), [Nature-based Solutions](#), and [Natural Climate Solutions](#) are similar concepts describing slightly different approaches of using nature or natural infrastructure to tackle the climate problem. The focus on Ecosystem-based Adaptation stems from the experience gathered in the country and alignment with current national policies. The terms, however, may be used interchangeably to describe the ancestral practices being recovered through the project, which have adaptation benefits and mitigation co-benefits.

⁵² For more information on the description of the project design please see the activity sheets in section 8.6 of the Feasibility Study (Annex 2a).

Figure 10. Overall project Overview



B.3.1. Outcome 1: Puna ecosystems are restored, conserved and better managed through the implementation of EbA measures complemented by CRVC:

94. This outcome will promote resilient Puna ecosystems and value chains by financing and co-financing the implementation of climate-focused investments led by IPLCs at local landscape level. By implementing community-based investments on the ground, the aim is (i) to maintain or improve the provision of Puna ecosystem services for climate resilience of the Indigenous Peoples and Local Communities and (ii) to strengthening Climate Resilient Value Chains that are dependent on and impacting on those ecosystems. iii) to monitor and evaluate the impact of the implementation of EbA measures iv) to nurture the dialogue between and within Indigenous Peoples and Local Communities to recover ancestral knowledge from older women and men to younger generations and to disseminate it and scaled up into Local Initiatives. A series of technological packages, trainings, information materials and communities' exchanges will be implemented to co-produce knowledge and foster local innovation that will result in investment on the ground. The outcome will enhance access to information obtained at local level, which is expected to trigger a change in behaviour towards conservation of ecosystems and their services and the potential of developing Climate Resilient Value Chains, through two main outputs:

- Investments for EbA measures and Climate Resilient Value Chains are implemented at the local landscape level
- The use of EbA knowledge is recovered and disseminated, and local monitoring committees and observation systems are implemented.

Activities in this outcome will address the barrier of limited economic opportunities by supporting the development of Climate Resilient Value Chains and enhanced entrepreneurship in the SHAP region.

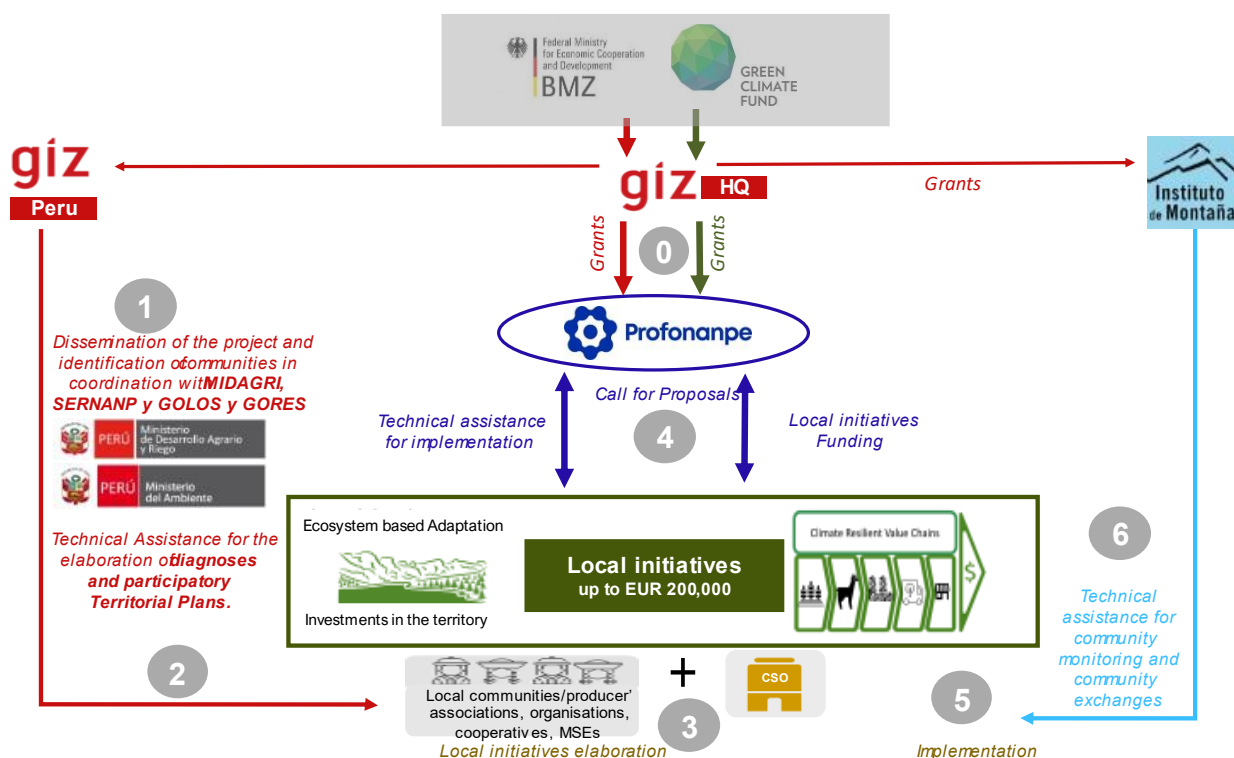
Output 1.1. Investments for EbA measures and Climate Resilient Value Chains are implemented at the local landscape level:

95. The project aims at supporting investments led by IPLCs for resilient ecosystems at local landscape level and strengthening Climate Resilient Value Chains with access to markets. This output will address the urgent need of direct finance to reduce the degradation of ecosystems and reduced services that result from climate change and unsustainable land management practices and to redress the lost linkage between ancestral practices and EbA. At the same time addresses weak associativity, limited economic opportunities and low access to finance and markets for Climate Resilient Value Chains in the SHAP region. Therefore, this output will cover the whole cycle from setting up the basis for the financing and implementation of EbA and Climate Resilient Value Chains (CRVC), through the provision of financial support and up to the provision of technical assistance for the implementation EbA measures and CRVC. Part of the financial and technical support provided in this output will be implemented through a call for proposal (launched by a, through this project established, Facility (the Puna Facility, for more information please see sub-activity 2.1.1.1.), for the implementation of integrated Local initiatives with an a)

Ecosystem based adaptation component and a b) CRVC component⁵³. The Local initiatives supported within this output will therefore focus mainly on investments in EbA measures meant to protect and restore the puna ecosystem and therefore re-establish its ecosystem functions and services to enhance its resilience against the effects of climate change and in investments in CRVC specifically investments aimed at increasing the resilience of Puna relevant value chains against the effects of climate change. The financial support will be complemented by technical support to project beneficiaries who are primarily IPLCs in its different forms of organizations (e.g., peasant/local communities, producers' associations, cooperatives (formalized and or in the process of formalization), MSE, community enterprises, producers' organizations (hereafter project beneficiaries) who implement the Local initiatives in order to add value to their products and facilitate access to markets. In addition, public resources for public investment projects and to provide technical assistance will be provided by different MIAGRI's programs.

96. Figure 11 below shows the cycle of the Local initiative implementation which will be implemented within Output 1.1.

Figure 11. Local initiative implementation cycle



Activity 1.1.1. Setting up the basis for financing and implementing EbA measures and climate resilient value chains (CRVC):

97. This activity will focus on creating a communication strategy to develop awareness among the local stakeholders about the project and create the basis for the implementation of IPLCs led Local Initiatives that comprises EbA measures and CRVC at landscape and/or micro water basin levels that will be financed by the Puna Facility. In this regard, IPLCs in their different forms of organizations (communities, community enterprises, associations, MSEs, cooperatives) will be able to receive technical support in order to participate in the call for proposals. The process of selecting beneficiaries has two important stages. The first is described in sub-activity 1.1.1.1 in which the districts of the prioritized project area (58 districts) will be visited to identify potential beneficiaries interested in participating in the Call for Proposals mechanism. Those IPLCs and their different forms of organizations willing to participate then will receive technical support in sub-activity 1.1.1.2 to develop participatory community diagnostics and plans and they will need to propose Local Initiatives where they prioritize EbA measures and Climate Resilient Value Chains that contribute to their resilience. In sub-activity 1.1.2.1, the potential beneficiaries will apply their Local Initiatives to the Puna Facility (managed by Profonampe) to be evaluated based on a list of eligibility criteria that will be established in the Puna Facility Operations Manual. This activity includes the following sub-activities:

98. **Sub-activity 1.1.1.1. Inform and identify communities, associations, cooperatives and Civil Society Organisations interested in participating in the project (EE: GIZ):** This sub-activity aims to inform local authorities and ILPCs about all project activities and specially Puna facility criteria (refer to step 1 of Figure 11)

⁵³ The selection of the EbA measures and CRVC will be based on the eligibility list as shown in table 6 of the Operations Manual (Annex 21). And Pub+Ishi

within the 58 prioritised districts to identify potential project beneficiaries interested in participating. A gender and culturally sensitive strategy for beneficiary engagement and communication will be developed by a consulting service in coordination with the project's communication team, MIDAGRI, Profonanpe, Instituto de Montaña (IdM), and SERNANP. This strategy will comply with the different plans (Environmental and Social Management Plan (ESMP), Gender Action Plan (GAP) and Indigenous Peoples and Local Community Engagement Plan (IPLCEP)) developed based on the GCF Social and Environmental Safeguards and Gender and Indigenous Peoples Policies, as well as, on Free, Prior and Informed Consent (FPIC) principles to allow IPLCs to actively participate in the design, implementation and monitoring of their own Local Initiatives. Before engaging with communities, regional government and district municipality authorities will be contacted to identify specific local needs and priorities regarding climate change adaptation and avoid isolated or nonaligned actions. It will be also important to actively engage local authorities from the beginning to promote support and increase awareness for EbA funding and inclusion in planning instruments (sub-activity 3.1.1.3). To this end, the communication strategy will be in place, and it will be coordinated with municipalities, NGOs and local radio stations. At this stage, the eligibility criteria to apply to the Puna Facility (described in Operations Manual - Annex 21) will be disseminated and potential beneficiaries who meet the eligibility criteria will be encouraged by hired GIZ local promoters to prepare ideas. To promote the coordination with the water MERESE, awareness-raising actions carried out in the areas of water MERESE, explaining the benefits they bring to local development and the synergy with the Project. All Indigenous Peoples and Local Communities and their forms of organization such as local communities, producers' associations, cooperatives, community enterprises and/or MSEs who agreed to participate in the project will present to GIZ their signed community agreements giving their informed consent to participate in the project and an attached form for a Local initiative idea (including a simplified explanation of the climate problem and a proposal on how to solve the climate change problem/s based on the eligibility list of EbA measures and CRVC as shown in table 6 of the Operations Manual (Annex 21)). Once GIZ receives all the Local initiatives ideas, a team of experts from GIZ, MIDAGRI, SERNANP and local authorities will evaluate and select the best proposed ideas. Then, in activity 1.1.1.2, the territorial diagnostics and intervention plans will be developed to continue preparing the Local initiative proposals for the Puna Facility. It is expected that after implementing the sub-activity at least 100 Communities, associations, cooperatives and/or micro and small enterprises have been selected to continue the support in sub-activity 1.1.1.2, in order to start the process of preparing them to access funding to implement EbA measures and CRVC to recover ecosystems and ensure climate resilient livelihoods.

99. **Sub-activity 1.1.1.2. Development of site-specific climate diagnostics and preparation of participatory intervention plans (EE: GIZ):** This sub-activity aims to support project beneficiaries to identify and prioritise through a participatory process resilient measures and Climate Resilient Value Chains in their territories through comprehensive climate diagnostics and intervention plans (refer to step 2). In this context, Indigenous Peoples and Local Communities and their other forms of organization such as MSEs, community enterprises, cooperatives, producer's associations and/or organizations whose idea were selected (sub-activity 1.1.1.1) will receive support from local promoters or CSOs in the development of participatory site-specific climate diagnostics to not only better understand climate and non-climate threats to ecosystem and livelihoods but also social and economic aspects. IPLC's own concepts and perceptions will provide an empowering and culturally appropriate way for communities to assess climate change impacts on their territories and plan adaptation solutions that build on their traditional knowledge and may be combined with modern scientific findings. When developing diagnostics, it will be also important to identify the causes of vulnerability in the specific sites such as communal organization, gender roles and inequalities over resources uses and access to basic services, etc. As well as understand which are their livelihood strategies and how ecosystems and their services in its site support them. The specific areas in each locality to be intervened with EbA measures and CRVC will be defined in this sub-activity and will depend on case-by-case (local consultations, specific site diagnostics and viability). For the development of an intervention plan per community, prioritization of EbA and CRVC will be held through a comprehensive and participatory process, where respective members of each IPLCs will be broadly consulted, ensuring active participation of women, young and elderly people during design phase, which will need to be ultimately approved by the respective decision-making body. Along with this, the interested project beneficiaries in applying to the Puna Facility will already have the basis to prepare their Local initiatives to apply to the call for proposal. The interested project beneficiaries will receive in this regard support to be prepared to apply to the Puna Facility (refer to step 3). To continue fostering the participatory process, potential IPLCs beneficiaries must designate "local researchers" (composed by women and men), who will lead and facilitate the diagnostics and planification process inside its community and encourage the intergenerational dialogue among community members to recover men and women traditional knowledge, discuss their concerns and identify solutions together (activity 1.2.1). Finally, local researchers will establish the community monitoring committees supported in activity 1.2.2. In addition, in this sub-activity, interventions in EbA and CRVC by different public and private actors (e.g., MIDAGRI through its programmes, SERNANP, water utilities and others) will be identified in order to determine early on whether a coordination/synergy with the potential Local initiatives is needed in the case the Initiative is financed by the Puna Facility. The rationale behind is to seek for complementarity and avoid duplication in the same territory.

Activity 1.1.2. Financing and implementing of EbA measures and Climate Resilient Value Chains:

100. This activity will focus on promoting climate resilient Puna ecosystems and CRVC through the implementation of climate-focused Local initiatives at local landscape level. The activity will provide financial support for Local initiatives through the Puna Facility (established under Activity 2.1.1) and through public resources (MIDAGRI's programmes). The two types of financial support are complementary and contribute to a general objective of improving the climate resilience of the IPLCs in the target territories.
101. **Sub-activity 1.1.2.1. Implementation of Local initiatives financed by the Puna Facility⁵⁴ (EE: Profonanpe):** This sub-activity has the objective to channel GCF and BMZ funds through a competitive selection process (open call for proposals (see step 4 in Figure 11)) in the high Andes of Peru involving investments in EbA measures and CRVC. The disbursement for each Local initiative will be differentiated according to three sub-windows: (i) Non-Repayable Grant Sub-Window: for IPLCs with their different forms of organization (local communities, producers' associations and cooperatives in the process of formalization), applying with the support of local CSOs trusted by communities; (ii) Results-based Repayable Grant Sub-Window: for Micro and or/ Small Enterprises, community enterprises and cooperatives permitted by law with commercial capacities and clear target markets; and (iii) Agroideas Matchmaking Sub-Window: for formalized producers' organizations (small and/or medium) with commercial capacities and clear target markets, that are eligible for the MIDAGRI support programme Agroideas⁵⁵. The selection process for the Local initiatives from preparation and approval for call for proposals to the signing of Local initiatives Agreements is described in the Operations Manual (see Annex 21) section 5. Within the process of evaluation and selection, the Local initiative selection criteria and scoresheet will include a gender representation criterion weighted with 20% and a gender quota of at least 30% of the selected Local initiatives should either support a women's association, cooperatives, organizations and/or a women led enterprise in order to ensure the equal access between men and women of the funds. It is expected that after implementing this sub-activity, 127 Local initiatives Grant Agreements will have been signed (with a total value of up to EUR 14.4 million), for the three sub-windows and a total of 23,914 hectares would have been restored and/or conserved (see step 5 in Figure 11. The beneficiaries are expected to provide contributions of at least 25% of the value of the Local initiatives, which may be contributed to labour or from another source, such as water utilities.
102. The management and governance structure⁵⁶ of the Facility will include: Profonanpe as the main EE who will have the responsibility to manage the Puna Facility through the Puna Facility Management unit (established within sub-activity 2.1.1.1.), an Independent Technical Assessment Committee (in charge of shortlisting Local initiatives evaluation), the Project Steering Committee (PSC) (composed by MIDAGRI, MINAM and GIZ) and the Project Management Committee (PMC) who will revise and approve amongst others the final list of Local initiatives evaluated according to the established processes and criteria. More information on the Puna Facility in regard to the different sub-windows (including repayment conditions) and sustainability can be found in B.4.5 and B.6.1 respectively.
103. **Sub-activity 1.1.2.2. Financial support through MIDAGRI (EE: MIDAGRI):** In this sub-activity, MIDAGRI, through its programmes (AGRORURAL, Sierra Azul, Irrigation Subsector Program (PSI), (National Institute for Agrarian Innovation) INIA, Agroideas and National Forestry and Wildlife Service (SERFOR) and Agroperu Fund), will finance additional interventions in the SHAP area that in some cases will complement the 127 Local Initiatives financed by the Puna Facility (as in the specific case of the Agroideas Matchmaking Sub-Window) in the same territory and will contribute to increase the resilience of High Andean populations to climate change. These will comprise three types of public interventions: (i) Public investment projects, which, on the one hand, include adaptation interventions such as the recovery of degraded High Andean grasslands, sowing and harvesting of water and technical irrigation to improve the efficiency of water resource application. These interventions are implemented directly by MIDAGRI's programs: AGRORURAL, Sierra Azul and PSI. On the other hand, INIA and SERFOR will implement public investment projects that will contribute to research and technology transfer in High Andean livestock and sustainable use of the vicuña, among others. It is expected that at least eight (8) public investment projects are implemented directly by MIDAGRI. (ii) Agroideas Program Non-reimbursable funds for business plans, which require co-financing from producers, to strengthen the adaptive capacities of climate change resilient value chains including guinea pigs, camelids and High Andean crops. This includes the acquisition of agricultural technology (e.g., machinery, infrastructure, vehicles and tools) in order to reduce costs and improve production systems and agricultural productivity. They contribute to increasing the resilience of value chains. The

⁵⁴ For more information on the establishment of the Puna Facility please see sub-activity 2.1.1.1. and for the operationalization Annex 21 Operation Manual.

⁵⁵ The Agroideas Matchmaking Sub-Window is meant to finance through a complementary grant support the development of more advanced CRVC businesses that also qualify for the existing Agroideas government program from MIDAGRI. Agroideas promotes agricultural development (via promoting cooperatives, strengthening of business management and adoption of sustainable environmental agricultural technologies) through non-reimbursable grants covering 60-80% of the investment required by an eligible agro-business, with a cap of ~EUR 315.000.

⁵⁶ For more information, please see chapter 3 Institutional arrangements and responsibilities for managing the financial mechanism of the Operations Manual (Annex 21).

producers' organisations who can prove to have been selected by Agroideas will be also eligible to apply to the third sub-window (Agroideas Matchmaking) of the Puna Facility to complement the financing of their business plans with financing from the Facility for Ecosystem based Adaptation measures. (iii) Individual credit lines and producer organizations credit lines provided by MIDAGRI through the Agrarian Development Bank (Agrobanco). The credits offered by MIDAGRI through Agrobanco possess certain features that complement the financing provided by the Puna Facility. One of the key features is their repayment period of 18 months, making it more suitable for business with rapid turnover. For instance, in the alpaca value chain, these credits can serve as working capital for alpaca cooperatives to gather and process fiber, resulting in a higher market price compared to individual sales from farms. Simultaneously, these same alpaca breeders can access financing from the Puna Facility for investments in EbA. The benefits of these investments are realized in the long term and contribute to mitigating climate change risks.

Activity 1.1.3. Technical Assistance for implementing EbA measures and Climate Resilient Value Chains at local landscape level:

104. The activity will be three folded. It will provide Technical Assistance (TA) complementary to sub-activity 1.1.2.1 for the refinement of proposals within the call for proposals for the Puna Facility and during Local initiative implementation (including TA for EbA and CRVC implementation, business development and access to finance). Complementary to sub-activity 1.1.2.2. it will also include the TA provided by MIDAGRI within their programs (which includes hiring extensionists and commercial coordinators, development of courses and events, among others). In addition, the project will also enhance organizational and entrepreneur skills, formalization and competences to access markets and develop economic opportunities for CRVC including EbA. Finally, the project will ensure women's access to technical assistance including financial education, and access to markets.
105. **Sub-activity 1.1.3.1. Technical assistance for the refinement of proposals and Local initiatives implementation (including EbA implementation, business development and access to finance) through the Puna Facility (EE: Profonanpe):** The objective of this sub-activity is to provide Technical Assistance (TA) that will accompany the financial support provided through the Puna Facility (sub-activity 1.1.2.1). The Puna Facility will offer Technical Assistance (TA) through a service provider hired by Profonanpe and in the case of the Non-repayable Grant Sub-Window optionally in conjunction with the CSO supporting the applicant (see step 6). The TA packages will be differentiated between:
 - i) TA for the **refinement of the local initiative proposals**, provided by the TA provider: This support will include a series of workshops held at regular intervals before the final proposal submission. The workshops aim to help applicants refine their local initiative proposals and address any concerns raised by the Facility Management Unit during the evaluation process. The workshop content will be tailored to the specific proposal types and FMU feedback. Further technical studies (e.g., soil studies) required to refine technical proposals will be supported.
 - ii) TA for the **implementation of the Local initiatives** provided by the TA provider and optionally in conjunction with the CSO in the case of the Non-repayable Grant Sub-Window: After signing the agreement with applicants, TA during the implementation will focus on three key areas: 1) Implementation of EbA and CRVC measures, including participatory approaches for planning, implementing, monitoring and reporting using the community monitoring approach (related to sub-activity 1.2.2.1, as well as recovering un upscaling ancestral knowledge) 2) Business development, and 3) Access to finance, including financial literacy.
106. **Sub-activity 1.1.3.2. Technical assistance through MIDAGRI (EE: MIDAGRI):** In this sub-activity, MIDAGRI, through its programmes will provide technical assistance and technology transfer for the implementation of EbA measures and value chains planned by MIDAGRI's programmes ((INIA, SENASA, Agrorural, Sierra y Selva Exportadora and SERFOR). This includes hiring extensionists and commercial coordinators, development of courses and events, among others. The TA topics foreseen include amongst others (1) commercial technical assistance; (2) technology transfer in value chains prioritised by the project, such as guinea pigs, pasture and forage, roots and tubers, Andean grains and legumes, and South American camelids; (3) technical assistance for the improvement of sustainable camelid breeding, including animal health actions etc. Through this sub-activity an integrative result in the territory will be achieved. Moreover, around 9,647 producers are expected to be trained.
107. **Sub-activity 1.1.3.3. Promote market access and economic opportunities for climate resilient and sustainable value chains (EE: GIZ):** This sub-activity will focus on supporting project beneficiaries to have a better market access for their Climate Resilient Value Chains . The project will support commercial connections between producers and buyers for the more prominent businesses, including women led business (e.g., participating in local markets or fairs, connections with gastronomic routes, etc.), as well as support with the development of market studies and market strategies for specific products. Under this sub-activity, the project team will also explore the potential for a partnership with the SIPAM project, and the online sales platform tool that was developed "Kusikuy". Potential partnerships with MIDAGRI programs such as Agrorural and Sierra y Selva Exportadora will also be considered. It is expected that through this sub-activity a digital platform for sale of puna products is operating. This will facilitate the access to market opportunities for project beneficiaries for their Climate Resilient Value Chain products.

Output 1.2. The use of EbA knowledge is recovered and disseminated, and local monitoring committees and observation systems are implemented:

108. This output 1.2 will complement output 1.1 establishing local monitoring committees and observation systems within the Local Initiatives that were selected by the Facility while recovering ancestral knowledge and practices, to empower high Andean communities to take a proactive role for increased participation in building climate resilience and to monitor their own progress in enhancing overall resilience. Through its activities it will induce a behavioural change and address the limited access to relevant information to support decision-making in the SHAP region, which will also be essential to recover disseminate and scale up adaptation ancestral practices and knowledge.

Activity 1.2.1. Recover, innovate and scale up ancestral knowledge and practices:

109. This activity seeks to develop and strengthen the capacities of communities, local experts (called *yachachiqs*⁵⁷ or *kamayocs*), and technical experts from national and regional institutions to increase the implementation of ancestral water and land management practices and technologies during project implementation and after project finalisation. This activity will also encourage the use and exchange of ancient knowledge and practices from both women and men in the Local Initiatives financed by the Puna Facility. In addition, it will promote dialogues on ancient practices and innovation within members of communities, building intergenerational memory (e.g., children, youth and adults) and fostering discussions on gender roles as well on women knowledge at the community level. In this sense, the project will support the recognition and appreciation of people's own values, knowledge and practices, improving their self-esteem and strengthening their identity with the territory. It will also expand on gender roles to reflect on their place in and relationship with ecosystems. For this to be effective, studies and communication materials (printed, radio spots, videos, etc.) in native languages (Quechua and Spanish) will be produced to capture lessons learned and promote replication of success stories. Exchange between communities will be financed by the project to encourage peer-to-peer learning and ownership of implemented solutions. Of particular importance is the transfer of experience already gathered from recuperating ancient practices and promoting participatory agreements in the Nor Yauyos Cochas landscape reserve. Special attention to ancestral practices that contribute to hydrological ecosystem services, generating contributions to the water MERESE.

110. **Sub-activity 1.2.1.1. Capacity building of local experts for the transfer of ancestral and innovative knowledge linked to EbA and CVCR measures (EE: GIZ):** This sub-activity aims to strengthen the capacities of local experts (*yachachiqs*⁵⁸, *kamayocs*, *talentos rurales*⁵⁹, rural extensionists, among others) to increase the application and replication of ancestral practices and technologies linked to EbA measures and CRVC in other communities in the territory. The role of local experts is to teach and transfer knowledge, using a peasant-to-peasant methodology, to implement and manage ancestral and innovative techniques for small irrigation systems, installation of greenhouses, animal breeding, cultivating pastures, and growing crops. This sub-activity will identify local experts in all of the 91 districts of the project, including i) entities at national and subnational level, ii) local CSOs working in the territory, and iii) community and/or association leaders that can potentially collaborate in the transfer of local knowledge (*yachachiqs*, *kamayocs*, rural talents). A capacity building programme for local experts will be developed, tailored to the profile (e.g., level of education and technical capacities) of the identified local experts and ancestral and innovative knowledge. It is expected that after implementing the training program, the program achieves to strengthen the capacities of local experts (a) within the project beneficiaries and (b) within the technical teams of national and subnational institutions. In this context, the project aims to strengthen the capacities of at least 910 local experts.

111. **Sub-activity 1.2.1.2. Production and dissemination of information materials on lessons learned from ancestral practices and innovation in a context of change (EE: Instituto de Montaña⁶⁰):** The objective of this sub-activity is to produce studies and communication materials (printed (e.g., brochures with infographics), digital, radio spots, videos, etc.) in Quechua and Spanish to capture lessons learned and promote the replicability of successful implementation of EbA and CRVC experience. This sub-activity will be based on identifying and systematising successful past experiences and lessons learned from communities that combine ancestral and innovative practices when implementing EbA measures and climate resilient agricultural practices. This will include successful experiences from the Nor Yauyos Cochas Landscape Reserve. Other existing systematisations of

⁵⁷ A *yachachiq* is an indigenous expert who obtained his practical knowledge on agricultural and livestock practices by an ancestral ways and transfers its knowledge to others

⁵⁸ Indigenous experts who obtained their knowledge to manage agricultural and livestock practices in ancestral ways.

⁵⁹ The "*talentos rurales*" are men and women who live in peasant communities and are known for their capacities and abilities to experiment, adapt, develop and teach good intercultural practices and rural innovations.

⁶⁰ Instituto de Montaña will place special emphasis on facilitating the implementation of quality participatory processes through coordination with local CSOs supporting beneficiary communities, producers' associations and cooperatives in the process of formalization during Local initiative implementation. These processes are intergenerational dialogue on ancestral practices and innovation, knowledge sharing with a gender perspective, and community monitoring to estimate the impacts of EbA measures. Capacity building of trainers (local CSOs and other stakeholders) will facilitate the training of villagers on the causal relationship between implemented activities and their impacts, and the benefits obtained. Capacity building will ensure a common understanding of concepts and working methodologies to facilitate participatory planning, implementation and monitoring processes that promote ownership and thus sustainability of EbA measures.

ancestral practices will be revised such as the ones developed by MIDAGRI⁶¹, ENCONTRAR Platform⁶², Encuentro de Saberes Small Grants Program-GEF⁶³. The systematisation will also include new experiences generated by IPLCs in general, and with a special focus on the ones generated by women groups during the project. As a result, this sub-activity will produce documents and communication materials in Quechua and Spanish on lessons learned from ancestral practices and innovation and disseminate these lessons through events and/or dissemination spaces. This sub-activity will be implemented in coordination with MIDAGRI (through their programs), MINAM and SERNANP, Regional Governments, Profonampe, and Sectoral and/or Territorial governance platforms as the good governance platform.

112. **Sub-activity 1.2.1.3: Knowledge sharing among beneficiary communities with a gender perspective (EE: Instituto de Montaña):** This sub-activity aims to support the recognition and appreciation of IPLCs ancestral practices regarding to the management of agriculture, livestock, water and ecosystems, contributing to promote the participation and leadership of Indigenous Peoples and Local Communities in recovering ancestral knowledge and encouraging a collective reflection on gender roles. The project will promote and facilitate exchanges between local communities who have been selected to receive funds from the Puna Facility (through sub-activity 1.1.2.1 and 1.1.3.1) or are being supported by MIDAGRI (through sub-activity 1.1.2.2. and 1.1.3.2) to foster peer to peer learning and to share innovations and practices developed between communities. Particular attention will be placed on recognizing diverse gender roles and expertise related to agriculture, livestock, water, and ecosystems management. This focus aims to prevent the exclusion of women participants and acknowledge their valuable expertise and knowledge. Additionally, it seeks to highlight their significant contributions towards building climate resilience within communities. At least one exchange among IPLCs women from different communities will also be promoted, (e.g agropastoralist, artisan women networks, etc) who possess substantive knowledge and skills in these different fields. These exchanges can work as space for women empowerment and inspiration. As part of this sub-activity, a community knowledge sharing plan with a gender perspective will be developed. This plan will define the implementation strategy, both interregional and intraregional exchanges, and the methodology for selecting experiences and means of exchange (meetings, workshops, internships, and others). This sub-activity will be implemented in coordination with MIDAGRI (through their programs), SERNANP, Profonampe, Local Governments.
113. **Sub-activity 1.2.1.4: Facilitating intergenerational dialogue on ancestral practices and innovation in a changing context (EE: Instituto de Montaña):** This sub-activity aims to build intergenerational memory (children, youth and adults) on ancestral practices and innovation in a context of change (climate, markets, and others) in selected beneficiaries of each Puna Facility call. An intergenerational dialogue on ancestral practices and innovations strategy and implementation plan will be prepared, containing the methodology, means and instruments (videos, books, games, others). Local researchers (composed by women and men) in each community will be trained to facilitate the intergenerational dialogue on site with the participation of women, children, youth, adults, and older adults. The spaces to carry out these dialogues could be schools or farmer field schools. Intergenerational dialogues will serve to recover and considerate different gender perspectives, roles, knowledge embedded within communities during the design, implementation and monitoring phases of the Local Initiatives that will be implemented in sub-activity 1.1.2.1. The intergenerational dialogue plan will be implemented through workshops, surveys, and other activities, depending on each selected community design. The results and achievements of the sub-activity will be systematised and disseminated. Relevant institutions involved in this sub-activity are: MINEDU (through local schools) and, MIDAGRI through its attached institute INIA, and SERNANP (in the case of Natural Protected Areas).

Activity. 1.2.2. Implement community monitoring and observation systems to measure the impact of EbA measures and provide feedback for regional and national policies:

114. This activity will be implemented in alliance with the IPLCs implementing Local initiatives and supporting local CSOs. The EE Instituto de Montaña will conduct a Training of Trainers (ToT) for local CSOs that will support beneficiary communities that were selected in Puna Facility's Non-repayable Grant Sub-Window and will provide technical assistance to facilitate the participatory design of indicators and protocols for participatory community monitoring. This activity will build upon Community Monitoring Committees (CMC), which applicants to the Puna Facility are required to create when applying their Local Initiatives. These Committees that will be composed by "local researchers" (composed by women and men) (designated previously in sub-activity 1.1.1.2) will lead the participatory monitoring process designed based on each community knowledge and interest. Then CMC will inform about the progress of the Local Initiative to the community members. Within this activity the capacities of local researchers and committees will be strengthened through facilitation, dialogue, training and other capacity-building measures. This will enable knowledge weaving and co-production to understand how traditional practices and

⁶¹ repositorio.midagri.gob.pe

⁶² Plataforma EnconRAR - Agricultura familiar andina (encontrarandes.org)

⁶³ PPD Perú – Programa de Pequeñas Donaciones (ppdperu.org)

scientific findings can explain the ecosystem functions and the effectiveness of EbA. Target populations will establish climate vulnerability social and biophysical indicators based on local knowledge as the observation of fauna and/or flora species as indicators of the ecosystem health, and other indicators based on men and women community member's interest. This also includes establishing and monitoring rough estimates on the potential for carbon storage and sequestration, water storage and climate regulation, as well as the provision of fodder, fiber and food, among other relevant ecosystem services. Committees will learn and discuss how to develop local indicators and technical protocols to assess changes in ecosystem service provision over time as the EbA measures are implemented and mature. The data produced by the local monitoring system will feed regional and national institutions studies and research on impacts and effectiveness of EbA measures, thus contribute to policy decision making processes. Follow-up on this activity will be coordinated at all levels of intervention, including from the local (community) level, in coordination with the national actors responsible for M&E systems and the NDCs.

115. **Sub-activity 1.2.2.1: Training and implementation of Community Monitoring Committees (CMC) to develop local monitoring systems (EE: Instituto de Montaña):** This sub-activity seeks to make project beneficiaries aware of the impact of climate change and poor landscape management on high Andean ecosystems, as well as the benefits obtained and the causal relationship between the implementation of EbA measures and their impacts on the recovery and conservation of their ecosystems and ecosystem services. Local Researchers (composed by women and men) designated in sub-activity 1.1.1.2 will form Community Monitoring Committees (CMC). CMC will collect information about the impact of the implemented EbA measures in a context of climate change through indicators proposed together with community members. Local researchers will lead and facilitate the community monitoring process since the selection of indicators to assess socio-economic, ecosystem and ecosystem service impacts. In the areas of water MERESE communities, the monitoring will seek to contribute to water utilities monitoring systems. This sub-activity will allow beneficiary communities to take a proactive role in monitoring their own progress and building climate resilience. CMC will participate in trainings and dialogue for selecting sites for monitoring, setting a baseline information for their Local Initiative and collecting data using an app that will be developed in sub-activity 1.2.2.2. Collected data will then feed into a data-driven information management system that will be also developed in sub-activity 1.2.2.2. This system will be connected to the Puna Facility monitoring system and to the project monitoring system. The information produced by the system will be used to report to MIDAGRI and SERNANP national monitoring systems that contributes to the NDCs. Therefore, discussions among all executing entities need to be conducted to agree guidelines and indicators for community monitoring. During this sub-activity, community monitoring protocols will be discussed with local researchers in each community in the frame of the agreed guidelines and provided to the information system responsible (sub-activity 1.2.2.2). Thus, a capacity building plan will be prepared by Instituto de Montaña and implemented to monitor the planned EbA measure's impacts in the selected recipients of the Puna Facility. The work will be carried out together with fund recipients under the Repayable and Agroideas Matchmaking Sub-Windows and the local CSOs that are supporting local communities, producers' associations and cooperatives under the Non-repayable Grant Sub-Window. Local CSOs will provide support in the territory and will bring permanent assistance to the community monitoring committees.
116. **Sub-activity 1.2.2.2. Development and implementation of a local data-driven information management system using digital technologies innovation measures (EE: GIZ):** This sub-activity aims to develop a Cloud based system that will store the data and information provided by the community monitoring committees (sub-activity 1.2.2.1). This information will be related to the baseline condition of the ecosystems and their ecosystem services (prior Local Initiative implementation); and the outcomes of the selected EbA measures and CRVC. The materials and equipment to be used for monitoring will be procured by GIZ and will be provided to the community monitoring committees when needed. The Cloud based system must be later hosted in one of the existing MIDAGRI's and/or SERNANP's national monitoring system. This system will be identified during implementation (sustainability condition). This sub-activity will include data collection tools and protocols for processing and sharing information. For data collection it will be developed an app to facilitate data gathering by the community monitoring committees. Special consideration will be given to hire a consultancy from a research specialized institution such as universities or other scientific centres at the beginning of this activity to support the establishment of this information gathering process that will be hosted in an existing national monitoring system. The collected data will be processed and can be used to provide feedback on local, regional and national policies or as a source for analysis linked to sub-activity 1.2.2.3.
117. **Sub-Activity 1.2.2.3. Analyse collected information on the impact of EbA measures (EE: GIZ):** This sub-activity aims to improve evidence of EbA effectiveness while at the same time increasing the knowledge and understanding of EbA measures for future decision-making processes (i.e., increasing community awareness and values about EbA, donor investment in EbA and support to strengthen and further mainstream the integration of EbA across relevant ministries, water utilities and public investments). Therefore, this sub-activity will develop studies and research using data generated by the community monitoring committees, thus contributing to the dissemination of coupled indigenous and scientific knowledge. Main topics to analyse will include impact measurement of EbA (carbon sequestration, ecosystem recovery and water regulation), cost-benefit analysis, cost

risk reduction in business operations, cost effectiveness and the type of ecosystems to be investigated (e.g., peatlands, wetlands, grasslands, others). The generated evidence about the EbA impact will be used to increase awareness and the engagement of potential investors in the Puna Facility (sub-activity 2.1.1.2) and the integration of EbA measures into policy and budgetary processes (sub-activities of component 3). In addition, a methodology for assessing carbon stock changes resulting from the implementation of EbA measures in the Puna ecosystem will be developed in coordination with MIDAGRI and MINAM. This will create the basis to prepare the national and local actors to improve the carbon inventory in the high Andean zone, especially the basis for considering how the implementation of adaptation NDCs that contribute to the conservation/maintenance of Puna ecosystems also contribute to climate change mitigation. Under this sub-activity, the methodology will be tested in at least one area within the Natural Protected Area System with committed communities that obtain conservation agreements signed (under the Puna Facility in sub-activity 1.1.2.1). Alliances with research centres, scientific and university entities will be sought for this purpose and the results of studies and research will be disseminated through the corresponding channels such as webpages, events, emails, journals, seminars, regional platforms according to the project objective.

B.3.1. Outcome 2: Public and private financing for EbA measures and climate resilient livelihoods are accessible and being actively used by vulnerable communities in the Puna ecosystem:

118. This outcome will align and leverage public and private financing for EbA measures and CRVC at different and coordinated levels. It will focus on the alignment, leveraging, mobilization and upscaling of private and public funds to finance EbA and CRVC activities on the ground. On one hand will support the establishment of an innovative financial mechanism called Puna Facility that will finance Local Initiatives through a call for proposals that will be operationalized in sub-activity 1.1.2.1 at the same time that mobilizes private and other donors' funds. On the other hand, the activity will aim at enhancing the offer and access to micro green financial products and increase the access to public programmes from MIDAGRI (e.g., Agroideas, Sierra Azul and Agrorural). This outcome has one main output, which is "Financial mechanisms for the implementation of EbA measures and improvement of Climate Resilient Value Chains are established and accessible to vulnerable communities in the Puna ecosystem".

Output 2.1. Financial mechanisms for the implementation of EbA measures and improvement of climate resilient livelihoods in the Puna ecosystem are established:

119. This output will focus on mobilizing different public-private mechanisms to finance EbA measures and CRVC, through the implementation of three main activities which will include: activity 2.1.1 to establish the Puna Facility for long-term financing of EbA measures and Climate Resilient Value Chains; activity 2.2.1 to strengthen capacities to develop and implement innovative mechanisms for EbA in high Andean ecosystems including hydrological payment for ecosystem services (MERESE), activity 2.1.3 to facilitate access to greener microfinance products and services that promote EbA measures and Climate Resilient Value Chains. These activities will address the limited medium to long-term financing for ecosystem-based climate solutions, while creating incentives to finance Climate Resilient Value Chains, as well as the lack of alignment and coordination of financing interventions on the ground and support beneficiaries to tackle limited economic opportunities in the SHAP.

Activity 2.1.1. Establish the Puna Facility for long-term financing of EbA measures and Climate Resilient Value Chains:

120. This activity aims to unlock investment for EbA measures in Puna ecosystems and Climate Resilient Value Chains through the "Puna Facility", managed by Profonanpe. The Puna Facility will be composed of the financial window that will have three sub-windows (linked to activity 1.1.2): i) Non-repayable Grant Sub-Window, ii) Results-based Repayable Grant Sub-Window, iii) Matchmaking Agroideas Sub-Window. In addition, to engage with the private sector and other international donors, a sustainable financial strategy will be developed by Profonanpe with the support of the technical advice of GIZ during the first years of implementation to mobilise resources from the private sector and different donors to secure amongst others the sustainability of the Puna Facility. In this regard, the sustainable financing strategy will include criteria for selecting interested companies (e.g., companies committed to social and environmental responsibility, without potential conflicts, linked to CRVC and key ecosystems), criteria for selecting demonstration sites, and an effective communication and marketing strategy to engage with the private sector. The sustainable financial strategy will be validated with the different private stakeholders and then approved by the Resilient Puna Project Steering Committee. Finally, to increase alignment and use of public funds for actions related EbA and CRVC, the activity will seek to increase the mobilisation of resources from public programmes for the High Andean population and their associated ecosystems. To this end, access to and alignment of investments from MIDAGRI and SERNANP and other government programmes will be promoted for EbA and CRVC measures through the development of dissemination material (e.g., Flyers) to facilitate the access to public financing.

121. **Sub-Activity 2.1.1.1. Preparation and continuous improvement of the allocation of resources of the Puna Facility (EE: Profonanpe):** To unlock investment in Puna ecosystems and CRVC, an innovative financial mechanism will be established and will be called "Puna Facility". The name "Facility" is based on its role as facilitator

in leveraging, mobilising and channelling different financial resources. In this context, the Facility is at the centre of the financial exit strategy (for more information please see B.6.1.), opening the possibility of channelling additional resources from multiple sources. This facility will target local communities, producers' associations, cooperatives, SMEs, community enterprises and producer' organizations and will finance Local initiatives with EbA measures and CRVC that enable adaptation to climate change. The operationalization of the Puna Facility from the awarding of Local initiatives and the provision of technical support to the selected applicants will be carried out in component 1 through sub-activity 1.1.2.1 and sub-activity 1.1.3.1 respectively. Through the sub-activity 2.1.1.1. with the technical support of GIZ, Profonanpe as EE will establish a Facility Management Unit which will be responsible for coordinating the Puna Facility under the supervision of Profonanpe's Monitoring and Evaluation Department. It will also coordinate with the Administration and Finance Management to grant funds and report on Local initiatives, and with the Innovation and Strategic Management Directorate to coordinate the leveraging of funds for the facility. The FMU will also coordinate the establishment of the Independent Technical Evaluation Committee, which will involve the selection and hiring of its members. It will also manage the selection and contracting of the technical assistance provider organization. Next to the management structure of the Facility, the Governance structure of the project through the Project Steering Committee and Project Management Committee will also play an important role in addition to their roles and responsibilities related to the overall project. After the call cycles are finalized, lessons learned will be systematized and if required, processes will be improved for the subsequent calls.

122. **Sub-Activity 2.1.1.2. Development and implementation of an innovative strategy to mobilize resources from the private sector and different donors (EE: GIZ):** This sub-activity aims to facilitate and secure funding through leveraging, mobilising and channelling financial resources from the private sector and other donors for EbA investments in Puna ecosystems and CRVC as part of the project's financial exit strategy. For this sub-activity, it is necessary to develop a sustainable financing strategy for the mobilization of future funds through Profonanpe. Therefore, the engagement of potential investors such as, private sector actors and different donors will be crucial to ensure the sustainability of the Puna Facility and the continuation of financing for EbA measures and CRVC. In order to elaborate the financing strategy, first a mapping of potential private sector actors and different donors will be conducted as it will be key to have a portfolio/directory of potential donors and their projects or initiatives that integrate EbA and CRVC measures in the High Andean areas. In addition, for the selection of companies to be targeted by the strategy, consideration will be given to those businesses interested based on Corporate Social Responsibility and that depend on the ecosystem services provided by the Puna ecosystems. For instance, those that benefit from the camelids fibre produced in the upper parts of the watersheds or those who benefit from water provision and regulation ecosystem services in the lower parts of the watersheds. Other type of businesses can be those who are part of the value chains promoted by the Puna Facility. For example, businesses that buy products to small producers. In addition, a narrative will be developed based on results from Puna Facility's Local initiatives systematizations and on the results of the studies developed in sub-activity 1.2.2.3. This will show how EbA measures reduce costs and risks in business operations and provide opportunities and benefits to the operation of various industries (e.g., hydroelectric, cement, water, mining, agro-industrial, etc.). In addition, events will be organized to attract the interest of several companies and/or international donors and to disseminate communication materials to inform on the effectiveness and impact of EbA measures. Travel to present good practices will be also coordinated. The strategy will also include a communication strategy. If the interests of the companies do not match with the Puna Facility, the project will facilitate the investments directly with potential beneficiaries.
123. **Sub-Activity 2.1.1.3. Promote access and alignment of investments and other government programs for EbA and Climate Resilient Value Chains (CVCR) measures (EE: GIZ):** The objective of this sub-activity is to increase the mobilisation of resources from public programmes related to sub-activity 1.1.2.2 and other government programmes for the High Andean population and their associated ecosystems. In coordination with MIDAGRI this sub-activity will develop simplified and user-friendly materials (e.g., flyers) out of guidelines for potential beneficiaries to access the different support options provided by MIDAGRI programmes and services; and other relevant government programmes that contribute to the implementation of EbA measures and/or CRVC. These materials will also support local governments in their alignment with the national government. In addition, there will be workshops for producers and local⁶⁴ governments to disseminate the information on the different public programmes. Promoters will be hired (the same as for 1.1.1.1 sub-activity) to support beneficiaries explaining the requirements of different public programs and how to access them. This will be carried out in coordination with Profonanpe, the General Directorates, programmes and funds of MIDAGRI (Agroideas, Sierra Azul, Sierra y Selva Exportadora, AgroBanco, Catastrophic Insurance, Rural and Indigenous Women's Fund), Regional Government

⁶⁴ In some cases, government programmes are designed to work with local governments to serve producers.

(ProCompite) and MIDIS (social programmes)⁶⁵. In addition, support will be provided for establishing the required coordination mechanisms to integrate the implementation of water MERESE with other government programs.

Activity 2.1.2. Strengthen capacities to develop and implement innovative mechanisms for EbA in high Andean ecosystems:

124. The activity will seek to overcome the barriers faced by public water utilities that have established or are willing to establish a mechanism for payment for ecosystem services (MERESE) schemes. Therefore, activities will be focused on strengthening stakeholders' capacities to identify and overcome bottlenecks and gaps in the implementation of MERESE. Local water utilities will receive technical support in planning, evaluation and selection of intervention projects; development of agreements with communities; and analysis of gender aspects to be included in MERESE's processes. During the trainings for water utility officers the participation of women will be secured. The related sub-activities will be developed in close coordination with the Ministry of Environment and the National Superintendence of Water Supply and Sanitation of Peru (SUNASS). In addition, the activity will include technical assistance for the establishment of a methodology to calculate carbon in the Puna ecosystems and provide capacity building for local actors (e.g., communities, local government, academia) aligned with national legal and institutional arrangements for carbon markets.

125. **Sub-activity 2.1.2.1. Technical assistance to improve existing hydrological MERESE processes in the project areas (EE: GIZ):** This sub-activity aims to strengthen water utilities capacities to identify and overcome bottlenecks and gaps for the implementation of the hydrologic PES (which in Peru it is called Retribution Mechanism for Ecosystem Services-MERESE) and thus mobilise financial resources not only for the conservation of High Andean ecosystems but also to increase water availability. The objective is to engage with men and women officers of water utilities that implement MERESE in key watersheds of the project as a complementary intervention to the Resilient Puna project and other public-private initiatives. The sub-activity will first include technical assistance for hydrologic MERESE which are already underway, such as the MERESes in: Arequipa (1), Lima (1), Cusco (2) and Apurimac (1). In the project target area five water utilities have hydrologic MERESE in place including EMUSAP Abancay, SEDAPAR Arequipa, EMAPA Cañete, EMSAPA CALCA in Cusco, SEDA Cusco. For more information on these water utilities see Table 25 in Annex 23. Given the gradual implementation of the hydrological MERESE in the target area, this sub-activity will provide technical assistance that will be focus on the a) establishment and/or strengthening hydrologic MERESE's good governance platforms, b) development of methodological tools for implementation, c) provision of technical information for the identification of water MERESE interventions, d) contributing to the design of the monitoring system, e) supporting the capacity development for the implementation of different implementation modalities as described under Section 3.1 in Annex 2d, f) establishment of contractual agreements with communities, g) development of EbA public investment projects, h) analysis of gender aspects to be included in MERESE's processes and systematization of learning experience at the end of the process. To consolidate the implementation of MERESE in each region, the specific technical assistance package will be assessed on a case-by-case basis according to the needs of each MERESE.⁶⁶

Activity 2.1.3. Support the greening of microcredits to promote EbA and Climate Resilient Value Chains:

126. This activity will focus on two main topics: (1) Supporting the greening of microcredit lines and business models and (2) supporting the dissemination of green financial products and services. In order to increase the supply of microfinance products and services to finance integrated EbA and CRVC initiatives, the project will support either the development of specific financial products or the greening of existing financial products for EbA measures and CRVC (aligned with the lists of EbA measures and CRVC developed in sub-activity 1.2.1.1) in Puna ecosystems. This will include the integration of gender-sensitive aspects in financial products to promote EbA and Climate Resilient Value Chains. The design will include a diagnostics identifying gender gaps and needs. Microcredit institutions officials will receive training to increase the capacity of MFIs to finance businesses with EbA measures with a gender approach. In addition, a communication strategy to disseminate the green financial products will be developed and implemented, raising awareness regarding the requirements and conditions, with a focus on the beneficiaries of the project.

127. **Sub-activity 2.1.3.1: Support to financial institutions in greening microcredits (EE: GIZ):** This sub-activity aims to develop and introduce greener microcredits for financing not only Resilient Puna beneficiaries but also

⁶⁵ In particular, the National Action Platforms for Social Inclusion Programme (PAIS), which is located in the most remote localities of the project and is designed to facilitate and coordinate the provision of services for programmes, projects and activities in social, economic and productive areas of public and private entities, in order to improve the quality of life of the rural and dispersed rural population.

⁶⁶ Also, the operationalization and diversification of MERESE schemes to leverage funds from beneficiaries of these services downstream to local communities upstream is one of the economic benefits of the project. EPSs will be supported to operationalize the MERESE scheme in their watershed with a potential of mobilizing USD 5-6 million by following EPS, EMAPA CAÑETE S.A., EMPSSAPAL S.A., EMUSAP ABANCAY S.A.C., SEDA CUSCO S.A.).

beneficiaries in other regions interested in implementing EbA measures and CRVC (aligned with the lists of EbA measures and CRVC developed in sub-activity 1.2.1.1) within the Puna ecosystems. Microfinance institutions (MFIs) with a presence in the project area and with interest on expanding their credit portfolio to small-scale farmers for adaptation measures will be selected (e.g., Caja rural los Andes, Caja Cuzco, Microfinance BBVA Foundation-Confianza), and technical assistance will be provided to either adjust existing products (including existing financial products for women) or develop new ones, incorporating the EbA approach in the financial risk analysis for prioritised value chains. In order to achieve this, first, MFIs' portfolio will be analysed, taking into account data disaggregation by gender and employing a cross-cutting approach. The goal is to identify the value chains prioritised in the project with the greatest exposure to climate risks, as well as to assess the exposure to credit risk based on gender. Then mitigation activities (EbA measures) will be proposed and included in the value chain business model to reduce these climate risks and thus reduce the negative economic impacts that climate events have on agricultural producers, men and women, and the credit portfolio of microcredit institutions. Moreover, through working sessions (peer-to-peer exchanges) and studies, we will seek lessons learned from current processes related to green financial products and financial products specifically designed for women. This review will encompass crucial stages, spanning from credit approval to loan disbursement and subsequent follow-up. The overarching objective is to integrate these insights into the development of microcredit greening activities. In parallel, market demand studies will also be carried out including diagnostics for identifying gender gaps and needs for accessing to microcredits. Demonstrating the viability of the proposed EbA and CRVC measures business model is essential. Therefore, concrete and tailor-made business models will be developed using the information generated and analysed in activity 1.2.2. In addition, demonstration sites will also be identified to showcase good practice and serve as models to be replicated and scaled up. Based on the results of the analysis, the microcredits to be adapted will be identified, as well as gender aspects to be included. In particular, to improve women's accessibility to credit, taking into account the diversity of women's technical, financial, digital and entrepreneurial capacities, including adapted services to women's specific challenges. During the adaptation phase, management tools (including digital tools, guidelines, and protocols) will be modified to include the EbA approach. This will include aspects such as sustainability and climate risk management criteria and development of tools for credit analysis based on agro-climatic and risk reduction information. The option of developing new green microcredit lines will be assessed by each microfinance institution based on their product development strategy. Relevant institution involved in this activity will be the microfinance institutions.

128. **Sub-activity 2.1.3.2. Raise awareness of greener microcredits to promote EbA and Climate Resilient Value Chains (EE: GIZ):** This sub-activity aims to promote the greener microcredits (either adapted or newly developed in the previous sub-activity) and includes the development and implementation of an awareness campaign for potential clients, with a focus on the project beneficiaries selected by the Puna Facility. In order to achieve this, the sub-activity will capacitate field personnel of financial institutions, which will promote the greener microcredits including trainings on requirements and conditions. Financial institution front office and field personnel will also be trained to effectively market and disburse loans to beneficiaries in remote rural areas in the High Andean zone. In addition, risk management of financial institutions will be strengthening, with a special focus on addressing climate risks within their portfolio. Credit officers from these institutions will receive training on the EbA approach, climate risks, gender considerations for rural financial market and innovative strategies to promote greater sustainability in businesses. Subsequently, the greener microcredits will be disseminated through different channels. This will include a launch event and an awareness campaign for the population of the project areas, especially for the beneficiaries of the Puna Facility. This will provide them other sources for financing their present and future EbA measures and CRVA.

B.3.3. Outcome 3. EbA and climate resilience are mainstreamed into multilevel landscape governance instruments:

129. This outcome will promote integrated landscape planning and policy improvement and coordination, by fostering dialogue and improving coordination among stakeholders that intervene in the landscape (local, regional and national governments, local communities, producers' organizations, watershed committees, and MIDAGRI extension services, among others). The corresponding activities proposed in this outcome will directly address identified barriers regarding i) lack of understanding of the causal relations of EbA practices to increase climate resilience, ii) poor coordination among national and subnational governments on the ground, including MIDAGRI programs, SERNANP and other sectors, iii) weak governance structures for a sustainable management of the landscape and its natural resources and iv) limited local capacities to adapt to climate change, v) Insufficient and lack of policies and regulatory framework enforcement for implementing the EbA approach and Climate Resilient Value Chains.
130. **Output 3.1. Multilevel landscape governance is improved through strengthening of national capacity, regulatory frameworks and M&E systems** The approaches that will be promoted by Resilient Puna require

progressive development of capacities of governmental Executing Entities⁶⁷, local actors, and subnational governments to promote the replicability and sustainability of the changes introduced by the project. Therefore, in order to promote the mainstreaming of EbA and climate resilience approaches at landscape and water basin levels into the national, regional and local planning processes (i.e., development plans, including budgets and further public investment project planning), women and men's official capacities at different levels (national, regional and local) will be strengthened and gender, intergenerational and interculturality cross-cutting approaches will be considered as well as the valuation of ancestral knowledge and practices. The output will also provide technical support to MIDAGRI and SERNANP to improve the regulatory framework for the promotion and implementation of EbA measures and CRVC in the High Andean areas of the country.

Activity 3.1.1. Strengthen the capacities for territorial planning and governance processes integrating EbA and climate resilience:

131. The activity aims to strengthen participatory sectoral and territorial platforms⁶⁸ so that EbA measures and CRVC developed by project beneficiaries can be replicated and scaled up. Through this activity the project will enable that lessons learned from the community's experiences in EbA and CRVC planning, and implementation are considered and scaled up in the agenda of government platforms⁶⁹ and in local and regional governments plans. In order to strengthen territorial planning processes, the activity aims to: (1) Strengthen the capacities of relevant government stakeholders for the incorporation of EbA and CRVC measures, integrating the gender perspective in their processes and interventions in the territory, (2) Strengthen participatory sectoral and territorial platforms for the articulation and scaling up of EbA measures and CRVC within the scope of the project by promoting IPLCs participation. Special emphasis will be given to encouraging women's participation in decision-making and planning processes. (3) Integrate EbA measures and CRVC in territorial planning and management instruments. Lessons from local planning processes, especially those derived from monitoring activities, should be integrated into territorial planning in the target watersheds (watershed management plans).

132. **Sub-activity 3.1.1.1. Strengthen the capacities of relevant national, regional and local government stakeholders for the incorporation of EbA and CRVC measures, integrating the gender perspective⁷⁰ in their processes and interventions in the territory (EE: GIZ):** This sub-activity aims to strengthen the capacities of relevant public officials working in thematic and geographical areas of the project at the national, regional, and local levels. The identification of specific participants will be based on pre-assessed capacity needs (for more information see Annex 2h) defined in through a participatory approach with SERNANP as part of MINAM and MIDAGRI. Personnel (both women and men) from both entities in charge of national programmes and rural extension services will undergo training. This training aims to enhance the incorporation and reinforcement of EbA and CRVA at landscape and water basin levels. It also aims at promoting the coordination with water MERESE (if any in the area), promote traditional practices and integrate gender consideration into agricultural extension programs. These efforts will enable officials to be able to recognize the diverse needs and roles of men and women farmers in agriculture. As a result, it will progressively encourage more involvement in women's roles, thus contributing to the reduction of gender inequalities in the agricultural sector. Training topics will include: i) EbA and CRVC: Climate risk assessment, territorial, landscape and water basin approach, ecosystem monitoring and evaluation, prioritisation of adaptation measures, MERESE, identification of indicators that match with the respective monitoring system. ii) Gender awareness and intergenerationally: Improve gender and intergenerational awareness among public officials and develop tools for gender and intergenerational mainstreaming in their processes and interventions in the territory. In order to provide capacity building, as a first step, content and materials will be developed for online and/or presential training modules, based on lessons learned generated by other project experiences and those generated during project implementation (based on the data generated in activity 1.2.2.). The relevant institution involved in this sub-activity are Instituto de Montaña as they will provide inputs from their past and present experience to develop the contents of the training modules, MIDAGRI through its Directorates-General, programmes and affiliated public bodies will contribute and participate in de design and implementation of training modules, SERNANP as they will contribute and participate in de design and implementation of training modules, and Subnational governments as they will participate of training modules.

⁶⁷ Please see Annex 2c were the capacities of the two governmental EEs MIDAGRI and SENRANP where assessed.

⁶⁸ By platforms is meant spaces officially formed to bring together different public and private stakeholders to discuss issues that are of interest to them and then coordinate, decide and plan concrete actions to address them. These topics can be based on the management of a specific territory such as the Natural Protected Areas Committees, or based on water management such as River Basin Councils, or based on the coordination of the agricultural sector in the territory such as the CGRAs, or based on the planning and coordination of the MERESE in a specific sub- basin such as the platforms of good governance of the MERESE.

⁶⁹ Platforms refer to formally established spaces where diverse public and private stakeholders convene to discuss about issues relevant to them.

⁷⁰ By gender perspective is meant being aware of and taking into account the different outlooks, roles and behaviours of me and women (may be also boys and girls – depending on their involvement).

133. Sub-activity 3.1.1.2. Strengthen participatory sectoral and territorial platforms for the articulation and scaling up of EbA measures and Climate Resilient Value Chains within the scope of the project (EE: GIZ):

This sub-activity has three specific objectives: (i) Promote the participation of IPLCs in decision-making processes at the local and regional platforms⁷¹, (ii) disseminate successful Local initiatives experiences financed in component 1 for scaling up EbA and CRVC measures, specially the one led by IPLCs women organizations and (iii) coordinate public and private interventions in EbA and CRVC measures at landscape and water basin level of each territory integrating a gender approach. The first step within this sub-activity involves identifying sectoral and/or territorial platforms that hold official recognition and social acceptance in each territory. These platforms will serve as forums where diverse public, private, and civil society organizations in the project area convene to foster dialogue, address concerns, and collaborate on coordinated efforts related to EbA investment and CRVC, both at the landscape and water basin levels. Potential platforms include Regional Committees for Agriculture Management (CGRA), Good Governance platforms of MERESE, Natural Protected Areas Management Committees, the Council of Basins, and others. The project will support the participation of IPLCs in its different forms of organizations (local community and producers' associations representatives) to bring to the platforms their best practices and concerns when implementing EbA measures and Climate Resilient Value Chains. Participation of women in these spaces of dialogue and decision making will be supported and facilitated. In addition, methodological guides for the analysis of the social context for the identification of actors and promotion of Good Governance Platforms will be developed and implemented. The project will facilitate the exchange of available data and information produced by each institution (e.g., climate, hydrological, finance, etc.) and will foster the inclusion of climate risk vulnerabilities in their planning processes, prioritisation of adaptation solutions, and promote agreements among platform members for future funding. Finally, successful EbA and CRVC experiences that have emerged because of the implementation within the project at landscape and water basin level (through the implementation of activities in component 1 will be disseminated.

134. Sub-activity 3.1.1.3. Integration of EbA measures and Climate Resilient Value Chains in territorial planning and management instruments (EE: GIZ): This sub-activity aims at increasing influence on subnational governments (regional and local) through their planning and budgeting instruments in order to leverage and increase public finance in EbA measures and CRVC in the medium and long term. The specific tasks assigned to this sub-activity will vary based on the level of intervention, encompassing local and regional levels, as outlined below. Technical assistance will be provided to local and regional government officers interested in developing or improving their development plans and budgeting instruments by incorporating EbA measures, gender and CRVC in

- i)** Local and/or Regional Concerted Development Plans: The technical assistance for local and regional officers will be focused in improving their participatory process and the prioritisation and planification of adaptation measures at landscape and water basin level. Additionally, climate risk analysis will be developed in territorial plans. Based on prior identification, the project will support local and/or regional governments in developing climate risk and vulnerability studies as part of the preparation of their local, regional or basin development plan. Including strengthen territorial planning with watershed approach for the adequate coordination, synergies and alignment of water MERESE.
- ii)** Local Climate Change Plans and/or Regional Climate Change Strategies: Technical support to local or regional technical officers will be focused in developing participatory Regional Climate Change Strategies and/or Local Climate Change Plans to prioritize and integrate EbA and Climate Resilient Value Chains measures into those climate change instruments.
- iii)** Regional and local budget & investment projects: Technical support to local and regional technical officers to prioritize and design EbA and Climate Resilient Value Chains within the local and regional public budgets, including budget for EbA M&E
- iv)** Natural Protected Areas' Master plans: the technical assistance for SERNANP will be focused in integrate the EbA approach in the participatory process of updating master plans. Support to update their climate risk and vulnerability studies will be done.

Activity 3.1.2. Strengthen regulatory frameworks and M&E systems at national level:

135. The activity aims to provide technical support to MIDAGRI and SERNANP to improve the regulatory framework for the promotion and implementation of EbA measures and CRVC in the High Andean areas of the country, as well as ensuring the participation of IPLCs in resolutions related to their participation in decision making process

⁷¹ Ibid

for the management of the territory and its ecosystems. In addition, it will provide technical support for the national long-term M&E systems of the different government entities to measure impact of EbA interventions and ensure data disaggregation by gender. Partnerships will be established with local universities and research institutes to support the monitoring of progress resulting from the interventions implemented by the different actors in the territory. This activity will also support MIDAGRI in improving its agricultural policy monitoring and evaluation efforts while complying with 4 Peru's NDC targets to which the project will contribute to: i) (AGRI17- Natural grassland management to secure feed for livestock and reduce vulnerability to climate change; ii) AGRI11 - Management of wild South American camelids (vicuñas) considering the effects of climate change; iii) AGRI16.- Implementation of business strategies that incorporate risk and opportunity management in the face of climate change; iv) AGU2 - Implementation of interventions related to planting and harvesting water for agricultural water security in watersheds vulnerable to climate change.

136. **Sub-Activity 3.1.2.1. Improving national regulatory frameworks linked to EbA measures and Climate Resilient Value Chains (EE: GIZ):** This sub-activity aims to improve the regulatory framework at the national level (norms, laws, budget, and investment guidelines) for the promotion and implementation of EbA measures and CRVC in the High Andean areas of the country. The sub-activity will develop proposals or draft standards based on the assessment of legal barriers related to ecosystem restoration and conservation, agricultural productive uses, sowing, and harvesting of water, IPLCs participation, water MERESE, among others. It will also include studies required for explanatory statements. Subsequently, consultancies to improve norms, plans, strategies will be made and will work in close coordination with the Regulatory offices in MIDAGRI, MINAM and SERNANP. As a third step, an Advocacy Plan will be developed for regulatory or legislative entities. In addition, a work plan will be developed in coordination with the Planning and Budget Offices of MIDAGRI and SERNANP to adjust or update budget programs and investment guidelines at national level. Then, these documents will be used by the ministries and regional and local governments to increase the allocation of public resources for EbA projects and CRVC.

137. **Sub-Activity 3.1.2.2: Improve coordination of EbA monitoring and evaluation (M&E) systems at the national level linked to project and NDC purposes (EE: GIZ):** This sub-activity aims to support the nationwide monitoring system with adequate protocols and measurement methodologies that allows MIDAGRI and SERNANP systematically collect, analyse data and information in order to detect the impacts of EbA interventions. This will allow policy makers to understand whether or in what extent EbA public investments are meeting their results, as well as monitor EbA effectiveness and evaluate progress towards the national adaptation climate contributions (NDCs). To this end, technical advice will be provided to MIDAGRI and SERNANP for the integration of the EbA approach and aspects of NDCs in its monitoring and evaluation processes and consider data disaggregation by gender. This sub-activity will be closed coordinated with the General Directorate of Climate Change and Desertification of MINAM that manages the Monitoring System for Mitigation and Adaptation measures (SIMOM). Community monitoring systems supported in activity 1.2.2 will also contribute with data and information to the national systems. This will be an exercise of exchanging information (data, tools for collecting data, interpretations, etc) from the local perspective and knowledge to national level and vice-versa. The project will also support MIDAGRI in the improvement of its reporting system for the NDCs in the agricultural and water NDCs to which the project will contribute to, by the development of adequate protocols and methodologies to gather field information and measure the impacts of MIDAGRI interventions and report MIDAGRI's NDCs progress implementation. To this regard, close coordination with the newly MIDAGRI's Agriculture and Irrigation Commission on Climate Change (CSARCC)⁷² created by Ministerial Resolution N° 0187-2023-MIDAGRI will be promoted. All actions mentioned above will be mainly coordinated with the General Directorate of Climate Change and Desertification of MINAM. Potential synergies with other relevant institutions such as SENAMHI, ANA, INAIGEM, SUNASS, and others for potential synergies will be promoted. Relevant entities involved in this activity will be Instituto de Montaña that will contribute to the design of the protocols to integrate community monitoring data M&E systems, MIDAGRI through its technical Directorates implementing Monitoring and Evaluation systems for ecosystems, and MIDAGRI's Agriculture and Irrigation Commission on Climate Change, SERNANP will contribute to the design of protocols in the of Natural Protected Areas, MINAM and affiliated entities such as SENAMHI and INAIGEM will contribute in the design of protocols and SUNASS will contribute to the design of protocols in the case of EPS target areas.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

B.4.2 Project management, implementation and execution structure

Role of GIZ as AE:

138. The Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH is a German-based private, limited liability, non-profit company owned by the Federal Republic of Germany; it is the Accredited Entity (AE) of this project, while the same GIZ with its management structure in Peru will operate as an Executing Entity (see next

⁷² The aim of this CSARCC is to monitor the implementation processes of climate adaptation and mitigation measures under responsibility of MIDAGRI, as well as develop annual reports regarding the advances of MIDAGRI's NDCs.

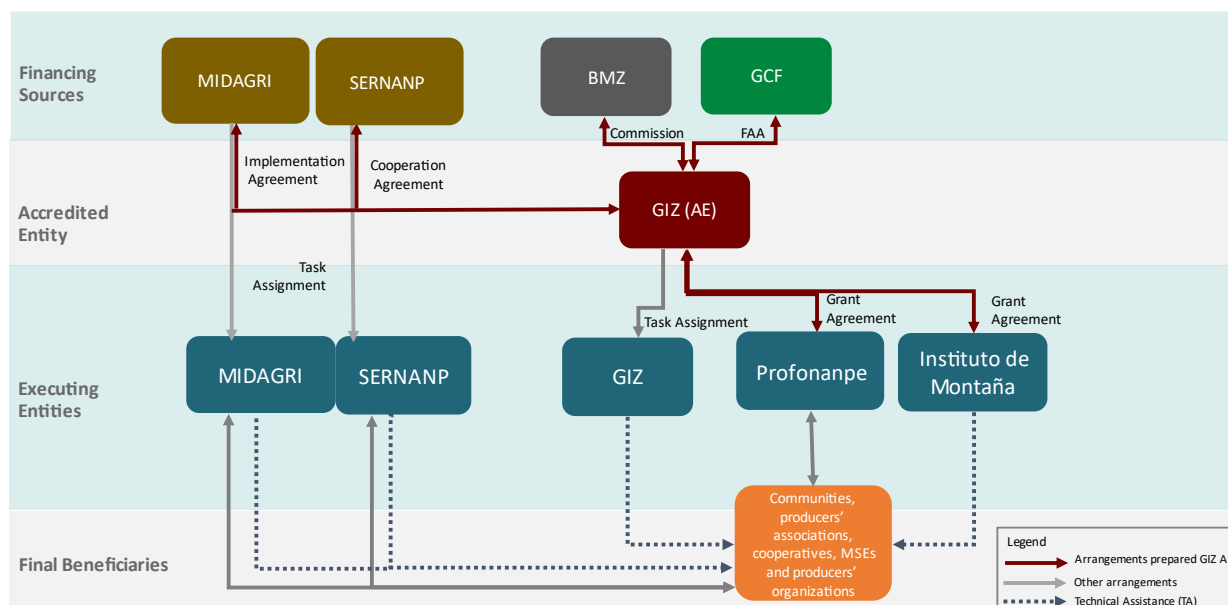
section). To avoid conflicts of interest, these two functions - accreditation and implementation function - are strictly separated, with different management structures within GIZ.

139. As the AE, GIZ will assume oversight responsibility of the project, as defined in the Accreditation Master Agreement (AMA) between GCF and GIZ (AE). As AE, GIZ will administer project funds on behalf of the GCF and will provide oversight guidance and quality assurance of Profonanpe and Instituto de Montaña as EEs receiving GCF funds through its relevant head office units.

140. In order to implement the Project, GIZ will need to establish legal arrangements with MIDAGRI, SERNANP, Profonanpe and Instituto de Montaña - see Figure 12. Legal below):

- The German Federal Ministry for Cooperation and Development (BMZ) will commission GIZ with the implementation of the GCF project (amended commissioning agreement). The GCF will transfer funds based on the Funded Activity Agreement (FAA) to the Accredited Entity GIZ.
- GIZ (as EE) will receive an internal task assignment from the AE for the implementation of the project.
- GIZ (AE) will amend an existing implementation agreement (i.e., subsidiary agreement), based on GIZ standard operating procedures with the MIDAGRI as the political partner of the project and Executing Entity executing activities with own funds (related to the BMZ commission and signed between GIZ and MIDAGRI).
- SERNANP as an Executing Entity executing activities with own funds will sign a cooperation agreement (i.e., subsidiary agreement), based on GIZ standard operating procedures with GIZ (AE).
- Finally, GIZ (AE) will sign with Profonanpe and Instituto de Montaña grant agreements (i.e., subsidiary agreements), based on GIZ standard operating procedures. These subsidiary agreements establish the legal basis on which GIZ makes the GCF Proceeds available to Instituto de Montaña to implement project activities and Profonanpe to set up, manage and operate grant disbursement through the Puna Facility, in accordance with the AMA and FAA.

Figure 12. Legal arrangements



GIZ as Executing Entity [Executing Entity-GCF funds]:

141. GIZ has been active in Peru since over 50 years and currently employs approximately 210 staff members, most of them Peruvian nationals. Specifically, GIZ Peru has been working on climate change and biodiversity issues in Peru since 2003 and current technical assistance in the sector amounts to approx. EUR 60 million.

142. In its capacity as an EE, GIZ will lead, and provide overall management of, the Technical Assistance (TA) to the project at national and subnational levels. It will be responsible for:

- Liaising with government institutions for overall coordination and alignment of objectives, managing technical assistance activities and ensuring full compliance with the expected project results and that adequate monitoring and evaluation procedures are implemented.
- Ensuring compliance with the Social and Environmental Safeguards, the Environmental and Social Management Plan (ESMP), the Gender Action Plan (GAP) and the Indigenous Peoples and Local Communities Engagement Plan (IPLCEP).

- Providing technical assistance to MIDAGRI and SERNANP to mainstream Ecosystem based Adaptation (EbA), climate resilience and gender aspects in all their programmes and to improve coordination between their operational units and collaboration with other sectors.
 - GIZ Peru, will also provide technical assistance to Profonanpe to improve their capacities to sustainable manage and promote the Puna Facility.
 - Technical assistance for the promotion of the "Mechanism for Remuneration for Ecosystem Services (MERESE)" schemes will also be under the responsibility of GIZ Peru and in close coordination with MINAM.
143. GIZ Peru, will also act as the technical secretariat within the Project Steering Committee and as the executive coordination within the Project Management Committee as well as the Implementation Unit, including the territorial teams.
144. In their role as technical secretariat GIZ Peru will keep the Steering Committee and the Management Committee informed in a timely manner about the main decisions or recommendations made by the GCF in the framework of the project implementation.
145. GIZ as an EE will be responsible for a GCF budget of EUR 19.315 million.

Profonanpe [Executing Entity-GCF funds]

146. Profonanpe is a non-profit private law institution organized under the laws of the Republic of Peru in accordance with Decree Law No. 26154. It is the only environmental fund in Peru and a direct access entity accredited before the GCF, with extensive experience in the management of environmental funds (Regional Water Fund in Piura, MERESE for Arequipa, etc.). Its mandate is to provide stable, long-term funding and to develop and implement innovative strategies for the conservation and management of protected areas. Its local, national and international partnerships with donors, its experience in managing environmental funds, and its strong fiduciary standards will ensure efficient and effective delivery of results.
147. Profonanpe will be responsible for the management of the "Puna Facility" (which will be set up under activity 2.1.1.) a competitive fund, which will provide through sub-activity 1.1.2.1. non-repayable and repayable grants, through calls for proposals aimed at promoting Local initiatives to implement Ecosystem based Adaptation measures and Climate Resilient Value Chains. In addition to the financial support, the Puna Facility will also provide Technical Assistance (through sub-activity 1.1.3.1) through a service provider for the implementation of the Local initiative, access to finance and business development.
148. In order to manage the Facility, Profonanpe will set up a Facility Management Unit (FMU). Among its functions⁷³ will be to:
- Ensure that all activities promoted and developed by the Puna Facility are in accordance with the strategic guidelines established by the Project Steering Committee.
 - Ensure submission of information to the Project Management Committee in case any modifications or updates are required at the operational or strategic level of the Puna Facility.
 - Develop the terms of reference for the calls for proposals.
 - First review of applications and pre-selection according to the terms of reference and established criteria and proposal for evaluation by the Project Management Committee (shortlist).
 - Negotiate Local Initiative Grant Agreements.
 - Monitor the execution of funded Local initiatives.
149. Under the supervision of GIZ AE, Profonanpe will ensure that all GCF and GIZ fiduciary standards are followed, complying with the stipulations of the Grant Agreement, always within the legal framework of the project (AMA, FAA).
150. Profonanpe will also participate in the Project Management Committee (PMC) and will be part of the Territorial Implementation Units (TIU).
151. In addition, it will liaise and coordinate with other project partners to guarantee the integral fulfilment of the expected results of the project.
152. Profonanpe as an EE will be responsible for a GCF budget of EUR 19.175 million.

The Ministry of Agricultural Development and Irrigation of Peru (MIDAGRI) [Executing Entity-Own contributions]

153. MIDAGRI is the Peruvian government institution in charge of the agricultural sector. It was created by Law 9711 on January 2, 1943. Its main function is to supervise and regulate the country's agricultural sector. The main objectives of MIDAGRI are: i) Strengthen producers' organizations and promote their integration under watershed and production chain management approaches; ii) Promote technological innovation and training linked to the business management of agricultural producers, providing technical assistance; iii) Establish an agricultural information system to enable economic agents to make efficient management decisions; iv) Facilitate agricultural producers' access to legal, administrative, management, financial, technical assistance, health and other services

⁷³ For more information, please see Annex 21 Operations Manual chapter 3.

that will enable them to improve their management capacity; v) To facilitate the linkage of small-scale agriculture with the market economy through the establishment of policies for the appropriate use of natural resources.

154. MIDAGRI as the main political partner of the project will also act as an Executing Entity executing activities through own funds through following relevant programs Agrorural, Sierra Azul, SERFOR, PSI, SSE, INIA, SENASA and Agroideas.
155. In addition, amongst his function in the project MIDAGRI will:
- Chair the Project Steering Committee and participate in the Project Management Committee and Territorial Implementation Unit.
 - Coordinate with the project partners to guarantee the integral fulfilment of the expected results of the project.
 - Ensures the efficient execution of its co-financing in its sector through the aforementioned programs and contribute to the achievement of the project's activities.
 - Lead coordination with aforementioned programs and the attached public bodies: SERFOR and INIA within the project structure and in the territory.
 - Internally, MIDAGRI will set up a sectoral commission to guarantee the coordination and harmonization of the interventions and co-financing of its units with the implementation of the project.

The National Service of Natural Protected Areas by the State (SERNANP) of the Ministry of the Environment (MINAM) [Executing Entity- Own contributions]

156. SERNANP is a specialized technical public agency attached to the Ministry of the Environment, in charge of directing and establishing technical and administrative criteria for the conservation of Natural Protected Areas (NPAs) and ensuring the maintenance of biological diversity. SERNANP is the governing entity of the National System of Natural Areas Protected by the Peruvian government (SINANPE), and as the technical-normative authority, it carries out its work in coordination with regional and local governments and landowners recognized as private conservation areas. It leads SINANPE with ecosystem, integral and participatory perspective with the aim of sustainably managing its biological diversity and maintaining the ecosystem services that provide benefits to society.
157. SERNANP will act as an Executing Entity executing project activities with own funds and amongst its functions it will:
- Coordinate with the project partners to guarantee the integral fulfilment of the expected results of the project.
 - Participating in both the Project Management Committee (PMC) and Territory Implementation Units (TIU).
158. In particular, SERNANP will co-finance and participate in the execution of the activities implemented within the Natural Protected Areas that are part of the project and together with MIDAGRI in the buffer zones.

Instituto de Montaña (IdM) [Executing Entity-GCF funds]

159. IdM is a non-profit organization that works for the conservation of the natural, cultural and spiritual values of mountain peoples and ecosystems. They have been working in the Andes since 1995 and in Peru they have programs in the highlands of Ancash, Piura, Junín and Lima.
160. It will act as Executing Entity, by contributing and scaling up its experience in the implementation of EbA measures in the Nor Yauyos Cochas Landscape Reserve to the other regions of the project. It will use participatory tools for participatory design, implementation and monitoring of EbA measures, which contribute to the ownership and sustainability of the Local initiatives supported by the project.
161. Amongst its functions in the project IdM will:
- Participate in the Project Management Committee and in the territory as part of the Territorial Implementation Units
 - Liaise and coordinate with other project partners to guarantee the integral fulfilment of the expected results of the project.
162. IdM as an EE will be responsible for a GCF budget of EUR 1.5 million.

B.4.2. Summary of the Capacity Needs Assessment

163. In the frame of the Funding Proposal development a Capacity Needs Assessment (see Annex 2h) was developed in order to assess the capacity needs and institutional gaps from the EEs (besides GIZ) for project implementation. This included evaluating EEs current capacities in relation to the project components, identification of gaps, and providing recommendations for capacity development to address these gaps effectively.
164. In this regard, the Capacity Needs Assessment draw two key conclusions. Firstly, in some cases, capacities exist, but enabling conditions are weak, at the same time the implementation process of EbA, climate resilience and gender approaches are making progress as it is the case of MIDAGRI for example, even though it has the capacities (established functions) to develop specific guidelines, methodologies and tools to incorporate a gender and climate resilience approach into sector planning, budgeting and investments; however, these are insufficient. Secondly, there is the need to enhance existing capacities and establish new ones where gaps exist as in the case

of MIDAGRI, even though it has several agricultural support programmes, there is little promotion of these in the high Andean zones.

165. A list of recommendations for each EE, based on the gaps identified can be found in section 5 of the Capacity Needs Assessment (Annex 2h).

B.4.3. Project Governance structure

166. The project Governance structure (shown in

167. Figure 13) was extensively discussed, developed and validated by the project partners. Following bodies will be established:

Project Steering Committee (PSC)

168. The highest level of governance will be overseen by a Steering Committee consisting of GIZ, MIDAGRI and MINAM.

169. MIDAGRI through the Vice Minister of Family Agriculture Development and Agrarian Infrastructure will chair the PSC meetings. The substitute is the General Director of the General Office of Planning and Budget of MIDAGRI. Specifically, PSC participating members will include:

- MIDAGRI Vice-Minister of the Vice-Ministerial office for Family Farming Development and Agrarian Infrastructure.
 - MINAM Vice-Minister of the Vice-Ministerial office for Strategic Development of Natural Resources, which will provide the required climate guidance and its alternate will be the General Directorate of Climate Change and Desertification.
 - Representative of GIZ
 - The PSC will provide policy and strategic guidance to the Project Management Committee (PMC), while ensuring compliance with the Funding Proposal approved by the GCF, the legal agreements, policies and requirements of the GCF, and the climate and national socio-economic development objectives.
170. As part of the responsibilities the PSC will:
- Acknowledge proposed tools, directives and/or standards presented by the PMC and advocate for their inclusion in public policy.
 - Encourage national authorities to take ownership of actions addressing climate issues and ensure project coherence within the international and national contexts.
 - Provide guidance for the fulfilment of legal agreements.
 - Monitor compliance with the project's commitments to social and environmental safeguards, the Indigenous Peoples policy, and the objectives and implementation of the respective plans (ESMP, IPLCEP and GAP).
 - Ensure effective inter-institutional and inter-sectoral coordination, as well as collaboration with civil society.
 - Resolve any disputes that arise within the Project Management Committee and holds decision-making authority for their resolution.
 - Annually approve the project Annual Operating Plan (AOPs), evaluate and provide approval for the execution of the AOPs (technical and financial execution reports).
 - Provide strategic guidance for project management and provides feedback on the AOP, based on systematic reporting of outcome and impact indicators.
 - Oversee the timely and adequate delivery of the respective contributions of the partners to the project.
 - Supports process transparency and the harmonization of working approaches in the territory.
171. The Project Steering Committee (PSC) will establish a Technical Secretariat, operated by GIZ Peru. The Secretariat will provide technical information to the members of the PSC and will serve as a bridge between the PSC members and the Project Management Committee (PMC). In addition, it will coordinate (before, during and after) all related aspects of the PSC sessions.
172. The PSC will meet at least 2 times a year and more frequently, when necessary.
173. PSC members have the option to organize extended PSC meetings where non-permanent PSC members can participate, for example for accountability and/or consultation with the NDA (Ministry of Economy and Finance), as well as to seek inputs on project-related matters with other stakeholders such as national, regional, and local authorities, donors, community and producer organizations, women's groups, Indigenous Peoples representatives, NGOs, private sector representatives, and thematic experts.

Project Management Committee (PMC)

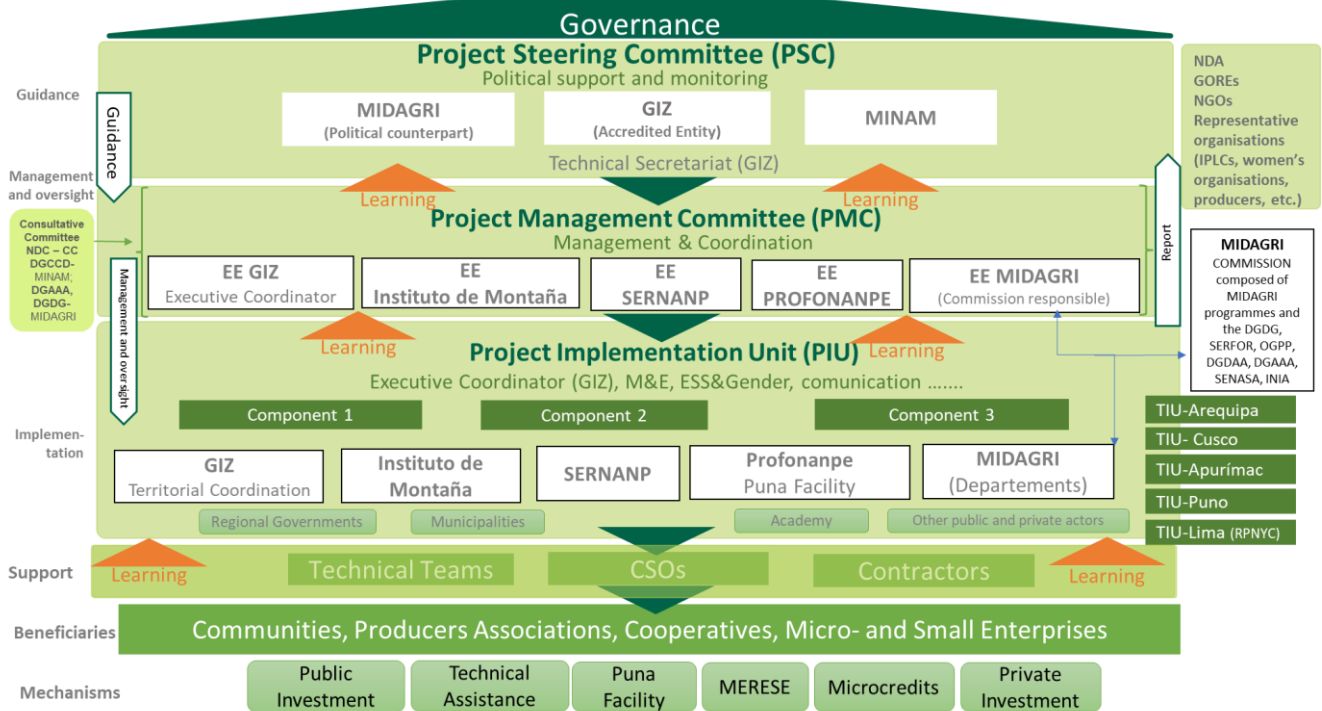
174. The Project Management Committee (PMC) will be responsible for overseeing and coordinating the project's execution among the Executing Entities and ensuring its implementation within the designated territory. The PMC will also provide regular reports to the Steering Committee.

175. It is the decision-making body for the management and implementation of the project. The PMC will be coordinated by GIZ Peru and will have its offices within MIDAGRI. If necessary, thematic working groups could be

created to coordinate cross-cutting activities in depth (e.g., on safeguards, gender, financial management or EbA/value chains, etc.).

176. The PMC will additionally be responsible to:
 - Review, reflect, and systematize lessons learned from the project, proposing their integration into tools, directives or standards to be considered by Executing Entities.
 - Generate proposals in response to changes, effects, or events in the project areas that may be attributable to climate change, based on intersectoral reviews with experts and agreements with the GCF.
 - Mobilize timely technical expertise from EEs and other stakeholders when project development issues are identified.
 - Timely share guidelines with the PSC and offer feedback to enhance project coherence on both national and international levels, while also facilitating the flow of information from the territories to the PSC.
 - Supervise the implementation and compliance with Social and Environmental Safeguards and Gender policies (i.e., ESMP, GAP and IPLCEP).
 - Ensure compliance with project management directives, guidelines, and information flow among Executing Entity members.
 - Coordinate and ensure the timely implementation of activities by each Executing Entity in the designated territory outlined in the Annual Operating Plan, systematically informing the PSC and promoting a common understanding among Executing Entities regarding the theory of change and its mainstreaming across all project activities.
 - Foster inter-institutional and inter-sectoral coordination and collaborate with civil society.
 - Oversee technical and budgetary project progress, aligning with approved plans, budgets, and indicators related to territorial needs outlined in Annual Plan documents.
 - Provide technical feedback on Terms of Reference (ToRs) and other selection processes for regional coordinator positions in Territorial Implementation Units.
 - Lead mid-term and final evaluations of the project.
177. The PMC will meet at least 2 times a year and permanent PMC members will include according to the entity:
 - MIDAGRI as an EE with a technical personal and financial person designated by MIDAGRI.
 - GIZ as an EE with the project manager, technical manager and financial manager.
 - Profonampe as an EE with a technical manager and financial manager.
 - SERNANP (MINAM) as an EE with a technical manager and financial manager.
 - Instituto de Montaña as an EE with a technical manager and financial manager.
178. Likewise, a Consultative Committee will be established. It will be invited to participate in the Project Management Committee (PMC) meetings through following members: the General Directorate of Climate Change and Desertification (DGCCD) of MINAM; the General Directorate of Agricultural Environmental Affairs (DGAAA) and General Directorate of Livestock Development (DGDG) of MIDAGRI, each with directorate and technical representatives; providing technical assistance on the project issues related to climate change, within the framework of its institutional competencies, including those corresponding to the implementation, monitoring, evaluation and reporting of nationally determined contributions (NDC). The Management Committee will issue the corresponding meeting invitation and agenda.
179. When necessary, extended Management Committee meetings can also be held, including the following MIDAGRI's programs, MINAM's General Directorates, National Institute for Glacier and Mountain Ecosystem Research (INAIGEM), National Service of Meteorology and Hydrology of Peru (SENAMHI). Other public or private entities, women's groups and/or Indigenous Peoples and Local Communities representatives, according to the subject to be addressed at the respective meeting.

Figure 13. Governance structure



Coordination of the Project Implementation Unit (PIU):

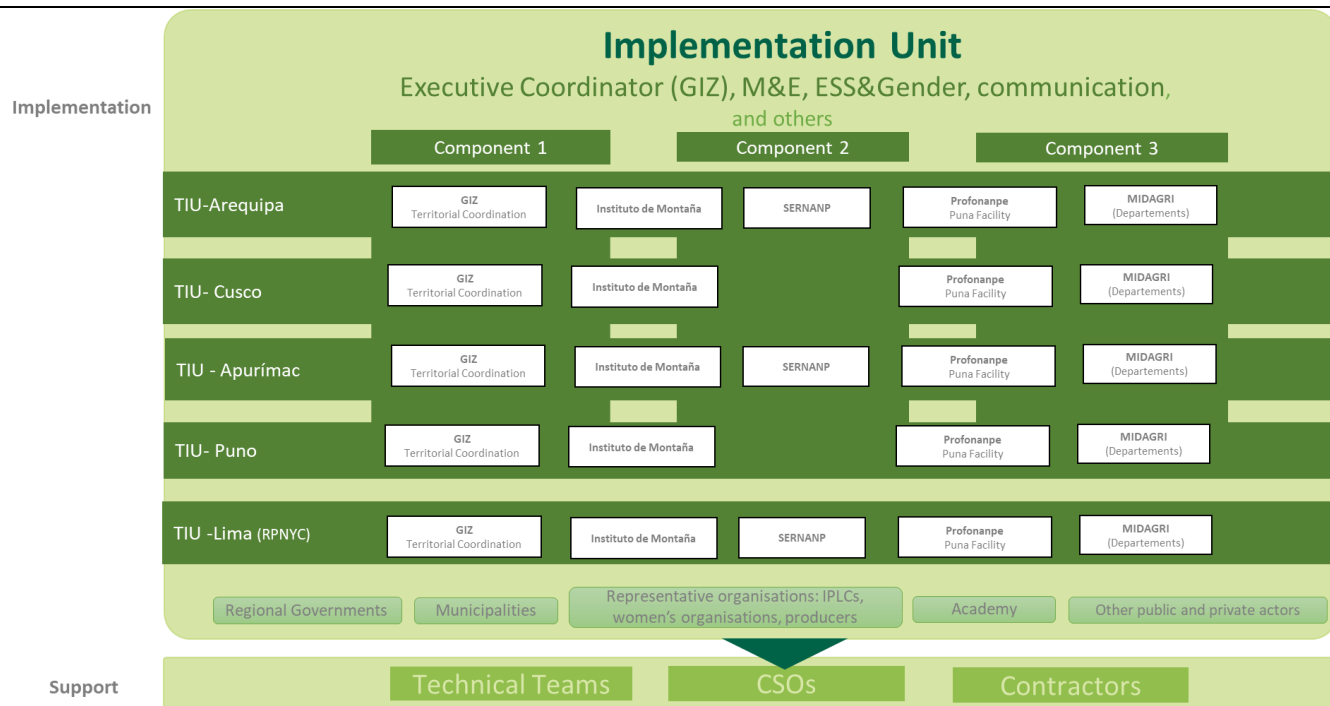
180. The governance structure will also include a Project Implementation Unit. The Coordination of this level will be in charge of GIZ, and its responsibilities will include:

- Ensuring consistency, providing feedback, and validation of learnings made by territorial teams, local communities, organizations and institutions around ecosystem management, conservation and restoration, territorial governance and the coordination of collaborative efforts at institutional and social levels.
- Systematizes knowledge about changes/events occurring in the intervention areas that could be attributable to climate change and the effects they generate in populations and ecosystems.
- Transmits in a timely and appropriate manner PSC orientations regarding to the national and international context and, where appropriate, follows up on their implementation by the territorial teams.
- Executes the recommendations of the PMC and ensures that the recommendations of the Territorial Implementation Units (TIUs) are discussed and addressed.
- Ensures compliance with the Social and Environmental Safeguards and Gender policies (i.e., ESMP, GAP and IPLCEP).
- Promoting a common understanding of the theory of change and mainstreaming it in all project activities by the territorial teams to ensure adaptive project management in order to foster information exchange and synergies among project experts and advisors.
- Guaranteeing the harmonized work with stakeholders in each territory and an adequate identification of the tasks and additional requirements that must be addressed in connection with the project.
- Ensuring the implementation of the project in accordance with the project proposal approved by the GCF.
- Coordinating and integrating the Territorial Annual Operating Plan into the general Annual Operating Planning of the project to be reviewed by the PMC and approved by the PSC.
- Coordinating the M&E system and monitoring the technical and budgetary progress of the project, in accordance with the approved Annual Operating Plan and budget. Including monitoring of a timely implementation in the territory and systematically informs the PMC in this regard.
- Preparation of project reports.
- Overseeing the progress of the indicators in charge of each EEs in the monitoring system and ensures that the implementation progress reports are consistent with the reports to be provided to the GCF.
- Ensuring exchanges among territorial teams to assess the project progress, challenges and lessons learned in each territory.

Territorial Implementation Unit (TIU):

181. For an efficient, coordinated and articulated implementation in the territory, the executing entities will be formed into implementation units in each region of the project area, coordinated by the GIZ. Part of the responsibilities of the TIU will include:
- Generate knowledge and learning regarding the project implementation, and the engagement with beneficiaries, public-private stakeholders, and other issues related to project implementation.
 - Identify, record, report changes/events occurring in the intervention areas that could be attributable to climate change and the effects they generate in populations and ecosystems.
 - Implement PMC guidance regarding the national and international context in project actions.
 - Follows PSC and PMC guidance.
 - Comply with the Social and Environmental Safeguards and Gender policies (i.e., ESMP, GAP and IPLCEP).
 - Participate in the Regional Agricultural Management Committees (CGRAs) meetings to inform, and coordinate issues of interest for the region and the Project.
 - Participate/promote dialogue/working groups or other coordination bodies, when needed.
 - Promote strategic alliances with local governments, regional governments, private sector, civil society organisations, to generate synergies during project implementation.
 - Coordinate the implementation of project activities in each territory, according to the Project's management instruments.
 - Prepare territorial Annual Operating Plan to feed the project Annual Operating Plan, in accordance with the approved Logical Framework. Report the progress of goals, indicators, bottlenecks, etc. to the PIU Coordinator,
 - Promote the participation of beneficiaries in the different public-private financing mechanisms and identify the needs of actors in the territory linked to the project's themes.
 - Supervise and monitor the implementation of project activities in the territory
 - Identify other local and regional initiatives that may have potential synergies with the interventions planned by the project and inform the PIU Coordination.
 - Identify the potential private sector financing in the territories and report to the PIU Coordination.
182. Part of the members of the Territorial Implementation Unit:
- GIZ: With regional technical coordinator.
 - MIDAGRI: With regional technical specialists of Agroideas, Sierra Azul, Agrorural, Serfor
 - SERNANP: With regional technical specialists.
 - Profonampe: With regional technical specialists.
 - Instituto de Montaña: With regional technical specialists.
183. Depending on the specific approaches and developments in the implementation of the project in each of the territories, TIUs may occasionally invite other participants such as representatives of Indigenous Peoples and Local Communities, women's groups, producers' associations cooperatives, civil society organizations, local governments and regional governments to participate in the meetings. At the request of a specific TIU and depending on the fulfilment of the TIU's work plan, the TIU coordinator may authorize the regular inclusion of one or more of these representatives.

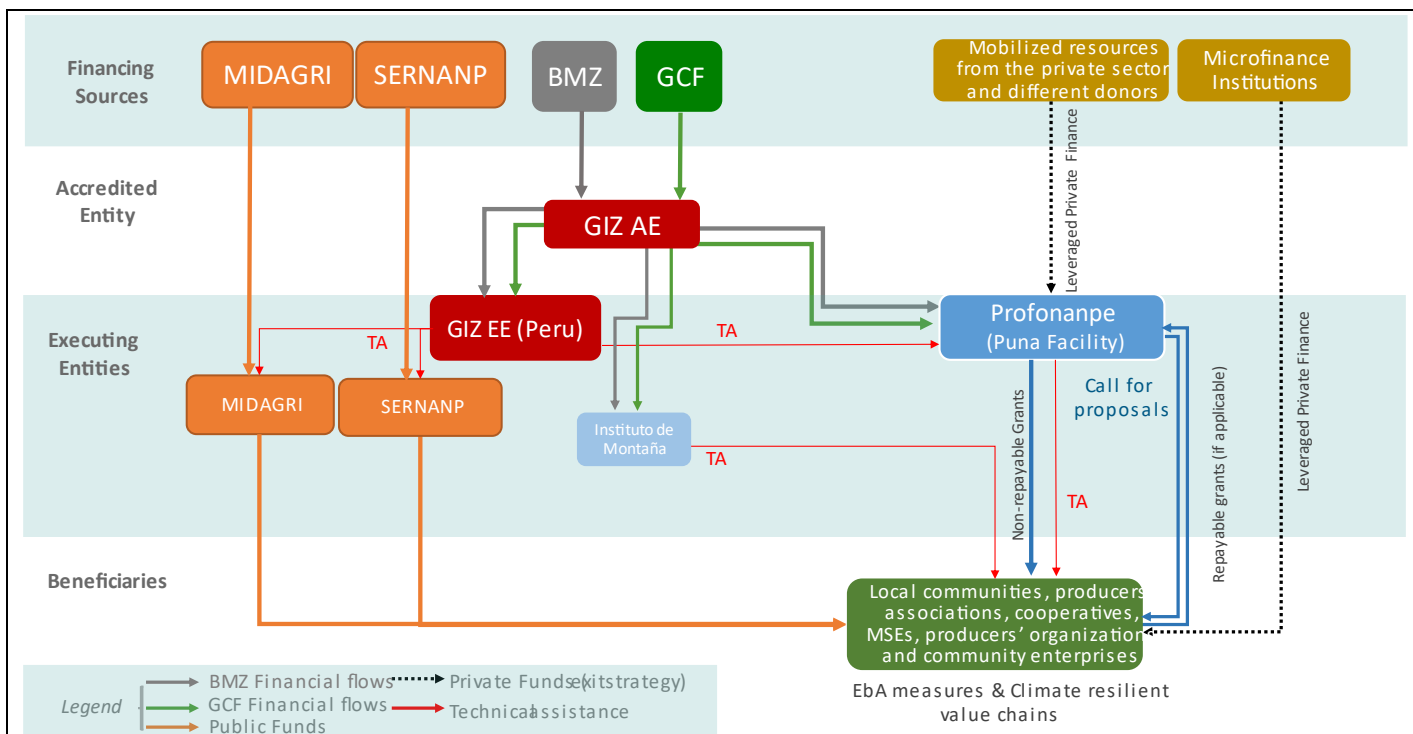
Figure 14. Structure of the Territorial Implementation Unit (TIU)



B.4.4 Flow of Funds

184. The following Figure 15 depicts the overall flow of funds for the project. Funds from GCF will be transferred to GIZ as AE, who will then transfer funds through the Puna Facility (manage by Profonanpe (in its role as EE) to the beneficiaries and directly to GIZ Peru and Instituto de Montaña (in their respective role as EEs). The Government of Peru through SERNANP and MIDAGRI will provide co-financing.
185. Additional funds either from Microfinance Institutions (MFIs) (as a result from the provided TA within the activity 2.1.2) in form of loans or mobilized resources in form of grants from the private sector and different donors (as a result of the TA provided to Profonanpe within sub-activity 2.1.1.2) could be leveraged. These funds will either flow directly through the MFIs to the project beneficiaries or through the Puna Facility to continue financing Local initiatives. For the later, initial discussions have already started with the Government of Canada who has expressed their interest in channelling grants through the Puna Facility.

Figure 15. Flow of funds



B.4.5 Puna Facility Funding Windows

186. As described above, GIZ will make GCF proceeds available to Profonampe to then channel funds to Indigenous Peoples and Local initiatives in their different form of organization (e.g., local communities, producers' associations, cooperatives (formal and in the process of formalization), Micro and/or Small Enterprises (MSEs)⁷⁴, community enterprises and producers' organizations⁷⁵) to finance Local initiatives combining EbA measures and the development of Climate Resilient Value Chains based on predefined eligible EbA measures and CRVC (as shown in table 6 of the Operations Manual (Annex 21). Further detailed procedures are outlined in the Operations Manual (Annex 21).
187. Profonampe, as described in the Operations Manual (Annex 21), will be responsible for the management (with guidance of the PSC and PMC) and transfer of GCF funds to beneficiaries, including all related financial monitoring and auditing and ensuring compliance with the GCF ESS and Gender requirements. Detailed responsibilities and the role of PSC and PMC are stated under chapter 3 Institutional arrangements and responsibilities for managing the financial mechanism in the Operations Manual.
188. Grant disbursements channelled through the Puna Facility will be through call for proposals (up to three within the project implementation) and will be differentiated by three sub-windows (see
189. Figure 16) based on the recipient of the funds. Mainly:
- Non-Repayable Grant Sub-Window** for formal/legal Indigenous Peoples and Local Communities in their different forms of organization (communities, producers' associations and cooperatives⁷⁶ in the process of formalization)⁷⁷ (from here on IPLCs), applying with the support of local CSOs.
The Non-repayable Sub-Window is meant to finance a maximum non-repayable grant amount per Local initiative activities (including both EbA and CRVC interventions) of up to EUR 100,000.
 - Results-based Repayable Grant Sub-Window** for micro and or/ small enterprises community enterprises and cooperatives permitted by law with commercial capacities and clear target markets. Hence:
 - Formal/legal micro and/or small enterprises
 - Formal/legal community enterprises according to the law N° 24656

⁷⁴ Micro and/or small enterprises are defined according to the [Legislative Degree 1086](#). Micro enterprises have from 1 to 10 employees and annual sales up to 150 taxable tax units (Unidad Impositiva Tributaria), while small enterprises have from 1 to 100 employees and annual sales up to 1700 taxable tax units (Unidad Impositiva Tributaria).

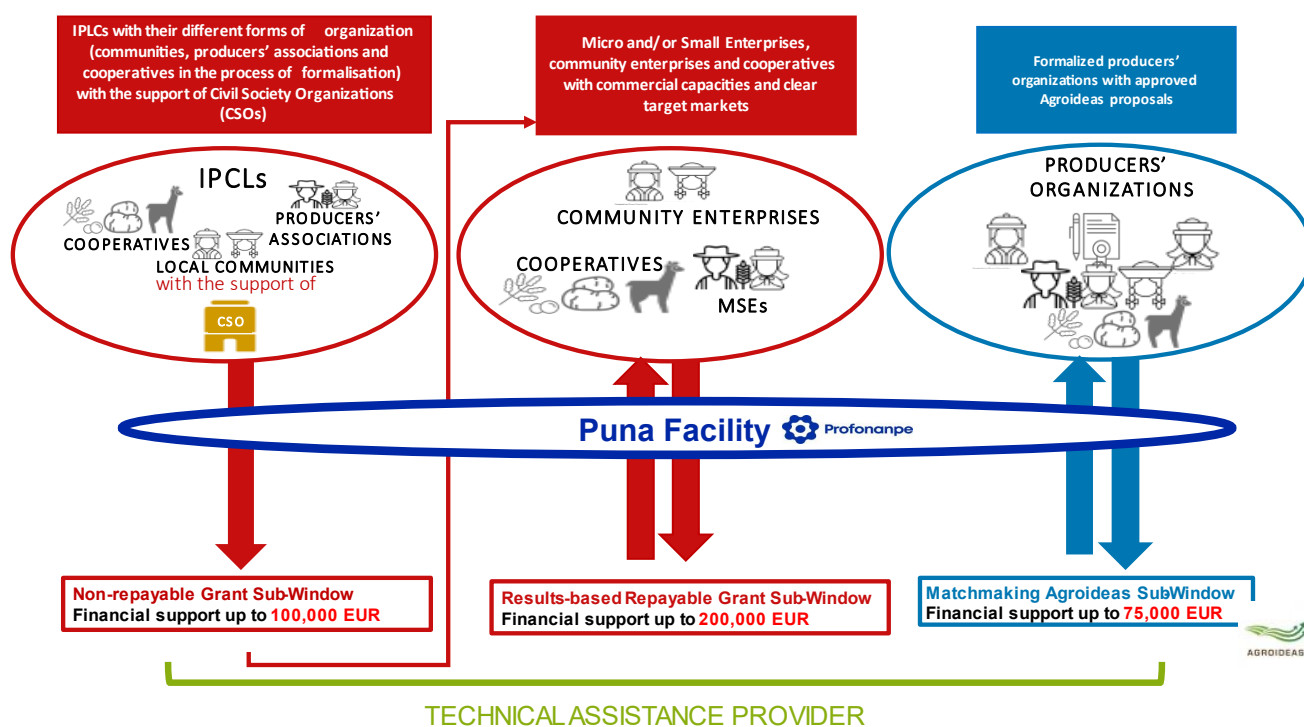
⁷⁵ Small and/or medium agricultural producers under any organizational form permitted by law: associations, native communities, peasant communities, limited liability companies, public limited companies, cooperatives and other forms allowed by law. The definitions of small and medium agricultural producers as well as the forms of organizations are defined by the Agroideas eligibility criteria that can be accessed [here](#).

⁷⁶ Cooperatives working on the topics related to the project.

⁷⁷ The Law No. 31335, law on the improvement of the association of agricultural producers in agricultural cooperatives was recently regulated on July 21, 2023. From this regulation some associations will migrate to cooperatives, regime in which they will be able to commercialize and have net benefits in an associative way, with which in the future they will be able to demonstrate the generation of income, but the process will involve some requirements that may not allow them to apply to sub-window 2 in the first call.

- Formal/legal Cooperatives according to the legislation in force.
- The Results-based Repayable Sub-Window is meant to finance a maximum repayable grant amount per Local initiative activities (including both EbA and CRVC interventions) of up to EUR 200,000.
- c. **Agroideas Matchmaking Sub-Window** for formalized producers' organizations with commercial capacities and clear target markets, that are eligible for the MIDAGRI support programme Agroideas. Hence:
- Small and/or medium agricultural producers organized under any organizational form permitted by law: associations, native communities, peasant communities, limited liability companies, public limited companies, cooperatives and other forms allowed by law.
- The Agroideas Matchmaking Sub-Window is meant to finance a maximum results-based repayable grant amount per Local initiative activities (including only EbA measures) of up to EUR 75,000.

Figure 16. Operation of the Puna Facility: Sub-Windows



190. The principle underlying the differentiation between the different sub-windows is related to the segmentation of the potential beneficiaries which are reflected in the differences between their a) current internal organizational and commercial capacities and b) the level of barriers they face (please see section B.2a.2. for more information). In this regard the Non-Repayable Grant Sub-Window is meant to benefit local communities, producers' associations and cooperatives in the process of formalization without commercial capacities and clear target markets and the Results-based Repayable Grant Sub-Window is meant to benefit Micro and/or Small Enterprises, community enterprises and cooperatives with commercial capacities and clear target markets. Moreover, the Agroideas Matchmaking Sub-Window (also for Results-based repayable grants) – in line with the guidelines of the Agroideas programme – is meant to benefit formalized producers' organizations with commercial capacities and clear target markets conforming to the rules and regulations of the Agroideas programme.
191. Reflecting the fact that Non-repayable Grant Sub-Window beneficiaries lack the required level of preparedness and ability to submit and implement Local initiative proposals on their own, applicants of the Non-repayable Grant Sub-Window will partner with qualified local CSOs to submit and implement approved Local initiatives (while ensuring that beneficiaries are proactively involved in the process and their capacity is built). **Error! Reference source not found.** in the Operations Manual (Annex 21)**Error! Reference source not found.** lists the eligible recipients for the three Sub-Windows in more detail.
192. Up to 100% grant repayments (if applicable for Sub-Window 2 and 3) will be based on specific performance triggers established case-by-case and agreed including but not limited to:
- i. Percentage or absolute increase in revenues compared to the revenues in the year before application to the Puna Facility, or the average revenues over a certain number of years before application;
 - ii. Percentage or absolute increase in profits over the same reference period;

- iii. Percentage or absolute increase in cash flows or cash balances over the same reference period;
- iv. Increase in business volumes, such as volumes of crop sold, alpaca fiber sold, alpaca heads in herd or similar indicators;
- v. Other initiative-specific triggers, such as winning a large sale contract or gaining a new large client (e.g., commitment by alpaca fiber processor to buy large volumes).

193. Upon achieving the performance trigger, grant repayment will follow a specific repayment schedule tailored to each Local initiatives to address the particularities of the EbA and CRVC interventions. The Puna Facility will aim to achieve full repayment within five years from the last grant tranche disbursement, however it is possible that repayment could take longer for certain local initiatives, depending on the timing of the trigger achievement and cashflow profile of the initiative. More information on the repayment conditions can be found in the Operations Manual chapter 6 and 7. Grants repaid will flow back to the Puna Facility. More information on the use of repaid grants and sustainability of the Puna Facility can be found in section B.6.1.

194. The Financial Window aims to support up to 127 local initiatives over 5.5 years for a total disbursed grant amount of up to EUR 14.4 million. The preliminary breakdown⁷⁸ of the funds between the sub-windows will be as indicated in the table below.

Table 6. Preliminary Grant Breakdown per Sub-Window

Sub-Window	Funding instrument	Cost per project in EUR up to	Estimated # of Local initiative	Estimated Grants in EUR per Sub-Window up to	Funding source
1. Non-repayable Grant	Non-repayable grants	100,000	75	7,500,000	GCF and BMZ
2. Results-based Repayable Grant	Results-based repayable grants	200,000	24	4,800,000	GCF and BMZ
3. Matchmaking Agroideas	Results-based repayable grants	75,000	28	2,100,000	GCF and BMZ
Total			127	14,400,000	GCF and BMZ

195. In parallel to financial support, the Puna Facility will offer Technical Assistance (TA) through a service provider hired by Profonanpe and in the case of the Non-repayable Grant Sub-Window optionally in conjunction with the CSO supporting the applicant. The TA packages will be differentiated between:

- a. TA for the refinement of the local initiative proposals, (please see Step 5 in Section 3.3. of the Operations Manual (Annex 21) provided by the TA provider and
- b. TA for the implementation of the local initiatives (please see Step 10 in Section 3.3.) provided by the TA provider and optionally in conjunction with the CSO in the case of the Non-repayable Grant Sub-Window

196. The total amount of TA that will be provided to the beneficiary by either the TA service provider or (in applicable) the CSOs will be of up to EUR 3.17 million.

B.4.6 Regulation, taxation and insurance

Privileges, immunities and taxation

197. The Federal Republic of Germany has concluded Framework Agreements on Technical Cooperation under international public law with the Government of Peru (dated 25th February 1998), which provide for certain privileges and immunities to be applied in projects and programmes of technical cooperation, including exemptions for taxes, customs, duties and fees. GIZ will endeavour to reach arrangements to have these privileges and immunities also applied to the Project, including GCF proceeds. However, formal agreements will only be entered into after the project has been approved.

Approvals, permits, licenses and land

198. At the time of submitting this Funding Proposal, GIZ is not aware of the project requiring specific approvals, permits, licenses or land to allow for the implementation of planned activities.

⁷⁸ The distribution of the grants per sub-window could change during project implementation based on the development and lessons learned from the first and second call cycle.

Currency

199. The local currency in Peru is the Peruvian Sol (code: PEN). The project's local transactions will use PEN. The AE will manage GCF proceeds in Euro. GIZ will apply its standard rates for the conversion of currency.

Insurance

200. GIZ, as the AE, will ensure that project activities are adequately insured as per GIZ standard operating procedures and common practices. GIZ standard operating procedures require contractors to ensure required insurance cover. GIZ policies provide insurance cover for GIZ staff. At the time of submitting the Funding Proposal, GIZ does not anticipate additional insurance to cover special risks in this project.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

B.5.1. Grant justification – Concessionality

201. While the government has increased its focus on climate finance, efforts to channel budgetary or other public resources to climate resilience in the high puna are still limited and scattered. The government's annual spending on climate change has been below USD 300 million for the past decade. For comparison, public investment on infrastructure has averaged USD 5.4 billion over the past five years and total public investment USD 10.6 billion.⁷⁹ The World Bank estimates at USD 45 billion Peru's investments needs in climate change mitigation and adaptation to 2030, with the biggest priorities identified in the transport, water and energy sectors. The government has increased public spending aimed at managing natural resources sustainably, from 1.9% of the total national budget in 2008 to 5.4% in 2018 (IDB, 2020). However, national resources remain insufficient due to the global environmental and climate crises, as shown in national environmental performance indicators (ranked from 0 to 1, where 1 signifies the highest possible score), for example the agriculture sector scores 0.3 with a regional average in LAC of 0.45, on the other hand the forest sector scores 0.26 with a regional average in LAC of 0.57 (IDB, 2020).
202. The required investment for climate adaptation needs cannot be met with the national budget alone. With the economy expected to slow down, inflation risk, relatively high unemployment, public debt on the rise and persistent political instability, it will be difficult for the government to rise to the climate investment challenge. In this context, the government is making a substantial effort to co-finance the proposed GCF project, matching the GCF contribution almost dollar for dollar (~1:1 co-finance ratio). GCF resources are crucial to achieve innovative coordination between private and public actors to scale the impact of interventions at the ecosystem and local community levels. The resources will also allow overcoming barriers to accelerate the implementation of existing but underdeveloped financial instruments (i.e., MERESE) and will generate capacities in rural population to access available financing (public and private). The proposed financial and technical support mechanisms, including business development, microfinancing and increased participation in specific Climate Resilient Value Chains (CRVC) will improve production systems and make them more resilient.
203. Available resources from the international community are also limited. Peru received USD 923 million in climate-focused official development assistance (ODA) in 2021, the majority in mitigation. Of this, only USD 13 million was ear-marked by the OECD to the agriculture (excluding forestry) and livestock sub-sectors.⁸⁰ It is possible that other ODA was channelled to these sectors via financial institutions and ear-marked as ODA to the financial sector. Still, agriculture and livestock draw a small minority of climate ODA to Peru. In the high puna, ODA directed at the project's target beneficiaries has been extremely limited.
204. Private sector financing of CRVC in the puna has been almost absent to date (see section 7.2 in the Feasibility Study (Annex 2a)). Universal and municipal banks in Peru focus primarily on large enterprises and/or urban borrowers. MFIs focus on MSME lending but only one (Confianza) has a meaningful presence in the target areas, as do a few small rural banks. Lending to CRVC is constrained by several barriers: (i) the inherent high risk of agricultural and livestock activities, exacerbated by climate change; (ii) logistical difficulties in reaching borrowers in the high puna (long distances and remote locations); (iii) limited financial literacy of prospective borrowers; (iv) limited understanding – by both borrowers and lenders – of the climate change impact on high puna economic activities, and how EbA and CRVC can mitigate such impact and improve creditworthiness; and (v) prevalence of informal businesses with limited collateral. As a result of these barriers, loan supply to the target beneficiaries is extremely limited and loan terms are inadequate to EbA measures and CRVC (short maturities, very high interest rates, inflexible repayment schedules). For example, capex loans offered by MFI Confianza have a maximum maturity of 3 years only and interest rates in the 60-70% range. One state-backed financial institution, Agrobanco, offers at least on paper concessional loans (with “low” rates of 20-30%) to the project's target beneficiaries, but in practice the organization is severely undercapitalized and operationally dysfunctional.
205. A significant portion of the GCF grant (EUR 21.6 million) is for technical assistance and capacity building activities that do not lead directly to financial reflows for beneficiaries. The use of grants for these activities is justifiable on a public-good basis. The remaining EUR 19.217 million portion of the GCF grant is to fund the establishment and operation of the Puna Facility, which will disburse repayable and non-repayable grants to

⁷⁹ World Bank (November 2022). *Peru: Country Climate and Development Report*.

⁸⁰ OECD climate ODA data, 2021.

revenue generating Local initiatives. A detailed financial analysis has been conducted (see Annex 3) to determine the financial IRR of Local initiatives in the three windows of the Facility, with and without GCF grants. The analysis is based on prototype Local initiatives in the alpaca and native potatoes sectors representative of the majority of Local initiatives to be funded by the Facility. Detailed assumptions were developed both for business-as-usual scenarios (not incorporating EbA and CRVC practices) and climate-resilient scenarios where beneficiaries invest in and implement the recommended mix of EbA measures and CRVC practices. The transition from business-as-usual to EbA/CRVC generates positive FIRR without grants in all prototype's Local initiatives, ranging from 8.4% (for alpaca breeding in the Non-repayable Grant Sub-Window) to 25.8% (native potatoes in the Results-based Repayable Grant Sub-Window). Still, grants are considered essential for the widespread adoption of EbA/CRVC among target beneficiaries, for the following reasons:

- i) In the **Non-repayable Grant Sub-Window**, (i) the beneficiaries live in poverty to the extent of often abandoning the high puna and moving to urban areas in search of jobs (see section D.4.2). (ii) They have virtually no savings, and certainly not to the tune of the EUR 125,000 and EUR 139,000 per beneficiary group required to implement the transition to EbA/CRVC (in native potatoes and alpacas, respectively). (iii) Beneficiaries have limited financial literacy and almost no access to finance; when finance is available, it is offered by micro-finance institutions at prohibitive rates and for short maturities that are unsuitable to the timeline of CR transition (the payback period for the EUR 125,000 investment is approximately 8 years). (iv) Beneficiaries do not have the knowledge to implement EbA measures nor the financial resources to pay for technical services provided by private experts in order to gain that knowledge. GCF non-repayable grants not only cover a significant portion (80%) of the EbA/CRVC investment but fund the technical assistance necessary to correctly implement the measures and put EbA/CRVC-adopting businesses on a sustainable path of cash generation, building their future bankability.
- ii) In the **Results-based Repayable Grant Sub-Window**, (i) beneficiaries have limited savings, but certainly not to the tune of the EUR 257,000 and EUR 283,000 per farmer group required to implement the transition to EbA/CRVC. (ii) They have limited financial literacy and almost no access to finance; when finance is available, it is offered by micro-finance institutions at prohibitive rates and for short maturities that are unsuitable to the timeline of transition. (iii) Beneficiaries do not have the knowledge to implement EbA and have limited financial resources to pay for technical services provided by private experts in order to gain that knowledge. Considering the attractiveness of the business model over the long term, however, non-repayable grants would be too concessional. To minimize concessional, a repayable grant is therefore introduced, with the benefits that (a) beneficiaries start building a track record in managing repayments that enhances their future bankability with MFIs and other financial institutions, (b) repayment terms are negotiated at Local initiative approval to suit the Local initiative's cashflow profile (unlike MFIs' rigid repayment schedules), (c) there is no interest charge (unlike the MFIs' prohibitive rates) and (d) repaid amounts flow back into the Facility, which will use it to support additional future Local initiatives. In addition to the repayable grant, beneficiaries also receive technical assistance. The combination of financial and technical assistance will put them on a path to full commercial bankability.
- iii) In the **Agroideas Matchmaking Sub-Window**, for which only existing formal businesses qualify, a very significant investment is required, to the tune of EUR 245,000 in our prototype Local initiative. While the applicants are formalized and may have some access to finance, it is not for this sort of amounts and would carry the prohibitive conditions imposed by MFIs. To alleviate the funding gap, the government's Agroideas program will provide non-repayable grants covering 80% of the value chain portion of the investment (e.g., construction of greenhouses, cultivated pastures and alpaca shearing equipment). The GCF grant will be repayable and therefore less concessional. It will cover 80% of the EbA portion of the investment, will carry no interest and will have repayment schedules tailored to the individual Local initiatives. Financial and technical assistance provided by Agroideas and GCF will put these businesses on a solid footing, increase their creditworthiness and likelihood of sourcing affordable commercial loans in the future.

- 206. The project strongly contributes to the GCF identified sectoral paradigm shift enablers. The government of Peru sees GCF financing in the form of grants as essential to achieve changes in the geographical and temporal scale required to face the challenges of accelerated change in climate and the loss of ecosystems and their climate services. In this regard, GCF funding is required to contribute to the GCF sectoral paradigm shift enablers.
- 207. At the project level, in the business-as-usual scenario, MIDAGRI will contribute with EUR 28.67 million that it has pledged in co-financing, focusing on the short-term productivity aspects, neglecting the value of the ecosystem services as a long-term solution for the high Andean communities and for other beneficiaries of puna ecosystem services. It would focus on isolated investments and activities financed from different programs (Agrorural, Agroideas or Sierra Azul) without looking for synergies and impact programs. Participation of local communities on the decision-making process would be limited and CSOs that have gained the trust and engagement of communities from experience working with them would not be involved. The MERESE (payment for ecosystem service) scheme would continue being innovative on paper but non-functional on the ground and little investment would reach target communities. Other climate-relevant puna ecosystem services such as carbon sequestration would not be

considered nor valued. GCF financing would fill an important gap in climate finance for the most vulnerable populations in Peru and promote a sector transformation towards conservation of Puna ecosystems and corresponding services.

B.6. Exit strategy (max. 500 words, approximately 1 page)

B.6.1. Replicability

208. **The proposed interventions are already replicating successful experiences**, such as i) the incorporation of ancestral EbA practices and experience on local implementation in the high Andes from Nor Yauyos Cochas⁸¹; Agrorural and Sierra Azul, Helvetas and Forest Trends; ii) the strategy developed by IFAD, GEF projects and Agroideas on business development with focus on specific value chains (South American camelid fiber and Andean agricultural products) in an economically-linked area (SHAP); and iii) experience by Profonanpe in managing environmental and climate funds (including GCF funds) and engaging with private stakeholders for PES schemes. This ensures ownership from the national EEs MIDAGRI, SERNANP and Profonanpe.
209. The model for replication is completed by i) focused intervention on priority landscapes/watersheds where information, lessons learnt and best practices will be obtained, ii) support via financial windows accompanied by technical assistance, iii) increased investment from private sources and iv) improved planning and coordination at landscape level considering communities participation. v) capacity building and experience exchanges among beneficiaries for increasing the application and replication vi) produce communication materials in native languages to capture lessons learned and promote replication of successful experiences. This approach can then be transferred to other high priority areas with similar Puna ecosystems and climate vulnerability conditions (e.g., Junin, Huancavelica, Ayacucho) at lower costs.
210. In addition, the project will serve as an example of a successful initiative aimed at promoting and facilitating the adoption of EbA measures and Climate Resilient Value Chains. By enhancing governance mechanisms at multiple levels within the landscape and fostering stakeholder engagement in the project's focus area, it will foster the ability to encourage replication and knowledge exchange for comparable endeavours not only within Peru but also across similar ecosystems in the broader regional context.

B.6.1. Sustainability

211. Improved ecosystem services, climate resilient livelihoods and increased access to public and private finance will ensure that target populations in the SHAP develop rural Climate Resilient Value Chains (CRVC) and local economies. The resources that will be requested from the GCF will enable expected results. The results will ensure the sustainability of the proposed interventions by creating an enabling environment in policy, technical and financial terms.
212. **The exit strategy is embedded into the project design.** The project, through its various components, will support both the financial sustainability (component 2), and **behavioural changes** (component 1 and 3).
213. **Financial sustainability** will be ensured **at the Local initiative level** through the following:
- Crucially, all Local initiatives financed by the Puna Facility will be profitable in the long-term and generating positive financial IRRs without grants. In the Local initiatives prototype analysed in Annex 3, these IRRs range from ranging from 8.4% (for alpaca breeding in the subsistence window) to 25.8% (native potatoes in the Results-based Repayable Grant Sub-Window). While grants are needed during the Puna Facility implementation for the reasons described in section B.5, once beneficiaries have implemented EbA measures and CRVC they will continue to do because it makes financial sense to them and significantly increases their incomes.
 - In addition, the initial investments required for EBA measures and CRVC will not need to be repeated in their entirety in the future. EbA measures, once implemented, only need regular and relatively low maintenance costs. Equipment for CRVC, on the other hand, has a limited useful life (e.g., 10 years for greenhouses) and will need to be replaced in the future, but it represents a minority of the overall EbA and CRVC investment. Enhanced creditworthiness of the beneficiaries and access to finance (see below) will be such that beneficiaries no longer need grants at the time of capex replacement but will be able to use commercial sources of finance, as well as accumulated savings resulting from profitable EbA and CRVC initiatives.
 - Local initiative applicants will be inherently more creditworthy and therefore bankable after the successful implementation of EbA measures and CRVC, not only because they are on a sounder financial footing, but also because their businesses will be formalized (including proper accounting and tax compliance) and the businesses that benefited from repayable grants under the applicable windows will have a track record of compliance with financial obligations, which financial institutions will view positively.
 - The technical assistance provided by the Puna Facility, in addition to the financial incentives, will be critical in this respect. All selected Local initiatives, including beneficiaries without commercial capacities and target markets (i.e. eligible under the Non-repayable Grant Sub-Window), will receive comprehensive

⁸¹ Implemented by the Instituto de Montaña, IUCN, UNDP and UNEP in partnership with SERNANP.

technical assistance on access to finance, in areas including: developing a fund-raising strategy; analysis of the positive financial implications of EbA and climate resilient business investments to reduce the perceived risk for prospective lenders; preparation of a pitch deck to present business to prospective lenders; introduction to MFIs active in the region; explanation of the terms of any loan products offered by MFIs and support with loan applications; and, basic financial literacy training including the importance of repayment to create a strong credit track record.

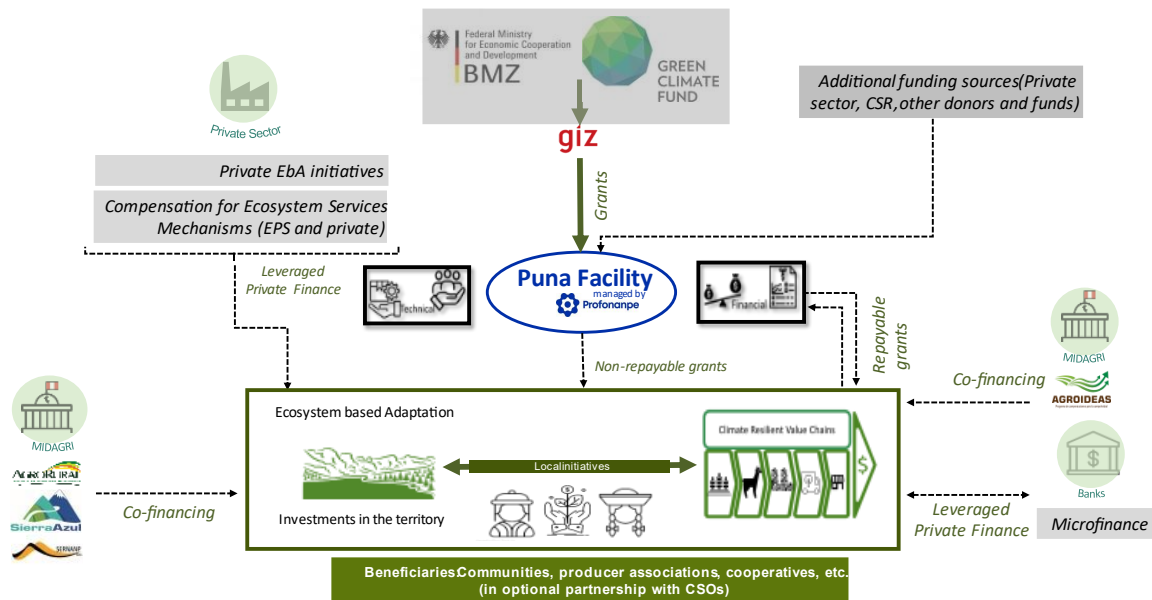
- v. The project will also work on the capital supply side to enhance the ability of MFIs (e.g., Caja rural los Andes, Caja Cuzco, Microfinance BBVA Foundation-Confianza) to increase the supply of microfinance products and services to finance integrated EbA and CRVC initiatives and among others better serve the beneficiaries of the Puna Facility, through technical assistance in the following areas: (i) introduction to concrete and tailor-made business models based on information generated and analysed from Local initiatives implemented during the project duration; (ii) support the modification or design of existing microcredits or tailored EbA and CRVC loan products; (iii) modification of management tools (including digital tools, guidelines and protocols) to integrate EbA approach; (iv) training of credit officers on each EbA and CRVC measure with particular focus on the costs required and their impact on beneficiary communities' cashflows over a 10-year period or longer; (v) training of credit officers on marketing and explaining the terms of the greener microcredits to prospective borrowers; (vi) training of credit officers on how the adoption of EbA and sustainable business practices alters the credit profile of individuals or associations, by generating an additional and more climate-resilient income stream, and interactive credit assessment exercises. The MFIs have expressed their interest in leveraging private finance by either adapting existing microcredits or developing new microcredits for the financing of EbA measures and/or CRVC. Options on how this interest will be formalized (e.g., signing of MoUs) will be assessed individually with the respective MFI.

214. **Financial sustainability** will be ensured **at the Puna Facility level** through the following:

- vi. Grants repaid from the Results-based Repayable Grant and Matchmaking Agroideas Sub-Windows will flow back into the Puna Facility most likely towards the end of the GCF project implementation and/or after project completion and will support further cycles of calls for proposals for the Non-repayable Grant Sub-Window⁸², after the completion of the Resilient Puna project. Each Results-based Repayable Grant Local initiative Agreement will require full 100% repayment and beneficiaries will be expected to comply with that full repayment. Conservatively, for the purpose of estimating reflows into the Facility, it is assumed that 50% of the overall repayable grant amount disbursed under the two sub-windows (i.e., Agroideas Matchmaking Sub-Window and Results-based Repayable Grant Sub-Window) will be actually repaid. Such repayment ratio could be the result of some Local Initiatives repaying the whole amount, some not repaying at all and some repaying in part – the exact combination is impossible to forecast at this stage. With the prototype Local initiative portfolio assumed in Annex 3, this would lead to a total of EUR 3.4 million being repaid. Considering the call-for-proposal and Local initiative implementation timing, most repayments are likely to occur towards the end of the GCF project period or later. In Annex 3, it is assumed that the Facility would “recycle” the repaid amount once these are paid back to support another 24 Local Initiatives substantially increasing its reach after the GCF project period.
- vii. Profonanpe has also expressed their commitment in contributing to the Puna Facility after project ends with EUR 1 million which are foreseen to cover operative staff costs.
- viii. In addition, Profonanpe intends to attract additional resources from donors and the private sector to capitalize the Puna Facility after and in addition to the GCF grant, through sub-activity 2.1.1.2. Profonanpe, as a national institution deeply involved in the project's design phase and accredited as a Direct Access Entity to the GCF, will oversee the operations of the Puna Facility, drawing upon its extensive experience in managing environmental funds. Evidence of success in attracting applicants through calls for proposals and supporting the implementation of EbA and CRVC Local initiatives under the GCF project will facilitate Profonanpe's raising of additional donor funds for the Puna Facility.

Figure 17. Sustainability approach of the Puna Facility

⁸² The feasibility of including results-based repayable grants in the subsequent cycles will be evaluated by Profonanpe taking into consideration the performance during the project implementation.



215. **Behavioural changes** (component 1 and 3) will facilitate the medium- and long-term sustainability. Project beneficiaries will be trained on implementation and maintenance of EbA interventions so they can continue providing benefits long after the project has ended. They will also be trained on improved production practices and Climate Resilient Value Chains (CRVC) to increase self-dependence and overall climate resilience. This will in turn conserve puna ecosystem services in the long term. Related to planning processes and institutions, the project aims at mainstreaming EbA and strengthening coordination capacity within MIDAGRI and between other sector and stakeholders in governance platforms that will continue operating and will then integrate climate change and enhanced coordination without additional cost.

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	40.79			million euro (€)		
GCF financial instrument	Amount	Tenor		Grace period	Pricing	
(i) Senior loans	40.79					
(ii) Subordinated loans						
(iii) Equity						
(iv) Guarantees						
(v) Reimbursable grants						
(vi) Grants						
(vii) Results-based payments						
(b) Co-financing information	Total amount			Currency		
	37.53			million euro (€)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
BMZ (implemented through GIZ)	Grant	7.91	million euro (€)			
MIDAGRI	Grant/In kind and loans ⁸³	28.67	million euro (€)			
SERNANP	In kind	0.95	million euro (€)			
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	78.32			million euro (€)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Project Co-finance ⁸⁴ : The following provides an overview of the co-financiers for the proposed project: MIDAGRI will contribute with EUR 28.67 million through grants (EUR 2,411,938), in kind (EUR 16,877,658) and loans (EUR 9,385,386) and SERNANP through in-kind contributions with EUR 0.95 million to the implementation of the project.					
	Key government inputs from MIDAGRI will be through a) Public investments projects ⁸⁵ (implemented by AGRORURAL, Sierra Azul, PSI, INIA and SERFOR), b) operational activities (implemented by programmes such as Agroideas) and c) other budget provisions, comprising individual credit lines to producer organisations (such as Directorate of Insurance and Promotion of Agricultural Financing (DSFFA)). Key inputs from SERNANP will be in-kind in the form of staff time contributions in order to coordinate project interventions in National Protected Areas.					

⁸³ Please note that this refers to the DSFAA of MIDAGRI.

⁸⁴ Co-financing commitment letters are provided in Annex 13, which includes Annex 13a Co-financing letter from MIDAGRI, Annex 13b Co-financing letter from BMZ and Annex 13c Co-financing letter from SERNANP (Carta N°30-2023-MIDAGRI-DVDAFIR and Letter N°091-2023-SERNANP-J).

⁸⁵ Please note that Sierra Azul, INIA and SERFOR contributions are investments and activities.

	BMZ financing: The total BMZ financing for the project amounts in grant to EUR 7.91 million. EUR 6.26 million will be implemented through GIZ as an Executing Entity in Peru. GIZ (Executing Entity) will finance the implementation of activity 1.1.1, 1.1.2, 1.1.3, 1.2.1, 1.2.2, 2.1.1, 3.1.1, 3.1.2; EUR 1.5 million will be forwarded to Profonanpe for the Puna Facility and EUR 150.000 to Instituto de Montaña. Other financing (e.g., leveraged private finance): The project will mobilize approx. EUR 6 million from international donors. Through sub-activity 2.1.1.2 the project will support in developing and implementing of an innovative strategy to mobilize resources from the private sector and different donors to support Profonanpe in ensuring the sustainability of the Puna Facility. In this regard, the Government of Canada has already expressed their interest in contributing to the project with approx. EUR 6 million. Of particular relevance, they expressed their interest in financing gender and monitoring related activities and in contributing to the Puna Facility in order to continue financing Local initiatives.
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C.2. Financing by component

Component	Output	Indicative cost million euro (€)	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount ⁸⁶ Options	Financial Instrument	Name of Institutions
1. Resilient ecosystems and communities	1.1 Investments for EbA measures and Climate Resilient Value Chains are implemented at the local landscape level	<u>52.68</u>	<u>21.65</u>	<u>Grants</u>	<u>31.03</u>	<u>Grants/In kind and loans⁸⁷</u>	<u>BMZ</u> <u>MIDAGRI</u>
	1.2 The use of EbA knowledge is recovered and disseminated, and local monitoring committees and observation systems are implemented	<u>6.27</u>	<u>4.77</u>	<u>Grants</u>	<u>1.50</u>	<u>Grants/In kind</u>	<u>BMZ</u> <u>SERNANP</u>
2. Public and private investment for EbA aligned and leveraged	2.1 Financial mechanisms for the implementation of EbA measures and improvement	<u>7.17</u>	<u>6.05</u>	<u>Grants</u>	<u>1.12</u>	<u>Grants</u>	<u>BMZ</u>

⁸⁶ Please note that the calculations of the contribution of the Government of Peru (MIDAGRI and SERNANP) were done in Soles and were converted to Euros using the SBS exchange rate dated April 2023 namely EUR 1= 4.1 Sol.

⁸⁷ Please note that this refers to the Irrigation Subsector Program of MIDAGRI

	of climate resilient livelihoods in the Puna ecosystem are established						
3. Institutional capacity strengthened to integrate EbA into territorial planning	3.1 Multilevel landscape governance is improved through strengthening of national capacity, regulatory frameworks and M&E systems	<u>5.57</u>	<u>3.93</u>	<u>Grants</u>	<u>1.64</u>	<u>Grants</u>	<u>BMZ</u> <u>SERNANP</u>
(b) M&E		<u>2.00</u>	<u>1.64</u>	<u>0.36</u>			
(c) Contingencies		<u>800,000</u>	<u>800,000</u>	<u>0</u>			
(d) PMC		<u>3.81</u>	<u>1.93</u>	<u>1.88</u>			
Indicative total cost (USD)⁸⁸		<u>85.26</u>	<u>44.40</u>	<u>40.86</u>			

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

216. In collaboration with the local communities, MSEs, community enterprises, cooperatives, producers' associations, producers' organizations and local CSOs implementing Local initiatives, Instituto the Montaña⁸⁹, will enhance the capabilities of stakeholders, promote the alignment of approaches, and facilitate the formation of community monitoring committees. The project will empower these committees by providing training, facilitating dialogue, and employing other capacity-building methods. This will enable the exchange of knowledge and collaborative efforts to better comprehend the connections between traditional practices, Ecosystem-based Adaptation (EbA), and climate resilience.

217. Income-generating alternatives will result from Climate Resilient Value Chains, where technological packages for climate resilience, increased productivity and market development (e.g., qochas, grassland conservation and restoration, infiltration ditches, organic fertilizer production, livestock management, alpaca fiber production, and agroecological products) will also be financed.

⁸⁸ Please note that the used exchange rate is based on the European Commission's official monthly accounting rates for the euro ([Exchange rate \(InforEuro\) \(europa.eu\)](https://www.ecb.europa.eu/press/pr/exchange_rates/index.en.htm)) for the month of September 2023 namely 1 EUR = 1.0886 USD.

⁸⁹ See role in section B 2.4 below

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential (max. 500 words, approximately 1 page)

D.1.1 Impact Potential - Adaptation

218. **Contribution to increased climate-resilient sustainable development: The project has an integrated vision of promoting EbA** at landscape level through: i) On-the-ground activities in Component 1 that will ensure the long-term provision of essential puna ecosystem services, such as water provision and regulation for upstream, middle stream, and downstream communities, fodder production for South American camelid husbandry and soil formation and retention for agricultural practices despite increased risks of drought and reduced water availability due to climate change; ii) the establishment of a financial facility with technical assistance in Component 2 complementing the funding from and increase the access to existing public MIDAGRI programs executed under Component 1, while providing a framework for scaling up the deployment of EbA interventions also after the project has ended (exit strategy); and iii) Integration of EbA into territorial planning and consensus-building processes through participatory dialogue platforms adjusted to local customs and gender needs that will promote the coordination of on-the-ground activities of the different stakeholders intervening in various ecological floors of the watershed and benefiting a range of interest groups (regional governments, municipalities, local communities, producers' associations, cooperatives, MSEs, community enterprises and producers' organizations).
219. In addition, the hydrological modelling carried out for this proposal (see Annex 2g) allowed estimating the potential impacts of this project resulting from the implementation of EbA interventions. In general, the analysis concludes that the EbA interventions results in improving the water availability in the district areas, for agriculture and livestock as well as for the ecosystems that extract water from the surface layers of the soil (more details on the results can be found in Annex 2g).

D.1.2 Ecosystems improved

220. The project is estimated to conserve and restore 23,914 hectares of Puna ecosystems that High Andean livelihoods depend upon. The project will promote direct interventions to maximize the landscape and ecosystem connectivity within and between watersheds to ensure the highest impact in terms of recovery of ecosystem services as an adaptation and mitigation strategy. The local impact will be monitored by community committees and at the national scale via a long-term monitoring and evaluation framework, as part of the support provided to strengthen intra- and intersectoral collaboration among MIDAGRI and MINAM programs and their contribution to the NDCs and other climate-related strategies. Both local and national results will be integrated into existing monitoring systems in the ministries which are being hosted by MIDAGRI's programs and SERNANP.

D.1.3 Direct and Indirect Beneficiaries

221. The total estimated number of project **direct beneficiaries** are 60,715 people (of which 30,088 are women) from 91 districts, 69% of the total beneficiaries have Quechua as their mother tongue and 31% have Spanish. Considering the numbers, the proposed GCF project is targeting Quechua Indigenous Peoples and Local communities, which are a significant part of the most vulnerable population within the SHAP region, which face significant climate-related challenges. In addition, this population are part of the most economically disadvantaged in the country with 73 percent of SHAP having a low or very low Human Development Index (HDI), aligning with the broader trend in Peru where higher altitudes correlate with lower HDI levels (PNUD, 2019)
222. Following the GCF IRMF the number of direct beneficiaries were calculated including individuals who will receive i) targeted support from the GCF-funded interventions and ii) a measurable adaptation benefit from a GCF-funded intervention. The estimation of these direct beneficiaries' accounts therefore for:
- a) Individuals that implement Local initiatives that will be financed and supported by the Puna Facility, multiplied by the average number of members of the groups/legal entities supported and the average members per household in the region., representing 31,750 people.
 - b) Individuals that will be supported by MIDAGRIs co-finance programs, multiplied by the average members number per household in the region., representing 26,315 people.
 - c) Individuals that will receive technical assistance from the GCF- funded intervention (e.g., trainings, support to develop site-specific climate diagnostics and preparation of participatory intervention plans) including.
223. The project will also **indirectly benefit** 2,011,856 people (of which 1,005,928 are women) this includes those that do not receive targeted support from the GCF-funded intervention but will benefit from water

generated by the improvement of ecosystems that contribute to water regulation in the upper and lower parts of watersheds which involve main cities such as Cusco, Arequipa, Abancay and Cañete.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

224. Changing the paradigm of highly threatened Puna ecosystems requires a transformation of behaviours and practices in the agriculture and husbandry sector. The project aims at shifting the reality of farming communities in the SHAP from a state characterized by climate vulnerability, lack of opportunities, and ecosystem degradation to a state characterized by conservation of the natural capital, climate resilience and improved economic productivity. This proposal incorporates the transformational pathway of promoting resilient agriculture and husbandry practices in High Andean communities through the implementation of EbA solutions with an mitigation co-benefit in priority landscapes to conserve essential ecosystem services sustaining the livelihoods of neighbouring farming communities and benefiting thousands of people who rely on the water generated by the improvement of ecosystems that contribute to water regulation in the basins. The combined results of improved farming and husbandry practices - by promoting and incorporating EbA measures and climate-resilient approaches in their productive activities, technical assistance, business development, access to finance and improved planning and coordination - will increase the overall resilience of High Andean communities while improving water regulation and maintaining important stocks for global carbon regulation.

D.2.1 Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

225. Climate innovation will be catalysed through improved ecosystem management and target Climate Resilient Value Chains (e.g., South American camelid fiber and Andean grains, among others complementary) supported by public and private finance. An inter-sectorial transformational change is supported. So far, conservation and production activities are carried out in isolation by SERNANP and MIDAGRI although many times they are in the same landscape. The project, hence, seeks to enhance coordination of public sector natural resources management in the territory. Moreover, the project aims to transform vulnerable communities' businesses and their usual production system into more sustainable and climate-resilient production systems. Locally generated and targeted innovation for market transformation through MIDAGRI's funds from Agroideas to strengthen producers' organizations and their business ideas, will be complemented by the Puna Facility in providing funding for EbA measures on a repayable basis and technical support to develop complementary interventions to the program in order to create climate-resilient livelihoods. It is important to highlight the strengthening of government institutions, will also support MIDAGRI in meeting its productivity and competitiveness commitments while complying with Peru's NDC targets.

D.2.2 Potential for upscaling and replication

226. The main approach to achieve upscaling and replication is using mobilization of investments at scale. The project aims at aligning and upscaling public and private investment for EbA and Climate Resilient Value Chains under Components 1 and 2. Specially, the Puna Facility will mobilize EbA finance at scale, aiming at putting in place the enabling conditions for additional long-term financing under Component 2 and 3. Sustained investment in the future will be enabled by laying the groundwork for valuing ecosystem services of the Puna ecosystems that will add to the conducive transformational environment. The technical support provided by the project will increase the access to Agroideas and Sierra Azul public programs that promote nature-based solutions' investment projects, business development, product transformation, and value addition in target value chains (i.e., alpaca and vicuña fibers, potato, Andean grains). The project will also offer the option of leveraging public funds from MERESE schemes and subnational governments to the areas where investments are needed to conserve ecosystem services. The capacity building provided to relevant stakeholders (Profonanpe, MIDAGRI, microfinance institutions) will enhance the potential for upscaling and replication.

D.2.3 Potential of knowledge and learning

227. Knowledge and learning will be promoted, enhanced and scaled through several alliance at different levels of intervention. These alliance will support the experiences replication of the project approach in other High Andean communities. Therefore, at local level, transfer of knowledge between target areas, integration of local experiences into larger planning processes, and the improved coordination between actors on the ground will set the stage to expand the EbA approach to communities with similar environmental and development characteristics. The project will focus on strengthening multisectoral institutional capacities and

regulatory frameworks. MIDAGRI and MINAM personnel, in charge of national programs and rural extension services, will be trained to include sustainable management and to promote traditional practices.

D.2.2 Contribution to the creation of an enabling environment

228. The main approach towards the creation of an enabling environment relies on transformational planning and programming to scale up EbA. This will be achieved by integrating community-responsive technologies, ancestral practices, and landscape interventions (EbA) into local, regional and national planning processes to enhance the provision of key ecosystem services for the local population and communities downstream. Communities and local leadership, including women and youth, will be strengthened. An increased collaboration between public and private sectors will contribute to transformational planning and programming and support the construction of the desired enabling environment. The project will support MINAM in its efforts to establish a methodology for calculating carbon on the Puna ecosystem and to lay the groundwork for carbon accounting and credit systems for future trading.

D.2.4 Contribution of the regulatory framework and policies

229. The project will facilitate the coordination within MIDAGRI and with other key sectors and actors in the territory. In this context, the project activities will support the mainstreaming of EbA and gender considerations within MIDAGRI'S and MINAM's daily operations. Improved interinstitutional coordination and collaboration will be sought within MIDAGRI and SERNANP, to ensure synergies in the same landscape. The project will contribute to the improvement of regulatory instruments to enhance EbA and climate resilient approach in the agricultural sector and the territory, and will strengthening of the national long-term monitoring and evaluation systems of the different government entities in the high Andean ecosystems and populations, in order to have monitoring protocols and measurement methodologies consistent with the impacts of EbA interventions (including carbon sequestration, climate information and its impacts on water and agriculture). It will also support monitoring systems that provide information on EbA interventions on water and agriculture for decision-making on adaptation responses, such as those developed by MIDAGRI, SENAMHI, ANA, INAIGEM and SUNASS.

D.3. Sustainable development (max. 500 words, approximately 1 page)

D.3.1 Environmental co-benefits

230. The project will have two environmental co-benefits. First co-benefit is the maintenance of ecosystem services in the area benefited through the conservation, restauration or sustainable management of ecosystems. The main ecosystems in the high Andes are peatlands, wetlands and grasslands, which provide a range of ecosystem services that benefit populations throughout the watershed (see Annex 2a, section 13.3 Environmental co-benefits for more details). The Puna ecosystems services include: i) Provisioning services which consist of fiber, fuel, food and fresh water; ii) Regulation services that include climate regulation, water regulation, water purification and erosion protection, resulting in the regulation of greenhouse gases, climatic processes, water storage, groundwater recharge and discharge, retention, recovery and removal of excess nutrients and pollutants and peat blanket protecting the underlying soils from erosion; iii) Supporting services provided by these ecosystems include biodiversity, soil formation, nutrient recycling and carbon storage; iv) Cultural services present in the SHAP are educational, recreational/, landscape beauty, spiritual and inspirational. The project will seek to ensure the long-term provision of these services through the three project components: i) implementation of EbA alternatives and training on sustainable business options and monitoring of ecosystem conditions; ii) increased access to technical assistance and finance, to invest in EbA measures, and iii) mainstreaming of the EbA approach into extension programs and improved participatory planning at the local and basin level. It is expected that 23,914 ha of Puna ecosystems will be conserved, restored or sustainably managed.

231. As second co-benefit, the project recognizes the reduction of GHG emissions, in this context, GHG emissions will be avoided through the conservation, restoration and sustainable management of ecosystems. The proposed intervention measures aim to prevent emissions from existing carbon sources, inappropriate management practices in livestock and crops, and contribute to carbon sequestration in soil and vegetation, through grassland management, forest restoration, Integrated soil fertility management, agroforestry, and wetland restauration and conservation. GHG reductions are calculated using the methodology presented in the IPCC Good Practice Guidance for LULUCF, applied though the tool EX-ACT (version 9.4) developed by FAO. The overall mitigation impact as a co-benefit from the project is a reduction of 407,657 tCO₂e in the 15 years analysed, which include five years of implementation phase and ten years of capitalization phase.

It is measured against a scenario without project that would have increased by 93,905 tCO₂e the GHG emissions in the areas of intervention. These numbers correspond to the impact during the 15 years considered for analysis in a total area of intervention of 23,914 ha. The details on the calculations can be found in the supplementary document GHG calculation sheet (please see Annex 22b for the calculation of the GHG emissions reduction as a co-benefit from the project and Annex 2a, chapter 13 for further discussion on the climate change mitigation co-benefits).

232. The environmental co-benefits will contribute to the fulfilment of the following Sustainable Development Goals (SDG): SDG 6 Ensure availability and sustainable management of water and sanitation for all (Target 6.6); SDG 13 Take urgent action to combat climate change and its impacts (Target 13.1 and 13.2) and SDG 15 Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss (Target 15.1 and 15.4).

D.3.2 Social co-benefits including health co-benefits

233. The project will have as a social co-benefit the awareness raising and capacity building of local stakeholders. Awareness raising and capacity building in EbA and gender approach and CRVC are mainstreamed into all components to strengthen capacities to integrate climate change adaptation into planning and actions. In component 1, activities focus on the beneficiaries - they will be trained and supported through the activities 1.1.1 and 1.1.3. Ancient knowledge and local expertise will be integrated through the activities 1.2.1. Local reflecting capacities, community monitoring and reporting will be trained and supported in activity 1.2.2. Institutional capacities will be reinforced through Component 3 and Component 2. In activities 2.1.2 EPS officials will receive technical assistance and training and in activity 2.1.3. Microfinance company officers also will be trained. Within Component 3, activity 3.1.1 public officials will be trained and in activity 3.1.2 support will be provided to strengthen monitoring systems so that lessons learned can be incorporated into decision-making processes.

234. The social co-benefits will contribute to the fulfilment of the following Sustainable Development Goals (SDG): SDG 13 Take urgent action to combat climate change and its impacts (Target 13.3).

D.3.3 Economic co-benefits

235. The activities of the project will support the achievement of the co-benefit 3 "Improvement of local economy through better economic opportunities and poverty reduction". The indicator designed to measure this co-benefit will be "Number of producers, of whom 50% are women have confirmed an improvement in their income from Climate Resilient Value Chains". This co-benefit will be done through implementation of the following activities: 1) Activity 1.1.2 will contribute to reduce climate risks supporting the implementation of Local initiatives involving EbA and climate-resilient value chains, helping to maximise economic sustainable opportunities; 2) Activity 1.1.3 will give technical assistance for implementation and promote market access for Local initiatives and integration of climate change adaptation skills into their livelihoods and 3) Activity 2.2.3 will contribute to reduce the barriers to finance for EbA and CRVC.

236. The economic co-benefits will contribute to the fulfilment of the following Sustainable Development Goals (SDG): SDG 1 End poverty in all its forms everywhere (Target 1.5) and SDG 2 End hunger, achieve food security and improve nutrition and promote sustainable agriculture (Target 2.3), SDG 10 Reduce Inequality within and among countries (Target 10.1) and SDG 5 Achieve gender equality and empower all women and girls (Target 5.5).

D.3.4 Gender-sensitive development co-benefit

237. The mainstreaming of gender approach within all the project activities will promote gender equality and enhance living conditions and economic opportunities of Andean women. This is achieved through associative means, promoting Local initiatives (including EbA measures and CRVC) with a gender-sensitive approach, integrating gender issues in platforms of knowledge exchange for ancestral practices and increasing women participation in watershed dialogue decision-making platforms. For this purpose, the project includes a gender action plan with specific activities and indicators that will measure the application of gender approach in a cross-cutting manner. Co-benefit 5 "Gender responsive approaches mainstreamed in EbA and CRVC measures" will be measured through the indicator "Number of innovative Local initiatives (including EbA and CRVC) with a gender-sensitive approach implemented".

238. The impact of designing local initiatives with a gender-sensitive approach will reflect on the equally enhancing for men and women of benefits in the local economy. This will be measured in co-benefit 3

“Improvement of local economy through better economic opportunities and poverty reduction”, where the indicator designed to measure this co-benefit will be “Number of producers, of whom 50% are women have confirmed an improvement in their income from climate resilient value chains”.

239. The gender co-benefits will contribute to the fulfilment of the following Sustainable Development Goals (SDG): SDG 5 Achieve gender equality and empower all women and girls (Target 5.5).

D.4. Needs of recipient (max. 500 words, approximately 1 page)

D.4.1 Vulnerability of the SHAP region

240. **The vulnerability in the SHAP is increased by a high occurrence of drought due to climate change combined with unsustainable practices.** The vulnerability of High Andean communities is enhanced by their intrinsic characteristics including i) weak organizational structures, ii) geographic dispersion, iii) use of wood, dung, and peat as energy sources, iv) poor infrastructure and availability of basic health, education and sanitation services, resulting in high infant morbidity and malnutrition and v) weak technical capacity in agriculture and husbandry practices, leading to the deterioration of the Puna ecosystems due to overgrazing, expansion of the agricultural frontier, poor management of grasslands and degradation of peatlands and wetlands. The degradation of High Andean grasslands is a widespread phenomenon: more than 40 percent of grazing areas are in poor conditions and acutely deteriorating. Without improved management, an additional 20 percent of grazing areas will be degraded by 2070. With dryer conditions, high-mountain peatlands, which have an important role in water regulation and carbon storage, are more susceptible to fragmentation in the long term ([Bury et al., 2010](#); [Polk et al., 2017](#); Cooper et al., 2019). Although poorly understood, rough projections using the scenarios of precipitation decrease in Southern Peru ([Urrutia & Vuille, 2009](#)) suggest strong wetland degradation, locally of about 25 to 100 percent ([Otto & Gibbons, 2017](#)). Indeed, in the 23 years between 1987 and 2010, wetland ecosystems in the Huascan National Park decreased from 12,402 to 6,821 hectares, a 45 percent net loss ([Polk et al., 2017](#)). In certain areas, higher temperatures partly contribute to land degradation and land-use change by allowing the introduction of sheep and cattle for extensive grazing in natural grasslands, where previously only lower-impact South American camelids were roaming. In order to address the main factor of local vulnerability, the project will promote actions to increase (i) water availability, (ii) resilience to climate change impacts on agricultural activities, (iii) sustainable land management through a landscape approach, (iv) conservation of Puna ecosystems and (v) overall resilience of the local population.

D.4.2 Economic and social needs of the beneficiaries

241. **Target populations are highly vulnerable to climate change.** The main direct beneficiaries of project interventions are local communities in the SHAP, including producers' associations, MSEs, cooperatives, community enterprises and producers' organizations and subnational governments, which are very vulnerable to climate change ([MINAM, 2016](#)). These local communities are among the most impoverished in the country (73 percent of the SHAP districts have a low or very low Human Development Index (HDI), confirming the general rule that in Peru the higher the altitude the lower the HDI ([PNUD, 2019](#)). This means that target local communities lack basic services such as health, potable water, sanitation, education and financing. The overall consequence is a general lack of capacity to sustainably manage the natural resources that support their livelihoods, which adds to their climate vulnerability. To increase local income, the project will mobilize public and private funding towards the high Andean population and their associated ecosystems. In this context, the project will enhance training of local communities, including producers' associations, MSEs, cooperatives, community enterprises and producers' organizations on business development and access to finance (including financial literacy) to ensure successful alternative and sustainable income-generating activities.
242. **Lack of access to markets.** Many communities in the SHAP face difficulties in accessing markets to sell their products ([Fan & Salas Garcia, 2018](#)). Therefore, developing and strengthening market linkages, facilitating commercial networks, and supporting local communities, cooperatives, MSEs community enterprises, producers' associations and organization will support creating opportunities for economic growth. In order to address the need of better market access, the project will strengthen local value chains through sustainable ecosystem management practices, enhanced associative processes, Climate Resilient Value Chains and improved access to finance and business development (including technical assistance in marketing, branding, customer outreach and certification of products). Technical support to local

communities, cooperatives, MSEs community enterprises, producers' associations and organization will add value to their products and facilitate access to national and specialized markets.

D.4.3 Financial needs

243. **Limited access to finance.** Financial institutions perceive the investment risk of the agricultural sector as too high and rarely develop products and services adapted to the needs of these vulnerable populations. When available, access to finance is at exorbitant costs via informal lenders named "drop by drop" or "express loans" with low or no requirements but daily rates of 10 to 40 percent ([Prestamype, 2023](#)). Rural microfinance institutions in Peru only represent 3.44 percent of the financial system in Peru ([Asomif, 2020](#)). In the SHAP there are some of them that give micro loans to rural people and small-farmers i.e. Caja Rural Los Andes and Caja Rural Inka Sur, furthermore, there are only one cooperative (Fonde Surco) with some small green loans for agriculture and livestock. Also, access of SHAP communities to matching public-private finance mechanism, such as Agroideas, for the development of sustainable agri-business plans is limited. For instance, according to information provided by Agroideas for the 2016-2020 period, only 4.4 percent of the total budget executed for approved business plans towards technological adoption have been channelled to the South American camelid fiber value chain. To address the financial needs, the project will directly support target communities in accessing funds earmarked by MIDAGRI for EbA activities (Agroideas, Agrorural and Sierra Azul) by i) developing simplified material out of programs guidelines to facilitate rural population to access the government support and ii) capacitating producers and local governments to disseminate the information on the different public programs. The project will on the one hand also provide technical assistance to MFIs in order for them to expand their offers in green products and on the other hand capacitate on business development and access to finance (including financial literacy) to local communities, cooperatives, MSEs community enterprises, producers' associations and organization that will allow target populations to access microfinance products and services.

D.4.4 Institutional needs

244. **Access to MIDAGRI programs requires improved associativity.** Producers' associations and communities organizations must become legally established associations to be eligible to access public programs ([Perea, 2014](#)). The project will therefore promote and support associativity (through technical assistance provided by the Puna Facility for beneficiaries eligible for the Non-repayable Grant Sub-Window) and strengthen business management for the adoption of climate resilient and environmentally sustainable agricultural technologies, animal husbandry and breeding management practices. Producers' associations, cooperatives, communities and other legally established organizations will be sensitized on the importance of EbA, trained on proper implementation and maintenance practices and supported in the development of Climate Resilient Value Chains.

245. **Public institutional capacity and coordination strengthened to mainstream EbA.** In Component 3 the project focuses on building the capacity of different entities, including MIDAGRI and MINAM to carry out EbA policy implementation in a more coordinated manner, mainstreaming EbA, gender, intercultural and intergenerational considerations in all its programs and to streamline its interventions to increase efficiency and impact towards climate resilience and reduced emissions in the SHAP. Improved coordination with other ministries and government entities at all levels will support the country in meeting its NDC targets.

D.5. Country ownership (max. 500 words, approximately 1 page)

D.5.1 Country prioritizing of the project

246. The project has been prioritized by the NDA to be presented to the GCF and it has the full support of the Peruvian government, through the Ministry of Economy and Finance which is the NDA, the Ministry of Agricultural Development and Irrigation which is the project leading ministry and the Ministry of Environment which leads climate change action in Peru. More recently, during the national message on the Celebration of the Peruvian Independence Day on the 28th of July 2023, the president mentioned the Resilient Puna project as part of the relevant programmes that the country is boosting to support climate change adaptation in the High Andes areas of the country.

D.5.2 Coherence with existing National Climate Strategy, policies and commitments

247. The project will be a major contribution to identified in Peru's first [NDC](#) targets. The project will support the fulfilment of adaptation 4 NDCs measures related to **agriculture and water**. NDC measures related to **agriculture** include: i) AGRI7 Natural grassland management to secure feed for livestock and reduce vulnerability to climate change, ii) AGRI11 Management of wild South American camelids (vicuñas)

considering the effects of climate change, iii) AGRI16 Implementation of business strategies that incorporate risk and opportunity management in the face of climate change. NDC measures related to **water** include: iv) AGU2 Implementation of interventions related to planting and harvesting water for agricultural water security in watersheds vulnerable to climate change. Since this project have a call for proposal mechanism the exact contributions to the NDCs will be obtained during the implementation phase. However, a contribution estimation to the four identified NDCs are indicated in Section 5.7 Table 21 of Annex 2a.

248. The project is fully aligned with the [National Adaptation Plan \(NAP\)](#). The proposed adaptation activities will follow the approach laid out in Peru's NAP, focusing on ecosystems, hydrographic watersheds, restoration and conservation of ecosystems, recover traditional knowledge and territorial planning. In addition, microfinancing for the promotion of EbA practices has been integrated in the NAP as a means to increase participation of the private sector and to provide very much needed access to financing for vulnerable populations.
249. The project is fully aligned with the [National Agrarian Policy 2021-2030](#) (Supreme Decree 017-2021 MIDAGRI) which aim is to improve access to markets for small farmers, improve natural resource management, improve technical and commercial capacities of agricultural producers and their incomes, promote producers associativity and entrepreneurship.
250. The project contributes to the risk management and climate areas lands at highest risk to drought were taken from the Plan for Risk Management and Adaptation to Climate Change in the Agricultural Sector Period 2012-2021 ([Plan GRACC-A](#)) to select target districts.
251. The project aligns with the [Road Map for Green Financing](#) and the general guidelines for the National Strategy for Green Growth (ENVA). The ENVVA aims to promote sustainable economic development while considering environmental protection and climate change mitigation. In particular, the project aligns with the following main objectives of the ENVVA: (i) Sustainable use of natural resources, (ii) Environmental mainstreaming in economic sectors, specifically promote environmentally friendly practices, technologies, and policies that minimize negative environmental impacts, (iii) Conservation and restoration of ecosystems, (iv) Strengthening environmental governance and institutions and (v) Access of communities to environmental services and information.
252. According to the NDA requirement, this project is also in line with the objectives of the National Competitiveness and Productivity Plan 2019-2030 which seeks to a) promote environmental sustainability in economic activities operations and b) boost local and external financial mechanisms.
253. The project is aligned with the GCF's 2020-2023 strategic programming and contributes to sectoral paradigm shift enablers. For Ecosystems and Ecosystem Services, the programme is aligned with Pathway 1: *Ecosystem-based management of terrestrial and freshwater systems*, to achieve a paradigm shift in the landscape management of the SHAP region. The programme uses *transformational planning and programming* via the fostering of an enabling environment for EbA interventions for improved ecosystem services. The project also *catalyses climate innovation* through the support of the development of new mechanisms to finance EbA and resilient livelihoods. It *mobilises finance at scale* through securing public and private financing through the Puna facility, MERESE schemes, and other new financial products.
254. Finally, it is important to highlight, that the project embraced an integrative approach by merging three initiatives. At the request of the Peruvian Government, through the Ministry of Agricultural Development and Irrigation (MIDAGRI), GIZ was assigned to integrate three concepts emerging from different sectors with focus on high Andean communities and ecosystems from slightly different perspectives. This shows the felt need within Peruvian institutions to address climate change vulnerabilities in the SHAP. The present proposal is therefore bringing together previous experience on Natural Protected Areas, focus on South American camelid fiber value chains, and finance at scale to implement ecosystem-based adaptation, thus facilitating complementarity and coherence according to country needs.

D.5.3 Capacity of AE and EEs to deliver

255. **GIZ** with its headquarters in Germany serves as AE, and GIZ with its management structure in Peru as one of the EEs. GIZ is one of the largest international providers of capacity development and technical assistance on climate change worldwide. GIZ as AE already developed and submitted several funding proposals to GCF, with 6 approved proposals under implementation (FP117, FP132, FP103, FP059, FP198 and FP200).
256. In addition, MIDAGRI, SERNANP, Profonampe and Instituto de Montaña will also serve as EEs.
257. **MIDAGRI** has extensive experience of channelling loan and grant resources for national and sub-national rural development projects from international financial institutions. It is currently implementing several projects from IFAD (e.g., [Avanzar Rural](#)), World Bank (e.g., [Integrated Water Resources Management in Ten Basins](#)) and IADB (e.g., [Agricultural Health and Agrifood Safety Development Program Phase II](#)).

258. **Profonanpe** as one of the main EEs has been accredited by the GCF since 2016. For its accreditation process, Profonanpe received technical support from a GCF Readiness Project with GIZ as delivery partner to become a Direct Access Entity. Profonanpe is implementing two GCF projects: 1) *Building the Resilience of Wetlands in the Province of Datem del Marañón* (FP001) with Profonanpe as the EE managing a total amount of \$6,240,000 of GCF grant; and 2) *Peruvian Amazon Eco Bio Business Facility* (Amazon EBBF) (FP193) with Profonanpe as the Direct Access Entity managing a total amount of \$8,972,400 of GCF grant. In addition, Profonanpe has gathered experience in designing and managing several environmental funds will be key, namely a Regional Water Fund in Piura, a competitive fund aimed at PES in high Andean ecosystems, and a MERESE fund for Arequipa. This experience will be key for their role in managing the Puna Facility established for this project and in bringing in their experience in managing PES and MERESE funds for the exit strategy.
259. **SERNANP** is the Government entity, attached to the Ministry of Environment (MINAM) in charge of ensuring the conservation of National Protected Areas (NPAs). As the governing entity of the SINANPE, it carries out and has experience on project implementation in the NPAs. In the past, it has implemented projects like the *EbA Lomas* with the support of UNDP. SERNANP will be in charge of the execution of the project activities within the NPAs and buffer zones, ensuring its efficient execution and the articulation of those activities within those areas.
260. **Instituto de Montaña** is a non-profit organization that works for the conservation of the natural, cultural and spiritual values of mountain peoples and ecosystems. They have been working in the Andes since 1995 and in Peru they have and are currently implementing programs in the highlands of Ancash, Piura, Junín and Lima.
261. Additional information on the AE and EEs' capacities to implement the project are provided in Section B.4.

D.5.4 Role of the Ministry of Economy and Finance (NDA)

262. The Ministry of Economy and Finance, as Peru's NDA, has been involved in the process of development and approval by the Government of Peru of the Concept Note and Funding Proposal. MIDAGRI as project proponent, along with Profonanpe, MINAM and GIZ as the AE have introduced the project idea to the NDA before starting the project pre-feasibility and concept note phase. The NDA has been also closely involved during the Funding Proposal development stage participating for example in a validation workshop held in June of 2023 in Lima Peru and will be continued to be involved following the national mechanisms for project selection established by the NDA.

D.5.5 Engagement with civil society organizations and other relevant stakeholders

263. Several consultations in the target areas have been carried out by GIZ in close coordination with project partners during the different stages of Concept Note and Funding Proposal development. A summary of the consultations held can be found in Annex 7.
264. **During Concept Note stage, engagement have taken place with different stakeholders.** At the national level, meetings were held with national government programs that operate in the territory, such as MIDAGRI's Programs, Textile Camelids Innovation Institute, as well as, with the SERNANP, INAIGEM and the SENAMHI. The project development team engaged for example with cooperatives, with other international cooperation projects such as Swiss Cooperation (Resilient Andes and Andean Forest Program), PNUD (Small Grants GEF program) to integrate lessons learned and best practices into the project design. In addition, consultations were held with NGOs that work in the puna region, such as Water Competence Centre (CCA), DESCOSUR and CONDESAN (Natural Infrastructure for Water Security Project) to understand their perception about communities' necessities in a context of climate change to incorporate it into the project design.
265. **More recently, during the Funding Proposal stage,** consultations with national, regional and local stakeholders were carried out in October 2022, November 2022 and May 2023 by organizing different workshops and meetings in target project departments in order to present the proposal, identify gender gaps and potential project risks and mitigation measures. As part of these consultations (May 2023), the project and relevant activities were presented and validated. Feedback provided by different stakeholders were further integrated into the project design. In general, the project was welcomed by the stakeholders, at subnational, district and community level. In total, the project has consulted 783 people (38% women and 62% men) out of which 65% were members of the main potential direct project beneficiaries (e.g. local communities, producers' associations and cooperatives) most of them Quechua population (see Annex 7). Consultations were conducted and closely coordinated and organized by MIDAGRI and SERNANP with local stakeholders to ensure the participation of communities, producer associations, women, youth, and elderly people.

266. During project inception and implementation, the project will follow a Stakeholder Engagement Plan (see Annex 7) in order to ensure continuous consultations with the stakeholders. The main actions will include i) development and implementation of a culturally adequate consultation and communication strategy, ii) development and implementation of a Grievance Redress Mechanisms and actions during iii) project implementation and monitoring. In addition, in table 8 a list of main actions to engage with the different stakeholders is being presented which includes different stakeholder categories (e.g., local communities, national, local governments, private sector etc.).

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

D.6.1 Efficiency and effectiveness

267. Ecosystem-based Adaptation has been widely viewed as a cost-effective approach to address the challenges communities face when dealing with climate change. A recent study that valued the ecosystem services of peatlands in the Yanque district of Arequipa, concluded that their regulation services could be valued at 450 USD/ha (Mango, 2017). Rolando et al., (2017) analysed the Puna ecosystems' impact by land use and climate changes and proposed "a set of technologies, practices and policies to preserve (or restore) the provision of these key ES: long fallowing, soil amendments, conservation tillage, rotational grazing, grassland ecological restoration, conservation of agrobiodiversity, modern irrigation and water harvesting, plant breeding, climate change mitigation schemes and payment for ecosystem services, and adapted traditional technologies". Most of these are proposed project interventions as described in Annex 2a, section 9. In the cost-benefit analysis for the management of native grasslands in Chanchayllo and Miraflores, Peru, realized under the Global Mountain EbA project, the benefit-cost ratio was 2.2 and the breakeven point at 10-15 years (Reid et al., 2017). Gammie and De Bievre (2015) show that a subset of proposed EbA interventions for the Lima water supply was deemed as a cost-effective option when compared to grey infrastructure for water provision, with prices below USD 0.25/m³.

268. The GCF investment represents a cost-effective efficient integrated landscape approach to comprehensive obtain adaptation benefits and mitigation co-benefits. The project will support 60,715 people directly, (50 % women) in increasing adaptive capacities and improving overall climate resilience and 1.9 million people indirectly (50 % women) through the maintenance of ecosystem services that sustain economic activities and populations downstream. In addition, through project activities 23,914 ha of Puna ecosystems will be sustainably managed, equivalent to a co-benefit of 407,675 t CO₂eq of avoided emissions.

D.6.2 Financial viability

269. For the in-depth justification of the GCF grant, please refer to section B.5. As noted there and in Annex 3, the financial IRR of Local initiatives supported by the Puna Facility is positive even without grants; GCF grants further increase the financial IRR (see table below). However, grants remain essential for the implementation of the Local initiatives for reasons including: subsistence nature of the beneficiaries in the Non-repayable Sub-Window and generally low income for other beneficiaries; no or limited savings to self-fund the initial EbA/CRVC investment; no or limited access to finance to fund EbA measures and CRVC; prohibitive interest rates and other terms of MFI loans; limited financial literacy; lack of beneficiaries' resources to hire technical experts in lieu of the Puna Facility's technical assistance. Importantly, the concessionality of Puna Facility grants is significantly minimized in the Results-based Repayable and Agroideas Matchmaking Sub-Windows through the repayment feature. All Local initiatives funded by the Puna Facility will be financially self-sustaining in the long-term, with no need for further concessionality. This will make them more bankable in the eyes of lenders – bankability will be enhanced also through technical assistance to both Puna Facility beneficiaries and, on the supply side, MFIs. Repayable grants will also instill financial discipline in the recipients, making them more attractive future borrowers.

Table 7. FIRR by Local initiative

Local initiative type	FIRR without grant	FIRR with grant
Non-repayable – Native potatoes	12.6%	121.8%
Non-repayable – Alpacas	8.4%	63.7%
Repayable – Native potatoes	25.8%	84.3%
Repayable – Alpacas	17.1%	48.1%
Agroideas – Alpacas	19.5%	56.8%

270. The economic IRR of the project is 8.2%, which is more than 2 percentage points above the 6% Peru social discount rate estimated by the World Bank and the Ministry of Economy and Finance. The economic NPV, based on the 6% social discount rate, is a positive EUR 11.9 million. Sensitivities have been run on two assumptions: (i) downside sensitivity to the already conservative CO₂ shadow prices and (ii) change in GHG reduction volume and economic benefits of Local Initiatives during the initial 3 calls for proposals. The analysis shows that the EIRR is positive even with extremely low CO₂ prices and with a 30% drop in emission savings volume and income generated by the initial calls for proposals. The EIRR remains above the social discount rate of 6% and the economic NPV is therefore positive if the emission savings volume and economic benefits do not drop more than 10%, even at carbon prices substantially lower than the base case assumption of EUR 60/t.

D.6.3 Application of best available technologies and practices

271. The promotion of sustainable landscape management practices, soil amendments, ancestral and modern technologies for water and soil conservation, exclusion zones and communal reserves, among others, will ensure resource conservation for future generations. In this context, the project catalyses climate innovation through the investment in Climate Resilient Value Chains, which integrate resilient practices and technologies. For example, the implemented EbA measures will improve water harvesting during the rainy season for distribution in times of drought by supplementing water from rainfall. This, together with efficient irrigation technologies, will improve and make sustainable use of water resources as for example for fodder production.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Currently, public and private financing for investing in EbA measures and Climate Resilient Value Chains in the puna ecosystem are limited. Although MIDAGRI has promoted support to the High Andean zones through different programs, they have not been sufficient to reach the most vulnerable populations to climate change in the region. The reasons for the insufficient public	<u>Low</u>	The target scenario with project implementation foresees a paradigm shift to promote the development and maintenance of financial, technical and regulatory support to the implementation of EbA measures and Climate Resilient Value Chains within the puna ecosystem, promoting the enhancement and recognition of ecosystems and their inhabitants. During project implementation, 60,715 direct beneficiaries within 91 districts and 2,011,856 indirect beneficiaries living in the lower parts of the watersheds of the project area will be reached, but project activities will also work on building the conditions to support mainstreaming of EbA and climate resilience, as well as strengthening institutional capacities in Peru. This will enable the development and scale up of financial options and	Through the establishment of the Puna Facility (Outcome 2 / Output 2.1) the project will permit the scaling up of financial incentive for local communities to implement EbA measures and Climate Resilient Value Chains. The Puna Facility will work with different sub-windows to provide financial support for the implementation of EbA measures and Climate Resilient Value Chains (CRVC) to support local most vulnerable communities and improve ecosystem services in the region. Besides providing financial support, through Outcome 3, the project will work on strengthening technical, social, political and institutional capacities of local actors, as well as instruments and norms for the management of ecosystems and their services. Likewise, the project will promote better governance in the prioritized areas, seeking that communities and local governments plan

	and private financing are related to the insufficient valuation of the contributions of the territories and their inhabitants to local and national development, this is based on and feeds back on the little development of enabling conditions, favourable environments and institutions for support local adaptation to climate change in the region, specifically those aimed at reaching the most vulnerable communities.		technical support for local communities to improve its climate resilience and promote climate change adaptation in the region. The successful experiences, the development of capacities and the favourable regulatory frameworks promoted by the project will allow the scale up of similar activities in a broader scope, as well as in the territory itself.	and manage their territories in close coordination. Through the dissemination of successful experiences, the installed capacities and favourable institutional frameworks, the project will make scaling up possible.
Replicability	The Peruvian High Andean region occupies 28% of the national territory and is home to 32% of the country's population. The insufficient and/or non-existent enabling conditions,	<u>Low</u>	The project will be an example of a successful initiative to promote and facilitate the implementation of EbA measures and Climate Resilient Value Chains. Through the improvement of multilevel landscape governance instruments and stakeholder engagement in the project target area, the project will be able to promote replicability and knowledge sharing for	Replicability is closely related to the regulatory, budgetary, and incentive context that favours the reproduction of successful experiences. In this sense, the project will contribute to the replicability of actions through the strengthening of regulatory frameworks, institutional arrangements, development of instruments and procedural mechanisms, as well as the capacity development that will enable local stakeholders

	favorable environment and institutions that promote and guarantee the livelihoods of this part of the population limits the development of EbA measures. Also, limited communication and low knowledge sharing between organizations and stakeholders working with similar initiatives currently hinders the adoption and replicability of similar initiatives. Ancestral and best practices about EbA are not disseminated and replicated enough in the region due to process of emigration of those who know the practices, lack of appreciation and recognition of this		similar initiatives not only in Peru, but in similar ecosystems regionally. As the project will be implemented in a total of 91 districts, of which 58 will be part of the target area of the Puna Facility and 33 will benefit only from the capacity building initiatives, the project will still have the capacity to expand its activities within other districts in the region.	to reproduce and improve EbA investments in the territory. The activities under project's Outcome 3 (output 3.1) will engage in improving governance and capacity development of relevant stakeholders. By implementing activities related to strengthening capacities, participatory platforms and their mechanisms of action, as well as improving coordination and regulatory frameworks, the project will be able to link the results achieved on the ground to national climate change goals, e.g. the country's NDC.
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	knowledge and the adoption of modern techniques that are not conforming with the region and are not sustainable.			
Sustainability	Local financial and technical structures to promote the implementation of EbA measures and Climate Resilient Value Chains still face challenges to be implemented and become sustainable, among them: the few financial mechanisms that incorporate an adaptation approach to climate change; the limited regulatory framework, the absence of planning tools that incorporate EbA. All of which hinders the transfer and adoption/adaptation of EbA measures by	<u>Low</u>	The establishment of the Puna Facility, with the aim to adopt in a long-term a facilitation role on leverage, mobilization and channelling of different financial resources will enable the continuous implementation of EbA and Climate Resilient Value Chains in the region after the conclusion of the project. The additional financial instruments that the project will support will provide the continued financing needed. By the end of the project's activities local relevant stakeholders will have the needed technical capacity and ownership of the project's results to ensure its continuity and self-sustainment. For example, the Puna Facility will be run by Profonanpe, a national institution that has been involved since the design phase of the project, is accredited as a Direct Access Entity to the GCF and has ample experience in the management of environmental funds.	Through the implementation of technical assistance and capacity building for local relevant stakeholders (outputs 1.1, 1.2 and 3.1), the project will be able to support its sustainability beyond the project's period. All the project's outcomes foresee activities and elements related to capacity building and strengthening of local systems that contribute to a sustainable enabling environment for the implementation of EbA measures and Climate Resilient Value Chains. Above all, sustainability will be achieved by the strengthening of regulatory and institutional frameworks that guarantee the sustainability of current and new EbA measures, as well as self-financing mechanisms for the maintenance and implementation of these measures. Activity 2.1.1 will ensure that all the elements needed for the self-sustainment of the Puna Facility are in place before the end of the project period. This will include the development of plans for continuous improvement of the allocation of resources for the Facility.

	the high Andean populations.			Activity 3.1.2 will work to strengthen regulatory frameworks at national level to ensure the sustainability after the project implementation period.
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁹⁰	
<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Annual reports from the Puna Facility and Executing Entities (GIZ, SERNANP, MIDAGRI). Project monitoring and evaluation reports quality assessed by the AE and EEs. Independent interim and final evaluation carried out on the project.	0	Total direct beneficiaries: 4,735 out of which 2,367 are women 156,924 indirect beneficiaries (of which 78,462 are women)	Total direct beneficiaries: 60,715 out of which 30,088 are women 2,011,856 indirect beneficiaries (of which 1,005,928 are women)	Direct beneficiaries of an adaptation intervention will include all individuals who will receive i) targeted support from the GCF-funded intervention and ii) a measurable adaptation benefit from a GCF-funded intervention, including: a) Individuals that implement Local initiatives that will be financed and supported by the Puna Facility, multiplied by the average number of members of the groups/legal entities supported and the average members per household in the region. b) Individuals that will be supported by MIDAGRIS co-finance programs, multiplied by the average members number per household in the region. c) Individuals that will receive technical assistance from the GCF- funded intervention (e.g trainings, support to develop site-specific climate diagnostics and preparation of participatory intervention plans) d) Public officials that will receive capacity building for adjusting regulations and norms.

⁹⁰ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

						Indirect beneficiaries of the project will be those that do not receive targeted support from the GCF-funded intervention but will benefit from water generated by the improvement of ecosystems that contribute to water regulation in the basin in the main cities of Cusco, Arequipa and Cañete. Mid-term target is expected to be 7,8% of the final target based on the project implementation plan.
<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options</u>	Annual reports from the Puna Facility (Profonanpe) and Executing Entities. (GIZ, SERNANP, MIDAGRI). Project monitoring and evaluation reports quality assessed by the AE and EEs. Independent interim and final evaluation carried out on the project.	0	Total direct beneficiaries: 4,735 out of which 2,367 are women	Total direct beneficiaries: 60,715 out of which 30,088 are women	Executing Entities (MIDAGRI, Instituto de Montaña, GIZ) will collect data on direct beneficiaries. Direct beneficiaries of the Puna Facility were calculated based on the following: a) 127 Local initiatives b) Average number per supported organization c) 5 members per households in High Andes Mid-term target is expected to be 7,8% of the final target based on the project implementation plan.
<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 4.2: Number of</u>	Annual reports from the Puna Facility (Profonanpe) and	0	690 Livestock Unit (LSU)⁹¹	8,854 LSU	The estimations of alpacas are derived by the hectares of natural grasslands that are expected

⁹¹ An alpaca is estimated as 0.25 LSU/head.

	<u>livestock brought under sustainable management practices</u>	Executing Entities (SERNANP, MIDAGRI). Project monitoring and evaluation reports quality assessed by the AE and EE. Independent interim and final evaluation carried out on the project.				to be restored through the Puna Facility, multiplied by the carrying capacity of the pastures. Additional alpacas will be supported through cultivated pastures and greenhouses for the production of fodder. Mid-term target is expected to be 7,8% of the final target based on the project implementation plan.
<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Annual reports from the Puna Facility (Profonanpe) and Executing Entities (SERNANP, MIDAGRI). Project monitoring and evaluation reports quality assessed by the AE and EEs. Independent interim and final evaluation carried out on the project.	0	Total direct beneficiaries: 4,735 out of which 2,367 are women 156,924 indirect beneficiaries (of which 78,462 are women)	Total direct beneficiaries: 60,715 out of which 30,088 are women 2,011,856 indirect beneficiaries (of which 1,005,928 are women)	For direct beneficiaries, agricultural production of local family farms will be improved through the implementation of project activities, contributing to food security. For indirect beneficiaries, water security benefits are expected, derived from the contribution to water regulation of the EbA measures implemented in the upper part of the basin. Mid-term target is expected to be 7,8% of the final target based on the project implementation plan.

<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.2: Beneficiaries (female/male) with improved food security</u>	Annual reports from the Puna Facility and Executing Entities (GIZ, SERNANP, MIDAGRI). Project monitoring and evaluation reports quality assessed by the AE and EEs. Independent interim and final evaluation carried out on the project.	0	3,546 out of which 1,773 are women	45,463 out of which 22,732 are women	Agricultural production of local family farms will be improved through the implementation of project activities, contributing to food security. The target under this indicator is expected to be 74.88% (percentage of the population estimated to be subsistence farmers) of the total direct beneficiaries supported under sub-activities 1.1.2.1, 1.1.2.2, 1.1.2.3, 1.1.1.2 and 1.2.1.1.
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	Annual reports from the Puna Facility (Profonanpe) and Executing Entities (SERNANP, MIDAGRI). Project monitoring and evaluation reports. Independent interim and final	0	1,865 ha	23,914 ha	Local communities will be interested in being part of the project and implement EbA measures and Climate Resilient Value Chains. Calculations will be based on an analysis of the area that will be brought under management practices restoration according to the area that the Local initiatives will cover in the 58 prioritized districts. Mid-term target is expected to be 7,8% of the final target based on the project implementation plan.

		evaluation carried out on the project.				
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under restoration and/or improved ecosystems</u>	Annual reports from the Puna Facility (Profonape) and Executing Entities (SERNANP, MIDAGRI). Project monitoring and evaluation reports. Independent interim and final evaluation carried out on the project.	0	1,865 ha	23,914 ha	Calculations will be based on an analysis of the area that will be brought under restoration according to the area that the Local initiatives will cover in the 58 prioritized districts. Local communities will be interested in being part of the project and implement EbA measures and Climate Resilient Value Chains. Mid-term target is expected to be 7,8% of the final target based on the project implementation plan. The area was calculated based on the geospatial analysis of EbA and value chains.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	Currently, EbA measures are not sufficiently mainstreamed into local institutional and regulatory frameworks in Peru. There is limited coordination between actors to reinforce the positive impacts of EbA	<u>medium</u>	In the target scenario institutional and regulatory frameworks in Peru will have EbA measures mainstreamed, with local communities' participation to ensure ownership. Local stakeholders will have	The project will actively contribute to the strengthening of institutional and regulatory frameworks through the activities under output 3.1. The objective will be to mainstream EbA	<u>Single sub-national area within a country</u>

	measures in the territory and the regulations do not consider the importance of protecting, restoring and conserving ecosystems. Moreover, local communities do not participate in the decision-making process at the local and regional level, which makes the institutional and regulatory frameworks weaker due to the lack of ownership and participation of local actors.		improved capacities and more awareness of the benefits of EbA measures and Climate Resilient Value Chains. This will enable the implementation of those measures at the local level, as well as the replication and scale up of the initiatives through the dissemination of knowledge and stronger regulatory and M&E frameworks.	approach and community participation into territorial planning instruments at the local and regional levels, and strengthen M&E systems at the national level.	
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	The knowledge about methodologies and good practices related to EbA and Climate Resilient Value Chains in the High Andean Region in Peru, including ancestral knowledge, is not well disseminated within local stakeholders. There is a limited overall generation and management of information that hinders the implementation of good practices. This is mostly due to the migration of those who know about the	<u>low</u>	The target scenario with the project implementation will include the support to generate, disseminate and deploy the best practices based on ancestral knowledge. This will enable other family farms, organizations and institutions based in districts that are not part of the project target area to access and disseminate the information, implementing the	The project will contribute to the effective knowledge generation and use of good practices by recovering and scaling up ancestral knowledge and practices related to EbA measures in the region through output 1.2. Moreover, under output 3.1 the project will work on the development of training and capacity building to local relevant stakeholders, which will support knowledge transfer to the most	<u>Single sub-national area within a country</u>

	practices, the lack of recognition of the importance of this knowledge by institutions and the adoption of technologies that are not suitable for the puna ecosystem and are not sustainable.		measures and achieving similar results.	vulnerable communities and other regions in the country.	
<u>Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	In the project target area, most of the agricultural units are family farmers and most of them are categorized as subsistence and extreme poverty. These farmers do not have the capacity to access markets and be included in the value chain due to the lack of financial and technical support in the region.	<u>low</u>	With the appropriate technical and financial support, family farmers in the puna ecosystem will be able to implement best practices and have access to financial instruments that will help them improve their production and secure their livelihoods. The target scenario will be to have Climate Resilient Value Chains implemented in the region, which would be supported through more demand for the specific products coming from the region, which would strengthen and transform the market at the national level.	The financial and technical support provided by the project's activities under output 1.1, 1.2 and 2.1 will enable the family farms in the region to improve their production and increase their resilience to climate change. The activities related to the development of Climate Resilient Value Chains under output 1.1 will support the inclusion of this improved production from family farmers in the region and will support the market transformation at the national level.	<u>National level (one country)</u>

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1. Puna ecosystems are restored, conserved and better managed through the implementation of EbA measures complemented by CRVC	Hectares of ecosystems restored, conserved and better managed through the implementation of EbA measures	Project monitoring and evaluation reports	0	1,865 ha	23,914 ha	The area was calculated based on the geospatial analysis of EbA and value chains.
Output 1.1 Investments for EbA measures and Climate Resilient Value Chains are implemented at the local landscape level	Value (EUR) of funds to support the implementation of EbA measures and CRVC at the local landscape level	Project monitoring and evaluation reports Signed grant agreements with Puna Facility Integrated financial management system (SIAF): information about public investment projects EMMA integrated management system (Profonanpe): disbursement reports MIDAGRI reporting (e.g., Agroideas, Agrorural, etc)	0	From the Puna Facility: EUR 1.12 million From MIDAGRI: EUR 7.8 million	From the Puna Facility: EUR 14.4 million From MIDAGRI: EUR 22.7 million	58 out of 91 districts prioritized will be part of the Puna Facility target area. Local communities are interested in implementing EbA measures and climate resilient value chains. Mid-term target for the Puna Facility is expected to be 7,8% of the final target based on the project implementation plan.

						For MIDAGRI programmes include Agrorural, Sierra Azul, Directorate for Insurance and Promotion of Agricultural Financing Agroideas, INIA, SERFOR and PSI. Mid-term target for MIDAGRI is expected to be 16,6% of the final target based on the project implementation plan
Output 1.2 The use of EbA knowledge is recovered and disseminated, and local monitoring committees and observation systems are implemented	Number of Local initiatives that implement EbA measures based on ancestral knowledge	Project monitoring and evaluation reports Field surveys results	0	10	100	58 out of 91 districts prioritized will be part of the Puna Facility target area. Local communities are interested in implementing EbA measures and climate resilient value chains.
	Number of local monitoring committees and observation systems established	Minutes of traditional assemblies/organizational assemblies that formalize the monitoring committees	0	10	100	

Outcome 2. Public and private financing for EbA measures and climate resilient livelihoods are accessible and being actively used by vulnerable communities in the Puna Ecosystem	Number of public and private financing modalities to promote EbA and CRVC established and being actively used by vulnerable communities in the Puna ecosystem	Project monitoring and evaluation reports MIDAGRI reporting (e.g., DSFFA) Integrated financial management system (SIAF): information about public investment projects Implemented by EPS with MERESE Agreements (e.g., MoU) signed between EPS and implementing institutions to implement EbA measures. Agreements (e.g., MoU) signed between financial institutions and the project Proof of disbursement of financial resources from the new financial modalities to the local communities in the Puna ecosystems or invested in EbA and CRVA (contracts, agreement, etc.).	0	1	3	58 out of 91 districts prioritized will be part of the Puna Facility target area. Stakeholders (e.g., rural banks, private microfinance institutions) are interested in developing and/or adjusting financing options for EbA measures and CRVC. Financial institutions such as rural credit banks are interested in developing and/or adjusting credit lines to support EbA and CRVC. Fondo Agroperú supports 14 programs and is interested in partnering with the proposed project. The project will consider as “actively used” the financial mechanisms that have been established and have disbursed financial resources to
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						the local communities in the Puna ecosystem or invested in EbA and CRVC.
Output 2.1 Financial mechanisms for the implementation of EbA measures and improvement of climate resilient livelihoods in the Puna ecosystem are established	Number of public and private financing modalities to promote EbA and CRVC available for vulnerable communities in the region	Project monitoring and evaluation reports MIDAGRI reporting (e.g., DSFFA) Integrated financial management system (SIAF): information about public investment projects Implemented by EPS with MERESE Agreements (e.g., MoU) signed between EPS and implementing institutions to implement EbA measures. Agreements (e.g., MoU) signed between financial institutions and the project	0	2	6	58 out of 91 districts prioritized will be part of the Puna Facility target area. Stakeholders (e.g., rural banks, private microfinance institutions) are interested in developing and/or adjusting financing options for EbA measures and CRVC. Financial institutions such as rural credit banks are interested in developing and/or adjusting credit lines to support EbA and CRVC. Fondo Agroperú supports 14 programs and is interested in partnering with the proposed project.
Outcome 3. EbA and climate resilience are mainstreamed into multilevel landscape	Number of regulatory frameworks or public instruments redesigned or	Project monitoring and evaluation reports	0	2	5	Willingness of the government in improving their regulatory frameworks

governance instruments	strengthened to include EbA and climate resilience	Public body decisions related to relevant regulatory frameworks Ministerial/directorial resolutions				
Output 3.1 Multilevel landscape governance is improved through strengthening of national capacity, regulatory frameworks and M&E systems	Number of government officials demonstrating improved capacity in EbA and climate resilience including gender aspects	Reports from pre and post capacity building training surveys	0	50% increase in self-reported awareness levels in population of 300 people capacitated	50% increase in self-reported awareness levels in population of 605 (182 women and 423 men) ⁹² people capacitated	Engagement of government officials to participate in the online and presential training provided by the project
Project/programme co-benefit indicators						
Co-benefit 1 Maintenance of ecosystem services in the project area	Area within the project target districts that are important for ecosystem services where EbA measures are implemented	Project reports	0	1,865 ha	23,914 ha	
Co-benefit 2 GHG emissions are avoided through the conservation and restoration of degraded ecosystems	Avoided GHG emissions in tCO ₂ e	ExACT tool	0	31,797 tCO ₂ e	407,657 tCO ₂ e	The overall mitigation impact as a co-benefit from the projects is a reduction of 407,657 tCO ₂ e in the 15 years analysed, which include five years of implementation phase and ten years of capitalisation phase. It

⁹² Based on the ratio of the participation of women and men during the public official engagement events as part of the consultations a 30% women's participation can be expected.

						is measured against a scenario without project that would have increased by 93,905 tCO₂e the GHG emissions in the areas of intervention. Mid-term target is expected to be 7,8% of the final target based on the project implementation plan
Co-benefit 3 Improvement of local economy through better economic opportunities and poverty reduction	Number of producers, of whom 50% are women have confirmed an improvement in their income from climate resilient value chains or climate-resilient practices.	Reports from MIDAGRI Pre and post workshop survey	0	0	70% of the local initiative beneficiaries	Implementation of 127 Local initiatives
Co-benefit 4 Awareness raising and capacity building of local stakeholders	Percentage of increase in self-reported awareness levels in population of 10,577 people capacitated	Pre and post workshop survey	0	50% increase in self-reported awareness levels in population of 825 people capacitated	50% increase in self-reported awareness levels in population of 10,577 people capacitated	Under capacitated people the project counts following: 9647 producers trained through MIDAGRI (activity 1.1.3) 910 local experts (273 women and 637 men)⁹³

⁹³ Based on reported information from Agrorural a 30% women's participation is expected.

						capacitated (activity 1.2.1) 20 Local actors capacitated (activity 2.1.2). Mid-term target is expected to be 7,8% of the final target based on the project implementation plan considering that the project implementation
Co-benefit 5 Gender responsive approaches mainstreamed in EbA and CRVC measures	Number of innovative Local initiatives (including EbA and CRVC) with a gender-sensitive approach implemented	Reports from Profonanpe	0	10	127 innovative Local initiatives (with EbA and CRVC) with a gender-sensitive approach implemented	Mid-term target is expected to be 7,8% of the final target based on the project implementation plan.
E.6. Project/programme activities and deliverables						
Activities	Description		Sub-activities		Deliverables	

Activity 1.1.1. Setting up the basis for financing and implementing EbA measures and Climate Resilient Value Chains	A communication strategy will be created to develop awareness about the project and create the basis for the implementation of EbA measures and CRVC through the Puna Facility. Technical support for local stakeholders (associations, CSOs, cooperatives) will be provided in order to participate in the call for proposals.	Sub-activity 1.1.1.1 Inform and identify communities, associations, cooperatives and Civil Society Organisations interested in participating in the project Sub-activity 1.1.1.2 Development of site-specific climate diagnostics and preparation of participatory intervention plans	• 1 Communication and engagement strategy developed • At least 100 Communities, associations, cooperatives and/or micro and small enterprises to be supported in 1.1.1.2 selected • 1 Training for project staff on concept standardization implemented • 1 Guideline (including participatory methodology) developed for site specific climate diagnostics and project intervention plans • At least 100 Local initiative intervention plans prepared • 3 Trainings for project staff on site-specific climate diagnostics implemented
Activity 1.1.2. Financing and implementing of EbA measures and Climate	This activity will focus on promoting climate resilient Puna ecosystems and CRVC through of the implementation of climate-focused Local initiatives at local landscape level. The activity will provide financial support for Local initiatives through the Puna Facility (established under Activity 2.1.1) and through public	Sub-activity 1.1.2.1 Implementation of Local initiatives financed by the Puna Facility Sub-activity 1.1.2.2 Financial support through MIDAGRI	• 127 Local initiative Grant Agreements signed

Resilient Value Chains	resources (MIDAGRI's programmes). The two types of financial support are complementary and contribute to a general objective of improving the climate resilience of the local communities in the target territories.		• At least 8 public investment projects implemented
Activity 1.1.3. Technical Assistance for implementing EbA measures and Climate Resilient Value Chains at local landscape level	This activity will focus on providing technical assistance for implementing EbA measures and CRVC through MIDAGRI and the Puna Facility and enhance the market access and economic opportunities for the Climate Resilient and Sustainable Value Chains.	Sub-activity 1.1.3.1 Technical assistance for the refinement of proposals and Local initiative implementation (including EbA implementation, business development and access to finance) through the Puna Facility Sub-activity 1.1.3.2 Technical assistance through MIDAGRI Sub-activity 1.1.3.3 Promote market access and economic opportunities for Climate Resilient and Sustainable Value Chains	• 127 tailored TA packages offered • 9,647 producers trained • 1 Digital platform for sale of Puna products operating
Activity 1.2.1 Recover, innovate and scale up ancestral knowledge and practices	This activity seeks to develop and strengthen the capacities of communities, local experts (called yachachiqs or kamayocs), and technical experts from national and regional institutions to increase the implementation of ancestral water and land management practices and technologies. The dialogue on ancient practices, knowledge and innovation will build intergenerational memory (children, youth and adults) at the community level on ancestral practices and innovation in a context of change (climate and markets, among others) and support the recognition and appreciation of people's own values, knowledge and practices, improving their self-esteem and strengthening their identity with the territory. It will also expand on gender roles to reflect on their place in and relationship with ecosystems. For this to be effective, studies and communication materials in native languages will be produced to capture lessons learned and promote replication of success stories. Exchange between communities will be financed by the project to encourage peer-to-peer learning and ownership of implemented solutions. Of particular importance is the transfer	Sub-activity 1.2.1.1 Capacity building of local experts for the transfer of ancestral and innovative knowledge linked to EbA and Climate Resilient Value Chains (CRVC) measures Sub-activity 1.2.1.2 Production and dissemination of information materials on lessons learned from ancestral practices and innovation in a context of change	• At least 910 local experts (273 women and 637 men)⁹⁴ capacitated • 2 sets of communication materials in native languages on lessons learned from ancestral practices and innovation are published • At least 5 dissemination events about lessons learned from ancestral

⁹⁴ Based on reported information from Agrorural a 30% women's participation is expected.

	of experience already gathered from recuperating ancient practices and promoting participatory agreements in the Nor Yauyos Cochas landscape reserve.	Sub-activity 1.2.1.3 Knowledge sharing among beneficiary communities with a gender perspective Sub-activity 1.2.1.4 Facilitating intergenerational dialogue on ancestral practices and innovation in a changing context	practices and innovation are carried out • 1 Knowledge exchange plan with gender perspective developed • 1 Report on community exchanges with gender perspective developed • 1 Intergenerational Dialogue Plan developed • 1 Systematization report on conducted intergenerational dialogues developed
Activity 1.2.2 Implement community monitoring and observation systems to measure the impact of EbA measures and provide feedback on regional and national policies	Instituto de Montaña, partnering with CSOs and local stakeholders, will reinforce stakeholders' capacities to support the establishment of community monitoring committees (based on the participatory action research approach, and previously identified in sub-activity 1.1.1.2). These committees will be responsible for community impact monitoring activities based on local knowledge and for liaising with the community for different project activities. The community monitoring committees' capacities will be strengthened, enabling knowledge weaving and coproduction on EbA, traditional practices and climate resilience. Target populations will establish climate vulnerability social and biophysical indicators based on local knowledge, complementing the local climate diagnostics. They will prepare participatory Local	Sub-activity 1.2.2.1 Training and implementation of Community Monitoring Committees to develop local monitoring systems Sub-activity 1.2.2.2 Development and implementation of a local data-driven information management system using digital technologies innovation measures Sub-activity 1.2.2.3 Analyse collected information on the impact of EbA measures	• 1 Community monitoring guideline developed • 1 Baseline report developed • 1 Final report of community monitoring progress developed • 1 Protocol to feed existing governmental Information Management

	initiative intervention plans under sub-activity 1.1.1.2, and a baseline in their own communities, including conditions and trends of ecosystem services in the surrounding landscapes. Committees will develop local indicators and technical protocols to assess changes in ecosystem service provision over time as the EbA measures are implemented and mature. The data produced by the local monitoring system will feed regional and national institutions studies and research on impacts and effectiveness of EbA interventions, thus contribute to policy decision making processes. Technical assistance will be provided for the establishment of a methodology to calculate carbon in the Puna ecosystems and provide capacity building for local actors (e.g. communities, local government, academia) aligned with national <i>legal and institutional arrangements for carbon markets</i>.		Systems from a Cloud Database with local data developed • 1 Existing governmental Information Management Systems fed with local data • At least 1 study on the impact of EbA measures completed • Methodology for the quantification of the carbon stock change cause by the EbA measures developed • At least 20 local actors capacitated
Activity 2.1.1 Establish the Puna Facility for the long-term financing of EbA measures and Climate Resilient Value Chains	This activity will establish the Puna Facility (linked to the operationalisation of the Facility in sub-activity 1.1.2.1.), managed by Profonanpe, to unlock investment for EbA measures in Puna ecosystems and climate resilient agrobusiness. The Puna Facility will be composed of one financial window (linked to sub-activity 1.1.2.1.) to receive funds from private sources and finance implementation of EbA and CRVC Local initiatives. This window will be supported by technical assistance (linked to sub-activity 1.1.3.1). The project will develop and implement other innovative financial mechanisms to mobilise resources from the private sector and different donors. This will be achieved through the design of a	Sub-activity 2.1.1.1 Preparation and continuous improvement of the allocation of resources of the Puna Facility Sub-activity 2.1.1.2 Development and implementation of an innovative strategy to mobilize resources from the private sector and different donors	• Puna Facility Management Unit established • 1 Report on systematized lessons learned developed • At least two MoUs to secure new funding sources for the Puna Facility are signed

	Private Sector Engagement Strategy, including criteria for selecting the companies in which advocacy actions will be deployed (for example, companies committed to social and environmental responsibility, without potential conflicts, linked to CRVC), criteria for selecting demonstration sites, and an effective communication and marketing strategy to engage the private sector. Finally, in order to increase the use of state funds for actions related to climate change adaptation, the activity will seek to mobilise resources from public programmes for the High Andean population and their associated ecosystems. To this end, access to and alignment of investments from MIDAGRI and MINAM and other government programmes will be promoted for EbA and CRVC measures through the development of improved instructions for access to public financing.	Sub-activity 2.1.1.3 Promote access and alignment of public investments and other government programmes for EbA and Climate Resilient Value Chains (CRVC) measures	• Sustainable financing strategy for Profonampe to access private and international financing developed • Dissemination material (e.g., Flyers) developed • 1 Final report about the increased execution of public funds for actions/measures related to climate change adaptation developed
Activity 2.1.2 Strengthen capacities to develop and implement innovative mechanisms for EbA in high Andean ecosystems	The activity will seek to overcome the barriers faced by public water utilities that have established or are willing to establish a mechanism for retribution for ecosystem services (MERESE) schemes. Therefore, activities will be focused on strengthening stakeholders' capacities to identify and overcome bottlenecks and gaps in the implementation of MERESE. These activities will be developed in close coordination with the Ministry of Environment and the National Superintendence of Water Supply and Sanitation of Peru (SUNASS).	Sub-activity 2.1.2.1 Technical assistance to improve existing hydrological MERESE processes in the project areas	• 1 Capacity Building Program for water utilities personal developed • Technical assistance (e.g., EbA approach and public investment) implemented in at least three MERESE
Activity 2.1.3 Support the greening of microcredits to promote EbA	This activity will focus on two main topics: (1) Supporting the greening of microcredit lines and business models and (2) supporting the dissemination of green financial products and services.	Sub-activity 2.1.3.1 Support to financial institutions in greening microcredits	• At least 2 business model EbA and CVCR by specific chain (according to the Project

and Climate Resilient Value Chains	In order to increase the supply of microfinance products and services to finance integrated EbA and CRVC initiatives, the project will support either the development of specific financial products or the greening of existing financial products for EbA measures and CRVC (aligned with the lists of EbA measures and CRVC developed in sub-activity 1.2.1.1) in Puna ecosystems. This will include strengthening the capacities of the officials who will offer the financial products. In addition, a strategy to disseminate the green financial products will be implemented, focusing on the beneficiaries of the project.	Sub-activity 2.1.3.2 Raise awareness of greener microcredits to promote EbA and Climate Resilient Value Chains	catalogue) elaborated • At least 2 adapted microcredits for EbA and CVCR business models developed and disseminated with microfinance institutions • 1 Communication strategy for microfinance institutions developed • At least 2 specific financial products for EbA and CVCR business models disseminated with microfinance institutions
Activity 3.1.1 Strengthen the capacities for territorial planning and governance processes integrating EbA and climate resilience	The new approaches that will be promoted by Resilient Puna require progressive development of capacities of implementers, local actors, and subnational governments to promote the replicability and sustainability of the changes introduced by the project. Therefore, in order to mainstream EbA and climate resilience approaches into planning processes (i.e. local and regional development plans, including budgets and investment project planning), official capacities at different levels (national, regional and local) will be strengthened and gender perspectives will be considered.	Sub-activity 3.1.1.1 Strengthen the capacities of relevant national, regional and local government stakeholders for the incorporation of EbA and Climate Resilient Value Chains measures, integrating the gender perspective in their processes and interventions in the territory Sub-activity 3.1.1.2 Strengthen participatory sectoral and territorial platforms for the articulation and scaling up of EbA measures	• Online and/or presentational training modules on how to mainstream Gender and EbA/CRVC approaches are developed • At least 605 (182 women and 423 men)⁹⁵ relevant

⁹⁵ Based on the ratio of the participation of women and men during the public official engagement events as part of the consultations a 30% women's participation can be expected.

	The activity aims to strengthen participatory sectoral and territorial platforms so that the project EbA measures and CRVC developed by project beneficiaries can be replicated and scaled up. Lessons learned from the community's experiences in EbA and CRVC planning, and implementation will be integrated.	and Climate Resilient Value Chains within the scope of the project Sub-activity 3.1.1.3 Integration of EbA measures and Climate Resilient Value Chains in territorial planning and management instruments	officials from the national (SERNANP and MIDAGRI officials), regional (5 regional government officials) and local (at least 50 local governments) government trained • Participatory mechanisms to involve communities/associations/cooperatives in the sectoral or territorial platform decision making processes are developed • Framework planning and projects documents at the regional and local levels with an EbA/CRVC and gender approach are developed
Activity 3.1.2 Strengthen regulatory frameworks and M&E systems at national level	The activity aims to provide technical support to MIDAGRI and MINAM to improve the regulatory framework for the promotion and implementation of EbA measures and CRVC in the High Andean areas of the country. In addition, it will provide technical support for the national long-term M&E systems of the different government entities to measure impact of EbA interventions. Partnerships will be established with local universities and research institutes to support the monitoring of progress resulting	Sub-activity 3.1.2.1 Improving national regulatory frameworks linked to EbA measures and Climate Resilient Value Chains Sub-activity 3.1.2.2 Improve coordination of EbA monitoring and evaluation (M&E) systems	• Regulatory framework which includes the promotion EbA and CVCR interventions in the high Andean areas of the

	from the interventions implemented by the different actors in the territory. This output will also support MIDAGRI in meeting its productivity and competitiveness commitments while complying with Peru's NDC targets.	at the national level linked to project and NDC purposes	country are developed NDC Monitoring system in the national scope with adequate protocols and methodologies to measure the impacts of EbA interventions on water and agriculture is implemented
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E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

E.7.1. Monitoring, reporting and evaluation arrangements

272. Monitoring, reporting and evaluation arrangements (in addition to the ESMP, IPLCEP and GAP) will comply with the relevant GCF policies, as stipulated in the AMA, FAA and project-related Financing Agreements and Implementation Agreements with Executing Entities and final beneficiaries, which EEs forward funds.

273. The project will apply a customized results-based monitoring and evaluation (M&E) system. The M&E system will track project inputs, activities and sub-activities. Outputs and impacts, as well as associated financial flows across all outputs in all project interventions areas.

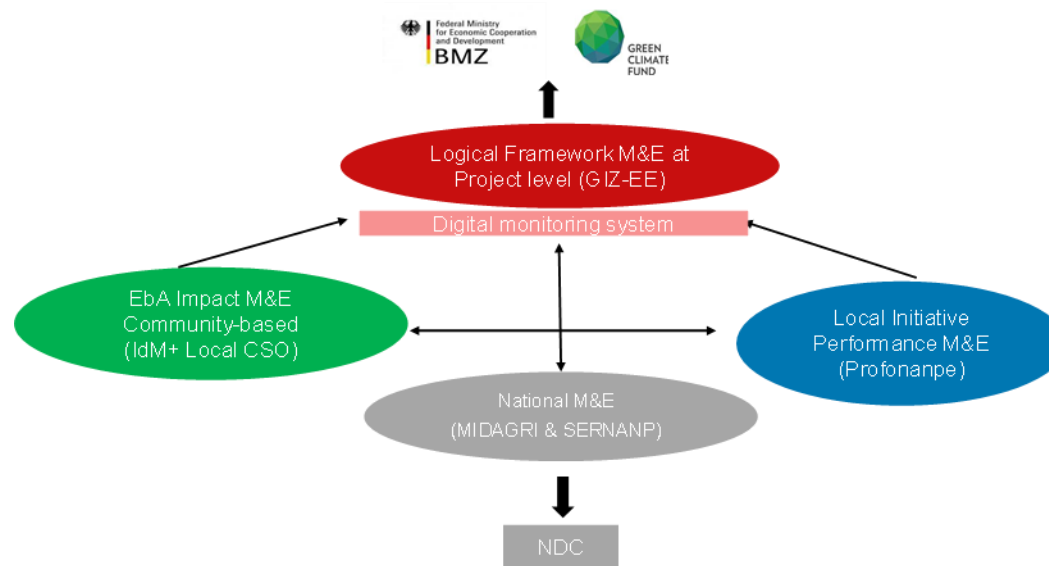
274. This system will be based on:

- GIZ Standard Operating Procedures ("GIZ's evaluation policy – principles, guidelines and requirements")
- Monitoring and evaluation plan (Annex 11)
- Integrated results management framework and logical framework
- Implementation timetable (Annex 5)
- Requirements of GCF's Annual Performance Report
- Local and national monitoring systems
- Procedures and requirements of project partners and stakeholders in Puna and at national level

275. The oversight for M&E and reporting lies within the GIZ GCF AE. On the other hand, GIZ Peru as EE will ensure that appropriate monitoring and evaluation procedures are implemented. A digital tool (e.g., dashboard) for the project M&E system that will follow-up the progress of all results, indicators Logframe, activities and deliverables will be developed. This dashboard will also include different modules to monitor ESMP, GAP and Stakeholder Consultation Plan. Access to all EEs will be provided to collect data generated by the implementation of their activities and sub-activities. Additionally, Profonampe as EE will supervise implementation and prepare periodic reports on the technical and financial achievements of the Local initiatives to be financed through the Puna Facility, as input to feed the project M&E system; Instituto de Montaña will support this work through the implementation and strengthening of community-based monitoring systems. The information collected at the local level will be integrated into the project monitoring system. Community monitoring committees will collect data in the field using an app that will be connected to a data-driven information management system which will also be a module of the project M&E

system. Finally, since the indicators set in the Logframe coincide with the national sectoral indicators, the information produced by the M&E project system will be used to report MIDAGRI and SERNANP national monitoring systems that contributes to the NDCs.

Figure 18. Project M&E system



276. The Project Management Committee (PMC) will be the body responsible for leading the preparation, monitoring and coordination of the Annual Operational Plans and the annual project budget. and discussing monitoring processes and results. Its work will be supported by Project Implementation Unit (PIU) that will coordinate and integrate the Territorial Annual Operating Plan into the overall Annual Operating Planning of the project. It will also oversee the M&E system, monitoring both the technical and budgetary advancements of the project in alignment with the approved Annual Operating Plan and budget. This includes ensuring the timely execution of activities within the territory and providing regular updates to the PMC on the progress in this regard.

E.7.2. Recruitment of M&E staff

277. Immediately when the project commences, GIZ's GCF AE will make available one expert (with gender expertise) to oversee and coordinate the project M&E and reporting routines. As soon as GCF project proceeds become available in Puna, the project will recruit a M&E specialist. The specialist will work closely together and support the management of the M&E system and provide on-the-job training for dedicated M&E experts from PMC and PIU and other stakeholder of the system where requested. In addition, the experts will maintain the Cloud Database, and additional specialist inputs may be required for certain technical areas (e.g., agriculture, ecosystems or livestock).

E.7.3. Independent monitoring and evaluation studies

278. **Mid-term and final evaluation:** GIZ AE will initiate a mid-term evaluation and a final evaluation of the project. GIZ evaluations are carried out by GIZs' independent evaluation unit. This unit is steered by a corporate unit that is separate from operational business and reports directly to the Management Board.

Moreover, evaluations are conducted with the support of external evaluators. GIZ applies the 'Evaluation criteria for German bilateral development cooperation' (2006, revised 2020) on a standard basis in project evaluations. These are based on the six evaluation criteria agreed by the Development Assistance Committee (DAC) of the Organisation for Economic Co-operation and Development (OECD): relevance, effectiveness, efficiency, overarching development results (impact), coherence and sustainability. Additionally, to the six OECD DAC evaluation criteria, the following five GCF criteria will be evaluated: Gender equity, country ownership, innovativeness in results areas, replication and scalability as well as unexpected results.

279. **GCFs' Evaluation Principle of Impartial, objective and unbiased and relevance, use and participation:** All provisions for upholding the GCF evaluation principles described in Chapter V (a-d) of the Evaluation Policy are followed (see DAC standards and DeGEval evaluation standards). In line with the UNEG norms, evaluation team members selected by GIZ must not be directly responsible for the policy setting, design or management of an evaluation project. An Interim Independent Evaluation Report and a Final Independent Evaluation Report will be submitted to the GCF as per timeline agreed in the funded activity agreement (FAA).
280. **GCFs' Evaluation Principle of Credibility and Robustness:** As a matter of principle, GIZ takes care to use an appropriate combination of quantitative and qualitative methods of empirical social research in evaluations. The mix of methods is adapted to the object of the evaluation and the evaluation questions. To ensure robust evidence, GIZ uses a theory-based approach as a minimum standard for its central project evaluations. The contribution analysis has proven to be a suitable design. The aim of contribution analyses is to collect empirical evidence on whether and to what extent contributions to observed changes can be associated with a project. Furthermore, the analyses should increase the understanding and knowledge of what works, what does not work and why.
281. Projects will be rated based on the six OECD/DAC and on the GCF evaluation criteria. Each of the OECD/DAC and GCF criteria is rated on a scale of 1 to 100. Using a points system serve to increase the transparency of ratings, whilst enabling better comparability between individual projects.
282. Results from the mid-term evaluation should be reflected within the project design, to influence upscaling, and improve overall project implementation (adaptive management).

E.7.4. Data collection and frequency

283. The RTGs will coordinate data collection for implemented activities through responsible executing partner GIZ, MIDAGRI, SERNNAP Profananpe and Instituto de Montaña in charge of leading the monitoring activities at local level. The PMC will supervise and guide the monitoring and evaluation. The PIU will aggregate monitoring reports based on RTG inputs at least every six months and make a summary report available to GIZ's AE unit at head office, again at least every six months.

E.7.5. Measurement and Evaluation (M&E) of implemented Local initiatives through Puna Facility to evaluate increase of resilience of local communities.

284. The M&E of implemented Local initiatives is an essential activity for monitoring the project's efficiency and effectiveness as well as for continued learning across the project. Instituto de Montaña will support in the implementation of the community level M&E (for more information please see activity 1.2.2.).

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

A detailed analysis of the project risks is provided in Chapter 10 of the Feasibility Study (Annex 2a). Environmental and social risks are considered in the Environmental and Social Impact Assessment (ESIA, Annex 6a), Indigenous Peoples and Local Communities Engagement Plan (IPLCEP, Annex 6c) and Environmental and Social Management Plan (ESMP, Annex 6b). Gender-specific risks are considered in the Gender Assessment and Gender Action Plan (Annexes 8a and 8b, respectively).

Selected Risk Factor 1: Insufficient financing may put at risk the long-term sustainability of the public or private investment

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>

Description

Insufficient public or private financial flows may put at risk the long-term sustainability of the project. Market intake for EbA measures and Climate Resilient Value Chains, especially those promoted via microfinancing may be insufficient. Fluctuations in the market value of cash products (e.g., South American camelid fiber, quinoa) may limit economic activity and entrepreneurship.

Mitigation Measure(s)

From project onset, activities will focus on increasing engagement of private entities (hydroelectric generation enterprises, agribusiness, cement, beverage, fiber and other industry) to leverage and mobilize resources for EbA and CRVC. A communication strategy will be developed to make the case for potential expected losses in these industries if Puna ecosystems are not conserved. Public funding will be increased by building official capacities in MIDAGRI and SERNANP and triggering operation and improving functionality of the hydrological MERESE scheme. The promotion of microfinancing for EbA will be done at both the offer and the demand side. Once microfinance institutions are trained in adjusting existing and/or developing microcredits to support EbA measures and/or Climate Resilient Value Chains they will continue promoting through their own channels. Local communities, producers' associations, cooperatives, MSEs, community enterprises and producers' organizations will be strengthened on business development and access to finance (including financial literacy) to ensure successful income-generating activities. Income diversification will be promoted through technical and financial support to avoid over indebtedness and stimulate investment.

Selected Risk Factor 2: Local stakeholders are not sufficiently organized, engaged and interested in participating in the project

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Low</u>

Description

Participation from local stakeholders in the project activities may not be as high as expected due to the lack of interest due to the lack of interest This could happen due to various factors, e.g., not seeing the value of EbA measures or the perception that the Puna Facility does not have an added value to the region. Moreover, there is a risk of 'consultation fatigue' in the districts where other initiatives may have taken place

Mitigation Measure(s)

The project has been designed in close coordination with key local stakeholders, and local consultations with potential project beneficiaries have been carried out to ensure that their interests and views are being addressed in the project. Continuous engagement with relevant stakeholders at all levels will help promote dialogue and ongoing capacity building to facilitate the engagement. The project will develop adequate communication and engagement strategies to ensure reaching out and continuously communicating to stakeholders and potential project beneficiaries about the project interventions throughout the planned project implementation. In addition, the project will emphasize documentation and dissemination of information to facilitate capacity building and knowledge exchange among local stakeholders, which will foster interest in participating in the project activities.

Selected Risk Factor 3: Coordination among regional government stakeholders is insufficient to sustain project results

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Low</u>

Description

Coordination among government stakeholders is insufficient and the lack of interest and ownership of the project may limit the sustainability of project results.

Mitigation Measure(s)		
The project is aligned with the priorities of the Government of Peru, and it has been developed in close coordination with Government institutions that will have a key role in its implementation, e.g., MIDAGRI and Profonampe. Extensive consultations and meetings with government actors and key stakeholders have been carried out during the project design phase. Component 3 of the project will be focusing on strengthening multilevel governance to ensure that the national institutions will have the technical capacity needed to carry out the activities after project completion.		
Selected Risk Factor 4: Risk of continuity due to political instability		
Category	Probability	Impact
<u>Governance</u>	<u>Low</u>	<u>Medium</u>
Description		
The project is constructed on the assumption of sustained commitment from the Government of Peru to fulfil its NDC goals and from MIDAGRI to continue its open and proactive engagement. Political instability in Peru may affect project continuity.		
Mitigation Measure(s)		
GIZ has carried out a consistent engagement process since it was requested to support the Peruvian government to access GCF financing for this project and will continue to do so throughout project inception and implementation. This approach provides a great strength that allows the project to persist despite the changes in government administration and ensures continuity, as it can be evidenced from the recent change in administration without affecting the project construction process.		
Selected Risk Factor 5: Maladaptation and degradation of ecosystems and social structures		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		
Improved economic opportunities in the SHAP may lead to increased pressure on ecosystems (e. g., number of cattle heads increases as a result of business-oriented approaches). Moreover, the support to certain Value chains such as crops may result in maladaptation if they contributed to ecosystems degradation (e.g., use of pesticides). Social structures may continue to deteriorate because of increased ecosystem degradation.		
Mitigation Measure(s)		
All proposed agricultural interventions are intended to increase productivity while ensuring that the carrying capacity of Puna ecosystems is not exceeded. The promotion of sustainable landscape management practices, soil amendments, ancestral and modern technologies for water and soil conservation, exclusion zones and communal reserves, among others, will ensure resource conservation for future generations. Business activities will focus on increasing economic output while decreasing environmental impact. The diagnostic for planning will start from local social characteristics and cultural values that could have an influence on risks and environmental dynamics. Social structures will be strengthened through Outputs 1.2 and 3.1., bringing local successful experiences into planning and decision-making at macro scales. Project beneficiaries will be trained on community-based monitoring processes and indicators. Ecosystem-monitoring capacities of the communities and government partners will be reinforced, integrating the best available science to determine limits in carrying capacity or intensity of agricultural activities.		
Selected Risk Factor 6: Difficulties and delays in unlocking technical and financial support		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Low</u>
Description		
Access to MIDAGRI programs as well as technical and financial support from the Puna Facility will be on a voluntary basis. This may limit the ability of the project to meet its targets or have sufficient concentration of interventions to show tangible results. Unlocking of public financing via MERESE schemes may require lengthy negotiations beyond the life of the project.		
Mitigation Measure(s)		
MIDAGRI has committed about 35 percent of project funds. Through the improved coordination obtained from component 3, a more streamlined approach for execution of MIDAGRI's funds will be operational. The project will provide support for project beneficiaries and MIDAGRI officers to facilitate the communication materials for the application process. Complementarity will be ensured with		

the Puna Facility through the project's Governance Structure as the Project Steering Committee and Project Management Committees will have an active role within the governance structure of the Facility. Participation of all relevant stakeholders (GIZ, Profonampe, MIDAGRI, MINAM) since project conceptualization is ensuring alignment throughout preparation, planning and execution of activities to meet targets. Lessons learned from project execution will be integrated into the procedures of the coordination unit. Local initiatives implementation will be supported by Technical Assistance and in some cases by the support of qualified CSOs and financed by the Puna Facility will be planned as to maximize concentration of positive impacts on ecosystem services. Concerning the MERESE mechanisms, the project will work in watersheds with potential of MERESE at different stages of evolution. Through a systematic and modular approach, the project will build capacities in EPS according to their needs. Those who are already collecting tariffs but do not know how to plan, design, develop, and implement interventions, will be invited to contribute to the Puna facility. For those MERESE where agreements between parties have not been reached, support will be provided to negotiate and finalize such agreements. All this effort will unlock funds for implementation of MERESE. EPS in earlier stages will take longer to fund activities but through the gained experience in the modular approach a speedier process will result.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

G.1.1 Environmental and Social Impact Assessment and Environmental and Social Management Plan

285. The Environmental and Social Impact Assessment and Environmental (ESIA) of the project is provided in Annex 6a. According to the ESIA the overall risk of the project has been rated with a category B (medium risk). In general, the project will have a positive environmental and social impact on the beneficiaries in the South High Andes of Peru. Nevertheless, potential adverse environmental and social impacts of the project will mostly site-specific, but due to the nature of the interventions these impacts will not be irreversible or complex in nature and will be easily remediable through preventive and mitigation measures. Therefore, respective plans including an Environmental and Social Management Plan (ESMP) (Annex 6b) and the Indigenous Peoples and Local Communities Engagement Plan (Annex 6c) were developed).
286. The ESIA primarily relies on primary data obtained through stakeholder consultations and secondary information researched. The Puna area has been subject to comprehensive study, resulting in a wide range of available literature concerning its ecosystems and social aspects. In parallel to the ESIA, stakeholder consultations have been conducted which are summarized in Annex 7 Summary of consultations and stakeholder engagement plan. This document provides next to an extensive mapping of stakeholders a summary of three missions which were carried out in the field. These three missions allowed the recollection of qualitative data that were used to better understand the potential risk from social and environmental point of view and the sensibility of the stakeholders, which were used amongst others for the development of the ESIA.
287. In general, stakeholders interviewed during fieldwork approved the project and its main components. Stakeholders met during fieldwork recognized perceiving the effects of climate change in their livelihoods. In addition, producers expressed their concern about a collective approach versus individual support, as most of the production is individual. They recognize that some actions must be carried out at larger scale when interventions such as building of qochas and/or irrigation investments are being considered. In particular, women are interested in transformation and commercialization of locally product raw material.
288. Furthermore, based on the analysis in general, the impacts of the project can be assessed as low to medium, these are mostly to the activities within component 1 (regarding activity 1.1.2 Implementation of EbA measures and Climate Resilient Value Chains) and component 2 (regarding activity 2.1.1. establish the Puna Facility for the long-term financing of EbA measures and Climate Resilient Value Chains), since the competition for funds could be a conflict driver, therefore and based on the results an ESMP and IPLCEP has been designed for the management of the identified risks (an overview of environmental and social (E&S) risks and impacts, and mitigation measure can be found in **Error! Reference source not found.** below, for ESS 7 Indigenous Peoples Table 9 details the respective mitigations actions). The implementation of the ESMP will be monitoring. This will involve tracking compliance with the Environmental and Social Standards (ESS) and assessing the advancement of ESMP implementation through bi-annual Results-Based Monitoring (RbM) workshops and the Environmental and Social Safeguards Management and Monitoring System (ESS MS system). The ESS MS system encompasses specific procedures, plans, organizational structures, planning activities, and resources geared toward establishing, executing, attaining, reviewing, and sustaining the Environmental and Social Requirements. Its development will take place during the inception phase, led by GIZ and in close collaboration with all EE ESS focal points.
289. In addition, ESS1 requires the establishment of an external communication and grievance redress mechanism (GRM) for the project to address grievances from affected communities and individuals, as well as external communications from stakeholders. At the GCF level, the independent Redress Mechanism handles complaints related to adverse impacts resulting from the GCF-financed activities' failure to adhere to GCF operational policies and procedures, including ESS standards. If a complaint is filed with the Independent Redress Mechanism, GIZ, as the Accredited Entity, will cooperate with both the Independent Redress Mechanism and the GCF. The GRM will be finalized during the project's inception phase and will be aligned with existing systems of Profonanpe and MIDAGRI. It will also incorporate cultural sensitivity aspects such as being available in the Quechua and will be presented in different forms to ensure that it is accessible to all stakeholders (e.g., online, physical in form of a complain box, etc.). The GRM will also include aspects related to Gender-based Violence (GbV) and Sexual Exploitation Abuse and Harassment (SEAH) in line with the Gender Analysis and Gender Action Plan, ensuring gender and cultural sensitivity and protecting the anonymity of complainants.
290. The table below shows the proposed mitigation measures E&S risks and impacts as shown in Annex 6b).

Table 8. Overview of environmental and social (E&S) risks and impacts, and mitigation measures

ESS	ESS Risks and impacts	Mitigation measures
ESS1	Environmental and social risks primarily pertain to the list of Ecosystem-based Adaptation (EbA) measures and proposed Climate Resilient Value Chains (CRVC). These risks could materialize in the absence of a comprehensive Environmental and Social Safeguards (ESS) review and monitoring process.	- Establishment of an ESS review and monitoring procedure for screening local initiatives in the Puna Facility. - If necessary, particularly for initiatives falling under risk category B, Profonanpe will provide support to beneficiaries in crafting ESS plans for their respective local initiatives.
ESS2	Accidents during construction and land work or contagion could occur, due to the requirement of extensive manual labour. Potentially occupational and safety risks during Local initiative implementation	- Implementation of health and safety measures based on a safety protocol.
ESS3	Risk of GHG emissions and climate change mitigation may arise if the proposed interventions for CRVC involve mechanization measures to boost productivity or for processing, especially when targeting specific overseas markets as part of the market access support. Additionally, the mismanagement of water resources in the watershed, potentially resulting from improperly designed irrigation and drainage systems, can have adverse consequences.	- Adequate following of eligible CRVCs as part of the selection criteria of the Puna Facility. - Promotion access to national markets.
ESS4	Mismanagement of water resources in the watershed through improperly designed of irrigation and drainage system could affect downstream users and have negative impacts on water, energy, waste and pollution prevention	- Adequate site-specific climate diagnostics and territorial intervention plan. - Offer of trainings regarding environmental risks.
ESS5	Potential sexual exploitation, abuse, harassment (SEAH) towards women's empowerment actions.	- Implement Gender Action Plan and adequate gender responsive GRM considering SEAH. - Strengthen women's empowerment through gender quota as part of the Puna Facility evaluation.
ESS6	Potential sexual exploitation, abuse, harassment (SEAH) towards women's empowerment actions.	- Develop free, prior and informed consent (FPIC) - Confirm ownership through mapping of the community, reviewing individual property and validating with community and interested persons. - Adequate review and eligibility documentation by Profonanpe. - Develop a Risk Management Plan by a risk manager.
ESS7	Risk of unexpected ecosystem degradation due to oil and gas leakage from vehicle, attendance overrun, overgrazing, introduction of alien species among others could occur. Potentially risks of increased frequentation, degradation of existing system/balance and pests and diseases.	- Please refer to measure 4
ESS8	Insufficient engagement and consultation of Indigenous Peoples and Local Communities could generate conflicts during project implementation.	- Implement Indigenous Peoples and Local Communities Engagement Plan (IPLCEP, Annex 6c).
ESS9	Potential risk of unexpected finding of some archaeological rest and artefact, possible unintended negative impacts on cultural heritage.	- Ensure by Profonanpe and MIDAGRI during implementation of Local initiative and public investment projects that unexpected findings are reported to MINCUL and local stakeholders.
ESS10	Risk of high expectations amongst Local initiative beneficiaries could occur due to improper development and organization of stakeholder engagement. Potential risk of insufficient stakeholder engagement and rise of high expectations amongst Local initiative beneficiaries.	- Adequately following and implementation by all EEs of the Stakeholder Engagement Plan (as part of Annex 7) developed for the proposed project.

G.1.2 Indigenous Peoples and Local Communities Engagement Plan (IPLCEP)

291. The project area is targeting mainly the Southern High Andes of Peru (SHAP Region), concentrating on four regions in Southern Peru: Cusco, Apurímac, Puno and Arequipa. The population of the local communities in the SHAP area are mostly part of the country's Quechua indigenous population. Nationally, around 5.1 million people consider themselves as part of the Quechua people. Most of the Quechua population lives organized politically and administratively in peasant communities. They are constituted by families linked by ancestral, social, economic, and cultural features. Almost the entirety of the population in the project area belong to the Quechua Indigenous Peoples Group. There is considerable uniformity in the cultural-socio-economic conditions within this area. As a general profile, the Quechua speaking people of the high Andes are subsistence level agriculturalists, living in poverty or extreme poverty reliant on livestock grazing and limited crop production. While most children attend primary school only a minority continue and complete their secondary education instead withdrawing to support the family's subsistence activities. The Peruvian state has made an initial identification of the Indigenous Peoples Groups (peasant and native communities) and has formally recognised 55 Indigenous or Original Peoples in the whole Peruvian territory. However, in the target area of the Resilient Puna project there is only one Indigenous Peoples Group present: the Quechua.
292. Based on the ESIA and ESMP (please see Annex 6a and b) and the stakeholders' consultations (please see Annex 7) it was shown that the main aspects for risks and impacts related to ESS 7 are: (i) conflicts may arise concerning the ownership and usage rights of the project site, along with the distribution of resulting benefits and employment opportunities., (ii) conflicts with downstream water user over interruption or alteration of water flows and water quality during and after the restoration works, (iii) opposition of landowners, land users and residents over restrictions or lost access to productive, grasslands and other natural resources collection areas and (iv) lack or poor gender consideration and no opportunities proposed to women.
293. The table below shows the proposed mitigation measures addressing ESS7 (for more information see Annex 6c).

Table 9. Proposed mitigation measures addressing ESS7

Risks and potentials concerning Indigenous peoples and local communities	Description of the Measure/Action:
Appropriately consider the local knowledge, especially in the context of ethnic culture diversity in Peru amongst project staff including ensuring the use of local languages (i.e., Quechua)	- Project capacity to implement IPLCEP - Hiring of staff to implement IPLCEP, including amongst others GIZ- ESS specialist with knowledge of Indigenous peoples and Local Communities and Promoters who can speak Quechua. - Hiring of Quechua interpreters to ensure that all communities understand information provided by the project.
Insufficient consideration and/or representation of Indigenous Peoples and local communities in project activities.	- Project guidelines and strategies (e.g., communication and engagement) will be design considering Indigenous Peoples and Local Communities. - Representatives of Indigenous Peoples and local communities will be given opportunity to participate in project activities (e.g., related to community monitoring, recover, innovate and scale up ancestral knowledge and practices).
Insufficient engagement and consultation of Indigenous Peoples and Local communities could generate conflicts during project implementation.	- Set up a consultation register. - Conduct meaningful consultation (e.g. decision-making body) and ensure transparent processes throughout project implementation to ensure a free, prior and informed consent before engaging in project activities. - Provide adequate information as well as engage with and seek the support of those who could be affected by conflicts throughout the whole Local initiative cycle ensuring active, free, effective, meaningful, and informed participation of individuals and groups throughout these processes. - Ensure equal rights for all different Indigenous Peoples and local communities, women, and men, while acknowledging differences between women and men and taking specific measures aimed at accelerating equality when necessary.
Insufficient participation and/or inclusion of Indigenous Peoples and Local communities in regional platforms, territorial planning, dissemination and monitoring process, and technical capacity building.	- Verify and confirm that there are quotas for the participation of Indigenous Peoples and Local communities and increase the number within the regulations of regional platforms. Including awareness raising about the significance of community involvement. - Prepare Indigenous Peoples and Local Communities for the participation in regional platforms. - Social safeguards specialist to ensure that communication and engagement strategy considers cultural responsiveness and social inclusion. Including on the strategies to project staff and Promoters and CSOs. - The project will pay special attention to culturally and gender sensitivity, considering indigenous groups and gender perspectives in the policy development process by engaging indigenous representatives.

	- Active engagement of traditional organizations' leaders, in capacity building, and development of territorial plans and other project supported measures.
Culturally inappropriate access and rights to agricultural, bofedales and grassland.	- in cases where the project suggests the restoration or protection of agricultural land, bofedales, grasslands, or other natural resources utilized by local community members (comuneros), careful consideration will be given to discussing and offering alternative access and livelihood options. - Efforts will be made to identify similar areas where poorer and landless households can access and depend on for their livelihoods. The project will employ participatory approaches, and project personnel, promoters, and Civil Society Organizations (CSOs) will receive training on gender equality and social inclusion.
Irrespective customary use of lands, territories and resources and rights related to cultural and spiritual heritage and values, traditional knowledge, resource management systems and practices, and livelihoods.	- Respect Indigenous People and Local Communities' spiritual/holy land sites and practices. - Respect customary land tenure rights and institutions.
Possible complaints from Indigenous Peoples and Local Communities, especially women, the youth and the elderly are not being considered.	- Respective trainings for GRM focal points. - GRM will be communicated to all relevant stakeholders, including the Indigenous Peoples and Local Communities as part of project engagement. - GRM developed considering culturally appropriate ways of handling the concerns of Indigenous Peoples and local communities in the project areas. - Use Indigenous People and Local Communities' interpreter for all Indigenous Peoples and Local Communities with low Spanish language skills. - Processing of complaints and documentation must include specific timeframe and commitments to resolve issues. - All complaint will be recorded (letter, email, record of conversation) and confidentially and safely stored together, electronically or in hard copy.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

294. The project's Gender Analysis (GA, see Annex 8a) and Gender Action Plan (GAP, see Annex 8b) align with international principles and commitments such as CEDAW, Beijing, Belém do Pará, and the SDGs, as well as national policies like the Gender and Climate Change Action Plan-PACC, National Policy on Gender Equality. These policies support gender equality and non-discrimination of women in various aspects of life, including social, political, cultural, climatic, and economic rights. Recognizing that the impacts of global warming affect men and women differently, the project aims to implement adaptation and mitigation measures that consider these distinct needs. The project's primary approach, Ecosystem-based Adaptation (EbA), emphasizes working with nature to address climate impacts while simultaneously reducing vulnerability and generating social and economic benefits. It's a collaborative process involving communities, public, and private organizations, aimed at creating sustainable opportunities for all parties at the landscape level.
295. Specifically, the gender analysis has identified the main factors that are hindering women's progress and resilience in the face of climate change. Overburdened care and unpaid domestic work; low participation in decision-making; lack of income; vulnerability of their housing and household environment (e.g., health, food, gender-based violence, poor housing) are the main gendered barriers to more equitable climate adaptation.
296. As mentioned in Section D.3, women in the SHAP area face more barriers than men that hinder the achievement of gender equity in climate change adaptation measures. Women face unequal access to resources (use, control) and receive lower income. They also have limited access to microfinance; less opportunities and time for education, training, and technical assistance/capacity to develop and implement nature-based solutions and climate-resilient agri-businesses. It is important not only to work to close these gender gaps but also to address the weaknesses in institutions and regulation policies, to EbA measures and Climate Resilient Value Chains, while also limited gender-sensitive information (data, systematization of experiences, lessons learnt, behavioural changes). For this, it is essential that the actions undertaken within the project do not result in an increased workload for women who already have a double burden between productive and reproductive work (domestic and care work). For this, the project must recognize women's knowledge, needs and expectations to generate effective responses in terms of building social and technical capacities, as well as opportunities for their economic autonomy, participation, and representation in decision-making.
297. Based on the identified gender gaps and recommendations, a Gender Action Plan (GAP) has been developed for the project. Table 10 includes some of the measures and targets that are integrated in the project's GAP (for more detailed information, please see Annex 8b):

Table 10. Overview of gender risks and actions integrated into the design

Risks and potentials concerning gender equality	Description of the Measure/Action:
Risk of not collecting disaggregated data	- Gender-sensitive M&E system to track female and male beneficiaries and ensure gender-sensitive project management.
	- Ensure increasing the participation and capacity building of women and youth in workshops offered by the project.
	- Establish cooperation between gender advisor and communication team to ensure that all project communication materials and guidelines include a gender equality and social inclusion (GESI) perspective. Including, when necessary, training on GESI for the communications team.
Beneficiaries not using the grievance mechanism for fear of harassment.	- Gender based Violence (GbV) and Sexual Exploitation Abuse and Harassment (SEAH) aspects are include in the grievance mechanism (GRM).
	- Ensure that identified beneficiaries (local communities, producers' associations, cooperatives, micro and small enterprises, community enterprises and producers' organizations) include female members.
	- Promoting a gender-sensitive communal planning process for investing in EbA measures and Climate Resilient Value Chains.
Designing EbA measures without considering women's needs, responsibilities and domestic burden	- Designed and implement of innovative measures for ecosystem-based adaptation with a gender-sensitive approach
Insufficient participation of women in technical trainings provided through the project	- Ensure women's access to technical assistance including financial education, access to markets, implementation of Local initiatives.
Women avoid disclosing ancestral knowledge	- Systematization of best practices of women's knowledge for replication. The project will promote the development of publications together with universities/public institutions, etc.
	- Dissemination of best practices of women's knowledge for replication. The project will promote the development of publications together with universities/public institutions, etc.
	- Ensure the participation of women in the observation, registration, and dissemination of information in the environmental community monitoring committees.
	- Gender criteria considered as part of the Local initiative evaluation.
	- Train water utility (EPS) officers of PES/MERESE to promote the inclusion of gender aspects into the MERESE's processes (agreements, governance platforms, project development, monitoring).
	- Support the design of gender-sensitive financial products to promote EbA and Climate Resilient Value Chains. The design will include a diagnostics identifying gender gaps and needs.
Planning processes do not include gender perspective	- Ensure gender perspective during the planning processes training public officials with an institutionalized training program led by the Peruvian ministry(s).
Women have difficulties in participation in planning processes	- Promote the participation of women in decision making planning processes
	- Strengthening the disaggregation of the M&E system with gender EbA approach at MIDAGRI.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

G.3.1 Financial Management

298. GIZ head office in Germany (GIZ HQ) will serve as the Accredited Entity (AE), administrating project funds and assuming responsibilities of oversight, guidance and quality assurance to the executing entities in accordance with the Accreditation Master Agreement signed between GCF and GIZ. The financial management of the project will follow GIZ's internal rules and regulations. GIZ has bank accounts with Deutsche Bundesbank and Commerzbank. GIZ will not open a specific bank account for the GCF project but will ensure that all funds provided are clearly identifiable and distinguishable from GIZ's other funds by setting up separate cost units exclusively for the funds disbursed by the GCF for each funded activity (ledger accounts). Funds received and expenditures incurred will be booked to the respective cost unit according to generally accepted accounting principles and procedures accepted by the German Government. As a general principle, GIZ disburses funds to recipients in accordance with the progress of the project. Executing Entities receiving GCF proceeds by GIZ (Profonanpe and, Instituto de Montaña) have to report on the proper use of funds and defined progress as a prerequisite for any further disbursement. Independent external auditors will perform annual financial audits of the project in line with International Auditing Standards.

299. Profonanpe will host the public-private Puna Facility and manage the different sub-windows and complementary TA under component 1 and 2. Under GIZ HQ supervision, Profonanpe will ensure that all fiduciary standards of the GCF are followed. The project Steering Committee will provide the necessary orientations to Profonanpe so beneficiaries (e.g., producers' associations, local communities, etc.) and in some cases in partnership with CSOs can apply to the call for proposal and be supported through TA throughout the implementation of Local initiatives. Funds received during, but mainly after the project, from the private sector or other international donors are planned

to be channeled through the Puna Facility to ensure the sustainability. Local initiatives will complement MIDAGRI's investments from Sierra Azul, Agrorural, and Agroideas (co-financing).

G.3.2 Procurement

300. In the case of procurement by GIZ, GIZ will follow its own Procurement Guidelines. GIZ is required to comply with the relevant contracting rules as established in the German Act against Restraints of Competition (GWB), the German Regulation on the Award of Public Contracts (VgV) and, if applicable, the Contracting Rules for the Award of Public Service Contracts (VOB, VOL/B and UVgO) when procuring services, construction work and supplies.
301. When awarding contracts for supplies and services (including consultancy services) to be financed in full or in part from the financing contract, the external Executing Entities will observe their own national regulation for public procurement and will in any case comply with the provisions mentioned in the Procurement Guidelines for projects funded by GCF/GIZ.
302. The Procurement Guidelines shall not contradict the applicable national procurement law and/or regulations for public procurement, which apply in the Executing Entities country. In principle, the regulations of the Executing Entities country are to be observed; the procurement procedures mentioned in the Guidelines are obligatory minimum standards (an overview of these minimum standards is available under Annex 10b). While implementing a project with public funds the Executing Entities should take reasonable account of economic efficiency as well as ecological and social aspects.
303. GIZ in its function as AE will have monitoring and oversight responsibility on the procurement activities carried out by the third parties. The monitoring and oversight procedures will follow GIZ's internal rules and regulations to ensure compliance with GIZ procurement guidelines which are contractually agreed and reviewed externally via annual audits.
304. The project's procurement plan is available in Annex 10a.

G.4. Disclosure of funding proposal

- ☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
- ☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☒ Environmental and Social Impact Assessment (ESIA) or
 - ☒ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☒ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☐ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☒ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☒ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☒ Annex 20 First level AML/CFT (KYC) assessment
- ☒ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁹⁶
- ☐ Annex X Other references

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.

⁹⁶ Annex 22 is mandatory for mitigation and cross-cutting projects.

No-objection letter issued by the national designated authority(ies) or focal point(s)

To: The Green Climate Fund ("GCF")

Lima, 15th November 2023

Re: Funding proposal for the GCF by GIZ, regarding Project "Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru"

Dear Madam,

We refer to the project titled Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru as included in the funding proposal submitted by Deutsche Gesellschaft Für Internationale Zusammenarbeit (GIZ) GMBH in Peru to us on 15 September 2023.

The undersigned is the duly authorized representative of Ministry of Economy and Finance, the National Designated Authority of Peru.

Pursuant to the GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

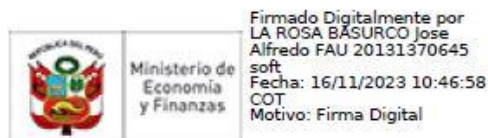
By communicating our no-objection, it is implied that:

- (a) The government of Peru has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Peru;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,



Mr. José Alfredo La Rosa Basurco
General Director, General Directorate of International Economic Affairs, Competition
and Productivity
National Designated Authority
Peru

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Peru, specifically in the departments of Apurímac, Arequipa, Cusco, Lima and Puno.
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, February 1, 2024
Language(s) of disclosure	English, Spanish, and Quechua
Explanation on language	Spanish and Quechua are official languages of Peru. Quechua has been identified as the indigenous language largely used in the project area mainly in Cusco, Apurímac, Puno and Arequipa.
Link to disclosure	English: giz2023-en-annex- 6a-environmental-social-impact-assessment.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5137884-evaluacion-de-impacto-ambiental-y-social-esia Spanish: giz2023-es-anexo-6a-evaluación-impacto-ambiental-social.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5137884-evaluacion-de-impacto-ambiental-y-social-esia Quechua: Ökosysteme in den peruanischen Hochanden an den Klimawandel anpassen - giz.de https://www.gob.pe/institucion/midagri/informes-publicaciones/5137884-evaluacion-de-impacto-ambiental-y-social-esia
Other link(s)	Peru - giz.de
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "ESIA" document (Annex 6a)

Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, February 1, 2024
Language(s) of disclosure	English, Spanish, and Quechua
Explanation on language	Spanish and Quechua are official languages of Peru. Quechua has been identified as the indigenous language largely used in the project area mainly in Cusco, Apurímac, Puno and Arequipa.
Link to disclosure	English: giz2023-en-annex-6b-environmental-social-management-plan.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138370-plan-de-gestion-ambiental-y-social-esmp Spanish: giz2023-es-anexo-6b-plan-gestión-ambiental-social.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138370-plan-de-gestion-ambiental-y-social-esmp Quechua: Ökosysteme in den peruanischen Hochanden an den Klimawandel anpassen - giz.de https://www.gob.pe/institucion/midagri/informes-publicaciones/5138370-plan-de-gestion-ambiental-y-social-esmp
Other link(s)	Peru - giz.de
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the "ESMP" document (Annex 6b).
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	Indigenous Peoples Plan (IPP), Gender Analysis (GA), Gender Action Plan (GAP) and Summary of Consultations and Stakeholder Engagement Plan (SEP): Thursday, February 1, 2024
Language(s) of disclosure	English, Spanish, and Quechua

Explanation on language	Spanish and Quechua are official languages of Peru. Quechua has been identified as the indigenous language largely used in the project area mainly in Cusco, Apurímac, Puno and Arequipa.
Link to disclosure	Indigenous Peoples and Local Communities Engagement Plan (IPLCEP) English: giz2023-en-annex-6c-indigenous-peoples-local-communities-engagement-plan.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138601-plan-de-involucramiento-de-pueblos-indigenas-y-comunidades-locales-iplcep Spanish: giz2023-es-anexo-6c-plan-involucramiento-pueblos-indigenas-comunidades-locales.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138601-plan-de-involucramiento-de-pueblos-indigenas-y-comunidades-locales-iplcep Quechua: Ökosysteme in den peruanischen Hochanden an den Klimawandel anpassen - giz.de https://www.gob.pe/institucion/midagri/informes-publicaciones/5138601-plan-de-involucramiento-de-pueblos-indigenas-y-comunidades-locales-iplcep Gender Analysis (GA) English: giz2023-en-annex-8a-gender-analysis.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138684-analisis-de-genero-ga Spanish: giz2023-es-anexo-8a-analisis-genero.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138684-analisis-de-genero-ga Quechua: Ökosysteme in den peruanischen Hochanden an den Klimawandel anpassen - giz.de https://www.gob.pe/institucion/midagri/informes-publicaciones/5138684-analisis-de-genero-ga

	Gender Action Plan (GAP) English: giz2023-en-annex-8b-gender-action-plan.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138739-plan-de-accion-de-genero-gap Spanish: giz2023-es-anexo-8b-plan-accion-genero.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138739-plan-de-accion-de-genero-gap Quechua: Ökosysteme in den peruanischen Hochanden an den Klimawandel anpassen - giz.de https://www.gob.pe/institucion/midagri/informes-publicaciones/5138739-plan-de-accion-de-genero-gap Summary of Consultations and Stakeholder Engagement Plan (SEP) English: giz2023-en-annex-7-summary-consultations-stakeholder-engagement-plan.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138757-resumen-de-consultas-y-plan-de-involucramiento-de-partes-interesadas Spanish: giz2023-es-anexo-7-resumen-consultas-plan-involucramiento-partes-interesadas.pdf https://www.gob.pe/institucion/midagri/informes-publicaciones/5138757-resumen-de-consultas-y-plan-de-involucramiento-de-partes-interesadas Quechua: Ökosysteme in den peruanischen Hochanden an den Klimawandel anpassen - giz.de https://www.gob.pe/institucion/midagri/informes-publicaciones/5138757-resumen-de-consultas-y-plan-de-involucramiento-de-partes-interesadas
Other link(s)	Peru - giz.de
Remarks	The IPP document is titled “Indigenous Peoples and Local Communities Engagement Plan (IPLCEP)” (Annex 6c).
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Thursday, February 1, 2024

Place	The ESIA/ESMP, the Indigenous Peoples Plan (IPLCEP), Summary of Consultations and Stakeholder Engagement Plan (SEP), the Gender Analysis (GA) and the Gender Action Plan (GAP) -are made available as hard copies at: The main office of MIDAGRI in Lima and regional offices within the project area: MIDAGRI Lima Jirón Cahuide 805, Torre Cahuide, piso 3 y piso 8 Distrito de Jesús María, Provincia Lima, Departamento Lima Zone Unit of Agro Rural Apurímac Jr. Andrés Avelino Cáceres N° 374, Distrito Abancay, Provincia Abancay, Departamento Apurímac Zone Unit of Agro Rural Arequipa Urbanización La Marina C 10, Distrito Cayma, Provincia Arequipa, Departamento Arequipa Zone Unit of Agro Rural Cuzco Av. Micaela Bastidas N° 310-314, Distrito Wanchaq, Provincia Cuzco, Departamento Cuzco Zone Unit of Agro Rural Puno Jr. Moquegua N° 264 Distrito Puno, Provincia Puno, Departamento Puno Zone Unit of Agro Rural Yauyos Calle Manco Cápac s/n, Distrito Yauyos, Provincia Yauyos, Departamento Lima The headquarters of SERNANP in the national protected areas within the project area: Nor Yauyos Cochas Landscape Reserve Headquarters Av. Huancavelica 3113 Urb. Covica el Distrito Tambo, Provincia Huancayo, Departamento Junín Ampay National Sanctuary Headquarters Urb. Santa Martha Mz M Lote 2, Distrito Abancay, Provincia Abancay, Departamento Apurímac Machu Picchu Historic Sanctuary Headquarters Avenida Oswaldo Baca N° 402, Urb. Magisterio 1ra. Etapa, Distrito Cusco, Provincia Cusco, Departamento Cusco +51 84 229297 Salinas y Aguada Blanca National Reserve Headquarters Urb. Independencia Americana B-28, Umacollo, Yanahuara, Arequipa +51 54 257461
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	Cotahuasi Sub-basin Landscape Reserve Headquarters Sede principal: Urb. Pampas de Aymaña F-12, Distrito Cotahuasi, Provincia La Unión, Departamento Arequipa	
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	GIZ office Peru Av. Prolongación Arenales 801, Distrito Miraflores, Provincia Lima, Departamento Lima +51 1 4229067 +51 1 4226188 giz-peru@giz.de	
	Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A	
Date of GCF's Board meeting	Monday, March 4, 2024	

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP226

Proposal name:	Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru
Accredited entity:	Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)
Country:	Peru
Project/programme size:	Medium

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project has a significant impact and sustainable development potential thanks to its combination of ecosystem conservation and sustainable development interventions along key agriculture and animal husbandry value chains vital to highly vulnerable, yet underserved communities in the Peruvian Southern High Andes	The Puna Facility will provide grants and reimbursable grants for investments with the potential for financial returns. GIZ has provided comprehensive justification on the need for grants (especially for unformalized entities with no market/finance access) and reimbursable grants (for entities with established capacities and markets). This should avoid concessionality capture and the crowding out of private investment, while delivering fund reflows to support additional investments
The project features a strong model to address the geographical, financial, knowledge and governance barriers that affect climate-affected value chains in the High Andes. The exit strategy will feature a combination of grants-based finance (Puna Facility) along with payments for ecosystem services and the development of microcredit products leveraging both public and private finance for ecosystem-based adaptation interventions	
The project is strongly country driven and will be executed through Profonanpe, the Ministry of Agricultural Development and Irrigation of Peru and the authority in matter of Natural Parks, which brings in a high level of buy-in and coordination, on-the-ground capacity and experience running similar projects	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVII, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Climate change and the unsustainable management of productive practices are degrading the puna ecosystems (peatlands, wetlands and grasslands) and the services they provide (provision and regulation of water; provision of fodder, food and fibre; nutrient and carbon regulation) in the Southern High Andes of Peru (SHAP). The Andean Indigenous Peoples and local communities (IPLCs) in the SHAP have been characterized throughout history by their capacity to adapt to environmental changes. Even though this traditional knowledge has been a key asset for Andean people when facing climate stress, nowadays the Andean population finds the increasing severity and unpredictability of extreme events, as well as the unpredictability of hydric regimes, quite challenging.

4. In this context, the project aims to enhance ownership and resources of Andean IPLCs to build their resilience to climate change by defining, implementing and monitoring their ecosystem-based adaptation (EbA) priorities to improve the puna ecosystems management and ensure the continuous provision of their services such as water supply that will be captured during the rainy season for their livelihoods during the dry season.

5. Overall, the project is expected to promote the sustainable management of 23,914 ha of puna ecosystems, directly benefiting 60,715 people and indirectly another 2,011,856 people. The mitigation co-benefit is estimated at 407,657 tonnes of carbon dioxide equivalent.

6. The accredited entity (AE) is the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), with the Ministry of Agricultural Development and Irrigation of Peru (MIDAGRI) as the leading partner. Profonanpe, the non-governmental organization Instituto de Montaña (IdM) and GIZ Peru will act as executing entities (EEs) receiving GCF funds, while MIDAGRI and the Peruvian National Service of Natural Protected Areas (SERNANP) will act as EEs that will contribute with financing to achieve the project's objective.

7. The project is submitted as environmental and social safeguards (ESS) category B.

8. It was developed with support from the Project Preparation Facility.

2.2 Component-by-component analysis

9. Overall, the project will support promoting a paradigm shift in the management of puna ecosystems and productive practices to increase the climate resilience of some of the most vulnerable groups in the country which include rural farming and herding communities that depend on these ecosystems for their livelihoods. To achieve this, the project consists of three components: (1) resilient ecosystems and communities, (2) public and private investments for scaling up EbA measures aligned and leveraged, and (3) multilevel territorial governance for the incorporation of EbA measures.

Component 1: Puna ecosystems are restored, conserved and better managed through the implementation of EbA measures complemented by climate-resilient value chains (CRVCs) (total cost: EUR 58,958,571; GCF cost: EUR 26,427,299; co-finance: EUR 32,531,272)

10. By implementing investments on the ground (through component 1), the aims are (1) to maintain or improve the provision of puna ecosystem services for the climate resilience of Andean IPLCs in the SHAP, and (2) to strengthen sustainable and productive value chains that are dependent on and which impact those ecosystems. This component will provide financial and technical support, through a call for a proposal mechanism for EbA and CRVC implementation on the ground.

11. Component 1 is structured in two outputs:

- (a) Output 1.1: Investments for EbA measures and CRVCs are implemented at the local landscape level; and
- (b) Output 1.2: The use of EbA knowledge is recovered and disseminated, and local monitoring committees and observation systems are implemented.

Component 2: Public and private financing for EbA measures and climate-resilient livelihoods are established and are actively used by vulnerable communities in the Puna ecosystem (total cost: EUR 7,172,425; GCF cost: EUR 6,054,159; co-finance: EUR 1,118,266)

12. Through component 2, the project will establish a facility implemented by Profonampe to deliver financing through a call for proposal for IPCL-led local initiatives oriented to increase climate resilience. The Puna Facility has been designed with three financing windows, to allow the flexibility to provide the adequate amount of support and concessionality to different groups of beneficiaries, depending on their organization status, commercial capabilities, and business plans. The second and third windows will provide reimbursable grants, with reimbursements triggered by various profitability and results-based triggers. This component will also align and leverage public and private financing for EbA measures and CRVCs at different and coordinated levels.

13. Component 2 comprises the following output and three activities, each focused on the establishment and development of complementary financial mechanisms:

- (a) Output 2.1. Financial mechanisms for the implementation of EbA measures and improvement of climate-resilient livelihoods in the Puna ecosystem are established;
 - (i) Activity 2.1.1: Establish the Puna Facility for long-term financing of EbA measures and CRVCs;
 - (ii) Activity 2.1.2: Strengthen capacities to develop and implement innovative mechanisms for EbA in High Andean ecosystems; and
 - (iii) Activity 2.1.3: Support the greening of microcredits to promote EbA and CRVCs.

Component 3: EbA and climate resilience are mainstreamed into multilevel landscape governance instruments (total cost: EUR 5,570,475; GCF cost: EUR 3,934,843; co-finance: EUR 1,635,632)

14. Through component 3, the project will promote integrated gender-sensitive landscape planning, governance platforms and policy improvement and coordination, fostering dialogue and improving coordination among stakeholders that intervene in the landscape (local, regional and national governments, rural communities, producer organizations, watershed committees, and MIDAGRI extension services, among others).

15. Component 3 consists of the following output:

- (a) Output 3.1. Multilevel landscape governance is improved through strengthening of national capacity, regulatory frameworks and monitoring and evaluation systems.

Project management costs (total cost: EUR 3,812,892; GCF cost: EUR 1,929,721)

16. This cost will support the management, coordination and implementation of the project activities, in compliance with the contractual obligations that are included or referred to in the legal agreements entered into with the AE. The GCF portion of the project management cost is less than 5 per cent of the grant portion of requested funding from GCF and is compliant with the GCF policy on fees.

Sources (EUR million)		Uses (EUR million) (in parenthesis, GCF)	
GCF (grant)	40.79	1.1 EbA investments	52.69 (21.65)
¹ BMZ/GIZ (grant)	7.91	1.2 EbA knowledge	6.27 (4.77)

¹ Germany's Federal Ministry for Economic Cooperation and Development

MINAGRI (grant + in-kind + loan)	2.41 + 16.88 + 9.39	2.1 EbA financial mechanisms	7.17 (6.05)
SERNANP (in-kind)	0.95	3.1 Landscape governance	5.57 (3.93)
Total	78.32	Monitoring and evaluation + project management cost	5.82 (3.57)

III. Assessment of performance against investment criteria

17. The Secretariat considers the proposal to be strongly aligned with GCF investment criteria.

3.1 Impact potential

Scale: Medium to high

18. The proposal aims to increase the resilience to climate change of 2,072,571 people living in southern Peru, including major cities such as Cusco, Arequipa, Abancay and Cañete, who will experience improvements in ecosystem services, most notably water quality and availability, generated by the improvement of ecosystems that contribute to water regulation in the upper and lower parts of watersheds, protecting against increased risks of floods and droughts as a result of climate change.

19. The project is expected to have 60,715 direct beneficiaries, particularly among Quechua-speaking Indigenous Peoples living in the highly vulnerable Southern High Andes region of Peru, will be directly supported by GCF funding, which will support (1) the strengthening of key value chains such as potatoes, quinoa and camelids, and (2) the implementation of EbA activities based on best practices and ancestral knowledge on an estimated 23,914 ha of puna ecosystems that are highly valuable to local communities, enhancing the provision of ecosystem services.

3.2 Paradigm shift

Scale: Medium to high

20. The project presents a strong theory of change responding to the threats to Puna ecosystems and the communities that depend on them. The theory of change focuses on transforming agricultural and animal husbandry value chains by addressing barriers related to access to finance, access to knowledge, and governance and institutional coordination.

21. The project will expand EbA approaches, so far piloted but not demonstrated at scale, by establishing financial mechanisms to channel public finance via the Puna Facility (with funding windows and instruments tailored to different sets of beneficiaries) and payments for ecosystem service mechanisms. Further, the project will give the capacity to community-level institutions to develop business plans, and the microcredit financial sector to develop tailored financial products for EbA approaches. Parallel development of both public and private finance sources is vital to the exit strategy and scaling-up of the project's results.

22. The project features a sound approach to knowledge management based on capturing ancestral knowledge, community monitoring, and learning lessons to identify and promote best practices in EbA. MIDAGRI and Ministry of the Environment personnel, in charge of national programmes and rural extension services, will be trained on sustainable management and traditional practices and tasked with knowledge dissemination to final beneficiaries.

3.3 Sustainable development potential

Scale: High

23. The programme has highly significant environmental, social and economic co-benefits, and is closely aligned with several of Peru's objectives in relation to the Sustainable Development Goals.
24. Environmental co-benefits are particularly relevant as a result of the project's ecosystem-based approach. The restoration and sustainable management of 23,914 ha will improve ecosystem connectivity and the provision of a broad range of ecosystem services – from biodiversity to water-flow management and soil protection – that are particularly critical to the livelihoods of target communities. The sustainable management of ecosystems will also result in a reduction of greenhouse gas emissions, estimated at 407,000 tonnes of carbon dioxide equivalent over 15 years.
25. Project activities are expected to yield significant social and economic benefits as a consequence of the strengthening of producer associations and value chains, community decision-making with a gender-mainstreaming approach, and the preservation of cultural values, most importantly through the preservation of ancestral landscape practices. The project is expected to contribute to job and income generation, with a target of increased incomes for over 42,000 beneficiaries.

3.4 Needs of the recipient

Scale: High

26. Despite Peru's status as a middle-income country, the project targets predominantly Indigenous Peoples' communities in isolated areas of the Southern High Andes (69 per cent of beneficiaries are Quechua Indigenous Peoples). In addition to severe vulnerability to climate change, particularly droughts and the availability of natural resources in fragile high-altitude ecosystems, the region ranks among the most impoverished in Peru, with a high incidence of poverty, low rates of access to basic services, and well-documented barriers to access to finance or markets.
27. As mentioned as part of the discussion on paradigm shift, the proposal takes on a comprehensive approach to tackling these needs, strengthening value chains, connectedness, governance and use of resources, and ultimately the Puna ecosystems which form the basis of the livelihoods in the region.

3.5 Country ownership

Scale: High

28. The programme is part of Peru's country programme and is well-aligned with both the Peruvian nationally determined contributions on agriculture and water, national adaptation plan, as well as sectoral policies on agriculture and green financing. The programme was developed in close consultation with beneficiaries, including Indigenous Peoples, and with a full range of stakeholders at national, regional and local levels.
29. The project will be implemented mostly through Profonampe, which is an AE to GCF and has ample experience in the operation of similar projects combining funding facilities and the provision of technical assistance, with community-based action in the areas of forestry and agriculture. The project will also benefit from the strong involvement as EEs of SERNANP (in charge of protected areas), IdM (non-profit focused on the conservation of the values of mountain peoples and ecosystems) and MIDAGRI, which will leverage strong on-the-ground capabilities and contribute substantial co-financing.

3.6 Efficiency and effectiveness

Scale: Medium

30. The funding proposal package presents a well-developed economic and financial analysis with clear definition of scope and well-referenced assumptions. It covers the economic and financial performance of the Puna Facility, including repayable, non-repayable and Agroideas window, which takes 47 per cent of the total GCF budget. The remaining 53 per cent of the funding is to be used for technical assistance and capacity-building, and therefore is not taken into account in the economic and financial analysis.
31. The economic analysis uses a social discount rate of 6 per cent, based on World Bank's report (2022) on Peru's climate investment. The results show the net present value of the project as EUR 11.9 million and the economic internal rate of return as 8.2 per cent over a reasonable 17-year time horizon, confirming that the proposed project is economically viable. However, if the discount rate is increased, for example to 8 per cent, which is used in other public sources such as the Ministry of Economy and Finance, the economic net present value is reduced significantly to EUR 0.8 million.
32. Financial adequacy and appropriateness of concessionality is demonstrated in the financial analysis. The AE has conducted a scenario analysis testing financial results with different financial instruments (loan). The financial model shows mixed results: it is still financially viable for the interventions under the repayable windows and the financial internal rate of return is above the weighted average cost of capital. The AE complements the results with a narrative justification for the need of grants. For the rest of the interventions, loans would not make the project viable. As cash shortfall is quite significant during the project implementation period, it would be difficult for beneficiaries to pay back during the project period. Grants would be the most appropriate instrument.
33. In terms of the financial sustainability beyond the project period, the key to the sustainability of the Puna Facility lies in the feasibility of mobilizing resources from the private sector, which is the purpose of subactivity 2.1.1.2. To demonstrate such feasibility, the AE was requested to conduct a cash flow scenario analysis, incorporating business models and estimated profits to be achieved by the end of the project, so that the private sector can be attracted and incentivized to get involved in the proposed initiative. The AE analysed each value chain under a business-as-usual scenario and climate-resilient (with-the-project) scenario, where all climate-resilient scenarios show revenue growth and net cash flows.
34. The repayable grants can also be an instrument for the financial sustainability of the project. The financial model estimates repayment rate as 50 per cent, assuming that different local initiatives may be able to repay in full or in part. The repayment stops in year 6, and after the project completion, the revolving funds may flow back into any of the windows under the Puna Facility, leaving uncertainty on how much funding would be reinvested, their impacts and sustainability beyond the project. The AE responded that Profonanpe will continue windows 2 and 3, under the same terms and conditions for the investment.
35. The cost-effectiveness is assessed to be medium. The project's adaptation cost is EUR 672/direct beneficiary and USD 20/total beneficiary. Cost per direct beneficiary is at the higher range of adaptation portfolio in the Latin America and Caribbean region, as the average cost/direct beneficiary in the region (single country projects only) stands at USD 348. The co-financing ratio is 1:0.92, with the sources entirely from the public sector.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

36. **Project background.** The project aims to align public and private investments and increase access to State programmes with a view to improving the climate resilience of High

Andean ecosystems and community livelihoods. Activities are intended to reduce the degree of losses from climate impacts, improve communities' relationship with ecosystems, increase their participation in the value chain and product transformation, improve their associative processes and market access, and monitor progress on decreasing their vulnerability. The project will finance the implementation of EbA measures and build CRVCs to restore and conserve the puna ecosystems. The project will also improve structural conditions that will enable rural communities in the High Andes to use and manage the Puna ecosystems in a more sustainable and climate-smart way, offering alternatives to reduce the unsustainable practices that are contributing to ecosystem degradation and magnifying climate change impacts. Among the environmental and social (E&S) co-benefits are the maintenance of ecosystem services in the project area, reduced greenhouse gas emissions, improved local economy and mainstreaming gender-responsive approaches in EbA and CRVC measures.

37. **Environmental and social risk category.** The AE assigned the project the moderate E&S risk category, which is considered equivalent of GCF category B. The Secretariat confirms the proposed E&S risk category, considering that the project will largely have positive E&S impacts on the ecosystems and the communities in the SHAP. Nevertheless, investments for EbA and CRVC measures (component 1) may have potential adverse E&S impacts, which will be site-specific, reversible or easily managed through standard mitigation measures. The Secretariat confirms that the assigned risk category is within the accreditation level of the AE.

38. **Safeguards instrument.** The AE has provided an environmental and social impact assessment (ESIA) and an environmental and social management plan (ESMP) with the funding proposal as the safeguards document. The document provides an overview of the national context for the E&S risk assessment, the potential E&S risks likely to be associated with the project activities and a range of EbA measures and CRVC interventions targeted by the project. As the project will mobilize public and private financing through the establishment of the Puna Facility to support local initiatives led by the IPLCs, site-specific ESIAs and ESMPs will be developed during the project's implementation.

39. **Compliance with the GCF ESS standards.** The following paragraphs provide a brief description of the programme's compliance with ESS standards.

40. **ESS 1: Assessment and management of environmental and social risks and impacts.** The ESIA has discussed the E&S baseline of the districts and regions in the SHAP targeted by the project and identified the potential E&S risks and impacts against the proposed EbA measures and CRVC interventions along with the indicative mitigation measures detailed in the project's ESMP. Most of the risks and impacts identified by the AE are of moderate scale and can be readily managed and mitigated through the appropriate procedures and plans as described in the ESMP. It is worth noting that the AE has identified that there may be pre-existing or project-induced conflict developing from the participatory process for engaging beneficiaries and supporting the design of intervention plans where EbA measures and CRVCs are prioritized for future local initiatives. Identified conflict risks are related to the exclusion of disadvantaged and vulnerable groups from the potential beneficiaries, competition for access to project grants, use of limited resources (including for downstream water users) and ownership and land tenure. To mitigate such risks, the AE will develop and implement an appropriate gender and culturally sensitive beneficiary engagement and communication strategy to ensure that the potential beneficiaries are properly identified. In the context of the Puna Facility, where financial instruments will be applied to support the implementation of local initiatives, because the potential E&S risks and impacts of such financing cannot be assessed in detail in advance, the ESIA outlines the general E&S due diligence principles and the E&S risk management procedures to be followed by activities supported under the Puna Facility. The facility-specific E&S due diligence and screening checklist will be developed during the project inception on the basis of the existing environmental, social and governance screening checklist established by the EE.

41. **ESS 2: Labour and working conditions.** The ESIA identifies that the project may involve potential low risks of accidents or contagion during the construction and land work of some EbA measures and CRVC interventions, where extensive manual labour may be involved. As no large-scale and long-lasting interventions are foreseen within the Puna Facility, the potential occupational and safety risks are expected to be low and limited to the phase of local initiative implementation. The AE will require site-specific labour management procedures, including health and safety measures and safety protocol, in each local initiative to ensure the provision of protective equipment, when necessary, following the code of ethics and implementation protocol of the EEs.
42. **ESS 3: Resource efficiency and pollution control.** The ESIA identifies that the project may involve low risks related to greenhouse gas emissions if the proposed CRVC interventions involve mechanization measures to increase productivity or processing, and accessing markets abroad. The AE also identifies a low risk of pollution concerning the mismanagement of water resources in the watershed in the case of incorrectly designed irrigation and drainage systems. The AE will mitigate such risks by following the eligibility criteria to select proper EbA and CRVC interventions, promoting access to national markets, conducting site-specific E&S due diligence, and providing technical assistance at the local landscape level on managing the project's environmental risks.
43. **ESS 4: Community health, safety and security.** The AE identifies that the project may expose local communities to low risks and impacts on their health and safety, including the exposure to potential accidents and accidents during the construction and land work and the potential conflicts over ownership and inappropriate selection or implementation of local interventions and initiatives. The project will require safety protocols specific to each funded local initiative to be developed and implemented to mitigate such risks. Since the project will be locally implemented at the community level and by the community members, with support from local civil society, it is likely that no specific security arrangements will be made for project activities.
44. **ESS 5: Land acquisition and involuntary resettlement.** The project does not include any activities that will require land acquisition through expropriation or other compulsory procedures. The EbA measures and CRVC activities that will be funded by the project will be designed by communities and community entities and the lands to be used for the funded activities will be in the possession of the community or used by the community. Nevertheless, the ESIA identifies the need to conduct due diligence on lands where any activity or civil works may take place to ensure that the land belongs to the community, the community is in agreement on the proposed use of the land, the land is clear of public/private ownership or use and the benefits of the activity or civil works to be conducted are distributed equitably across the community. In terms of the potential risks and impacts on involuntary resettlement, the ESIA has assessed that there is a risk that project activities could create restrictions on land use and/or access to natural resources. The ESIA also notes the need to conduct community consultations to understand the potential losses (and whether they fall disproportionately on some people in the community) and develop consensus on alternative access, use and/or livelihoods opportunities. The ESIA and the ESMP require that landowners' titles be confirmed and that the users of any selected local initiatives site to be funded by the Puna Facility ensure the consent of the entire community through the communal assembly and agreement signed.
45. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** The AE has identified low risks and impacts on biodiversity as the project aims to improve the conservation of natural resources and their climate- resilient use. The creation of improved road access may facilitate greater access to the public and thus accelerate the degradation of sensitive areas; there could be unexpected damage to wetlands due to an incorrect assessment of the situation; there could be overgrazing due to an incorrect choice on the CRVC component; agroforestry and reforestation may attract new pests and diseases into

the targeted area; increased tourist numbers through community ecotourism may impact conservation if incorrectly managed; exotic species could be introduced into natural protected areas; altered drainage patterns in *qochas* or infiltration ponds could affect adjacent lands; and the potential beneficiaries, such as members of communities, associations or cooperatives, may be involved in illegal mining activities. As four national protected areas fall inside of the eligible districts identified by the project, to prevent and mitigate risks on biodiversity and natural resources, the project will closely coordinate with SERNANP as EE. The AE will also coordinate with the chief of each decentralized office in the project's target areas and conduct a site-specific assessment on the compatibility of the proposed measures inside the target areas to avoid introducing prohibited species or risking the existing biodiversity.

46. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** The funding proposal seeks to support, through a grant facility and accompanying technical support, investments led both by Indigenous Peoples and by local communities for resilient ecosystems at the local landscape level and strengthening CRVCs with access to markets. It acknowledges the many Indigenous Peoples within the project area and the differential impacts of the project on these diverse groups. An IPLC engagement plan has been provided. There is potential for benefits to Indigenous Peoples, including to the promotion of their knowledge systems. In line with their roles and functions, the Indigenous Peoples Advisory Group is available to provide advice to the AE and EEs, who are also encouraged to share emerging good practices and success stories with them.

47. **ESS 8: Cultural heritage.** The project aims to restore ancient practices and promote the recovery of ancestral practices in the Puna system. The AE has assessed that the project's presence presents limited risks for cultural heritage sites, especially in the southern part of Peru as the centre of the ancient Inca Empire. To mitigate such risks, the project will request a Certificate of Non-existence of Archaeological Remains where needed. There is also a risk of unexpected findings of archaeological remains and artefacts emerging during land work. In such a case, the project will inform the Ministry of Culture and request it to assess the findings and define appropriate measures to avoid loss of access to cultural heritage. If a cultural heritage site is found in the local initiatives, the AE will ensure that unexpected findings are reported to the Ministry of Culture and local stakeholders and that chance find procedures are followed.

48. **Sexual exploitation, abuse and harassment (SEAH) safeguarding.** The ESIA has identified that the project poses potential gender-based violence and SEAH risks as part of the AE human rights due diligence and gender analysis in the context of project-supported activities and funded local initiatives. To prevent the project from exacerbating SEAH risks, the AE will implement a zero-tolerance policy on SEAH and operate a survivor-centred gender-responsive and culturally sensitive grievance redress mechanism (GRM) to redress SEAH allegations. Upon obtaining their consent, the project will promptly refer the survivor to the relevant gender-based violence services, including legal, psychosocial, medical care, safety and security related support. The AE will also ensure confidential reporting and monitoring of SEAH incidents. The AE is encouraged to incorporate a SEAH risk assessment as part of the E&S due diligence to be conducted for funded local initiatives under the Puna Facility and to ensure that all project stakeholders have access to the dedicated SEAH GRM as well as the GRMs established by the AE and EEs at the institutional level to redress the project-related SEAH allegations.

49. **Implementation arrangements.** The AE has identified the primary roles and responsibilities among the AE, the EEs and implementing partners in detail for the implementation of the ESMP. The AE, in its additional capacity as EE, will lead and provide overall management and technical assistance to the project at the national and subnational level. The AE will also be responsible for ensuring the project's compliance with the E&S safeguards standards, the ESMP, the gender action plan and the IPLC engagement plan. GIZ Peru will act as the technical secretariat within the Project Steering Committee and as the executive coordination within the Project Management Committee as well as the Territorial

Implementation Unit. Profonanpe, in its capacity of one of the project's EEs, will be responsible for the management of the Puna Facility under the supervision of the AE. Profonanpe will set up a Facility Management Unit to ensure compliance and alignment with the strategic guidance established by the Project Steering Committee. Profonanpe will participate in the Project Management Committee and the Territorial Implementation Unit to liaise and coordinate with other project partners to guarantee the integral fulfilment of the expected results of the project.

50. **Stakeholder engagement and information disclosure.** The AE has provided a stakeholder engagement plan with the proposal, which summarizes the consultations conducted during the design and development of the project and identifies key stakeholders and their respective roles and interests in the project, including the relevant ministries, local government agencies, non-government institutions, civil society institutions, representative organizations for Indigenous Peoples, other local organizations (such as cooperatives, productive organizations) and women's organizations. The stakeholder engagement plan outlines the strategy and processes and identifies specific actions to be taken by the AE and the EEs to ensure that stakeholders are continuously engaged throughout the project's life cycle, including the level-local initiatives funded by the project. The project will conduct participatory community monitoring and the scaling-up of best practices, including ancestral knowledge and practices. As the site-specific E&S due diligence will be conducted during the project implementation, the AE should disclose the ESS documentation of all category B subprojects funded by the Puna Facility in both English and the local language (if not English).

51. **Grievance redress mechanism.** The project will establish a project-level GRM during the inception phase that will allow any individual to register a complaint and will be harmonized with the existing GRMs established by the AE and the EEs at the institutional level. The project will also establish a dedicated SEAH-related GRM system to align with the requirements of the GCF Environmental and Social Policy, including to ensure accessibility to the Secretariat's Independent Redress Mechanism. The ESIA outlines the detailed steps to be followed during implementation to ensure the effectiveness of the GRM system to be applied by the project and its stakeholders.

4.2 Gender policy

52. The AE has provided a gender assessment and Gender Action Plan therefore complies with the requirements of the GCF Gender Policy.

53. The gender assessment was conducted through desk review of available data and fieldwork in which consultations were conducted with officials within project locations, while also engaging in discussion with communities with significant numbers of women and men.

54. The gender analysis has identified key factors that are hindering women's progress and resilience in the face of climate change. Although there is a formal institutional framework for gender mainstreaming in public institutions, the mechanisms to operationalize them are fragile. This is characterized by a high rotation of human capital and a type of management that usually lacks resources and capacities for the fulfilment of its functions. Despite these challenges, MINAM and SERNANP have shown a sustained interest in incorporating gender approaches in their actions that should be strengthened and made visible. Rural women are often in charge of collecting water for cooking and cleaning, using the land for livestock grazing and fetching firewood, among others. Women use and interact with natural resources and ecosystems on a daily basis and as a result are also among the first to feel the adverse effects of climate change. Rural women also face several barriers such as lower levels of education, inability to speak a language other than Spanish, low empowerment and leadership experience and the burden of domestic and productive work. Despite women's important roles in small scale agriculture and water harvesting, women's heavy domestic workload, along with social norms and resultant barriers, limit their availability, access to resources and opportunities for education, training,

paid employment, and participation in decision-making spaces. These translate into under-representation in community organizations, in water users' organizations, and in decision-making spaces. Women also have less access to and control of natural resources and decisions about land, water, production and sale of agricultural products.

55. The gender action plan provides actions that address the gender gaps identified through the gender assessment. Actions have corresponding targets, indicators, timeline, budget and the requisite gender expertise. The activities in the Gender Action Plan align with the broader theory of change and the three major components of the project: implementing investments on the ground (through component 1), establishing a facility implemented by Profonampe to deliver financing for IPCL-led local initiatives oriented to increase climate resilience (component 2), and promote integrated gender-responsive landscape planning, governance platforms and policy improvement and coordination, fostering dialogue and improving coordination among stakeholders that intervene in the landscape (component 3). The theory of change has also included co-benefit 5 "Gender-sensitive approaches mainstreamed in EbA and CRVC measures. Specific activities include training water utility officers of the Mechanisms of Remuneration for Ecosystem Services Program (MERESE) to promote the inclusion of gender aspects into the MERESE's processes (agreements, governance platforms, project development, monitoring). Supporting the design of gender-responsive financial products that improve access to finance for women and other traditionally marginalized groups (with a 50 per cent target going to women, women-led micro, small and medium-sized enterprise's and women-led initiatives) and promote EbA and CRVCs. Strengthen gender mainstreaming during the planning processes and training of public officials with an institutionalized training programme led by the Peruvian ministry(ies). Further, the activities support and promote the participation of women in decision-making planning processes through training and requirements for women's representation and gender expertise in decision-making bodies of the project. Capacity-building and communication initiatives will be designed to be accessible for women and address practical barriers to women's participation. Women's access to technical assistance will include financial education, access to markets, implementation of local initiatives and the collection and systematization of best practices for women's engagement and women-led EbA initiatives for sharing and replication.

56. Given the content and scope of documents submitted, the following recommendations are made to the AE:

57. The AE to include baseline data collection activity related to the number of women producers and for women-led initiatives at the inception phase. Such information on intent needs to be included in the Gender Action Plan along with a timeline, budget and responsibilities. Changes to the Gender Action Plan as a result of this action should be communicated to the Secretariat;

58. The Secretariat notes that each IPLC will voluntarily apply to the Puna Facility with tailored technical support to refine their original proposals. Addressing the challenges for women, women-led organizations or groups in voluntarily applying should be given due attention and strategies clearly outlined to ensure sufficient applications will be received from women. (The ability to volunteer might depend on many factors such as awareness, know-how and ability to engage (permissive society or the enabling environment within the home) in such voluntary action.);

59. The AE indicates that pre- and post-surveys will be given to recipients to measure increased capacity. There is a need to include indicators to measure this aspect.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

60. GCF is requested to provide a grant of EUR 40.79 million to improve the Puna ecosystem management through EbA in the Southern High Andes of Peru. BMZ via the AE is co-financing a grant of EUR 7.91. MIDAGRI is providing a total of EUR 28.67 million through grants (EUR 2.41 million), in-kind (EUR 16.88 million) and loan (EUR 9.39 million) and SERNANP is providing an in-kind support of EUR 0.95 million. The total co-financing is EUR 37.53 million.

61. The Puna Facility will be EUR 20.6 million in size, including GCF and co-financing, which will have three sub-windows targeting different beneficiaries with non-repayable and payable resources.

62. Peru has an investment grade credit rating and the government's strong balance sheet may support availing concessional debt from GCF. The AE has requested full grant financing based on the insufficient national resources to meet the climate change adaptation needs.

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

63. GIZ is the AE and co-EE of the project. It has been operating in Peru for over 50 years, over 20 years of which have been on climate change and biodiversity issues. The technical assistance in the sector currently amounts to approximately EUR 60 million.

64. There are four additional co-EEs, namely (i) MIDAGRI, (ii) SERNANP, (iii) Profonanpe and (iv) IdM). MIDAGRI and SERNANP are government entities, Profonanpe is a private non-profit institution of public and social interest, also an AE to GCF, and IdM is a non-government organization. The AE carried out the capacity assessment of the EEs and provided recommendations based on the gaps identified.

4.3.3. Project-specific execution risks (medium risk)

65. Lack of interest from the local stakeholders: successful implementation of some of the project activities (e.g. activities under output 1.1) lies on the participation of the local stakeholders on a largely voluntary basis. If there is lack of interest and motivation from the local participants, the project may be delayed in its implementation. The AE states that the main incentives are learning opportunities and community reciprocation for the participation and the low interest can be mitigated by a communication and engagement strategy which is part of the project activities.

66. The sustainability of Puna Facility will depend on multiple factors:

67. The successful implementation and continuous maintenance of the subprojects financed by the facility, currently the financial analysis assumes six to ninefold net income increase with the project support;

68. The repayment ratio by the beneficiaries, currently estimated at EUR 3.4 million, which accounts for approximately 50 per cent;

69. Profonanpe's additional contribution as indicated in the funding proposal (to be confirmed); and

70. Additional resources mobilized from the private sector as envisaged in Sustainable Financing Strategy (activity 1.2.2.1).

71. Monitoring responsibility: The community monitoring committees will play an important role in monitoring the impacts of projects by having a dialogue with the participating communities and reporting the progress made by the business and subprojects. The reporting obligation, including repayment trigger will be part of these monitoring efforts. Therefore, there could be a potential conflict of interest between the grant recipient groups and monitoring groups. The AE mentioned that one of the EEs (Profonanpe) will have regional monitors who

will validate the information with appropriate means of verification that will be established according to the nature of the individual local initiatives.

72. Supply and demand risk of leveraged financing from microfinance institutions (MFIs): leveraged financing from MFIs is not considered as part of the project impact. Instead, additional resources are anticipated to be necessary for the sustainability of the Puna Facility and local initiatives, therefore this project supports technical assistance for the MFIs to facilitate this process. The lack of interest from MFIs in attending such training and the risk of low demand from MFIs to develop a new financial product for the EbA measures can be mitigated by a providing a perspective of EbA being a new market for MFIs. Additionally, generating EbA-related data aims to create profit-oriented cashflows, contributing to value chains and portfolio diversification or restructuring. Regarding the risk of low demand from loan seekers, given that concessional financing from the Puna Facility is a one-time occurrence, it is anticipated that the new financial products will meet the needs of groups/beneficiaries in the future even when the interest rates are high.

73. Project viability and concessionality: The AE carried out the economic and financial analysis with the EUR 19.2 million of GCF grant portion to fund the establishment and operation of the Puna Facility, which has revenue generation potential. The benefit is calculated based on the financial and economic benefit from local initiatives financed under the Puna Facility and the full financial analysis shows financial internal rates of return before and after GCF support. Without the GCF grant, the analysis results in an internal rate of return (IRR) of 8.4 per cent to 25.8 per cent while with the GCF grant, IRR of 48.1 per cent to 121.8 per cent. Despite those high IRR results, the AE argues that the loan option is not feasible, and grant is still required due to (1) large negative cash flow in an earlier project stage; and (2) high interest rate and margins from MFIs given the credit risk of the target beneficiaries. The economic internal rate of return was positive with the sensitive analysis considering changes in shadow carbon price, greenhouse gas reduction volume and social discount rate.

4.3.4. GCF portfolio concentration risk (low risk)

74. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results area and single proposal is not material.

4.3.5. Compliance risk (medium risk)

75. This project is largely focused on addressing the resilience and sustainability of communities in Peru. As such, much of the work will be undertaken with local communities and institutions. Potential risks of money laundering or countering the financing of terrorism do not seem unusually great or significant. As an AE, GIZ has considerable experience in managing and supervising similar projects and has established policies and procedures in place for managing financial risks and compliance issues. Nevertheless, as capabilities of implementing partners and beneficiaries may vary, it is important that the AE ensures appropriate implementation of its internal controls and policies to address any risks and vulnerabilities as they may arise. Accordingly, based on the project description and the experience and policies and procedures of GIZ, the compliance risk here is rated as medium.

4.3.6. Recommendations

Summary risk assessment	
Overall programme	Medium
Accredited entity/executing entity capability to implement this programme	Medium
Project-specific execution	Medium
GCF portfolio concentration	Low

Compliance	Medium
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4.4 Fiduciary

76. GIZ serves a dual role, acting as both the AE and one of the EEs through its management structure in Peru. In its capacity as the AE, GIZ oversees the project in accordance with the accreditation master agreement (AMA) and administers GCF funds on behalf of the Secretariat, providing guidance to EEs (Profonanpe and IdM).

77. GIZ, as an EE, leads the technical assistance efforts at national and subnational levels, ensuring coordination, compliance with safeguards, and mainstreaming climate resilience and gender aspects.

78. In order to implement the project, GIZ will establish legal arrangements with MIDAGRI, SERNANP, Profonanpe and IdM to facilitate project execution.

79. Profonanpe, an EE, manages the Puna Facility with a budget of EUR 19.175 million, implementing EbA measures and CRVCs. IdM will act as an EE by contributing and scaling up its experience in the implementation of EbA measures in the Nor Yauyos Cochas Landscape Reserve to the other regions of the project. It will use participatory tools for participatory design, implementation and monitoring of EbA measures, which contribute to the ownership and sustainability of the local initiatives supported by the project. Lastly, MIDAGRI and SERNANP act as EEs with their own funds, coordinating activities and contributing to the project's objectives.

80. Funds from GCF will be transferred to GIZ as the AE, who will then transfer funds to Profonanpe, GIZ Peru and IdM (in their respective role as EEs).

81. The financial management of the project will follow GIZ internal rules and regulations. GIZ will not open a specific bank account for the GCF project but will ensure that all funds provided are clearly identifiable and distinguishable from other funds of the AE by setting up separate cost units exclusively for the funds disbursed by GCF for each funded activity (ledger accounts).

82. As a general principle, GIZ disburses funds to recipients in accordance with the progress of the project. EEs receiving GCF proceeds from GIZ (Profonanpe and IdM) must report on the proper use of funds and defined progress as a prerequisite for any further disbursement. Independent external auditors will perform annual financial audits of the project in line with international auditing standards.

4.5 Results monitoring and reporting

83. This is an adaptation project that contributes to GCF adaptation results areas 1, 2 and 4. In addition, it has a strong mitigation co-benefit. The AE provided a strong monitoring and evaluation approach as part of the proposal, which was further refined during the review process. The project is expected to reach 60,715 direct beneficiaries (0.2 per cent of the national population) and 2,011,856 indirect beneficiaries (6.8 per cent of the national population). The AE has provided a methodology for estimation of beneficiaries across different activities.

84. The theory of change is well developed and provides a logical link between the activities, outputs and outcomes under the project. It also lists five co-benefits likely to be generated under the project, which are adequately categorized. The assumptions articulated are relevant and realistic.

85. The logical framework provides a strong set of indicators that can be monitored under the project linked with the guidance contained in the Integrated Results Management

Framework. This includes outcome-level project indicators, listed in section E5 of the funding proposal. The targets provided are clearly explained, and any assumptions explicitly identified.

4.6 Legal assessment

86. The AMA was signed with the AE on 15 November 2017, and became effective on 18 January 2019. The first term of accreditation of the AE lapsed on 17 January 2024. The AE was re-accredited by the Board pursuant to decision B.37/18. The first amendment and restatement agreement amending and restating the AMA was signed on 3 December 2023 and became effective on 18 January 2024.

87. The AE has provided a certificate confirming that it has obtained all internal approvals, and it has the capacity and authority to implement the project.

88. The proposed project will be implemented in the Republic of Peru. The Secretariat provided a draft agreement on privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Republic of Peru on 9 September 2015.

89. The Heads of the Independent Redress Mechanism and Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF are made only after GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

4.7 List of proposed conditions (including legal)

90. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

91. Signature of the funded activity agreement in a form and substance satisfactory to the Secretariat within 180 days of the date of Board approval; and

92. Completion of the legal due diligence to the satisfaction of the Secretariat.

Independent Technical Advisory Panel's assessment of FP226

Proposal name:	Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru
Accredited entity:	Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)
Country:	Peru
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1. This funding proposal¹ for a medium-sized adaptation project in environmental and social safeguards category B is submitted by the Government of Peru through the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) as accredited entity (AE).
2. The five executing entities (EEs) are GIZ, the Ministry of Agricultural Development and Irrigation (MIDAGRI), the National Service of Natural Areas Protected by the State (SERNANP), the Peruvian Trust Fund for National Parks and Protected Areas (Profonanpe) and Asociación Instituto Andino de Montaña (IdM). MIDAGRI and SERNANP are Government of Peru entities, Profonanpe is a private non-profit institution of public and social interest and IdM is a non-governmental organization.
3. The total cost of the funding proposal is EUR 78.32 million, of which EUR 40.79 million is GCF finance and EUR 37.53 million is co-finance from the German Federal Ministry for Economic Cooperation and Development and from the Government of Peru through MIDAGRI and SERNANP.
4. The aim of the Resilient Puna project is to increase the resilience of communities, ecosystems and livelihoods in the southern High Andes, which are negatively affected by climate change, through implementation of ecosystem-based adaptation (EbA) measures, complemented by climate-resilient value chains, the establishment of public and private financing for investing in these measures, and support from multilevel landscape governance instruments. The project supports Quechua-speaking and other farming communities to revive ancestral knowledge on sustainable land and water management, to enhance productivity even in the face of climate change, while simultaneously maintaining crucial hydrological services for downstream water users and important stocks for global carbon regulation.
5. The project is structured into three main components, with the outcomes as shown in the table below.

Component	Outcome	GCF funding
1. Resilient ecosystems and communities	Puna ecosystems are restored, conserved and better managed through the implementation of ecosystem-based adaptation (EbA) measures complemented by climate-resilient value chains	EUR 26.42 million
2. Public and private investments for scaling up EbA measures	Public and private financing for EbA measures and climate-resilient livelihoods are accessible and being actively used by vulnerable communities in the Puna ecosystem	EUR 6.05 million

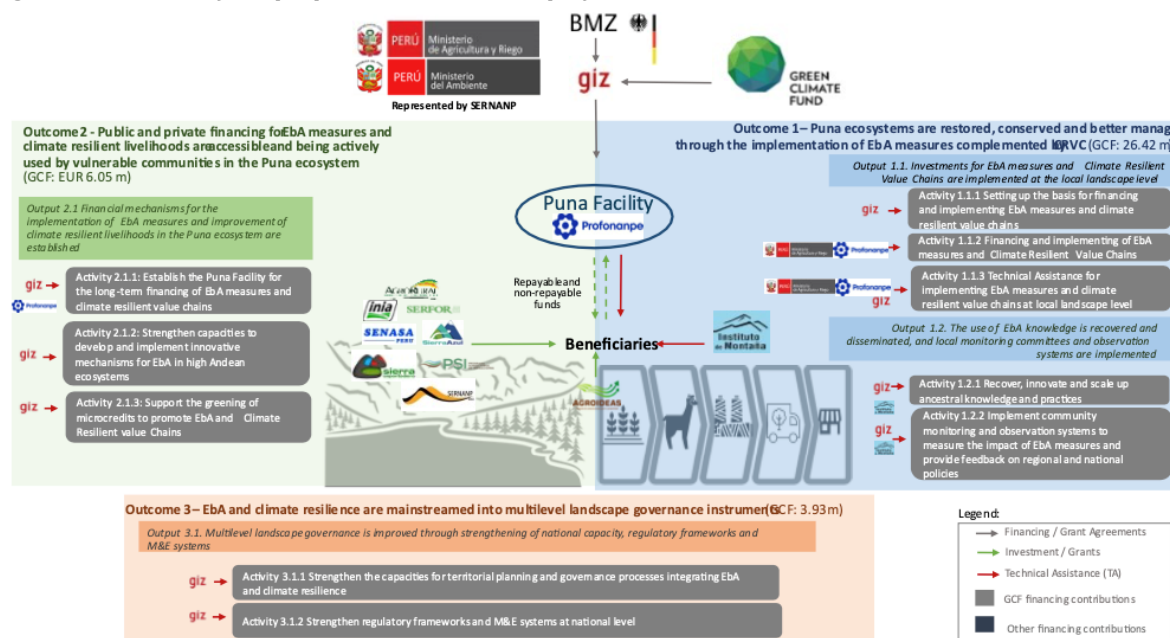
¹ Funding proposal package for "Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru", received from GIZ on 15 December 2023.



3. Multilevel territorial governance for the incorporation of EbA measures	EbA and climate resilience are mainstreamed into multilevel landscape governance instruments	EUR 3.93 million
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6. The project includes the establishment and operation of the Puna Facility, implemented by Profonampe, to deliver financing – through a call for proposals for local initiatives, led by indigenous peoples and local communities (IPLCs), that aim to increase climate resilience. The Puna Facility will fund EbA measures and climate resilient value chains, and will have three sub-windows: (i) Non-Repayable Grant Sub-Window: for IPLCs with various forms of organization (local communities, producers' associations and cooperatives in the process of formalization), applying with the support of local CSOs trusted by communities; (ii) Results-based Repayable Grant Sub-Window: for micro and small enterprises, community enterprises and established cooperatives; and (iii) Agroideas Matchmaking Sub-Window: for formalized producers' organizations with commercial capacities and clear target markets, that are eligible for the MIDAGRI support programme Agroideas².

Figure 1. Structure of the proposed Resilient Puna project



7. The outputs and activities leading to these outcomes are as shown in figure 1,³ which also indicates the EE or EEs responsible for executing each activity.

1.1 Impact potential

Scale: High

8. The project targets three adaptation results areas, enhancing the resilience of (i) most vulnerable people and communities; (ii) health and well-being, and food and water security; and (iii) ecosystems and ecosystem services.

² The Agroideas scheme promotes cooperatives, strengthening of business management, and adoption of sustainable agricultural technologies, through non-reimbursable grants covering 60-80% of the investment required, with a cap of ~EUR 315.000. The Puna Facility Agroideas Matchmaking Sub-Window will provide a complementary repayable grant to advanced businesses in climate resilient value chains.

³ Source: Figure 10 in the funding proposal.

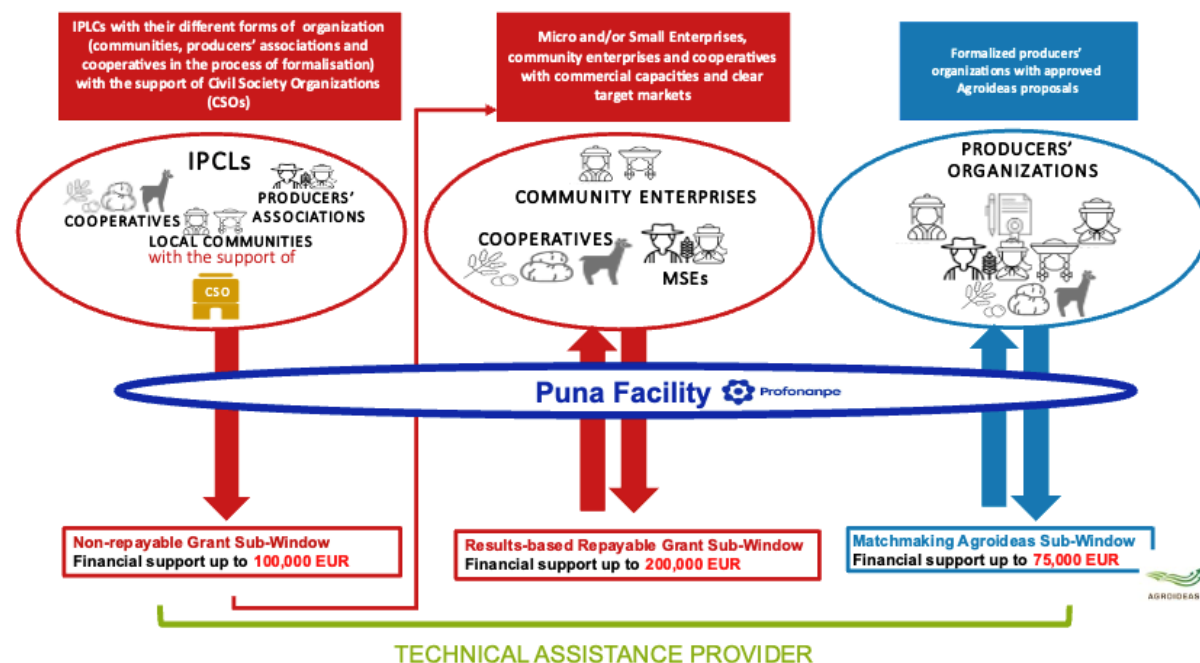
9. Climate change mitigation through avoided emissions of greenhouse gases (GHGs) as a result of project interventions, estimated at 407,657 tonnes of carbon dioxide equivalent (t CO₂ eq) over 15 years, is considered as a co-benefit of the project, and is discussed under section 1.3 below.
10. The funding proposal outlines the major climate change related hazards characterizing the southern High Andes of Peru. Increased temperatures have contributed to glacier melting, while changing rainfall patterns, including longer periods of drought, are affecting water availability in an area where 80 per cent of the population is heavily reliant on rain-fed agriculture. Ongoing climate change is combined with unsustainable productive practices that are degrading the Puna ecosystems (peatlands, wetlands or *bofedales*, and grasslands) and the services they provide (provision and regulation of water; provision of fodder, food and fibre; nutrient and carbon regulation).
11. The project will work in 91 districts in the regions of Puno, Cusco, Arequipa, Apurimac and Lima, covering 5,314,607 ha. These districts comprise high-altitude Andean areas (over 3,500 m above sea-level), or are in a buffer zone above 2,800 m, where vulnerable communities practice crop and livestock farming in Puna ecosystems with high or very high risks of droughts and other climate hazards, and including degraded lands and areas that have suffered melting of glaciers. Of the 91 districts, 58 will be eligible for the Puna Facility (covering 4,116,475 ha) and capacity-building activities, while 33 districts will benefit only from capacity-building activities. Four natural protected areas managed by SERNANP fall inside eligible districts: Nor Yauyos-Cochas Landscape Reserve in the Lima region; Cotahuasi Subbasin Landscape Reserve and Salinas y Aguada Blanca National Reserve in the Arequipa region; and Ampay National Sanctuary in the Apurimac region.
12. The project aims to directly benefit 60,715 people (of whom 30,088 will be female). This number is made up of:
 - (a) Individuals who implement local initiatives financed and supported by the Puna Facility, multiplied by the average number of members of the groups/legal entities supported and the average number of members per household in the region, representing 31,750 people;
 - (b) Individuals who will be supported by MIDAGRI co-finance programmes, multiplied by the average number of members per household in the region, representing 26,315 people;
 - (c) Individuals who will receive technical assistance from the GCF-funded intervention (e.g. training, support to develop site-specific climate diagnostics and preparation of participatory intervention plans), representing 1,740 people; and
 - (d) Public officials who will receive capacity-building for adjusting regulations and norms, representing 910 people.
13. The project will also indirectly benefit 2,011,856 people (of whom 1,005,928 are female) who will benefit from water generated by the improvement of ecosystems that contribute to water regulation in the upper and lower parts of watersheds, including residents of major downstream cities such as Cusco, Arequipa, Abancay and Cañete.
14. Site-level activities will engage local farming communities, including Indigenous Quechua Peoples, farm households, producer associations and agricultural cooperatives, micro, small and medium-sized enterprises, and local government. Adaptation benefits will accrue to vulnerable communities and downstream water users, through EbA and climate-resilient value chain interventions in component 1 of the project, to be sustainably financed and embedded in governance regimes through interventions in components 2 and 3 respectively. The on-the-

ground interventions will take place across 23,914 ha Puna ecosystems, contributing to the results targeted in the updated Strategic Plan for the GCF 2024–2027.⁴

15. **Ecosystem-based adaptation.** The funding proposal highlights seven key EbA measures and shows how each reduces vulnerability to specific climate hazards. The measures are conservation and restoration of *bofedales* (wetland ecosystems), improvement of family *qochas* (natural reservoirs), agroforestry, forest restoration with native species, restoration of *andenes* (agricultural terraces), integrated soil fertility management, infiltration ditches, contour farming, conservation agriculture and sustainable grassland management. Enhancing the likely impact of these measures in the landscape is the excellent resource in the feasibility study annexed to the funding proposal,⁵ outlining restoration techniques, costs and goals in great detail. The independent Technical Advisory Panel (iTAP) suggests that this could usefully be made into a manual for project implementers and producers, translated into Spanish and Quechua.

16. **Climate-resilient value chains.** Support to climate-resilient value chains includes a comprehensive approach to enhance the productivity and climate resilience of production systems (supported by the EbA measures), as well as technical assistance for value addition. Target value chains are alpaca and vicuña fibres, potato and Andean grains, with complementary chains including traditional Andean crops,⁶ breeding of small animals (guinea pigs), community-based tourism and traditional handicrafts. Technical assistance is provided to access and implement Puna Facility funded local initiatives (both EbA and climate-resilient value chains, business development and access to finance). The component also involves facilitating access to MIDAGRI technical assistance and the promotion of market access for new or expanded products.

Figure 2: Structure of the proposed Puna Facility



⁴ The updated Strategic Plan for the GCF 2024–2027 includes the targeted result “Ecosystems: Support for developing countries that results in 120 to 190 million hectares of terrestrial and marine areas conserved, restored or brought under sustainable management”.

⁵ See funding proposal, annex 2a, section 9.2, pp.184–226.

⁶ Traditional crops include Kiwicha (*Amaranthus caudatus*), Cañihua (*Chenopodium pallidicaule*), Tarwi, Oca (*Oxalis tuberosa*) and Mashua.

17. The iTAP assesses the impact potential of the project as high.

1.2 Paradigm shift potential

Scale: Medium to high

18. The proposed project is designed to bring about a paradigm shift away from the current situation of continuing degradation of Puna ecosystems of the southern High Andes region, with declining agricultural productivity and downstream water provision (and increasing GHG emissions), leaving communities increasingly vulnerable to climate change. Under the project, a new approach is developed in which communities, as stewards of the Puna, are supported to manage the ecosystems sustainably, adapt to a changing climate and break the cycle of poverty and degradation. This requires a transformation of practices by farming communities, and a new approach by government in supporting them to become productive and ecologically sustainable, even in the face of a changing climate.

19. **Sustainability of results.** Central to achieving a sustained effect, beyond the project interventions on the ground in component 1, is the establishment of mechanisms for long-term financing of EbA and climate-resilient value chains in component 2, through three approaches: (i) establishing the Puna Facility and working to ensure its long-term viability; (ii) leveraging public funds from existing payments for watershed services (MERESE) schemes;⁷ and (iii) increasing producers' access to Agroideas, Sierra Azul and other public programmes that promote investment in nature-based solutions, business development, product transformation and value addition in target value chains.

20. **Upscaling and replication.** The potential for replicating the project approach in other parts of the High Andes is strong, and is set up by component 3, enhancing the enabling environment for support to climate-resilient agriculture and value chains, and developing the policy and regulatory framework. The creation of an enabling environment relies on transformational planning and programming to scale up EbA, integrating community-responsive technologies, ancestral practices and landscape interventions into local, regional and national planning processes. Also important in this regard are the investments in public institutional capacity and coordination to mainstream EbA through national and regional government. Development of regulatory frameworks is achieved through mainstreaming of EbA and gender considerations within the daily operations of MIDAGRI, MINAM and SERNANP; improving norms, laws, budget and investment guidelines to enhance EbA and climate-resilient approaches in agriculture and land-use planning; and feeding into and strengthening the national monitoring and evaluation systems to allow the incorporation of EbA.

21. **Knowledge and learning.** Knowledge has a very central role in the project strategy, which emphasizes the value of traditional practices (combined with modern scientific knowledge). A key part of the EbA approach is consciously reviving and disseminating ancient practices that are crucial for adapting to climate change, for example (i) storing rainwater in micro water reservoirs (*qochas* in Quechua) for recharging aquifers; (ii) constructing stonework irrigation channels to bring water from high-altitude rivers and *qochas* to irrigate pastures or create watering troughs; and (iii) holding annual festivals for communal clearing of irrigation channels (as well as vicuña shearing and other communal activities). Although much knowledge has been lost through what the funding proposal describes as "low valuation of native cultures, cultural erosion, high rates of rural-urban migration",⁸ knowledge is still present, especially in some Quechua communities of Cusco, Apurimac and Puno, who retain forms of organization

⁷ Established through Law 30215, MERESE provides that water utility companies must make regular payments that must be used for the restoration and conservation of ecosystems that provide them with hydrological services.

⁸ Funding proposal, paragraph 19.

based on the *ayllu* concept,⁹ with strong cultural and spiritual values promoting conservation, equity and social cohesion at the landscape level.

22. The iTAP welcomes the inclusion of targeted activities in output 1.2 to make this knowledge revival possible. Sub-activity 1.2.1.1 builds the capacity of local experts for the transfer of ancestral and innovative knowledge linked to EbA and climate-resilient value chains measures, involving *yachachiqs*, *kamayocs*,¹⁰ *talentos rurales*¹¹ and rural extensionists, in a capacity-building programme and in transferring knowledge to communities and the technical teams of national and subnational institutions. This includes knowledge of ancestral and innovative techniques for small irrigation systems, installation of greenhouses, animal breeding, cultivating pastures and growing crops. Other sub-activities facilitate knowledge-sharing among beneficiary communities with a gender perspective, and disseminate information materials in accessible formats and language.

23. Sub-activity 1.2.1.4, “Facilitating intergenerational dialogue on ancestral practices and innovation in a changing context”, led by IdM and working through local civil society organizations, is an important intervention. It is surprising, however, that no other project activities focus on youth, or target them for training on climate-resilient value chains, to counter the trends of out-migration and erosion of knowledge. A full revival of traditional practices will not be possible without more systematic engagement of the next generation, and the iTAP recommends that youth who are present in the target regions, while not necessarily being active in farming, should be included in the EbA, climate-resilient value chains, entrepreneurial and financial training under sub-activity 1.1.3.1.

24. It is welcomed that the project will collect and analyse information on the impact of EbA measures, to improve evidence of EbA effectiveness and inform future decision-making processes, through sub-activity 1.2.2.3. Studies and research will be conducted using data generated by the community monitoring committees, for example on impact measurement of EbA (carbon sequestration, ecosystem recovery and water regulation), cost-benefit analysis, cost risk reduction in business operations, cost-effectiveness and the type of ecosystems to be investigated (e.g. peatlands, wetlands, grasslands). This is important as part of building the evidence base for the CGF global investment in EbA approaches.

25. Overall, based on the above, the paradigm shift potential of the project is assessed as medium to high.

1.3 Sustainable development potential

Scale: High

26. The proposed project is aligned with the United Nations 2030 Agenda for Sustainable Development, including significant contributions towards Peru’s efforts to meet Sustainable Development Goal (SDG) 1 (No Poverty), SDG 2 (Zero Hunger), SDG 5 (Gender Equality), SDG 6 (Clean Water and Sanitation), SDG 10 (Reduced Inequalities), SDG 13 (Climate Action) and SDG 15 (Life on Land). Specific co-benefits anticipated from project interventions are discussed below.

⁹ The funding proposal (para. 19) states that “In the pre-colonial times Andean worldview was represented by the *ayllu* concept which is understood as the form of social organization in Andean communities where individuals share same reciprocity norms and rituals with respect to humans, animals, rocks, spirits, mountains, lakes, rivers, pastures, crops. The main objective of *ayllu* was to give guide in the positive relationship between communities and their social and natural environment.”

¹⁰ These are Indigenous experts who obtained their practical knowledge on agricultural and livestock practices by ancestral ways and transfer this knowledge to others.

¹¹ These are men and women who live in peasant communities and are known for their capacities and abilities to experiment, adapt, develop and teach good intercultural practices and rural innovations.

27. **Environmental co-benefits.** As a climate change mitigation co-benefit, GHG emissions will be avoided through the conservation, restoration and sustainable management of Puna ecosystems across 23,914 ha. The proposed intervention measures aim to prevent emissions from existing carbon sources through inappropriate management practices in livestock and crop farming, and contribute to carbon sequestration in soil and vegetation, through grassland management, forest restoration, integrated soil fertility management, agroforestry and wetlands restoration and conservation. Anticipated GHG reductions are calculated using the methodology presented in the Intergovernmental Panel on Climate Change *Good Practice Guidance for Land Use, Land-Use Change and Forestry*, applied through the tool EX-ACT (version 9.4) developed by the Food and Agriculture Organization of the United Nations. The overall expected mitigation impact is a reduction in GHG emissions of 407,657 t CO₂ eq in the 15 years analysed (5-year implementation phase and 10-year capitalization phase), measured against a scenario without the project interventions (see funding proposal, annex 22b).
28. In addition to carbon sequestration, the project will enhance biodiversity and ecosystem functioning – protecting and restoring peatlands, wetlands and grasslands, which provide a habitat for a range of fauna, as well as ecosystem services – including soil formation, nutrient recycling, water regulation, water purification, local climate regulation, water retention and storage, groundwater recharge and discharge, recovery and removal of excess nutrients and pollutants, and a peat blanket protecting the underlying soils from erosion.
29. **Socioeconomic co-benefits.** Awareness-raising and capacity-building of local stakeholders, including women, on EbA and value chain strengthening will improve social capital and cohesion. Ancient knowledge and local expertise will be integrated through a participatory approach. Activities to enhance the productivity of farming and grazing should improve household income and nutritional status, and the iTAP welcomes the inclusion of an indicator in the logical framework to measure this co-benefit.¹² New livelihoods based on climate-resilient value chains are expected to bring about local economic development, with new jobs and markets being created. Reducing the barriers to access to finance for expanding production and agri-processing through the Puna Facility will also promote rural prosperity. The extent to which grants and repayable grants (flexible, zero-interest loans) will promote financial literacy and the development of a track record that can enable future access to microfinance remains to be seen (see section 1.6 below).
30. **Gender co-benefits.** Engagement of women in project livelihood opportunities and decision-making will promote women's empowerment. The mainstreaming of a gender approach within all the project activities will promote gender equality and enhance the living conditions and economic opportunities of Andean women. This is achieved through promoting local initiatives (including EbA measures and climate-resilient value chains) with a gender-sensitive approach, integrating gender issues in platforms of knowledge exchange for ancestral practices and increasing women's participation in watershed dialogue decision-making platforms. For this purpose, the project includes a gender action plan with specific activities and an indicator for co-benefit 5 "Gender responsive approaches mainstreamed in EbA and CRVC measures".¹³
31. The iTAP assesses the sustainable development potential of the project as high.

1.4 Needs of the recipient

Scale: High

32. Peru's southern High Andes region is highly vulnerable to climate change, and is expected to experience a continuing trend of increased maximum temperatures, decreased

¹² "Number of producers, of whom 50% are women, have confirmed an improvement in their income from Climate Resilient Value Chains".

¹³ "Number of innovative Local initiatives (including EbA and CRVC) with a gender-sensitive approach implemented".

minimum temperatures (with frost affecting crops), increased heavy rainfall events, longer dry periods between rains, and ongoing glacial retreat and melting. High-mountain peatlands and wetlands are particularly vulnerable to the increasing length of dry spells and drought conditions, and will become more susceptible to fragmentation, damaging their role in water regulation and carbon storage.

33. Climate change impacts are compounding already unsustainable farming practices by agropastoral communities, worsening vulnerability. A decline in the once-widespread traditional knowledge used to sustainably manage the natural resources that support Andean livelihoods, combined with continued lack of access to agricultural extension, finance and markets, are causing declining agricultural productivity and incomes, with young people leaving the mountains to seek work in cities. In some areas, this loss of labour has caused households to shift from crop farming to less labour-intensive cattle farming, accelerating the cycle of degradation of the native grasslands. This is compounded by higher temperatures allowing the introduction of sheep and cattle for extensive grazing in parts of the region where previously only lower-impact South American camelids were roaming.¹⁴ Today, the once biodiverse and highly productive Puna ecosystems are vulnerable to overgrazing, uncontrolled expansion of the agricultural frontier, poor management of grasslands and degradation of peatlands and wetlands (also causing GHG emissions).

34. The target agropastoralist communities in the southern High Andes of Peru region are highly vulnerable to climate change because of their reliance on rain-fed agriculture and extensive grazing. This is compounded by poverty and underdevelopment – 73 per cent of the southern High Andes of Peru districts have a low or very low Human Development Index. Local communities are geographically dispersed, lack basic services such as healthcare, potable water, sanitation education and financing, and rely on wood, dung and peat as energy sources. Poverty is compounded by limited access to existing government agricultural support programmes, and access to finance often only at exorbitant costs via informal lenders known as “drop by drop” or “express loans”, with low penetration by rural microfinance institutions.

35. Downstream communities that rely on the mountain catchments’ water regulatory services to maintain flow for hydropower, domestic and industrial use are also vulnerable to climate change worsened by poor catchment management. Increased temperatures have already contributed to the melting of 51 per cent of the region’s glaciers, while changing rainfall patterns and longer drought periods have affected water availability, with decreases in overall rainfall projected for southern Peru. In addition, huge stocks of carbon stored in the Puna ecosystems could be released into the atmosphere if they continue degrading, with global consequences.

36. Overall, based on the above, the needs of the recipients are rated as high.

1.5 Country ownership

Scale: High

37. The Resilient Puna project has been prioritized by the Ministry of Economy and Finance (MEF), as Peru’s nationally designated authority, for presentation to GCF. It has the full support of the Peruvian Government through MEF, MIDAGRI, as the ministry leading the project, and MINAM, which leads climate change action in Peru.

¹⁴ Research indicates that over 40 per cent of grazing areas across the High Andes are in poor condition and acutely deteriorating, with further 20 per cent at risk of being degraded by 2070. Cooper, D. et al. “Drivers of peatland water table dynamics in the Central Andes, Bolivia and Perú”, *Hydrological Processes* 33 (13) (2019), pp. 1913-1925.

38. **Policy and strategy alignment:** The project is closely aligned with the targets set out in Peru's nationally determined contribution,¹⁵ supporting the fulfilment of four adaptation measures related to agriculture and water: (i) AGRI7, "Natural grassland management to secure feed for livestock and reduce vulnerability to climate change"; (ii) AGRI11, "Management of wild South American camelids (vicuñas) considering effects of climate change"; (iii) AGRI16, "Implementation of business strategies addressing risks and opportunities in the face of climate change"; and (iv) AGU2, "Interventions on planting and water harvesting for agricultural water security in watersheds vulnerable to climate change". The iTAP welcomes the attempt by the AE to quantify the project's potential contribution to these four targets (see table 21 in annex 2a to the funding proposal), and hopes the actual results will be tracked and fed into the national monitoring and evaluation system for reporting to the United Nations Framework Convention on Climate Change secretariat on adaptation.

39. The project will support the implementation of Peru's national adaptation plan through focusing on ecosystems, hydrographic watersheds, restoration and conservation of ecosystems, recovering traditional knowledge and territorial planning, as well as microfinancing for the promotion of EbA practices. The project is also aligned with the National Agrarian Policy 2021–2030 and the Plan for Risk Management and Adaptation to Climate Change in the Agricultural Sector for 2012–2021. The establishment of the Puna Facility and the support to climate-resilient value chains are well aligned with the Road Map for Green Financing, the National Strategy for Green Growth and the objectives of the National Competitiveness and Productivity Plan 2019–2030.

40. **Roles of key entities.** The roles of the key entities in overseeing or implementing the project, and their relevant experience, capacity or mandate to play these roles, are set out below. It is noted that two separate entities of GIZ are involved in the project, playing distinct roles, which will not overlap. The roles are as follows:

- (a) GIZ is a German-based private, limited liability, non-profit company owned by the Federal Republic of Germany; It is serving as AE, providing oversight of the project. GIZ, as an AE to GCF since 2016, has already developed and submitted several funding proposals to GCF, with six approved proposals under implementation (FP117, FP132, FP103, FP059, FP198 and FP200) and a seventh starting soon (FP214). GIZ is one of the largest international providers of capacity development and technical assistance on climate change worldwide;
- (b) GIZ Peru, with its management structure based in Peru, is the main EE, and is responsible, either alone or with other EEs, for the execution of all the project activities except for activity 1.1.2. MIDAGRI, SERNANP, Profonanpe and IdM will also serve as EEs;
- (c) MIDAGRI has extensive experience of channelling loan and grant resources for national and subnational rural development projects from international financial institutions. It is currently implementing several projects from the International Fund for Agricultural Development, the World Bank and the Inter-American Development Bank. The project will enable target communities to access a number of existing MIDAGRI programmes, which form co-finance to the project (see section 1.6 below);
- (d) Profonanpe, a private non-profit organization, is the EE responsible for the implementation of activities 1.1.2 (with MIDAGRI), 1.1.3 (with GIZ Peru) and 2.2.1 (with GIZ Peru). Profonanpe's capacity is indicated by its accreditation as a national direct access entity by GCF since 2015, with two projects under implementation as AE and EE (FP001 and FP193). Profonanpe's experience in designing and managing several

¹⁵ Peru's first nationally determined contribution, as updated in 2020.

environmental and watershed funds in Peru will be key for its role in managing the Puna Facility and leveraging the MERESE funds for the exit strategy;

- (e) SERNANP is the Government entity, attached to MINAM, in charge of ensuring the conservation of national protected areas (NPAs). As the governing entity of the national protected areas system, it has experience on project implementation in the NPAs, and will be in charge of the execution of the project activities within the NPAs and buffer zones; and
- (f) IdM is a non-profit organization that works for the conservation of the natural, cultural and spiritual values of mountain peoples and ecosystems. It has been working in the Andes since 1995 and in Peru is currently implementing programmes in the highlands of Ancash, Piura, Junín and Lima.

41. **Engagement with stakeholders, including civil society.** Several consultations in the target areas have been carried out by GIZ in close coordination with project partners. At the national level, initial meetings were held with national government programmes operating in the project territory, such as MIDAGRI programmes, the Camelid Textiles Innovation Institute and, with SERNANP, the National Institute for Research on Glaciers and Mountain Ecosystems and the National Meteorological and Hydrological Service of Peru. The project development team also engaged with related donor-funded initiatives, to integrate lessons learned and best practices into the project design. In addition, consultations were held with non-governmental organizations that work in the Puna region, such as the Water Competence Centre, the Centre for Studies and Promotion of Development in the South, and the Consortium for the Sustainable Development of the Andean Ecoregion to understand their perceptions of communities' needs in the context of climate change,

42. During the development of the funding proposal, consultations were held with national, regional and local stakeholders in October and November 2022 and May 2023, with presentation and validation of activities through workshops in target project departments, after which feedback was integrated into the project design. In total, the project has consulted 783 people (38 per cent women and 62 per cent men) out of which 65 per cent were members of the potential direct project beneficiaries (e.g. local communities, producers' associations and cooperatives), most of them of the Quechua population (see annex 7 to the funding proposal). Consultations were conducted and coordinated by MIDAGRI and SERNANP, ensuring the participation of communities, producer associations, women, youth and elderly people. During project inception and implementation, the project will follow a stakeholder engagement plan, including "a culturally adequate consultation and communication strategy"¹⁶ and a grievance redress mechanism.

43. In response to a question from the iTAP about why consultations with Quechua-speaking Indigenous Peoples had been limited to individual communities rather than national bodies,¹⁷ the AE explained that the period of initial consultations coincided with a political crisis in Peru, making such consultations impossible, but that "the first consultations with representatives at national level took place in December 2023 with the National Agrarian Confederation and with the National Organization of Andean and Amazonian Indigenous Women of Peru", and indicated that annex 7 could be updated to reflect ongoing consultations. The iTAP also welcomed the news that the AE had already had an initial engagement with the GCF civil society organization network.

44. Country ownership of the project is thus seen to be high.

¹⁶ Funding proposal, paragraph 266.

¹⁷ See written questions from the iTAP (4 January) and responses from GIZ (11 January).

1.6 Efficiency and effectiveness

Scale: Medium

45. Of the project's total proposed budget of EUR 78.32 million, 52 per cent (EUR 40.79 million) is proposed to be a GCF grant, and 48 per cent (EUR 37.53 million) is made up of co-financing to the project, closely woven into the project logic, and broken down as follows:

- (a) MIDAGRI will contribute EUR 28.67 million through grants (EUR 2.41 million), in-kind contributions (EUR 16.88 million) and loans (EUR 9.39 million). These will cover (i) public investments projects (implemented by the Agrarian Productive Development Program, Sierra Azul, the Subsectoral Irrigation Program, the National Institute for Agricultural Innovation and the National Forestry and Wildlife Service; (ii) operational activities (implemented by programmes such as Agroideas, providing matching funding for agribusiness); and (iii) individual credit lines to producer organizations. Supporting target communities to meet the established technical, financial or institutional requirements for these MIDAGRI programmes and funds¹⁸ is central to the project's exit strategy;
- (b) SERNANP will contribute in-kind contributions worth EUR 0.95 million to the implementation of the project in the Arequipa, Apurimac and Lima regions, including staff time contributions in order to coordinate project interventions in NPAs; and
- (c) Co-financing from the German Federal Ministry for Economic Cooperation and Development for the project will be fully integrated into the project activities, with a total grant amount of EUR 7.91 million. EUR 6.26 million will be implemented through GIZ as an EE for eight activities; EUR 1.5 million will be forwarded to Profonanpe for the Puna Facility and EUR 150,000 to IdM.

46. In addition to the co-finance included in the funding proposal, the project aims to mobilize additional resources from international donors and the private sector to support Profonanpe in ensuring the sustainability of the Puna Facility (through sub-activity 2.1.1.2). The Government of Canada has already expressed interest in contributing approximately EUR 6 million, particularly for financing gender and monitoring related activities, and replenishing the Puna Facility to continue financing local initiatives, although this is still to be confirmed.

47. **Level of concessionality.** For the elements of the project unrelated to the Puna Facility, that is, investment in public goods for EbA and strengthening of the enabling environment, the 100 per cent grant from GCF seems justified. The GCF investment in implementing, financing and scaling up EbA represents a cost-effective and efficient integrated landscape approach to obtaining adaptation benefits and mitigation co-benefits. There is a growing body of evidence, including from High Andes interventions, on the cost-effectiveness of EbA approaches to addressing climate change, to which the proposed project will contribute.¹⁹ Other parts of the project involve private goods, for example Puna Facility investments in family crop farming or agribusiness micro, small and medium-sized enterprises, where profits will be made, although probably with relatively small margins (see discussion below on financial rate of return).

48. **Economic rate of return.** The economic internal rate of return (EIRR) of the project as a whole is 8.2 per cent, which is more than two percentage points above the 6 per cent Peru

¹⁸ These include Agroideas, Sierra Azul, Sierra y Selva Exportadora, AgroBanco, Catastrophic Insurance, Rural and Indigenous Women's Fund, Regional Government (ProCompite) and the Ministry of Development and Social Inclusion (social programmes).

¹⁹ For example, a cost-benefit analysis conducted for the Global Mountain EbA project, funded by the International Climate Initiative, on the management of native grasslands in Chanchayllo and Miraflores, Peru, showed a benefit-cost ratio of 2.2:1, with the break-even point at 10–15 years. Podvin, K. *et al.* "Climate Change Adaptation in the Peruvian Andes: implementing no-regret measures in the Nor Yauyos-Cochas Landscape Reserve" in Murti, R. and Buyck, C. (ed.) (2014). *Safe Havens: Protected Areas for Disaster Risk Reduction and Climate Change Adaptation*. Gland, Switzerland.

social discount rate estimated by the World Bank and MEF. The economic net present value, based on the 6 per cent social discount rate, is a positive EUR 11.9 million. A sensitivity analysis shows that the EIRR is positive even with very low carbon prices and with a 30 per cent drop in emission savings volume and income generated by the initial calls for proposals. The EIRR remains above the social discount rate of 6 per cent and the economic net present value is therefore positive if the emission savings volume and economic benefits do not drop more than 10 per cent, even at carbon prices substantially lower than the base case assumption of EUR 60/t.

49. **Financial rate of return and choice of financial instruments.** The financial rates of return on climate-resilient production of the key products of native potatoes and alpacas are estimated by the AE as shown in the table below.

Local initiative type	Financial internal rate of return without grant	Financial internal rate of return with grant
Non-repayable – Native potatoes	12.6%	121.8%
Non-repayable – Alpacas	8.4%	63.7%
Repayable – Native potatoes	25.8%	84.3%
Repayable – Alpacas	17.1%	48.1%
Agroideas – Alpacas	19.5%	56.8%

50. These potential rates of return, calculated for the financial analysis, appear to the iTAP to have been overestimated,²⁰ creating an apparent mismatch between the potential profitability of these key agricultural activities and the possibility of funding them through pure grants from the Puna Facility. The iTAP acknowledges that the principle underlying the three sub-windows is related to the segmentation of the potential beneficiaries by organizational and commercial capacities and the level of barriers they face.²¹ However, it seems short-sighted not to also take into account the potential profitability of activities to be invested in. Pure grants may indeed make sense for collective EbA investments in the landscape. But offering a pure grant to a producers' association that is expecting a high financial return on its activity, represents a missed opportunity – both to maximize the development of the association's financial track record and to generate a return for replenishing the facility.

51. The non-repayable grant window is currently proposed to form 52 per cent of the total funds in the Puna Facility²². The iTAP believes that the total volume of pure (non-repayable) grants should be as small as possible, with such grants reserved for local initiatives that are unlikely to generate significant profits. The iTAP notes that GIZ is not accredited with GCF for on-lending, but believes that the design of the facility to cover only pure grants and repayable grants (that are effectively zero-interest loans, with highly flexible repayment terms) means that the effect on promoting recipients' financial literacy and financial inclusion will be limited. Ideally, it would have been preferable for the facility to have included regular low-interest, favourable-term (long tenor, long grace period) loans, in order to maximize eligible participants' capacity to understand the principles of risk and repayment, develop their financial literacy, enhance their prospects of accessing future commercial microfinance, and help to break the cycle of poverty.

52. The iTAP recommends that the AE and Profonanpe should think strategically about the balance between pure grants and repayable grants offered through the Puna Facility. Ideally, the project should apply the principle of minimum concessionality, and avoid giving pure grants to

²⁰ See written questions from the iTAP (4 January) and responses from GIZ (11 January).

²¹ Funding proposal, paragraph 190.

²² Funding proposal, Table 5. Preliminary Grant Breakdown per Sub-Window.

productive investments identified as having a high financial internal rate of return even without grants – instead providing repayable grants in these cases. This would help to maximize the borrowing organizations' repayment track record, and replenish the fund to reach more people.

53. It is noted that the project explicitly addresses barriers to beneficiaries' future access to commercial finance through a sub-activity under output 2.1 on "Support to financial institutions in greening microcredits" – working with microfinance institutions with a presence in the project area and with interest on expanding their credit portfolio to small-scale farmers for adaptation measures (e.g. Caja Rural los Andes, Caja Cusco, and Financiera Confianza – an entity of BBVA Microfinance Foundation) and providing technical assistance to adjust existing products (including for women) or develop new ones, incorporating the EbA approach in the financial risk analysis for prioritized value chains. Another sub-activity aims to "Raise awareness of greener microcredits to promote EbA and Climate Resilient Value Chains" through an awareness campaign for potential clients, training for field personnel of financial institutions, and cooperation in strengthening risk management approaches. This needs to be complemented by in-depth entrepreneurial and financial training in sub-activity 1.1.3.1 – currently described only in broad terms.

54. **Replenishing the facility.** Although the project proponents anticipate some return on the repayable grants, this amount is expected to be only 50 per cent of the total disbursed, and applies to only 48 per cent (windows 2 and 3) of the portfolio as currently structured. In effect, much of the Puna Facility operates like a small grants facility, and seems unlikely to attract private sector investors who would expect a conventional return. The AE has acknowledged that a special strategy is required for replenishment, and has designed sub-activity 2.1.1.2 to address this: "Development and implementation of an innovative strategy to mobilize resources from the private sector and different donors". This activity focuses on accessing funding from the private sector and other donors, and includes mapping and engagement of potential private sector actors, both those interested in corporate social responsibility investment in the Andes and those whose businesses depend on the ecosystem services provided by the Puna ecosystems, such as hydroelectric, cement, mining, beverages and camelid fibre products, who might wish to invest in the facility.

55. One-off contributions from donors or companies, however, would not constitute a long-term replenishment strategy for the Puna Facility. Two more sustainable options exist and could be explicitly addressed in the strategy to be developed for the facility. One option, which may be implied in the funding proposal, is for Profonanpe to reach long-term agreements with companies dependent on upper catchment management for water provision such as hydropower or beverage industries – in effect a type of payment for watershed services scheme. A second option could be a long-term source of State revenue channelled into the facility by law – such as a portion of the national Tax on Consumption of Plastic Bags or the Aporte por Regulación contribution by mining, hydrocarbon, electricity and other companies to the Environmental Assessment and Control Agency. The iTAP recommends that a full business plan should be developed in sub-activity 2.1.1.2, explicitly targeting long-term agreements and State revenue sources.

56. **Payment for Watershed Services.** The efficiency and effectiveness of the project interventions is further enhanced by sub-activity 2.1.2.1, which will help to activate Peru's existing payment for watershed services scheme, MERESE, bringing additional resources into the target regions.²³ The project will provide technical assistance to water utilities for five existing MERESE schemes in Arequipa, Lima, Cusco and Apurímac (details can be found in annex 23 to the funding proposal). Technical assistance packages may include support in planning, evaluation and selection of EbA public investment projects; strengthening governance platforms; contributing to the design of the monitoring, gender and capacity development

²³ Not directly linked to the Puna Facility, given that MERESE schemes are clearly circumscribed in how they operate.

systems; developing methodological tools for implementation; and reaching contractual agreements with communities. A methodology will also be developed, in coordination with MIDAGRI and MINAM, to calculate carbon sequestration in the Puna ecosystems and provide capacity-building for local actors (e.g. communities, local government, academia) aligned with national legal and institutional arrangements for carbon markets.

57. The project's overall efficiency and effectiveness is rated as medium.

II. Overall remarks from the independent Technical Advisory Panel

58. The iTAP believes that the project "Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru" has been well planned, consulted and designed by GIZ and the Government of Peru. The project is an important one for reducing the vulnerability of High Andean agropastoral communities and downstream water users, and contributing to arresting an ongoing cycle of poverty and degradation that would otherwise lead to ecosystem collapse and significant GHG emissions.

59. The iTAP recommends that the AE undertake the following actions to strengthen the sustainability of project results:

- (a) Include youth who are present in the target regions, while not necessarily being active in farming, in the entrepreneurial training (including financial literacy) through the technical assistance facility for the Puna Facility (sub-activity 1.1.3.1);
- (b) Consider carefully the balance between pure grants and repayable grants offered through the Puna Facility, and apply the principle of minimum concessionality when supporting profit-making activities (sub-activity 2.1.1.1); and
- (c) Develop a full business plan for the Puna Facility, explicitly targeting long-term agreements with private sector partners, and long-term sources of revenue, including from national environmental taxes and payments (sub-activity 2.1.1.2).

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP226)

Proposal name:	Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru
Accredited entity:	Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)
Country:	Peru
Project/programme size:	Medium

Impact potential
Thank you for the positive assessment. No further comments.
Paradigm shift potential
Thank you for the positive assessment. We welcome iTAP's recommendation. No further comments.
Sustainable development potential
Thank you for the positive assessment. No further comments.
Needs of the recipient
Thank you for the positive assessment. No further comments.
Country ownership
Thank you for the positive assessment. No further comments.
Efficiency and effectiveness

Thank you for the positive assessment. We welcome iTAP's recommendations. No further comments.

Overall remarks from the independent Technical Advisory Panel:

Thank you for the positive assessment. We welcome iTAP's recommendations. No further comments.

Gender analysis for Resilient Puna: Ecosystem based Adaptation for sus- tainable High Andean communities and ecosystems in Peru

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Title of the project:	Resilient Puna: Ecosystem based Adaptation for sustainable High Andean communities and ecosystems in Peru	
Country/countries:	Peru	
Sector:	Climate change: ecosystems-based adaptation	
Commissioning party:	BMZ and the Green Climate Fund (GCF)	
Recommendation for gender policy marker ¹ :	If the marker has already been determined, this should be stated here. In case the recommendation based on the gender analysis deviates from the previously assigned marker, this should also be noted.	
Purpose for conducting the gender analysis	☒ for a new project ☐ for a follow-on project ☐ for a modification offer ☐ for the GIZ portfolio in a country ☐ other: Click here to enter text.	
Status of the gender analysis	☐ Provisional gender analysis ☒ Finalised gender analysis	
Method conducting the gender analysis	☐ Desk study ☒ Desk study + data collection (e.g. on-site appraisal mission, interviews etc)	
Involved Gender FP	More than one option can be selected:: ☒ at the project level: Enter name ☐ *at the country level: Enter name ☐ *at the level of the department or regional division: Enter name ☐ *of the responsible unit in GloBe: Enter name <u>and</u> unit ☐ of the sectoral department: Enter name <u>and</u> unit * Please note: The Gender FP must be copied into all emails sent to the S+G Desk. For bilateral projects, the Gender FP of the country office should be involved; for regional projects, either the Gender FP of the country or area where the project has its headquarters, or the Gender FP of the department; for sectoral and global projects, the Gender FP of the responsible unit in GloBe.	
Other persons involved	☐ officer responsible for the commission ☐ project member ☐ appraisal mission team member ☐ planning officer ☒ external consultant	
Author(s) of the gender analysis	GIZ	
Date	December 2023	

¹ According to the OECD DAC, all ODA-financed projects should be assigned a marker. The gender marker is requested for BMZ projects and EU co-financing arrangements. Other public German commission parties do not currently request it.

List of Abbreviations

ANA	National Water Authority
CEDAW	Convention on the Elimination of All Forms of Discrimination against Women
CONAMUCC	National Committee on Women and Climate Change
CSW	Commission on the Status of Women
EBA	Ecosystems based Adaptation
ECOSOC	UN Economic and Social Council
ENAHO	National Household Survey
FEMUCARINAP	the National Federation of Peasant, Artisan, Indigenous, Native and Salaried Women of Peru
GG	GIZ Gender Equality Marker Policy
INEI	National Institute of Statistics and Informatics
IPCC	Intergovernmental Panel on Climate Change
JNE	National Election Jury of Peru
MESAGEN	Gender Roundtable for International Cooperation in Peru
MIDAGRI	Ministry of Agriculture and Irrigation
MINAM	Ministry of Environment
MIMP	Ministry of Women and vulnerable populations
NDC	Nationally Determined Contribution
NGO	Non-Government Organisation
ONAMIAP	Organisation of Andean and Amazonian Indigenous Women of Peru
PAGCC	Gender and Climate Change Action Plan
PEMIG	Multisectoral Strategic Plan for Gender Equality
PNIG	National Gender Equality Policy
PROFONANPE	Peruvian Environmental Fund for the Conservation of Protected Areas
SDG	Sustainable Development Goals
SERNANP	National Service for Natural Protected Areas
UNFCCC	United Nations Framework Convention on Climate Change
UNFPA	United Nations Population Fund
WEF	World Economic Forum

Chapter 1: Introduction

The “Resilient Puna: Ecosystem-based approaches for sustainable High Andean communities and ecosystems in Peru” Project (hereafter “Resilient Puna”) aims to increase the resilience of Andean communities in the departments of Arequipa, Cusco, Puno and Apurímac, through the management, conservation and restoration of high Andean ecosystems; an increased access to public and private financing; and a stronger territorial planning towards Ecosystem-based Adaptation (EbA). It is organised into three components: (i) Puna ecosystems are restored, conserved and better managed to support climate resilient livelihoods, through the implementation of EbAs measures ; (ii) Public and private financing for EbA measures and climate resilient livelihoods are in place and accessible for vulnerable communities in the Puna Ecosystem; and, (iii) EbA and climate resilience are mainstreamed into multilevel landscape governance instruments. The project will invest in EbA and climate resilient value chains, providing technical support to communities and producer associations to achieve sustainable business plans; directly benefiting the conservation and restoration of 23,914 hectares of high Andean ecosystems.

Various international and national instruments have defined guidelines for gender equality in key climate change mitigation and adaptation actions². Not only from a social, but also from an economic and climate perspective. The link between gender, environment and climate encompasses a number of issues based on the socially constructed relationships between men and women, which vary across cultures, geographies and time. Gender gaps are still wide and trigger inequalities in access to their basic rights, opportunities and benefits throughout their life cycle, which are reflected in their health, education, employment, participation, decision-making power, access to and control of natural resources, capabilities, living a life free of violence, among others. Climate change represents a major driver of biodiversity loss, while ecosystem degradation undermines the resilience of people and nature to climate-related shocks and stressors. Persistent gender inequalities underpin both of these crises, which increase vulnerability to risks and reduce the options available to people, especially women, in maintaining their livelihoods and good living, and their access to rights, goods and services.

The gender analysis of the Resilient Puna (RP) Project is in harmony with these international principles and commitments (CEDAW, Beijing, Belém do Pará, SDGs), as well as with national ones (Gender and Climate Change Action Plan-PACC, National Policy on Gender Equality³, National Strategic Development Plan to 2050, National Plan against Gender Violence, among other policies) that support work in favour of gender equality and non-discrimination of women in the access to their social, political, cultural, climatic and economic rights and resources. According to the United Nations Economic Commission for Latin America and the Caribbean (ECLAC) (2017), the impacts of global warming that affect men and women differently, require adaptation and mitigation measures that recognise their different needs. It also implies promoting the mainstreaming of a gender perspective in public policies to address climate change, with the aim of observing, analysing and transforming inequalities and inequities in the condition, construction of roles and position of men and women in society (UN-Women, 2017).

The project has adopted as its main approach “Ecosystem-based Adaptation” (EbA), an approach that promotes working with nature to cope with gradual and extreme climate impacts, which affect people, their livelihoods and their economy. In this approach, two central characteristics are identified in order to better understand the gender approach:

- i) it is a people-based approach, as it seeks to reduce their vulnerability while at the same time generating social and economic benefits; and,

² Peru is committed to the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), the Beijing Declaration and Platform for Action and its follow-up through the Commission on the Status of Women (CSW), the Sustainable Development Goals (SDGs), the United Nations Framework Convention on Climate Change (UNFCCC) and the Conferences of the Parties (COP) to the convention, among other standards, commitments and recommendations, permeate national spheres.

³ “The public need for a National Gender Equality Policy aimed at correcting the problem of structural discrimination against women stems both from constitutional norms and the Peruvian State’s commitment to human rights treaties...”. PNIG, 2019.

ii) it is a collaborative process in which communities, public and private organisations, work together to have an impact at the landscape level and generate sustainable opportunities over time for all parties⁴.

Linking the gender EbA approaches is both challenging and enriching. It allows to move from analysing the resources, decisions, behaviours, roles and interrelationships of men and women with their environment, towards integrating ways and mechanisms to break down the barriers that have been hindering women's progress in the framework of their adaptation and resilience to climate change.

Specifically, the gender analysis has identified the main factors that are hindering women's progress and resilience in the face of climate change. Overburdened care and unpaid domestic work; low participation in decision-making; lack of income; vulnerability of their housing and household environment (e.g., health, food, gender-based violence, poor housing) are the main gendered barriers to more equitable climate adaptation. Although the response to this situation requires the involvement of both women and men, it is important to mention that the gender dimension addressed in the project defines actions that mainly strengthen the role of women, seeking to "levelling the playing field" for their real and effective involvement in the implementation of the project, as well as a better appropriation of the EbA issue and taking advantage of the opportunities and benefits provided by the project.

The project recognises that an adequate vision and inclusion of gender aspects in its different stages is one of the main pillars for the effectiveness and sustainability of the intervention. For this reason, both the Gender Analysis (GA) and the Gender Action Plan (GAP) of the Resilient Puna project focus on the mainstreaming of the gender approach from its formulation to its implementation, monitoring and evaluation. Therefore, the RP project will also conduct a monitoring strategy to evaluate and verify that no negative results are produced (under the principle of do no harm). The project will encourage women's participation and leadership to be promoted and recognised in the environments in which the project will be developed, in many of which they have been traditionally excluded. The central objective of the analysis and integration of the gender approach in the project is to make women's barriers visible to contribute to reducing asymmetric situations in the care, exploitation and use of natural resources, as well as the rights of people, especially women, within a natural ecosystem. In this sense, having a reading of the gender approach in this area leads us to build egalitarian processes that allow people to build climate resilience and exercise their rights, while protecting the ecosystems on which they depend.

Finally, the methodology to collect information to prepare this gender action plan involved a desk review of available data and fieldwork data collection techniques. Regarding desk review, it included analysis of available secondary data sources such as national policies and work of national mechanisms, as well as a comprehensive review of secondary literature that includes various research reports, articles and papers on the subject from both national and international organisations. Regarding fieldwork data collection, between August and November 2022, two fieldtrips were organised to potential districts of intervention in the departments of Arequipa, Cusco and Puno. During these fieldtrips, personnel from GIZ, MIDAGRI and FIAP conducted focus groups, in-depth interviews and surveys to analyse general perceptions about the Resilient Puna project, governance structures regarding management of natural resources, local livelihoods and gender gaps in rural communities. To address gender topics, 89 in-depth interviews were conducted, of which 40 were with women and 49 with men. The greater number of interviews with men was due to their greater willingness to be interviewed. Likewise, surveys were applied to build a baseline that gathers evidence and statistical data, for the definition of indicators of the Gender Action Plan of the project. Likewise, the baseline reflects as close reality of the situation of women and men at the territorial level as well as the actors involved in the context in which they develop.

⁴ UICN. Soluciones AbE. Información disponible en internet. Página: solucionesabe.org

Chapter 2: Information on gender dimensions in the country or the region

2.1 Meta level (norms)

Peru is the fourth country with the largest population in South America. In 2020, the population was 32.6 million inhabitants: 50.4% women and 49.6% men. This population lives mostly in urban centres (79%) and 21% in rural areas (INEI, 2020b). Peru also hosts great cultural diversity with 55 indigenous peoples, 51 from the Amazon and 4 from the Andes (Ministerio de Cultura, 2022). According to the last National Population Census (2017), 26% of the population identifies as part of an indigenous group, which makes around 6 million people (INEI, 2017).

Moreover, Peru is a country with high levels of inequality, in terms of income, property, access to basic social services, infrastructure and gender. In 2020, monetary poverty increased by 9.9 percentage points and reached 30% of the Peruvian population. This affected 46% of people living in rural areas (3.1 million people) and 26% of people living in urban areas (6.8 million people) (INEI, 2021b).

In terms of gender inequity, Peru has progressed launching different norms for the mainstreaming of a gender approach in public policy, but it is still insufficient. The most recent 2020-21 World Economic Forum (WEF) Global Gender Gap Index placed Peru 37th out of 156 countries in terms of gender gaps pertaining to economic participation and opportunity, education, health and survival, and political empowerment dimensions (WEF, 2020-2021) and 7th place in Latin America. Similarly, the Gender Inequality Index in 2019 placed Peru with 0.395 points, a position below the average of countries with high human development (0.340) (UNDP, 2022).

Peru is a state that prohibits discrimination and recognises equality as a fundamental human right (Political Constitution of 1993, Art. 2 numeral 2). Article 191 of the Constitution establishes the responsibility of the Peruvian State to remove obstacles that limit the right to equality or that constitute discriminatory acts. However, there is structural discrimination against women in Peru, which is systematic, generalised, and massive. This discrimination generates multiple oppressions for women, limiting their opportunities and systematically violating their rights. Women are affected by multiple discriminations that are evident in the persistence of limitations to their autonomy (physical, economic and decision-making), as well as the existence of gaps in different dimensions (reproductive health, empowerment and labour market) and areas (health, education, employment, participation, use of time, among others), with the situation worsening in the rural settings. Structural discrimination based on gender does not affect all women equally, who do not constitute a homogeneous group, since there are indigenous, Afro-Peruvian, disable, displaced, poor and non-poor women.

The supremacy of the masculine over the feminine is institutionalised in social relations, in public and private institutions and in politics, through discriminatory sociocultural patterns (beliefs, prejudices, dogmas, and customs), which are reproduced, transmitted intergenerationally and, therefore, perpetuated. The origin of these sociocultural norms lies in the differences in power between women and men and male privilege. Discrimination against women is rooted in norms and an institutional culture that reinforce women's subordination. Violence against women is one of the main instruments used to maintain women in a situation of subordination and as a means of disciplining them to prevent their access to rights. This not only prevents their participation in social, political, economic, and cultural decision-making processes, but also makes it impossible for them to transform these spaces that reproduce patriarchal structures of domination.

According to the National Social Relations Survey in 2019 (ENARES, for its acronym in Spanish), the index of social tolerance towards violence against women is 59% of the surveyed. When referring to attitudes that support such violence, 53% of male and female respondents

state that women should first fulfil the role of mother and wife, and only then can achieve their personal projects (INEI, 2019).

Gender inequality has an impact on development and the economy, but above all on families' wellbeing, where unpaid labour and care roles are still locked by stereotypes, and stigmas, in which women tend to have a heavier burden than men and fewer opportunities to achieve their wellbeing. In the context of confinement due to the Covid 19 pandemic, there was a difference of 19 percentage points between men and women, regarding the time spent on household chores (MIMP, 2020). This burden for women made it more difficult for them to re-enter the job market under similar conditions as their male counterparts. In 2022, 82% of women perform domestic care work as opposed to 16% of their male counterparts, showing that women spend 27 hours more on unpaid domestic work per week (Flora Tristan & IEP, 2021).

By positioning the gender approach with key partners throughout the project implementation process, it will be ensured that the impact of the project will contribute to promote the improvement of living conditions for the entire population in general, and for women in particular. To do this, it is necessary to strengthen the process of making visible the disadvantaged conditions in which rural women's participation in communities and decision-making spaces related to natural resources' management takes place. The political and economic empowerment of rural women has positive effects on household income, food security and nutrition, in general in families' wellbeing.

2.2 Macro level (policies and laws)

At the international level, Peru has ratified the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW, 2010). It has also ratified the Inter-American Convention on the Prevention, Punishment and Eradication of Violence against Women (Convention of Belém do Pará), which establishes international standards to prevent, punish and eradicate violence against women in the public and private spheres. It has also signed commitments such as the Beijing Declaration and World Platform (1995) and the 2030 Agenda, whose Sustainable Development Goal (SDG) 5 establishes gender equality and whose target 5.5 seeks to ensure women's full and effective participation and equal opportunities for leadership at all decision-making levels in political, economic and public life.

At the national level, in addition to the constitutional norms, the Peruvian State has laws, regulations and guidelines that are mandatory for all levels of government. There are different policies that promote equal opportunities between men and women in public policies and plans, the main one being the National Policy on Gender Equality, which establishes 6 priority objectives: 1. To reduce violence against women; 2. To guarantee women's access to and participation in decision-making spaces; 4. To guarantee the exercise of women's economic and social rights; 5. To reduce institutional barriers that hinder equality in the public and private spheres between men and women; and 6. To reduce the incidence of discriminatory socio-cultural patterns in the population. This policy has a Multisectoral Strategic Plan for Gender Equality (PEMIG, for its acronym in Spanish).

These policies stipulate the obligation to promote the adoption of egalitarian values, practices, attitudes and behaviours between men and women, to guarantee the right to non-discrimination and the eradication of violence against them. They seek to guarantee the full exercise of all their rights (civil, political, economic, social, and cultural). They also establish obligations to promote the physical, economic, and decision-making autonomy of women in society and in public administration. They also contemplate the intersectionality approach, paying special attention to the most vulnerable groups. With respect to gender mainstreaming in the State, it basically involves capacity building for civil servants, the generation of differentiated information, the participation of women in spaces for dialogue and decision-making, as well as the incorporation of this approach in management documents, programs and projects.

Main national instruments on gender

- Political Constitution of Peru (1993). Art. 2 inc. 2 (equality and non-discrimination based on sex). Art. 191 (State responsibility for removing obstacles to equality or that constitute discriminatory acts).

National Policies

- National Agreement (2002). Policy 11 (seeks to combat discrimination and promote equal opportunities between men and women).
- Supreme Decree N° 054-2011-PCM Bicentennial Plan: Peru to 2021 (throughout the text, non-discrimination, equity between men and women, affirmative actions and equal opportunities and universal access to basic services are proposed).
- National Agreement (2019). Vision of the country to 2050; proposes to eradicate discrimination and ensure equal access to opportunities and conditions, in a fair and inclusive society that leaves no one behind.

Laws and regulations of legal rank

- Legislative Decree N°635 Criminal Code (1991) amended by Legislative Decree N° 1323 (2017) and Legislative Decree N°1410 (2018); criminalises the delinquencies of discrimination and incitement to discrimination, aggravated in the case of public officials and harassment, sexual harassment, sexual blackmail and dissemination of images, audio visual materials or audios with sexual content and modifies the procedure for sanctioning sexual harassment.
- Law N°26772 (1997), prohibits discrimination, annulment or alteration of equal opportunities or treatment in job offers and access to educational training.
- Law N° 27942 (2003), to prevent and punish sexual harassment.
- Law N° 28983 (2007), to promote equal opportunities between women and men.
- Legislative Decree N°1098 (2012). Law of Organisation and Functions of the Ministry of Women and Vulnerable Populations - MIMP. Governing body for public policies for women and gender mainstreaming.
- Law N° 30314 (2015), to prevent and punish sexual harassment in public spaces.
- Law N° 30364 (2015), to prevent, punish and eradicate violence against women and members of the family group.
- Law N° 30982 (2019), amends Law 24656, General Law of Peasant Communities, to strengthen the role of women in Peasant Communities.
- Legislative Decree N°1470 (2020), establishes measures to guarantee the care and protection of victims of violence against women and members of the family group during the health emergency declared by COVID-19.
- Law N° 31030 (2020), modifies norms of the electoral legislation to guarantee parity and gender alternation in the lists of candidates.
- Law N° 30982, (2021) that amends Law 24656, General Law of Peasant Communities, which strengthens the role of women by establishing that at least 30% of the members of the Community Board, the body responsible for the government and administration of the community, must be women.
- Supreme Decree N° 010-2022-MIMP, promotes mandatory training on gender perspective for the prevention of violence and discrimination in public administration.

Supreme Decrees and Resolutions

- Supreme Decree N° 005-2017-MIMP: Public entities of the National Government and Regional Governments create a commission, committee or working group, for Gender Equality.
- Supreme Decree N° 002-2018-JUS: National Human Rights Plan 2018 – 2021; establishes four strategic objectives for the protection of women: (1) Reduce violence against women, (2) Promote the participation of women in decision-making spaces, (3) Guarantee the exercise of women's sexual and reproductive rights, (4) Guarantee the generation of women's own income.

- Supreme Resolution N° 129-2018-PCM, creates the multisectoral commission of a temporary nature against VAW, the fight against which constitutes a national priority.
- Supreme Decree N° 056-2018-PCM, approves the General Policy of the Government to 2021 and sets as a priority guideline to promote equality and non-discrimination between men and women, as well as to guarantee the protection of women against all types of violence.
- Supreme Decree N° 008-2019-MIMP, approves the National Policy on Gender Equality.
- Supreme Decree N° 014-2019-MIMP: Regulation of Law N° 27942, Law for the Prevention and Punishment of Sexual Harassment.
- Supreme Decree N° 002-2020-MIMP, approves the Multisectoral Strategic Plan for Gender Equality of the National Gender Equality Policy.
- Supreme Decree N° 012-2016-MINAM, approves Peru's Gender and Climate Change Action Plan.
- Chief Resolution N°187-2020-ANA conforms the Commission for Gender Equality of the National Water Authority.

Although these regulations are in place, the main challenge is their limited implementation. The persistence of inequality between women and men is reflected in the insufficient political will for the effective mainstreaming of the gender approach, the absence of adequate diagnoses, inter-institutional coordination, unified records, and a trained and sensitised civil service, so that they understand that if they are not committed to closing gender gaps and their adequate management, the actions carried out within the framework of their function are contributing to deepening them.

2.3 Meso level (institutions)

Peru has a governing body for public policies for women since 1996 (Legislative Decree N°866, then MIMDES, Organic Law 27779). Since 2012, it changed its name to the Ministry of Women and Vulnerable Populations (MIMP, for its acronym in Spanish), with Legislative Decree N° 1098. Today, this is the governing body for addressing national and sectoral policies on women's rights, as well as prevention, protection and care against violence against women, promotion and strengthening of gender mainstreaming in policies, plans, programs and projects of the state and public and private institutions. Although this ministry has the steering role, it needs to be strengthened, especially for the articulation with all sectors and levels of government. Its main weakness lies in the limited capacities, institutional resources, and budget to fulfil its role. This weakness expands with the continuous political instability, absence of long-term state policies, and political willingness of some key sectors of the government, like the current Parliament that avoids debates on key issues like the sexual rights for women and girls.

However, there is a strong social movement that defend women's and LGBTQ+ human rights, including the National Network for the Promotion of Women, the Stop Street Harassment collective, the Manuela Ramos Movement and the Flora Tristán Peruvian Women's Centre (a member of the Latin American Women and Habitat Network), among others. There is also a rural and indigenous women's movement with groups and organisations such as: the National Organisation of Andean and Amazonian Indigenous Women of Peru (ONAMIAP), the National Federation of Peasant, Artisan, Indigenous, Native and Salaried Women of Peru (FE-MUCARINAP), among others. All these organisations do influence the issuance of regulations and public policies favourable to women, carrying out awareness campaigns towards public opinion and defend emblematic cases before national and international judicial instances.

In Peru, the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH participates in the Gender Roundtable for International Cooperation in Peru (MESAGEN). This is a coordinating body for international cooperation in Peru. Its objective is to contribute to gender equality and the empowerment of women, girls and adolescents, within the framework

of national plans and policies, international commitments and obligations signed by Peru. Currently, MESAGEN is made up of diplomatic representations and international organisations of official cooperation (bilateral and/or multilateral) that work in gender equality and the promotion of women's human rights in Peru. Currently, GIZ is implementing a project on the prevention of gender violence against women, supporting the consolidation of the Specialised Justice System on Violence (SNEJ) for the protection and punishment of violence against women and family members.

2.4 Micro level (target groups)

In Peru, there are marked patterns of gender-based roles and/or behaviours, and historically established stereotypes and social expectations about women's and men's aspirations and opportunities persist and are clearly discriminatory in that they limit women's options for full development. In both political and economic participation, men have a large advantage over women, for example, in access to paid employment, in the quality of employment accessed, and in control over time available for work and training. According to the World Economic Forum (WEF, 2020), Peru is one of the countries in the world that shows only setbacks in the wage gap, ranking 128th out of 144 countries in the world.

Health: Andean and indigenous women are the population group with the least access to prenatal check-ups by a medical professional. During 2020, only 13.1% of women were attended by a medical professional compared to 39.5% of women living in urban areas. Likewise, in the same year, limited access to institutionalised deliveries were reported, with a difference of 15.8% between urban (96.9%) and rural (81.1%) institutional deliveries. In addition to the limited institutional care for deliveries, the deliveries that took place in a health facility were carried out by nurses and not by medical professionals (17.4% rural versus 4.2% urban) (INEI, 2021).

Regarding the birth rate and teenage pregnancy, although the departments prioritised by the project do not have the highest teenage pregnancy rates at the national level, the department of Apurímac presents a rate of 10.7% of teenage pregnancy, compared to 5% in the departments of Arequipa, Cuzco and Puno. Also, 5% of women reported unmet needs for family planning in the same departments, with Puno having the lowest percentage of modern contraceptive use of the four departments prioritised by the project (See Table N°1) (INEI, 2021).

Table N°1
Percentage of adolescent pregnancy, use of modern contraceptive methods, and family planning needs, in the prioritised departments

Department	(%) Adolescents 15-19 years ever pregnant	(%) women using modern contraceptive methods	(%) women with unmet needs for family planning
National	6.6%	78.1%	5.5%
Apurímac	10.7%	79.9%	5.9%
Arequipa	5.0%	80.7%	5.0%
Cusco	4.6%	80.1%	7.1%
Puno	5.7%	75.3%	5.1%

Source: INEI, 2021

In 2020, acute respiratory infections (ARI) in children under five years of age were higher in rural areas (7.2%) compared to 4.9% in urban areas. ARI in children is more prevalent in households where charcoal is used as fuel for cooking, 12.9% compared to 7.3% of cases where firewood, straw or agricultural waste is used. On the other hand, acute diarrhoeal disease (ADD) continues to be an important cause of mortality in children, with 9.4% of children under five years of age suffering from the disease, with a higher percentage of boys (9.9%) than girls (8.9%) (INEI, 2021).

Education: Illiteracy rates of women in the departments prioritised by the project exceed the national average (8% for women and 3% for men). In the case of Apurímac, 20% of women are illiterate compared to 5.8% of illiterate men. The same occurs in Cusco, 16% of women are illiterate and 3.8% of illiterate men; in Puno, 14.2% of women are illiterate and 3% of men. Arequipa has a lower percentage of illiteracy, with 4.7% of illiterate women (4.7%) and illiterate men (1.5%) (INEI/ENAO, 2019).

In 2021, the net school attendance rate was 91.4% for females and 91.1% for males (INEI 2022). Also, by 2021, Cusco had a higher net primary school attendance rate for girls than for boys, in contrast to Apurímac, Arequipa and Puno. On the other hand, Cusco has a higher net secondary education attendance rate for adolescent girls aged 12 to 16 than for boys in the same age group, while Puno has the largest gap in favour of adolescent boys (MINEDU, 2020).

At the primary level, the access gap between girls and boys has been closed at the national level (See Table N°2). However, at the secondary level, inequality reappears, to the detriment of adolescent girls. In 2020, 71.9% of females compared to 76.7% of males attended a secondary educational centre. Part of the explanation for the gaps at secondary level is due to dropping out of school, caused by economic or family reasons (45.8% rural, 43.9% urban); pregnancy or early marriage (24.9% rural and 16.8% urban), and graduation or studies at an academy (11% rural and 27.5% urban) (MINEDU, 2020).

Table N°2
Illiteracy, school attendance and dropout rates

Department	Illiteracy rate for people aged 15+ (a)		Net attendance ratio (*) in regular education (b)			Dropout rate in regular education (b)
	Women	Men	Early childhood Education	Primary Education	Secondary Education	Total
National	8.1%	3.0%	81.2%	94.4%	83.5%	9.1%
Apurímac	20.0%	5.8%	99.0%	97.7	87.0%	1.1%
Arequipa	4.7%	1.5%	82.9%	95.6%	90.2%	3.8%
Cusco	16.0%	3.8%	84.4%	98.8%	89.8%	4.7%
Puno	14.2%	3.1%	65.0%	99.3%	93.8%	2.3%

Source: MIMP Policy Monitoring and Evaluation Office

(a) INEI/ENAO, 2019

(b) MINEDU, 2020

(*): number of persons attending a given level of education (pre-primary/primary/secondary) who are in the age group set for that level, as a proportion of the total population of that age group (INEI, MINEDU)

Income and employment: Between 2010 and 2019, the department with the largest income gap was Apurímac, with 28.2% more women than men with no income of their own. It is followed by Cusco and then Arequipa and Puno, all below the national average. In 2020 during the pandemic, the income gap narrowed in all department, but with two different dynamics. In Apurímac, Cusco and Puno, while the percentage of men with no income of their own had risen, the percentage of women fell significantly. One possible explanation is that, faced with the loss of formal employment for men, women started new activities and enterprises to generate income. In Arequipa and at the national level, the reverse is true: confinement has affected many enterprises and employment. As a consequence, both percentages of people with no income of their own have risen sharply, men more than women, narrowing the gap between the two (INEI, 2020).

Table N°3
Women and men with no income of their own, 2010-2019 and 2020

Department	2010-2019 (%)			2020 (%)		
	Women	Men	Gap W-M	Women	Men	Gap W - M
National	31.3	12.1	19.2	36.0	19.7	16.3
Apurímac	39.6	11.5	28.2	32.8	12.1	20.7
Arequipa	28.4	12.0	16.4	38.1	22.3	15.8
Cusco	33.4	13.7	19.7	30.5	18.2	12.3
Puno	29.6	15.9	13.7	25.1	17.4	7.7

Source: INEI; Gender Indicators, 2020.

Regarding employment, in rural areas, 68% of women are engaged in agricultural work. Sales and service jobs represent 41.9% of job positions for women in urban areas and 18.7% in rural areas (INEI/ENDES, 2021). At the national level, 28.1% of women aged 15-49 who are self-employed have an agricultural job, while 36.2% have a non-agricultural occupation. In the case of women in agricultural occupations, 37.6% are unpaid family workers and only 46.3% of women in agricultural occupations received cash income. Of the total number of women who did not receive income in cash or in kind in return for their work, 32% were in agricultural occupations (INEI/ENDES, 2021).

Paid and unpaid domestic work: In general, in Peru, there is a higher percentage of women in informal sectors who carry a greater burden of unpaid domestic work, which reduces or prevents them from having a paid job. Domestic chores and care for family members, like children, the elderly, and sick/disable family members fall mainly on women. On average, women assume 23 hours more unpaid work per week than men (39 hours and 28 minutes vs. 15 hours and 54 minutes) (INEI/ ENUT, 2011). Moreover, analysing women's employment is challenging as some of the occupations they carry out are not perceived as work and are not reported, especially when it is agricultural work on family plots, family businesses or in the informal sector (INEI/ ENDES, 2021). While the burden of unpaid and care work at home has historically fallen on women, during the pandemic, the closure of schools and the saturation of the health system affected them doubly, placing a greater burden of unpaid work on them for the duration of their confinement at home.

Participation in decision-making: Since the implementation of the Parity and Alternation Law in 2021 for parliament Congress and sub-national authority candidates, there has been an increase in women's participation at these levels⁵. In 2021, 49 women and 81 men were elected for Parliament (See Table N° 4). Political participation of women increased from 27.7% in the previous period (2016 – 2021) to 37.30% today. However, women's subnational representation in the Parliament is still limited, with women representing half the number of male representatives from the departments of Apurímac, Arequipa and Cusco. For Puno, there is no representation of women in Parliament, compared to five male congressmen representing this department (ONPE, 2022).

Table N°4
Political participation, by prioritised department

Region	N° seats at national parliament 2021-2026	
	Women	Men
National	49	81
Apurímac	1	1
Arequipa	2	4
Cusco	2	3
Puno	0	5

Source: ONPE, 2022

Regarding female sub national authorities, out of 25 departments, only 2 are governed by women (See Table N°5). Of these two women, none of them correspond to the departments

⁵ In 2021, the Law No. 30996, of parity and alternation for elected positions became mandatory. This law stipulated that in the 2021 women should occupy 40% of the lists, in 2026 45% and in 2031, 50%.

prioritised by the RP project. In terms of local authorities, Cusco has 2 from 13 provinces with female mayors. Neither Apurímac, Arequipa, nor Puno have women mayors at the provincial level. Despite the parity and alternation rule applied on sub national elections on 2022, the position of women's political participation in the prioritised departments were critical (INEI 2022).

Regarding female local authorities, out of more than 1800 municipalities, only 90 municipalities have a female mayor. In the prioritised departments, Apurímac has two elected female mayors from 84 municipalities. In Arequipa, out of 109 municipalities, only 9 have a female mayor. In Cusco, out of 112 municipalities, only two are female mayors; in Puno, out of 109 municipalities, only four are women.

Table N° 5
Number and percentage of subnational government authorities

Department	Sub national governors		Provincial mayors		Municipal mayors	
	Women	Men	Women	Men	Women	Men
National	2 (8%)	23 (92%)	8 (4,8 %)	188 (95, 2%)	90 (4,9%)	1772 (95,1%)
Apurímac	0	1	0	7 (100%)	2 (2,4%)	82 (97,6%)
Arequipa	0	1	0	8 (100%)	9 (8,3%)	100 (91,7%)
Cusco	0	1	2 (15,4%)	11 (84,6%)	2 (1,8%)	110 (98,2%)
Puno	0	1	0	13 (100%)	4 (8,3%)	105 (91,7%)

In general terms, the situation of women's political participation reflects the situation of all forms of women's participation in decision-making, whether at national, sub-national or community level. This situation deserves special attention if the project aims at strengthening climate governance and adaptation from a gender perspective.

Gender Violence: In 2022, 55% of Peruvian women were victims of violence perpetrated by their husband or partner. Both physical violence and sexual violence were reported in higher proportions in rural areas, 28.6% and 7.2%, respectively; if ethnic self-identification⁶ is taken into account, the proportion of women who were victims of violence was higher in rural areas, 28.6% and 7.2%, respectively, women of native origin represent the highest percentage of violence victims with 54.4%⁷. In the case of psychological and/or verbal violence, it occurred more frequently in women in urban areas (51.5%). Of the four prioritised departments, Apurímac shows higher percentages of violence against women (total and psychological violence); the percentage of physical violence is higher in Cusco and sexual violence is higher in Puno (See Table N°6). Reasons stated by women victims of physical violence for not reporting nor seeking help was because *it was not necessary* (42.9%), followed by a *feeling of shame* (16.8%) and *lack of knowledge* (they did not know where to go or did not know about services, 11.0%), among other reasons was fear of being beaten or their children, fear of causing problems to the person who hit them, fear of divorce/separation (INEI/ENDES 2021). Women victims of trafficking registered with the National Police of Peru have also been reported in the departments prioritised by the project. In 2020, Cusco reported 19 trafficked women and Arequipa reported 13, the highest figures of the four prioritised departments (National Prosecution, 2020). Regarding feminicides; in 2022, 7 feminicides were reported in Cusco and 4 in Arequipa (MIMP, 2022).

Table N° 6
Percentage of family, physical and/or sexual violence, trafficking and feminicide

	Violence against women ever perpetrated by a husband or partner (a)	Physical and/or sexual violence by spouse or partner in the last 12 months (a)	Women victims of trafficking(b)	Number of feminicide (c)
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⁶ Quechua, Aymara, native of the Amazon, belonging to or part of another indigenous or aboriginal people.

Department	Total	Psycho-logical	Physical	Total	Sexual	Total	Total
National	54.9%	50.8%	26.7%	33.6%	5.9%	342	67
Apurímac	64.1%	60.7%	37.8%	43.3%	11.8%	4	1
Arequipa	60.7%	54.0%	36.3%	36.0%	9.5%	13	4
Cusco	62.2%	58.6%	38.7%	42.0%	11.4%	19	7
Puno	62.2%	57.8%	36.3%	38.5%	12.2%	1	4

Source: MIMP; Policy Monitoring and Evaluation Office.

(a) INEI/ENDES, 2021

(b) National Prosecution, 2020

(c) MIMP/ National Programme for the Prevention and Eradication of Violence against Women and Family Group Members; 2022

Conclusion 1 for the project

The formal right to equality faces multiple barriers that are expressed in the persistence of gender gaps at all levels and in all social spheres, constituting one of the main obstacles to sustainable development that leaves no one behind. In Peru, there is still no adequate guarantee for the full realisation of the right to equality and the context of the health emergency due to the presence of COVID-19 placed women in a situation of vulnerability. In spite of this, there is a fairly advanced regulatory framework that obliges all sectors of the State and all levels of government to mainstream a gender perspective in public policies. In this line, working with political partners and agriculture and water stakeholders and users provides an opportunity to contribute to closing the gaps through gender mainstreaming, making visible the need to carry out actions that, in the long term, collaborate in achieving a social and institutional transformation towards greater equality in highly masculinised sectors and in a post-pandemic context.

Chapter 3: Gender dimensions in the sector

3.1 Meta level

On July 28, 2022, the UN General Assembly adopted resolution A/RES/76/300, that recognises access to a clean and healthy environment as a human right, as well as the need for full implementation of environmental agreements. It also recognises that climate change, overuse of natural resources, air, land and water pollution, loss of biodiversity, among other factors, prevent the full enjoyment of this right. According to the Intergovernmental Panel on Climate Change (IPCC) in addition to impacting women and girls in different ways, climate change can aggravate gender inequalities and increase women's vulnerability.

One of the most important international instruments in the field of women's human rights is the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW, 1979). This instrument provides an understanding of the gender equality dimension in order to take measures against discrimination against women and to guarantee their access to the same rights and opportunities as men. In 2018, the CEDAW Committee highlighted the importance of reinforcing the collection and analysis of disaggregated data, ensuring policy coherence, improving coordination between sectors, promoting the role of subnational governments, and capacity building and access to technology. Similarly, with regard to rural women's rights, it recommended reinforcing rural economies, promoting local employment and livelihood opportunities for women.

The Declaration and Platform for Action of the Fourth World Conference on Women in Beijing, 1995, is another key international instrument. It was one of the first platforms to state the importance of gender mainstreaming in all areas, including the environment. This platform includes 12 prioritised axes, one of which is "Women and the environment", highlighting the need for women's participation in decision making spaces, the incorporation of their interests and

concerns in programmes and policies, and the need to establish mechanisms to measure the impact of environment and development policies on women's lives. In this framework, the Commission on the Status of Women (CSW), under the UN Economic and Social Council (ECOSOC), focused exclusively to gender equality and the empowerment of women and to the follow-up of the Beijing Platform for Action, stated that gender equality and the empowerment of women and girls is a key aspect of climate change and risk disaster reduction (CSW66). It also highlights and reaffirms the leadership of women and girls as key agents of change in safeguarding the environment and addressing the adverse effects of climate change. (E/CN.6 /2022/L.7. United Nations. Economic and Social Council. Commission on the Status of Women Sixty-sixth session).

The United Nations Framework Convention on Climate Change (UNFCCC) seeks to stabilise greenhouse gas concentrations at levels that prevent human intervention that puts the environment at risk. In 2014, the UNFCCC Conference of the Parties (COP) in Lima adopted the Lima Work Programme on Gender to promote gender balance and gender-responsive⁸ climate policy. In 2017, the Bonn Gender Action Plan (GAP) was agreed to advance gender mainstreaming in all climate action. In this framework, the Paris Agreement, a commitment by UNFCCC member states to limit global warming to less than 2 degrees Celsius, includes three lines of action linked to: emissions reduction, adaptation to climate impacts and financing of actions, including mentions linked to gender equality and women's empowerment, as measures that must be respected, promoted and taken into account in responses to climate change.

The 2030 Agenda and the Sustainable Development Goals (SDGs) include SDGs linked to gender equality, in addition to SDG 5, which aims to achieve gender equality and the empowerment of women and girls, as well as other SDGs that include gender-specific targets. For example, SDGs 6, 10, 13 and 15 are closely linked. Target 6.2 addresses equitable access and the need to pay special attention to women, girls and people in vulnerable situations; SDG 10 is about reducing inequalities and including all people in social, political and economic life; SDG 13 is about taking urgent action to combat climate change and its impacts; and SDG 15 is linked to the conservation and sustainable use of terrestrial ecosystems and includes the promotion of fair and equitable participation. In sum, the agenda highlights that gender equality is necessary to achieve the SDGs.

3.2 Macro level

At the national level, Peru was one of the first countries to have a Gender and Climate Change Action Plan (PAGCC, for its acronym in Spanish), aimed at state action to reduce gender gaps in the fight against climate change. This Plan focuses on eight priority areas: forests, water resources, energy, food security, health, education, risk management. The PAGCC emphasises the greater disadvantage of women in rural areas and indigenous populations. Its objectives are mainly focused on technical capacity building, generation and management of disaggregated information (and recognition of ancestral and local knowledge and practices), development of policies and instruments, and the development of mitigation and adaptation measures with a gender perspective, which are also included in the Regulations of the Framework Law on Climate Change.

Peru's National Policy on Gender Equality (PNIG) is in line with the main international declarations and agreements on the protection of women's rights and gender equality⁹, identifies

⁸ Please note that the GIZ working definition for gender responsive follows the OECD Gender Equality Continuum (2022): A gender-responsive approach not only considers gender differences, inequalities, ratios and aspects, e.g., by targeting and including a certain number of women in project activities but goes a step further in responding to gender specific needs and/or requirements, or by collecting gender-disaggregated data for monitoring purposes. Gender responsiveness aims to ensure that all people can engage in and/or benefit from certain measures regardless of their gender (e.g., trainings or strategies are developed that consider gendered needs, preferences, and constraints).

⁹ Like CEDAW, Beijing Declaration 1995, Belén Do Pará, among others.

structural discrimination against women as a public problem. This policy points out that discrimination against women is caused by socio-cultural patterns that privilege the masculine over the feminine, assigning them unequal social roles to the detriment of women, reinforcing inferior and subordinate treatment that restricts their rights, opportunities and economic, social, political and cultural benefits (MIMP, 2019).

In 2019, the National Policy on Gender Equality (PNIG, 2019) was approved, whose objective is to "address, from various angles, the causes and consequences of the persistence of structural discrimination against women", in accordance with the CEDAW and the Beijing Platform for Action. The priority objective 4 of the PNIG states "to guarantee the exercise of women's economic and social rights" and considers the guidelines 4.2: Increase women's access to and control of natural, productive and patrimonial resources, whose related services are: 4.2.3: Grant rights linked to water and its associated goods in rural areas, for domestic and productive use and 4.2.4: Management of the conservation, recovery and sustainable use of ecosystems and natural resources, with the participation of women, in a context of climate change; where it is expected to work on the formalisation of agrarian property, women's rights linked to water and associated goods, management of the conservation, recovery and sustainable use of ecosystems and natural resources, training, technical assistance and financing for women's enterprises. Priority objective 5 is also linked to the development of capacities for public management with a gender perspective and the elimination of institutional and socio-cultural barriers that hinder equality (services 5.1.1, 5.3.1, 5.4.2) and objective 6, associated with information management, such as the generation of disaggregated statistical information and follow-up information on public investment and expenditure with a gender.

Regarding spaces and mechanisms for the advancement of women and gender equality associated with adaptation to the impacts of climate change, the government of Peru has the Ministry of Women and Vulnerable Populations (MIMP) and the Ministry of Environment (MINAM). Both ministries are governing bodies for national gender equality and environmental policies, respectively. They are responsible for designing guidelines, policies, norms, programmes and budgets related to women, environment and climate change respectively.

At the subnational level, through the Supreme Decree 005-2017-MIMP, the government of Peru established a mechanism for Gender Equality in each of the entities of the national and subnational governments. The main purpose of this decision is to advance towards the mainstreaming of the gender approach within state institutions and the reduction of gender inequalities at the institutional level, but also at the level of goods and services of state bodies at the national, subnational and local levels of government, including the agencies involved in the implementation of the Resilient Puna Project.

There is also a National Committee on Women and Climate Change - CONAMUCC. This organisation was set up by MINAM and made up of representatives of more than 50 organisations to strengthen women's participation in climate action. The CONAMUCC was created in response to the need to support the presence and participation of women and indigenous peoples in environmental decision-making. This space was created in the framework of the development of the Regulations of the Framework Law on Climate Change, which included a prior consultation with indigenous organisations in Peru (including women's organisations), highlighting the need to work with a gender focus and cultural and linguistic relevance between sectors and levels of government, taking the PAGCC and the PNIG as a reference and considering it as a space to make indigenous women visible, the valuation of their knowledge and the promotion of education for women and girls. The Multisectoral Strategic Plan (PEMIG) operationalises the implementation of the PNIG in the 52 services prioritised for the 21 public institutions responsible. As the MIMP is the lead agency for gender issues, the sectors must report to it on their progress in implementing the PEMIG. It is important to note, as an opportunity, the current process of formulating the National Climate Change Strategy 2050, which proposes the inclusion of cross-cutting approaches (gender, interculturality, intergenerational)

and includes a participatory process that includes indigenous and women's organisations, among other key actors at the national level.

3.3 Meso level

Women in the Andes, due to their socially constructed roles and responsibilities, and the still existing conceptions of male superiority struggle with a position of inferiority regarding access and control over natural resources such as land, water, production and commercialisation of agricultural products. Moreover, their responsibilities in the reproductive sphere, like domestic labour and family care limit their availability of time and opportunities for education, training, paid employment and participation in decision-making spaces. This translates into under-representation in different type of organisations. Despite this, women actively participate in the management of key natural resources such as water and land. While men concentrate their participation and decision making in agriculture and husbandry, women are responsible for water supply for subsistence farming, food preparation, cleaning and hygiene of family members, including the ill and disable (Carrillo & Remy, 2022).

The livelihoods of rural communities in the Andes are scarce and quite vulnerable to climatic factors: sale of alpaca fibre and vicuña wool, sale of alpaca meat, sale of textile handicraft products, sale of natural salt and iodine. The quantities available for the sale of these products are highly variable because they depend on the quality of their productive resources (quality of pastures for livestock feed, access to water sources) and environmental factors (timely and sufficient rainfall), so that their monetary income is also highly variable. The harsh climate (strong winds, low temperatures) as well as the scarcity of water sources do not allow them to grow agricultural crops or high nutritional quality pastures for livestock. Due to this precarious livelihood situation, young and middle-aged women and men travel frequently to nearby urban centres to earn supplementary income. In the absence of husbands and young people, household chores and child rearing fall more heavily on women in rural households, who may also wish to migrate permanently to the cities if their husbands can find income opportunities to make and sustain this decision.

Family economy in rural households: During fieldwork, on the question of who contributes the most money to household expenses, women's answers reveal a new situation: 40% of women contribute the most money to the household; then 30% of women state that the contribution is equal from both sides: husband and wife; while men's answers are different: 46% think that husband and wife contribute in a similar way, but 42% state that it is men who contribute the most. Women interviewed mentioned six main products as income sources, the most important being textile handicrafts (33%), alpaca on foot (20%) and guinea pigs (13%). Raising and selling guinea pigs is the most important economic activity for women, as it allows them to obtain a greater income for their households. On the other hand, men control the income obtained from the sale of livestock and by-products (alpaca on the hoof, 38%; alpaca fibre, 19%). They show little or no participation in the commercialisation of textile products (2%).

Domestic labour: Interviews conducted in the field with rural women from Cusco and Puno (November 2022) indicate that 58% of rural women dedicate 4 hours of domestic work per day, and 20% more than 4 hours. In contrast, the hours reported by men are significantly lower: only 21% of rural men spend 4 hours on domestic chores, and only 9% spend more than 4 hours. In this respect, 18% of women mentioned the acquisition of improved cookers or liquefied gas cookers, as options to alleviate the time spent cooking. In the case of rural women, if we add 4 hours of domestic work to 9 hours of productive work, the total workload is at least 13 hours a day, and the time can be longer, because many of them declare that each day they start their work at 4 am and finish at 8 pm, which means a total of 16 hours. It is known that in many cases women do not recognise as "work" activities such as tending livestock, fetching water or firewood. On the other hand, 70% of rural men reported a domestic workload of less than 4 hours, and 52% a productive workload of more than 9 hours. This makes a total of less

than 13 hours of work per day. This imbalance is greatest in households where men are absent to work outside the community, and where women head the household (40%). In terms of cooking devices, more than 80% of the respondents stated that they do not own improved cookstoves. Most rural women have traditional cookers and use firewood and dung as fuel. Even in the case of rural households that have electricity, it is used only for internal lighting, and is avoided for electric cookers or cookers, because of the costs it could generate. Firewood and dung are used to heat houses to mitigate low temperatures in the highlands.

Participation and leadership: The interviews conducted in the field indicate that rural women do not have equal access to communal government positions compared to men. In the interviews, 55% of women report having held community government positions, 20% lower than that reported by men (75%). This gap is caused by several barriers faced by rural women, such as lower levels of formal education, speaking a language other than Spanish, low empowerment and leadership, and the burden of domestic and productive work, among others. To date, there is still a wide gap in the presence of women as decision-makers. As noted above, women's political participation in decision-making in the departments prioritised by the project represents 25% compared to 75% for men. This same gap is reflected in the decision-making spaces in the territories and in the community, participation spaces in the four prioritised departments.

Access to water: According to interviewed women during fieldwork, the availability of safe water is less than or equal to 50%. The other half of households obtain water from various sources, some unreliable in the sanitary sense. According to the 2017 National Demographic Census, 48% of rural households receive water supply from a public network inside the dwelling, 13% obtain water from a well, 9% from a public network outside the dwelling, and 8% from a river, ditch or spring (INEI, 2018). In the rural Andes, the main carriers of water for human and animal consumption are women (Carrillo & Remy, 2022). According to the field results, the task of carrying water from a spring to the household is not a burden assigned only to women in rural households. It is likely that families can easily access to springs that arise from the proximity of glaciers. Regarding, access to water for farming, 51% of interviewed persons during fieldwork indicated that they have the chance to irrigate their plots when there is no rainfall, using water from streams and brooks. However, there were almost no mention to available reservoirs to capture rainwater. Irrigation tasks in plots and pastures in highland areas are considered masculine tasks.

Health: In addition to the scarcity and distance of the health posts, a significant proportion of rural women and men distrust the medical personnel who provide healthcare. Only 43% of women have declared that they have confidence in the services of the health posts, but it is worth mentioning that this degree of acceptance is higher in the districts of Puno: Nuñoa (57%) and Santa Rosa (63%). Rural men show a higher degree of trust in health posts (50%), and even more so in the districts of Puno. The reasons for rural women's distrust are mistreatment and communication difficulties due to the use of a language other than Spanish. The main reason cited by men was language differences. Regarding domestic violence, 93% of interviewed women stated that this is a reality that they face frequently. 85% of men had the same opinion. In terms of where to report violence episodes and request protection, 44% mentioned that they appeal to peasant round (*ronda campesina*), mostly made up of men.

Education: Education among farmer population of the SHAP area shows very low rates (See Table N° 7). Cusco and Puno are the departments with the highest level of population that did not pass the primary level of education, with 50% or more of their population, about one third (31%) of the population over 25 years old had only primary education, while 17% had no or only initial education. The level of access to higher education in the Apurímac area stands out as 21%, while in the rest of the departments this level is around 8%. The gender gap across the four prioritised departments for primary and secondary education is between 9 and 17 points with a more dramatic situation in Puno.

Table N°7
Gender gaps in education among farmer population in prioritised departments

Department	Education level achieved (%)						Gender gap (W-M) (%)		
	At most primary		Secondary		Superior 1/		At most primary	Second-ary	Superior 1/
	Women	Men	Women	Men	Women	Men			
National	74.0	63.8	19.5	28.7	6.5	7.5	10.2	-9.2	-1.0
Apurímac	77.6	61.8	17.1	30.4	5.3	7.7	15.8	-13.3	-2.4
Arequipa	51.7	39.3	14.5	29.2	17.2	20.1	12.4	-14.7	-2.9
Cusco	74.6	62.9	19.8	30.6	5.6	6.5	11.7	-10.8	-0.9
Puno	72.3	54.9	23.7	36.9	4.0	8.2	17.4	-13.2	-4.2

1/ Includes non-university and university higher education.

Source : INEI/ IV Agrarian Census 2012.

For women and men aged 25 and over with secondary education, Puno and Apurimac are the two departments with higher gender gap, followed by Cusco (See Table N°8). Gender gaps in secondary education rates between 10 and 21 points. These three departments overcome the national average, except for Arequipa which is slightly below. For the RP project, acknowledging the gender gap in secondary and primary education is essential to strengthen capacity building for women in EbA.

In terms of the labour force, the gap between women and men aged 15 and over is also considerable, with Arequipa having the largest gap (-22.9%), above the national average (21.8%). The participation of women and men in Arequipa is also far below the other departments and the national level. Apurímac comes second (-14.6%), while Cusco and Puno have a much smaller gap, around 10%.

Table N° 8
Gender inequality in terms of education and employment of the population at regional and national level - 2020

Department	(%) Population 25+ with at least secondary education			(%) Labour force participation rate for population 15+		
	Women	Men	Gap W- M	Women	Men	Gap W - M
National	65.9	76.4	-10.4	55.4	76.2	-21.8
Apurímac	47.9	69.0	-21.2	73.2	86.8	-14.6
Arequipa	74.5	84.5	-10.1	49.2	71.1	-22.9
Cusco	54.1	69.9	-16.8	70.6	79.5	-9.9
Puno	51.4	75.1	-24.6	73.3	81.8	-9.5

Source: INEI – Gender Indicators, 2020

Participation of women and men in farming: Between the 1994 and 2012 agrarian censuses, women's participation in farming increased significantly in the four prioritised departments. Therefore, gender gap in this section reduced by about half, especially in Apurimac, Cusco and Puno (See Table N°9). These rates are similar to the national level (from 59.5% in 1994 to 38.4% in 2012), with the exception of Puno where the reduction of the gender gap was lower. Unfortunately, the increased participation of women in farming production is not necessarily linked to the improvement of their living conditions. Indeed, rural women have felt an increase workload in farming and reduced participation of men in this sector. The low expectation of men to improve their net income from subsistence agriculture and the low productivity led men to leave farming and migrate looking for better off jobs like mining, construction, and transport. In these scenarios women are usually left to take care of the children and the elderly, the house and the farm, with few prospects for a real increase in income, due to lack of time, technology, infrastructure, access to water, to training, linkage to markets, and funding.

Table N° 9
Women and men working in farming

Department	Agricultural producers (Thousands of people)		Participation in agricultural production (in %)	
	1994	2012	1994	2012

	Women	Men	Women	Men	Women	Men	Gap W-M	Women	Men	Gap W-M
National	346.0	1 360.9	691.9	1 554.8	20.3	79.7	-59.5	30.8	69.2	-38.4
Apurímac	12.9	54.7	26.1	56.6	19.1	80.9	-61.8	31.5	68.5	-36.9
Arequipa	11.0	32.7	19.9	37.8	25.3	74.7	-49.5	34.5	65.5	-31.0
Cusco	26.4	116.6	55.3	125.4	18.5	81.5	-63.1	30.6	69.4	-38.8
Puno	45.8	136.2	85.0	128.4	25.2	74.8	-49.6	39.8	60.2	-20.3

Sources: INEI, Gender Indicators, 2022/ Agrarian Census 1994 and 2012.

Farming land managed by women and men: On average women manage between 30% to 34% less farming land than men in the four prioritised departments (See Table N°10). Available non-farming land per capita (mainly grassland) in Arequipa and Puno is higher than in Cusco and Apurímac and the national average. These gender gaps are considerable, similar to the national level, which limits the prospects of raising women's income due to lack of innovation and diversification to increase land and labour productivity. In the area of influence of the RP project it will be necessary, in addition to knowing the areas managed by women and men, to verify the ownership of the land, its productive potential (arable and non-arable, with and without access to irrigation) and who uses it, considering that within the high Andean peasant communities it is very common to allocate communal land for grazing without possession titles, let alone formal ownership titles.

Table N° 10
Farming area managed by women and men

Department	Farming land						Gender gap (W-M)					
	(Per capita in hectares)						(Surface in hectares)			(in %)		
	Total		Farming		Non-farming		Total	Farming	Non-farming	Total	Farming	Non-farming
	Women	Men	Women	Men	Women	Men						
National	4.6	7.0	1.8	3.0	2.8	4.0	- 2.4	- 1.2	- 1.2	-34%	-40%	-30%
Apurímac	2.3	3.4	1.9	2.7	0.4	0.7	- 1.2	- 0.8	- 0.3	-34%	-31%	-46%
Arequipa	17.8	21.6	1.7	2.7	16.0	18.8	- 3.8	- 1.0	- 2.8	-18%	-36%	-15%
Cusco	4.6	6.7	1.6	2.2	3.0	4.5	- 2.1	- 0.6	- 1.5	-31%	-27%	-33%
Puno	9.2	13.3	1.3	1.7	7.9	11.6	- 4.1	- 0.4	- 3.7	-31%	-26%	-32%

Source: INEI, Gender Indicators/Agrarian CENSUS 2012

Cultivated area managed by women and men by type of irrigation: In all four prioritised departments, women manage between 25% and 35% of the irrigated arable area, while men, in between 65% and 76% of the irrigated arable area. In rainfed land (rainfall only) the gap is wider, with only 15% to 31% of women managing arable area, while men manage between 69% and 85% (See Table N°11). In terms of irrigated area, the gap between women and men in Arequipa is similar to the national level, far exceeding Cusco, Puno and Apurímac. In the rainfed gap, Apurímac and Cusco are well above Arequipa. Puno has the smallest gap. The availability of irrigation water all year round is a key factor to increase the productivity of family farming, particularly for families in the Andean region, that raise livestock, such as camelids and small farming animals (guinea pigs, poultry). Livestock raising is an activity mostly carried out by women and requires fodder in the dry season (from June to October or November) to maintain the size of the herds, their productivity and income. In order to establish the baseline for the RP project in selected communities and districts, in addition to quantifying the irrigated and rainfed arable land available to women and men, it will be necessary to collect disaggregated by sex information about grasslands with irrigation potential through sowing and water harvesting.

Table N°11
Cultivated farming area managed by women and men by type of irrigation

Department	Cultivated area by type of irrigation (Thousands of hectares)				Percentage of cultivated area driven by women and men by irrigation type					
	Under irrigation		Rainfed		Under irrigation			Rainfed		
	Women	Men	Women	Men	Women	Men	Gap W- M	Women	Men	Gap W- M
National	369.7	1 148.9	412.5	1 859.3	24%	76%	-51%	18%	82%	-64%
Apurímac	27.3	50.9	10.0	55.3	35%	65%	-30%	15%	85%	-69%
Arequipa	23.9	71.5	0.1	0.3	25%	75%	-50%	31%	69%	-38%
Cusco	28.0	57.8	32.9	115.2	33%	67%	-35%	22%	78%	-56%
Puno	3.7	7.3	48.2	105.0	34%	66%	-33%	31%	69%	-37%

Source: INEI, Gender Indicators/Agrarian Census 2012

Plots managed by women and men: In all four departments, women manage a lower number of plots (2.65 hectares in average) than men (3.12 hectares) (See Table N°12). In Andean family farming, the number of plots is related not only to the total cultivable area per person and the volume of production, but also to risk management: the greater the number of plots located on different ecological levels, the lower the agro-climatic risk (whether from frost, hailstorms, drought and pests or diseases) that the farmer has to face, and the higher the production expectation.

Table N°12
Average number of plots driven by women and men

Department	Average number of plots		Gap (W – M)
	Women	Men	
National	2.2	2.4	- 0.2
Apurímac	2.5	3.2	- 0.7
Arequipa	2.2	2.5	- 0.3
Cusco	2.4	2.8	- 0.4
Puno	3.5	4.0	- 0.5

Source: INEI, Gender Indicators/ Agrarian Census 2012

Farming land tenure for women and men: Although women manage fewer plots than men in all departments, women are more frequent owners than men in Apurímac, Cusco, Arequipa and at the national level, with the exception of Puno. Conversely, there are more male communal landowners than female communal landowners in these same regions, with the exception of Puno (Table N°13). Leasing is less common, except in Arequipa, and particularly by men, with a significant gender gap.

Table N° 13
Farming land tenure for women and men

Department	Tenure status of the plot of land											
	Owner			Community owner			Tenant			Possessor		
	Women	Men	Gap W- M	Women	Men	Gap W- M	Women	Men	Gap W- M	Women	Men	Gap W- M
National	76.5	71.1	5.4	11.6	15.8	-4.1	5.0	5.5	-0.5	3.5	4.5	-1.0
Apurímac	73.2	65.9	7.4	17.1	23.9	-6.8	4.6	4.2	0.3	4.0	4.9	-0.8
Arequipa	80.7	76.5	4.1	1.2	1.3	-0.1	9.2	13.3	-4.1	2.9	3.1	-0.2
Cusco	36.3	30.7	5.5	55.4	61.4	-6.1	3.0	2.5	0.5	3.6	4.0	-0.4
Puno	90.7	91.5	-0.8	5.1	4.8	0.3	2.0	1.4	0.6	0.9	1.3	-0.4

Source: INEI, Gender Indicators/Agrarian Census 2012

Farming area cultivated by women and men according to the destination of production: The largest area cultivated for commercialisation is in Cusco and Arequipa followed by Apurímac, with a gap in favour of women in Apurímac and Arequipa. In Puno, self-consumption and

animal feed production predominate, accounting for more than 80% of the cultivated area. For self-consumption, women cultivate more area than men in all four regions, particularly in Apurímac and Puno, but below the national average. Both results show the connection women producers have with markets and also their greater concern than men for feeding their families.

Table N°14
Farming area cultivated by women and men by destination of production

Department	Destination of production (%)											
	Sale			Self-consumption			Feed for their animals			Self-supply		
	Women	Men	Gap W- M	Women	Men	Gap W- M	Women	Men	Gap W- M	Women	Men	Gap W- M
National	58.1	61.2	-3.0	22.1	16.4	5.7	18.9	21.6	-2.7	0.9	0.9	0.0
Apurímac	48.3	44.4	3.9	39.0	35.2	3.8	12.5	19.9	-7.4	0.2	0.5	-0.3
Arequipa	65.6	63.9	1.8	7.0	5.6	1.3	26.9	29.9	-3.0	0.5	0.6	-0.1
Cusco	65.0	65.1	-0.1	23.6	22.8	0.8	11.0	11.1	-0.1	0.5	1.1	-0.6
Puno	12.6	15.6	-3.0	45.0	42.6	2.5	40.1	40.1	0.1	2.2	1.8	0.5

Source: INEI, Gender Indicators/ Agrarian Census 2012

Women and men who manage plots and received training, technical assistance or business counselling: With circa 15% of women and 25% of men receiving training, technical assistance (TA) or entrepreneurship assistance (EA), the gap between women and men varies between -4.4% and -10.2%, exceeding the national average in three of the four departments (See Table N° 15). The training, technical assistance and business advice received, when they are conducted timely and appropriate to the circumstances of producers, are key to improve producers' access to markets and to non-financial and financial services, resulting in higher productivity and income.

Table N° 15
Women and men who manage plots and received training, technical assistance or business counselling

Department	Total agricultural producers (x 1000 pp)		Received training, TA or EA 1/ (x 1000 pp)		Percentage receiving training, TA or EA 1/		Gender gap (W – M)
	Women	Men	Women	Men	Women	Men	
National	691.9	1 554.8	65.8	253.5	9.5	16.3	-6.8
Apurímac	26.1	56.6	2.3	9.1	8.7	16.1	-7.4
Arequipa	19.9	37.8	3.1	9.7	15.4	25.6	-10.2
Cusco	55.3	125.4	8.1	28.9	14.6	23.0	-8.4
Puno	85.0	128.4	8.3	18.2	9.8	14.2	-4.4

1/ Includes those receiving technical assistance (TA) and business advice (EA).

Source: INEI, Gender Indicators / Agrarian Census 2012

Access to credit for women and men working on farming: Of the total number of farming producers, only between 5% and 20% of men took out a loan; and for women it is even lower, between 4% and 12%, with a negative gap for women and a ratio of 1 to 8 among departments in both cases (See Table n°16). Arequipa and Apurimac are above the national average, Cusco and Puno below. For both women and men who took out a loan, the approval rate is high, between 89% and 95%, and the gap between the two is small, around + 1%. Unfortunately, after a loan is approved, there is no available public information on payment conditions and arrears.

The extreme shortage of capital of small producers does not allow them to invest in infrastructure (e.g. planting and harvesting water, sheds, greenhouses, road improvements), or to introduce new cultivation and grazing practices (e.g. sprinkler irrigation for pasture cultivation, mobile roosts to facilitate herd management) for the benefit of their families, and particularly women.

Table N° 16
Women and men working on farming that access to credit

Department	Total farming producers (x 1000 pp)		They managed credit					They granted (him/her) a loan				
			Thousands of people		% producers			Thousands of people		% producers who obtained a loan		
	Women	Men	Women	Men	Women	Men	Gap W - M	Women	Men	Women	Men	Gap W - M
National	691.9	554.8 ¹	44.2	161.2	6.4	10.4	-4.0	39.6	146.0	89.5	90.6	1.1
Apurímac	26.1	56.6	2.2	7.2	8.6	12.7	-4.1	2.1	6.8	93.1	94.1	1.0
Arequipa	19.9	37.8	2.4	7.7	12.1	20.5	-8.4	2.3	7.4	94.4	95.6	1.2
Cusco	55.3	125.4	3.9	11.1	7.0	8.8	-1.9	3.5	9.8	89.6	88.7	0.9
Puno	85.0	128.4	3.4	6.4	4.0	4.9	-1.0	3.0	3.6	89.6	88.6	1.0

Source: INEI, Gender Indicators/Agrarian Census 2012

Reasons why women and men farmers do not apply for credit: In Apurímac, Arequipa the main reason why women and men do not apply for credit is because they consider "not to need it" (Table N° 17). For all reasons for not applying for credit, the gap between women and men in the four departments is variable and less than or equal to 2%. Both, women and men, have the same reasons for not taking out a loan. For people who "think they will not get it", there is a positive gap in all departments, but less than 2%, which could reflect some degree of women's insecurity or fear of credit. As for "other reasons" (at least 3 very different reasons), the gap is slightly negative in all departments and at the national level.

Table N° 17
Reasons why women and men farmers do not apply for credit

Department	Reasons for not taking out credit (as a percentage of total who did not take out credit)																	
	No need			High interest rates			Lack of collateral			Believe will not get			Cumbersome formalities			Other reasons 1/		
	Women	Men	Gap W- M	Women	Men	Gap W- M	Women	Men	Gap W- M	Women	Men	Gap W- M	Women	Men	Gap W- M	Women	Men	Gap W- M
Apurímac	42.1	42.7	-0.5	30.1	29.5	0.7	10.5	12.2	-1.7	6.6	4.8	1.8	5.7	5.4	0.3	4.9	5.4	-0.5
Arequipa	38.2	38.3	-0.1	30.4	30.1	0.3	12.5	12.4	0.2	5.3	4.4	0.9	6.7	7.0	-0.3	6.9	7.9	-1.0
Cusco	35.4	33.4	2.0	38.2	39.3	-1.0	8.9	10.1	-1.1	5.2	4.3	0.9	7.4	7.5	-0.1	4.9	5.4	-0.6
Puno	21.1	20.7	0.4	46.0	46.2	-0.2	13.2	13.9	-0.8	8.0	6.2	1.8	6.7	6.5	0.1	5.1	6.5	-1.4
National	36.8	34.9	1.9	30.0	28.0	2.0	15.2	18.1	-2.9	6.5	5.9	0.6	5.6	5.8	-0.2	5.9	7.4	-1.5

Source: INEI, Gender Indicators/Agrarian Census 2012

Farming women and men that belong to an association: With more than 72% of women and men organised in associations, committees or cooperatives, Arequipa far exceeds the other departments, including the national level (See Table N° 18). It is followed by Cusco and Apurímac, with 25% or less, and lastly by Puno with less than 6% and 8%, respectively for women and men.

Table N° 18
Farming women and men that belong to an association

Department	If you belong to an association committee or cooperative (in %)		Gap gender (W-M)
	Women	Men	
National	22.0	23.3	- 1.3

Apurímac	19.8	23.3	- 3.5
Arequipa	72.5	72.4	0.2
Cusco	25.5	24.4	1.1
Puno	5.8	7.8	- 2.0

Source: INEI, Gender Indicators/Agrarian Census 2012

Membership in a producer organisation, be it an association or a cooperative, is key to enable both women and men to adapt to climate change. This membership provides manageable transaction costs in commodity markets, as well as non-financial and financial services to increase their resilience, based on the ecosystems in which they live. For this reason, although there is no apparent wide gap in any of the departments, this variable of women's and men's membership in a productive organisation is central to a project baseline, as well as to the Gender Action Plan to be implemented, in order to prevent a gap from opening up during project implementation. Finally, in addition to their participation as members, the occupation of leadership positions by women in the organisations will facilitate the closing of gaps, particularly those related to access to assets, training, technical advice and financing to improve their living conditions, both at the level of their families and their communities.

Conclusion 2 for the project

In general terms, women in the SHAP area face more barriers than men that hinder the achievement of gender equity in climate change adaptation measures. Women face unequal access to resources (use, control) and receive lower income. They also have limited access to microfinance; less opportunities and time for education, training, and technical assistance/capacity to develop and implement nature-based solutions and climate-resilient agribusinesses. It is important not only to work to close these gender gaps but also to address the weaknesses in institutions and regulation policies, to implement nature-based solutions and climate resilient value chains, while also increase availability of still limited gender-desegregated information (data, systematization of experiences, lessons learnt, behavioral changes). For this, it is essential that the actions undertaken within the project do not result in an increased workload for women who already have a double burden between productive and reproductive work (domestic and care work). For this, the project must recognize women's knowledge, needs and expectations to generate effective responses in terms of building social and technical capacities, as well as opportunities for their economic autonomy, participation, and representation in decision-making.

Chapter 4: Gender responsiveness and expertise, as well as gender equality in the partner organisation(s)

Public Partner Organisations

Peru is one of the first countries to have a Gender and Climate Change Action Plan (PAGCC, 2015), aimed at state action to reduce gender gaps in the fight against climate change. This Plan focuses on eight priority areas: forests, water resources, energy, food security, health, education, risk management. Among the main scopes designed, this policy proposes the strengthening of technical capacities, generation and management of disaggregated information (and recognition of ancestral and local knowledge and practices), development of policies and instruments and the development of mitigation and adaptation measures with a gender focus, which is also included in the Regulation of the Framework Law on Climate Change.

In 2019, the National Policy on Gender Equality was approved (PNIG, 2019)¹⁰ whose objective is to *"address, from various angles, the causes and consequences of the persistence of structural discrimination against women"*, in accordance with the CEDAW and the Beijing Platform for Action. Priority objective 4 of the PNIG states "Guarantee the exercise of women's economic and social rights" and considers the guidelines 4.2. Increase women's access to and control of natural, productive and patrimonial resources, whose related services are: 4.2.3: *Granting of rights linked to water and its associated assets in rural areas, for domestic and productive use, and;* 4.2.4: *Management of the conservation, recovery and sustainable use of ecosystems and natural resources, with the participation of women, in a context of climate change.*

In both cases, it is expected to work on the formalisation of agrarian property, women's rights linked to water and associated goods, conservation management, recovery and sustainable use of ecosystems and natural resources, training, technical assistance and financing for women's enterprises. Likewise, priority objective 5 of the PNIG is linked to the development of capacities for public management with a gender perspective and the elimination of institutional and socio-cultural barriers that hinder equality; and objective 6 of the same policy, associated with the generation and management of statistical information disaggregated by sex, and follow-up information on public investment and expenditure with a gender perspective. In the sense of the previous paragraph, through Supreme Decree 005-2017-MIMP, the Peruvian government established a mechanism for Gender Equality in each of the entities of the National and Regional Governments, including the attached bodies. The main purpose of these spaces is to advance towards gender mainstreaming within State institutions and the reduction of gender inequalities at the institutional level, as well as at the level of goods and services of State bodies at the three levels of government, including the sectors and agencies involved in the implementation of the Resilient Puna Project.

At national level, public agencies in Peru are required by regulation (Supreme Decree N° 005-2017-MIMP) to establish a gender instance (Committee, Commission or Working Group) in their institutions. Specifically, the Ministry of Women and Vulnerable Populations (MIMP) and the Ministry of the Environment (MINAM), are governing bodies for national gender equality and environmental policies, respectively. They are the main governing bodies for guidelines, policies, norms, programmes and budgets related to women, environment and climate change respectively.

MINAM is the ministry that has shown greater interest in incorporating the gender approach both at the institutional level and in its public policies. In 2017, MINAM created the Working Group for Gender Equality (Ministerial Resolution N° 225-2017-MINAM). MINAM created and installed the first National Committee on Women and Climate Change - CONAMUCC, made up of representatives of more than 50 organisations at the national level, whose objective is to strengthen women's participation in climate action. This committee responds to the need to strengthen the presence and participation of indigenous peoples and women in environmental decision-making. This space was created in the framework of the development of the Regulations of the Framework Law on Climate Change, which included a prior consultation with indigenous organisations in Peru (including women's organisations), highlighting the need to work with a gender perspective and cultural and linguistic relevance between sectors and levels of government, taking the PAGCC and the PNIG as a reference and considering it as a space with emphasis on indigenous women, valuing their knowledge and promoting the education of women and girls.

SERNANP as part of MINAM

¹⁰ Supreme Decree 008-2019-MIMP, April 2019. The PNIG-PEMIG Multisectoral Strategic Plan operationalises the implementation of the PNIG in the 52 services prioritised for the 21 public institutions responsible. At the same time, it also sets out the corresponding budget allocations and the executing units in charge. As the MIMP is the lead agency on gender issues, the sectors must report to it on their progress in implementing the PEMIG.

SERNANP, in 2018, launched the Working Group for Gender Equality through Presidential Resolution N° 204-2018-SERNANP. SERNANP has recently modified its regulations to promote and guarantee the full and effective participation of women and men in the Management Committees of each of the 70 Natural Protected Areas (NPAs), which are located in 22 departments of the country. It has also designed the tool *¿How are we doing?* to analyze the processes, progress, and priorities of these management committees. This tool promotes the dialogue, collaborative decision-making, and the implementation of joint actions, while integrating intergenerational, intercultural and gender approaches. Together with PROFONANPE, these institutions have organised two contests "*Entrepreneurs by nature and Entrepreneurs by Cotahuasi*". These contests aimed at strengthening private entrepreneurship that contribute to the sustainable use of natural resources associated with PNAs. The contests also seek to empower the role of women, promoting that 20% of the winning proposals must be led by women.

Moreover, SERNANP's Annual Welfare and Social Development Plan 2022 includes the need to address gender inequalities and address gaps. SERNANP has also to promote and measure women's participation in national protected areas. The organisation has a gender specialist responsible for reporting on the SERNANP indicators of the PNIG but has not been involved in the elaboration of the RP project concept note. As part of the Annual Staff Welfare Plan, training on gender, audio-visual pieces on violence, reporting and harassment and knowledge on reporting channels have been proposed. Different contributions that women can make in "traditionally" male activities are recognised, e.g. trail building, and work as park rangers. However, some signs, such as the internal work regulations, suggest that there is no comprehensive knowledge of gender equality and that there is a need to strengthen institutional knowledge of the institution's internal regulations and the projection of its functions. In the formulation of the people development plan (2022) the diagnoses of training needs and demands are gender neutral and do not show the differentiated needs and interests of women and men, considering gender roles, the use of time, the reconciliation of family, personal and work life, and aspects of the organisational climate that allow for egalitarian relationships.

MIDAGRI

MIDAGRI has a Commission for Gender Equality (Ministerial Resolution N°. 0347-2017-MINAGRI). In 2021, it launched the Directorate for the Promotion of Farming Women to articulate, propose and promote technical and regulatory instruments and mechanisms to increase the participation of women in agriculture interventions from the three levels of government. In 2022, MIDAGRI approved the Strategy "Entrepreneurship of Rural and Indigenous Women" to finance enterprises formed by rural and indigenous farming women and to provide them with technical advice in agriculture, forestry, husbandry, and handicraft. MIDAGRI intervenes in the study area through: Agrorural, the Sierra Azul Fund and the Agroideas Programme. Agrorural seeks to contribute to the competitiveness of agricultural production of small and medium producers through the promotion of associativity, the adoption of agricultural technology, among others. Interventions are mainly focused on improving cultivated pastures and sheds.

MIDAGRI has shown through different projects, its willingness to mainstream gender and/or equal opportunities between women and men. However, there is still a need to strengthen awareness and knowledge on the subject and to provide guidance on how the gender approach can be applied in programmes and projects throughout the entire cycle. During fieldwork, it was possible to recognise that knowledge on gender issues is dispersed and heterogeneous within MIDAGRI and the Agrorural and Agroideas, for example. Some representatives interviewed did not identify gender inequalities in their institutions. A few representatives even considered no need for a gender perspective in the work, as "there is no discrimination against women". Nor Agrorural or Agroideas have gender specialists.

For Agrorural, it is still necessary to consider the gender approach as an integral part of the work (not a separation between "technical" issues and the invisibilization of the "social" during

implementation). The same case is not measured for Agroideas. The projects do not include a gender perspective, it is not introduced, it is not monitored. In the few cases where women have been considered, at least in the narrative of the plans, no disaggregated goals or indicators are considered, and the differentiated needs of men and women are not taken into consideration in order to achieve equal benefits; these could not be measured either, given that, as has been mentioned, no disaggregated indicators (or initial analysis) are considered, nor do they include strategies that aim to change unequal power relations.

Concrete facts within the sector can strengthen/sensitise the need for gender work: Knowledge and evidence that demonstrates the existence of gender gaps, the different ways in which men and women can be affected by a given phenomenon, is a factor that contributes to the internalisation of gender work. For example, the case of the SODEGA system, which includes information disaggregated by sex and age, once it was identified that women are at a disadvantage when negotiating their products. The same case with successful experiences that can be used as a future reference, such as the case of FORMAGRO, which is present among the officials who learned about the case.

The creation of the Directorate for the Promotion of Women Agricultural Producers (DPMPA), as well as the strengthening/promotion of family farming, provide an opportunity to strengthen the work in favour of gender equality, particularly within the framework of the RP project. However, the DPMPA needs to be strongly involved in the design of the project and, in general, in connection with the other programmes and projects of MIDAGRI, in order to ensure their articulation and the strengthening of the inclusion of the gender approach in the initiatives, in accordance with the objectives of this directorate.

The publications on social media, as well as the calls for personnel reviewed, generalise the jobs as male, which from the invitation shows a gender bias. This demonstrates the need to strengthen gender awareness and knowledge across the entire sector. Considering these results, it can be recommended to strengthen gender knowledge at the organisational level as a starting point for the design of management documents and institutional policies and to overcome the challenges identified.

Finally, despite the efforts and initiatives in the public stream, gender instances are still weak mechanisms, characterized by high turnover and lack of resources and capacities for the optimal fulfillment of their functions. An evaluation conducted in 2017 of The National Plan for Gender Equality (PLANIG), points out that no institution reaches the 12 indicators proposed for the implementation of this plan, although MINAM complied with 10 of them, being in the group with more indicators fulfilled, while MINAGRI reached 8 indicators (Guevara, 2017).

Other Partner Organizations

PROFONANPE

PROFONANPE is responsible for mobilising, managing and channelling resources for the implementation of programmes and projects linked to the country's environmental sustainability. It coordinates and contributes to the development of proposals jointly with public and private institutions, as in the case of its participation in the elaboration of the Concept Note and the formulation process of the Resilient Puna Project. In this context, both in the development of proposals and in the management and implementation of the funds obtained, Profonanpe must ensure compliance with the administrative-financial regulations, as well as the social, environmental and gender standards (including policies to prevent harassment, violence and exploitation) of the donors. Profonanpe has a series of policies, strategies and tools compiled in an Environmental and Social Management System (ESMS), which responds to international standards to which the entity is accredited, such as the Green Climate Fund (GCF) and the Adaptation Fund (AF). The ESMS comprises two main elements: the Social and Environmental

Policies (SEP) and the Social and Environmental Safeguards Manual, which allows the SEP to be operationalised. SEP 7 refers to the mainstreaming of the gender approach, the differentiated effects on men and women in the interventions and the promotion of greater participation of women, as well as the contribution to maintaining equitable power relations. SEP 8 is linked to indigenous peoples, both SEPs linked to the RP project. Leadership positions are distributed between men and women; in the management area, women represent 67%. As for the Board of Directors (representatives of the public sector, NGOs, private sector/cooperation), women represent 25%. Profonanpe has an Equity, Diversity and Inclusion Policy for the areas of intervention of the organisation. It also has a Gender Policy and an Institutional Gender Plan for work within the institution. Yet, Profonanpe does not have a gender specialist within its team.

Instituto de Montaña (IdM)

IdM has expressed its willingness to mainstream gender and/or equal opportunities between women and men and has collected some good practices that reflect this work. Yet, no policies, plans or management instruments have been accessed in order to review the level of gender mainstreaming in these documents and how it is carried out in practice in the organisation. IdM counts with gender specialist to highlight the work done for women's empowerment and their contributions to environmental conservation. In addition, IdM has good practices in the empowerment of women in environmental projects. An interesting fact has been the promotion of the slogan for the Mountain Day in which IdM has participated: "Women who move mountains" and the different spaces for dialogue that have taken place around this theme. There are, however, spaces

Conclusion 3 for the project

Public institutions

Although there is a formal institutional framework for gender mainstreaming in public institutions, the mechanisms to operationalise them are fragile. This is characterised by a high rotation of human capital and a type of management that usually lacks resources and capacities for the fulfilment of its functions. Despite these challenges, MINAM and SERNANP has shown a sustained interest in incorporating the gender approach in their actions that should be strengthened and visible. In the case of MIDAGRI, some efforts have been made to promote gender equity. This is indeed an opportunity to better promote the incorporation of the gender approach into MIDAGRI actions.

Other partner organizations

PROFONANPE has the instruments and capacities to mainstream gender in its activities. In general, the Resilient Puna project has the potential to contribute to gender mainstreaming both in management and planning instruments and through greater participation of women in training, dialogue, and decision-making processes.

Chapter 5: Gender responsiveness and expertise in the project/ among project staff in the country

5.1 Equal opportunities in the team

The team members have not yet been appointed, but all of them will be trained in gender equity and interculturality approaches. Nevertheless, it is important to mention that the project will also count with a focal point for safeguards and gender. It is recommended, that this person should have expertise in the field of natural resources management, climate change, and a gender expertise, with clear know-how in mainstreaming gender in projects implementation from a cross-cutting perspective.

In addition, the project will be part of GIZ South Cone (*Cono Sur*), to which the offices of GIZ in Peru belong. GIZ South Cone has an Intercultural and Gender Group (GIG). This group follows the guidelines established by the Strategy as well as the BMZ Gender Action Plan (2014; 2016) and the GIZ Gender Strategy (2019). Additionally, GIZ South Cone in Peru has a Policy and Protocol against sexual harassment and gender discrimination in the workplace in accordance with Law N° 27942, as amended by Legislative Decree N°1410, and its regulation (Supreme Decree N° 014-2019-MIMP). In 2020, staff received a specialised training about institutional instances of counselling and/or listening and action against sexual harassment, such as the Sexual Harassment Intervention Committee (IHS). Twenty-five people participated, 18 women and 7 men from various GIZ programs and projects in Peru. Also, in 2020, a sensitisation process of six workshops was carried out for GIZ staff in Peru, with the participation of 25 employees in average, in each one of these sessions. An analysis of the staff shows that out of 152 employees, 56% are women (GIZ Peru, 2020). There are no differences in remuneration between male and female staff (GIZ Peru 2019). Currently, there are five colleagues with national contracts in the management team, three of whom are project managers.

5.2 Gender expertise of the team members

All GIZ South Cone staff in Peru receive a training in gender topics during their induction process. Therefore, new staff will be aware of the existence of an internal policy of zero tolerance to discrimination and gender violence. GIZ South Cone in Peru has a gender focal point, supported by the Gender and Interculturality working group (GIG), which allows for the exchange of experiences among colleagues from different projects and sectors, as well as collegial advice related to gender-responsive project design.

Conclusion 4 for the project

It is recommended that the team, once constituted, should have the necessary support to strengthen its knowledge and skills in gender mainstreaming, in order to ensure that project implementation is gender-responsive and careful not to generate negative impacts. Gender mainstreaming with intersectionality must be considered in all the activities to be carried out within the framework of the project.

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Gender Action Plan

GCF Funding Proposal

Resilient Puna Project

Resilient Puna: Ecosystem based Adaptation for sustainable
High Andean communities and ecosystems in Peru

14.12.23, v.04

Deutsche Gesellschaft für Internationale Zusammenarbeit – GIZ GmbH

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1. Introduction

The “Resilient Puna: Ecosystem-based approaches for sustainable High Andean communities and ecosystems in Peru” “Resilient Puna” project (hereafter “Resilient Puna”) aims to increase the resilience of Andean communities in the departments of Arequipa, Cusco, Puno, Lima (Yauyos) and Apurímac, through the management, conservation and restoration of high Andean ecosystems; to also promote an increased access to public and private financing and a stronger territorial planning towards Ecosystem-based Adaptation (EbA). It is organised into three components: (i) Puna ecosystems are restored, conserved and better managed to support climate resilient livelihoods, through the implementation of EbAs measures; (ii) Public and private financing for EbA measures and climate resilient livelihoods are in place and accessible for vulnerable communities in the Puna Ecosystem; and, (iii) EbA and climate resilience are mainstreamed into multilevel landscape governance instruments. The project will invest in EbA measures and Climate Resilient Value Chains, providing technical support to communities, producer associations and cooperatives to achieve sustainable value chains; directly benefiting the conservation and restoration of 23,914 hectares of high Andean ecosystems.

According to the GCF gender guidelines, one of the aims of combating climate change is to promote mechanisms to increase capacity for effective climate change planning and management in developing countries. This also includes a focus on women living in vulnerable communities. Specifically, the GCF proposes the need to develop a Gender Action Plan (GAP) that details the actions and measures for the empowerment of women, from the design of the intervention, the implementation and evaluation of the project, respecting and valuing the contribution of women and men. Climate change and adaptation requires the intersection of approaches that allow for the sustainable improvement of living conditions of people in coexistence with the ecosystem, especially women and girls who have been “left behind” in the attention of the main economic, climate and development policies. Therefore, key elements across this GAP are:

- (a) Supporting innovations and measures to combat climate change through a comprehensive gender perspective,
- (b) Stimulate climate investments that:
 - ✓ (i) Promote gender equality through climate change mitigation and adaptation measures.
 - ✓ (ii) Minimise social, gender and climate-related risks in all climate change action.
- (c) Contribute to reducing the gender gap in situations of social, economic and environmental vulnerability and exclusion, exacerbated by climate change, through GCF climate investments that mainstream gender equality issues.

The central objective of the GAP is to propose measures to break down barriers and reduce asymmetrical situations in the care, exploitation and use of natural resources, as well as the rights of people, especially women, within a natural ecosystem based on the Gender Analysis (see Annex 8a) conducted for the project. The design of gender measures and indicators are oriented towards developing and implementing gender-responsive¹ approach in order to increase women's well-being

¹ Please note that the GIZ working definition for gender responsive follows the OECD Gender Equality Continuum (2022): A gender-responsive approach not only considers gender differences, inequalities, ratios and aspects, e.g., by targeting and including a certain number of women in project activities but goes a step further in responding to gender specific needs and/or requirements, or by collecting gender-disaggregated data for monitoring purposes. Gender responsiveness aims to ensure that all people can engage in and/or benefit from certain measures regardless of their gender (e.g., trainings or strategies are developed that consider gendered needs, preferences, and constraints).

by taking into account gender-specific needs, preferences and overcome gaps to ensure that women's rights are advanced and that they will benefit from programme activities.

In Peru, although progress has been made, especially in the formal regulatory sphere, gender inequality persists, which can be clearly seen in the gender gaps and barriers to the full enjoyment of the human rights of women. These gaps are aggravated in the case of Peruvian women who face multiple factors of discrimination and exclusion (ethnic origin, disability, religion, class, sexual orientation, gender identity, place of residence, etc.). The persistence of gender gaps at all levels and in all social spheres constitutes one of the main obstacles to sustainable development that leaves no one behind. In the context post Covid pandemic, it is urgent to work on gender gaps that have been reinforced, partly because of women's increasing burden of domestic and unpaid care work.

Women, due to their socially constructed roles and responsibilities, and the persistence of conceptions of male superiority in the Peruvian society, are in an inferior position with regard to access to and control of natural resources and decisions about land, water, production and sale of agricultural products. Therefore, as they are not considered as relevant actors, their needs are invisible, and their contributions are not taken into account. Equally, the fulfilment of roles in the reproductive sphere (domestic chores and family care) as almost exclusive responsibilities, overloads their daily workday, limiting the availability of time resources and access to opportunities for education, training, paid employment, and participation in decision-making spaces, which translates into under-representation in community organisations, in water users' organisations, and in decision-making spaces. This discrimination is rooted in norms and an institutional culture that reinforce women's inferior or subordinate status, as well as in the unequal assignment of hierarchical roles. In this context, violence against women is one of the main instruments to keep women in a situation of subordination, and as a means of disciplining them to prevent their access to rights, as occurs in cases where they try to access decision-making positions. This not only prevents them from participating in social, political, economic and cultural decision-making processes and bodies, but also makes it difficult and hard to transform these spaces that reproduce patriarchal structures of domination.

This project “Resilient Puna” will address existing gender gaps, especially since women are more vulnerable than men to the effects of climate change. Indeed, rural women are often in charge of collecting water for cooking and cleaning, using the land for livestock grazing, fetching firewood, among others. Women use and interact with natural resources and ecosystems on a daily basis. They are also the first to feel the effects of climate change when they have to travel longer distances to find resources to look after their families. Gender inequality and social exclusion only increase the negative effects of unsustainable and destructive environmental management on women and girls.

Considering the gender gaps, the ongoing discrimination and violence against women in Peru, this Gender Action Plan (GAP) proposes interventions in all phases of the Resilient Puna Project cycle. The project will also conduct a monitoring strategy to evaluate and verify that no negative results are produced (under the principle of do no harm).

Gender impact and related risks

Considering that in the Andean region women play an important role in small-scale agriculture and water harvest but are still unrepresented in local level decision making. Women are exposed to a very high workload and a high degree of domestic violence. The project will promote women's active participation in the development of value chains and income-generating measures and their involvement in decision-making and territorial planning processes and therefore enhance living conditions and economic opportunities of Andean women.

Unintentional negative impacts could be that active participation of women might not be welcome by their partners/husbands/sons and might lead to conflicts within families and/or communities and in some cases lead to domestic violence. Another risk might be that decision making bodies in traditional communities and/or governmental entities allow women's participation but don't respect women's opinion and side-line them, which could lead to frustration and finally resignation of women. Similarly, women lead businesses could not thrive due to resistance by men lead business partners.

Methodology

The methodology of this Gender Action Plan involved a desk review and analysis of available secondary data sources such as national policies and work of national mechanisms, as well as a comprehensive review of secondary literature that includes various research reports, articles and papers on the subject from both national and international organisations. It is also based on fieldwork in Andean districts that are potential users of the project, conducted between August and November 2022. Other gender analyses carried out for projects implemented in Peru and Ecuador were also utilised in carrying out this study.

2. Priority areas for Gender Mainstreaming

2.1 Gender dimensions in country/ region/ sector

Gender institutional framework

Peru is a state that prohibits discrimination and has committed to close gender gaps by signing international agreements and treaties and launching different mechanisms nationally for mainstreaming the gender approach in public policy. At the international level, Peru is part of the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW, 2010) and the Inter-American Convention on the Prevention, Punishment and Eradication of Violence against Women (Convention of Belém do Pará); the Beijing Declaration and World Platform (1995); and the 2030 Agenda. In terms of climate change, Peru is also member of the Commission on the Status of Women (CSW) that declares that gender equality and the empowerment of women and girls are key aspects of climate change and disasters ´reduction (CSW66). It reaffirms the leadership of women and girls as main agents of change in safeguarding the environment and addressing the adverse effects of climate change. Peru has also signed the Lima Work Programme on Gender to promote gender balance and gender-responsive climate policy, launched in 2014, during the 20th UNFCCC Conference of the Parties (COP) in Lima.

At the national level, in 2015, Peru launched a Gender and Climate Change Action Plan (PAGCC) to reduce gender gaps in the fight against climate change. Moreover, the National Gender Equality Policy (PNIG) launched in 2019, identifies structural discrimination against women as a public problem. It includes priority objectives to increase women's access to and control over natural resources, capacity building and the elimination of institutional and socio-cultural barriers, information management and monitoring of information on investment and public spending with a gender perspective. To promote the gender approach in the fight against climate change with the civil society, in 2019, the MINAM installed the National Committee of Women and Climate Change (CONAMUCC) as an open space for the informed, effective, timely and continuous participation of women in Peru's comprehensive climate management. CONAMUCC hosts more than 50 women's organizations and NGOs at the national level to work on gender and climate change issues.

Despite these institutional efforts, there is structural discrimination against women which limits the state capacity at the local and subnational levels to actually implement these policies. As a consequence, women lack opportunities to fulfil their autonomy (physical, economic and decision-making), increasing gender gaps in different dimensions such as education, reproductive health, empowerment, labour market, political participation, use of time, among others. This situation worsens especially in rural settings including the south high Andes of Peru (SHAP area).

Gender barriers in the rural Andes and opportunities for gender mainstreaming

According to the conclusions drawn from the Gender Analysis conducted for the project (see Annex 8a) following Gender barriers are faced in the rural Andes:

Barrier 1: Economic and financial barriers based on conclusion 2

Women and men in de Andes have inequal access to loans. In 2012 only between 5% and 20% of men farmers took out a loan; and for women farmers it is even lower, between 4% and 12% (INEI 2012). Moreover, income generation is also unequal among Andean women and men. According to the National Agricultural Census (2012), there are more women with no income of their own. For example, in Apurimac 32.8% of women no have income in their own compared to only 12.1% of men. In the case of Arequipa women without income of their own (38.1%) while men (22.3%). In Cusco

women (30.5%) and men (18.2%). Finally in Puno, women 25.1% while men 17.4%. In the case of women in agricultural occupations, 37.6% are unpaid family workers and only 46.3% of women in agricultural occupations received cash income.

Although women's participation in farming production has increased in the Andes, this growth has not necessarily translated into improvement in women's living conditions. When men leave communities looking for better jobs, women have to take care of the household and the farm, with limited access to resources and prospects for better income due to lack of time, access to water, linkage to markets, and capital (workforce). Rural women raise domestic animals, process primary products or sell their small surpluses in local markets, while men's participation in productive activities is more closely linked to more competitive and profitable value chains². In addition, another important gender gap in the project's region is related to the limited access and control over natural resources due to socially constructed gender roles and male superiority. On average, women manage between 30%-34% less farming land than men in the SHAP area.

Barrier 2: Technical barriers based on conclusion 2

Women have limited opportunities for education, training, paid employment, and participation in decision-making spaces, perpetuating gender gaps. Despite this, women actively participate in managing key natural resources such as water and land. The availability of safe water in rural households in the Andes is less than 50%, with women being the main carriers of water for human and animal consumption, and while irrigation tasks are considered masculine, families can easily access water from springs. Regarding the family economy in rural households, breeding and selling guinea pigs is the most important economic activity for women, while men tend to control the income from livestock and by-products and show little or no participation in the commercialization of textile products.

Rural women also face other barriers (lower levels of education (indicated by a 23% illiteracy rate compared to 8% among men), speaking a language other than Spanish (as over 35% of women use Quechua as their mother tongue), low empowerment and leadership and the burden of domestic and productive work, among others) that prevent them from having equal access to communal government positions compared to men, leading to a significant gap in women's participation in decision-making inside the communities, but as well as in decision-making spaces local, regional, and national levels.

Finally, the incorporation of the gender approach in the territorial planification processes and investment projects (including measures acknowledging women's knowledge, needs, responsibilities and domestic burden) development is a pending and necessary task. It is necessary to include conceptual frameworks and formal tools of the Peruvian State that incorporate the needs and women's visions from the formulation, implementation and evaluation of projects.

Barrier 3: Knowledge and information barriers based on conclusion 2

Women tend not to participate in briefings outside their homes and also receive less training, technical assistance, and entrepreneurship assistance, which limits them on the one the one hand to have an informed decision in case they participate in decision-making platforms and on the other hand their prospects for increasing their income, and emphasizes the importance of providing timely and appropriate training to improve access to markets and financial services. There are also low loan rates among farming producers, particularly women. Women and men tend to be organised in associations, committees, or cooperatives in the same rate in Arequipa, Apurimac, Cusco but show poor association in Puno (less than 8%). In addition, gender traditional norms in the Andes often prevent women from achieving the same level of visibility as men, or benefit from the same opportunities and recognition, despite their deep knowledge of agriculture, water and ecosystems.

² retos y oportunidades de la agricultura familiar en Perú : care Perú

Barrier 4: Institutional barriers based on conclusion 1 and 3

In general terms, in Peru there is a fairly advanced regulatory framework at different levels of government requiring the mainstreaming of a gender perspective in public policies, however, there is still no adequate guarantee for the full realisation of the right to equality. There is a formal institutional framework for gender mainstreaming in public institutions, but the mechanisms to operationalise them are fragile. This is characterised by a high rotation of human capital and a type of management that usually lacks resources and capacities for the fulfilment of its functions.

Opportunities for gender mainstreaming

Based on the identified barriers in the Gender Analysis (as described above) the Resilient Puna with the project activities and measures (please see chapter 3) proposed has the potential to contribute to women empowerment and gender mainstreaming both in financial instruments and in territorial planning processes at community and regional levels to strengthen women's economic and adaptative capacity to climate change. This only could be achieved through greater access to financial resources and participation of women in training, dialogue, and decision-making processes. For this project, working with political partners, public staff, agriculture and water stakeholders and users' providers is an opportunity to contribute to closing the gaps and tackle the barriers through gender mainstreaming, making visible the need to carry out actions that, in the long term, collaborate in achieving a social and institutional transformation towards greater equality in highly masculinised sectors and in a post-pandemic context.

2.2. Gender expertise, gender responsiveness, and equal opportunities in the partner organizations

Although there is a formal institutional framework for gender mainstreaming in public institutions, the mechanisms to operationalise them are fragile. This is characterised by a high rotation of human capital and a type of management that usually lacks resources and capacities for the fulfilment of its functions. Despite these challenges, SERNANP have shown a sustained interest in incorporating the gender approach in their actions that should be strengthened and visibilized. In the case of MIDAGRI, some efforts have been made to promote gender equity. This is indeed an opportunity to better promote the incorporation of the gender approach into MIDAGRI actions. Profonanpe has a Gender Policy³ place to mainstream gender in its activities. In general, the Resilient Puna project has the potential to contribute to gender mainstreaming both in management and planning instruments and through greater participation of women in training, dialogue, and decision-making processes.

2.3. Gender expertise and responsiveness in the project team

All team members will be trained in gender equity and interculturality approaches. It is important to mention that some project members already have intermediate knowledge on gender and interculturality approaches based on previous academic and professional experience. The project will also count with a focal point for safeguards and gender. GIZ South Cone staff in Peru has been trained in gender topics during their induction process. Moreover, it has a gender focal point which allows for the exchange of experiences among colleagues from different projects and sectors, as well as collegial advice related to gender-responsive project design.

³ <https://profonanpe.org.pe/wp-content/uploads/2021/08/Politica-de-Genero.pdf>

3. Gender Action Plan and Indicators table

Guiding principle of this Gender Action Plan is as follows: Considering that in the Andean region women play an important role in small-scale agriculture and water harvest but are still unrepresented in local level decision making. Women are exposed to a very high workload and a high degree of domestic violence. The project will promote women's active participation in the development of value chains and income-generating measures and their involvement in decision-making and territorial planning processes and therefore enhance living conditions and economic opportunities of Andean women.

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ID	Component/ Output/ activity/barrier	Risks and potentials concerning gender equality (=derived from the barriers above)	Description of the Measure/Action: (=mitigation measures for risks =promotion measures for potentials)	Indicators	Baseline	Target and Evaluation Criteria (=indicators for successful implementation, quantitative and qualitative indicators are possible, where applicable sex aggregated)	Timeframe	Resources needed for Implementation (human and material resources)	Responsible entity for the implementation	Cost Estimation in EUR
1	All (Core Indicator 2)	Risk of not collecting disaggregated data	Gender-sensitive M&E system to track female and male beneficiaries and ensure gender-	M&E system elaborated and operating.	0	1 Gender- sensitive M&E system.	Y: 1	M&E systems and focal point of all EEs	GIZ- M&E specialist GIZ- Safeguards	M&E system 250,000 GIZ- M&E specialist 954,940

ID	Component/ Output/ activity/barrier	Risks and potentials concerning gender equality (=derived from the barriers above)	Description of the Measure/Action: (=mitigation measures for risks =promotion measures for potentials)	Indicators	Baseline	Target and Evaluation Criteria (=indicators for successful implementation, quantitative and qualitative indicators are possible, where applicable sex aggregated)	Timeframe	Resources needed for Implementation (human and material resources)	Responsible entity for the implementati on	Cost Estimation in EUR
			responsive project management.						and Gender advisor Profonanpe-Social and Gender Safeguard specialists	
2	All (cross-cutting)/Barrier 2	Failure to design communication materials and guidelines packages per activity which include a gender equality and social inclusion approach.	Establish cooperation between gender advisor and communication team to ensure that all project communication materials and guidelines include a gender equality and social inclusion (GESI) perspective. Including, when necessary, training	All project communication materials and guidelines packages per activity include a gender equality and social inclusion approach.	0	100% of the project communication materials and guidelines packages per activity include a gender equality and social inclusion approach.	Y: 2-6	Safeguards and Gender advisor Communications advisor Profonanpe-Gender specialist IdM- Gender specialist	GIZ-Safeguards and Gender advisor GIZ-Communications advisor Profonanpe-Gender specialist IdM- Gender specialist	GIZ-Safeguard and Gender advisor 144,000 GIZ-Communication advisor 300,300 Profonanpe-Social and Gender

ID	Component/ Output/ activity/barrier	Risks and potentials concerning gender equality (=derived from the barriers above)	Description of the Measure/Action: (=mitigation measures for risks =promotion measures for potentials)	Indicators	Baseline	Target and Evaluation Criteria (=indicators for successful implementation, quantitative and qualitative indicators are possible, where applicable sex aggregated)	Timeframe	Resources needed for Implementation (human and material resources)	Responsible entity for the implementation	Cost Estimation in EUR
			on GESI for the communications team.							specialist 197,813 IdM- Gender specialist 32,176 Communicatio n material 60,000 GESI training 6,000 ⁴ Capacity Building Strategy with a Gender Focus 37,519
3	All (Cross-cutting)	Beneficiaries not using the Grievance	Gender based Violence (GbV) and	Operative gender- and culturally	0	2 Operative gender- and culturally	Y:1-6	GIZ- M&E System	GIZ-M&E specialist	GRM 25,00

⁴ At least 1 training per year. Costs per training approx. EUR 1,000.

ID	Component/ Output/ activity/barrier	Risks and potentials concerning gender equality (=derived from the barriers above)	Description of the Measure/Action: (=mitigation measures for risks =promotion measures for potentials)	Indicators	Baseline	Target and Evaluation Criteria (=indicators for successful implementation, quantitative and qualitative indicators are possible, where applicable sex aggregated)	Timeframe	Resources needed for Implementation (human and material resources)	Responsible entity for the implementati on	Cost Estimation in EUR
		Redress Mechanism (GRM) for fear of harassment.	Sexual Exploitation Abuse and Harassment (SEAH) aspects are included in the Grievance Redress Mechanisms (GRM) of the project and financial mechanism (Puna Facility) which are gender and culturally sensitive to ensure anonymity of protection of complaint.	sensitive GRMs which includes GbV and SEAH aspects.		sensitive GRMs which includes GbV and SEAH aspects.		Profonanpe- M&E System Grievance Redress Mechanisms Trainings and workshops with local CSOs Communication materials	GIZ- Safeguards and Gender advisor GIZ-Staff in the regions Profonanpe- Monitoring specialists Profonanpe- Monitoring specialists Profonanpe- Social and Gender (see ID 2) Profonanpe- Environmental Safeguard specialists Profonanpe- Technical monitors in the regions	Trainings and workshops 6,000 ⁵ Profonanpe- Monitoring specialists 232,840 Profonanpe- Social and Gender (see ID 2) Profonanpe- Environmental Safeguard specialists 197,813 Profonanpe- Technical monitors in the

⁵ At least 1 training per year.

ID	Component/ Output/ activity/barrier	Risks and potentials concerning gender equality (=derived from the barriers above)	Description of the Measure/Action: (=mitigation measures for risks =promotion measures for potentials)	Indicators	Baseline	Target and Evaluation Criteria (=indicators for successful implementation, quantitative and qualitative indicators are possible, where applicable sex aggregated)	Timeframe	Resources needed for Implementation (human and material resources)	Responsible entity for the implementation	Cost Estimation in EUR
										regions (4) 381,517
5	Activity 1.1.1 Setting up the basis for financing and implementing EbA measures and Climate Resilient Value Chains /Barrier 2	Failure to design gender- responsive territorial local plans due to poor participation of women in the design of territorial plans, either because they do not have the time, they do not feel capable of participating or because it generates family conflicts o because men capture main leadership roles.	Promoting a gender-responsive territorial planning process for Climate Resilient Value Chains investing in EbA measures.	N° of gender- responsive territorial local plans for investing in climate change adaptation.	0	At least 100 gender- responsive territorial local plans for investing in climate change adaptation.	Y: 2-4	Promoters Local CSOs supporting communities, producers' associations and/or cooperatives Trainings for local promoters	GIZ and Instituto de Montaña coordinating with MIDAGRI and SERNANP	Included within the budget of activity 1.1.1. 2,199,853 IdM- Gender specialist 32,176

ID	Component/ Output/ activity/barrier	Risks and potentials concerning gender equality (=derived from the barriers above)	Description of the Measure/Action: (=mitigation measures for risks =promotion measures for potentials)	Indicators	Baseline	Target and Evaluation Criteria (=indicators for successful implementation, quantitative and qualitative indicators are possible, where applicable sex aggregated)	Timeframe	Resources needed for Implementation (human and material resources)	Responsible entity for the implementati on	Cost Estimation in EUR
6		Failure to collect, monitor and analyse disaggregated data due to lack of baseline information.	Collect information for baseline related to the number of women producers and for women led initiatives	Baseline report	0	1 Baseline report developed	Inception Phase	Secondary and primary information (e.g. surveys, desk review)	GIZ	Monitoring & Evaluation surveys EUR 59,000
7	Activity 1.1.2 Financing and implementing of EbA measures and Climate Resilient Value Chains (Co-benefit 5)/Barrier 2	Designing EbA measures without considering women's knowledge, needs, responsibilities, and domestic burden.	Designed and implement of innovative Local initiatives (including EbA and CRVC) with a gender-responsive approach.	N° of innovative Local initiatives (including EbA and CRVC) with a gender-responsive approach implemented.	0	127 innovative Local initiatives (with EbA and CRVC) with a gender-responsive approach implemented.	Y: 2-6	Technical experts Promoters Trainings Local CSOs supporting communities, producers' associations and/or cooperatives Profonanpe technical	GIZ- Safeguards and Gender advisor EbA Experts Profonanpe- Technical assistance providers	Included within the budget of activity 1.1.1 see ID 5, 1.1.3 9,461,507 and activity 1.2.2 2,955,526

ID	Component/ Output/ activity/barrier	Risks and potentials concerning gender equality (=derived from the barriers above)	Description of the Measure/Action: (=mitigation measures for risks =promotion measures for potentials)	Indicators	Baseline	Target and Evaluation Criteria (=indicators for successful implementation, quantitative and qualitative indicators are possible, where applicable sex aggregated)	Timeframe	Resources needed for Implementation (human and material resources)	Responsible entity for the implementation	Cost Estimation in EUR
								assistance providers Community monitoring		
8	Activity 1.1.3 Technical Assistance for implementing EbA measures and Climate Resilient Value Chains at local landscape level/Barrier 3	Insufficient participation of women in technical trainings provided through the project due to the family burden e.g., traditional gender roles which limit women to participate in trainings.	Ensure women's access to technical assistance including financial education, access to markets, implementation of Local initiatives	N° of female members of the beneficiaries of the Puna Facility that receive technical assistance for implementing EbA measures and Climate Resilient Value Chains, including financial education and	0	50% of female members of the beneficiaries of the Puna Facility receive technical assistance for implementing EbA measures and Climate Resilient Value Chains, including financial education and access to markets.	Y: 2-4	Profonanpe technical assistance providers Gender responsive trainings schemes (e.g., provide childcare, respect certain timeframes, provide assistance for travel or organize training in proximity to women homes)	Profonanpe	Included within the budget of activity 1.1.3 see ID 6.

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				access to markets.						
9	Activity 1.2.1: Recover, innovate and scale up ancestral knowledge and practices/Barr ier 2 & 3	Women avoid disclosing ancestral knowledge to the dismissed women's knowledge by men.	Innovative people centred approach for example through storytelling for systematization of best practices of women's knowledge for replication. The project will promote women's involvement in the development of publications together with universities/public institutions, etc.	N° of best practices in EbA and/or Climate Resilient Value Chains initiatives led by women systematized.	0	At least 6 best practices in EbA and/or Climate Resilient Value Chains initiatives led by women systematized.	Y: 4-6	Workshops Field trips Interviews Systematization	Instituto de Montaña (IdM)	Included within the budget of activity 1.2.1 2,707,713
10			Dissemination of best practices of women's knowledge for	N° of best practices and lessons learned in EbA and/or	0	At least 6 ⁶ best practices and lessons learned in EbA and/or	Y: 6	Publications Dissemination event	IdM- Communicatio n specialist	Included within the budget of

⁶ The target of 6 best practices led by women in EbA and/or CRVC were chosen by estimating an approximately 50% of the total eligible EbA measures and CRVA (i.e., 10 EbA measures and 2 CRVC) .

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			replication. The project will promote the development of publications together with universities/public institutions, etc.	Climate Resilient Value Chains initiatives led by women are included in publications for dissemination.		Climate Resilient Value Chains initiatives led by women disseminated in Quechua through publications such as brochures, pamphlets, plans, handbooks, radio spots and videos.			GIZ-Communications advisor	activity 1.2.1 see ID 8
11	Activity1.2.2. Implement community monitoring and observation systems to measure the impact of EbA measures	Failure to promote women's participation in community monitoring systems, because they do not have the time, they do	Ensure the participation of women in the observation, registration, and dissemination of information in the environmental community monitoring	N° women that participate in community monitoring committees.	0	At least 2 women out of the 6 members that participate in community monitoring committees.	Y: 2-5	Monitoring trainings Contracts between Profonanpe and beneficiaries of the Puna Facility	IdM Profonanpe	Included within the budget of activity 1.2.2 see ID 6 and 1.1.2 39,429,820

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	and provide feedback on regional and national policies/Barrier 2	not feel capable of participating or because it generates family conflicts.	committees under the approaches promoted by the project.							
12	Activity 2.1.1. Establish the Puna Facility for the long- term financing of EbA measures and Climate Resilient Value Chains /Barrier 1	Inability to find women-led associations, communities, cooperatives and/or organizations that accomplish formal requirements to apply.	Ensure the approval of Local initiatives proposed by women led associations, communities, cooperatives and/or organizations.	At least 30% of the selected Local initiatives should either support a women's association, cooperatives and/or organizations.	0	At least 30% Local initiatives approved should either support a women's association, cooperatives, organizations and/or a women led enterprise.	Y: 2-5	Facility Management Unit	Profonanpe	Included in the activity 2.1.1 4,203,150
13	2.1.2 Strengthen capacities to develop and	Inability to find 25% of female water utility	Train water utility officers of PES/MERESE to promote the	N° of water utility officers trained to promote the	0	30 water utility officers (at least 25% women) trained to	Y: 2-5	Trainings Consultants	GLZ- Safeguards and Gender advisor	Include in the activity 2.1.2 938,756

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	implement innovative mechanisms for EbA in high Andean ecosystems/Barrier 2	officers willing to participate.	inclusion of gender aspects into the MERESE's processes (agreements, governance platforms, project development, monitoring)	inclusion of gender aspects into the MERESE's processes		promote the inclusion of gender aspects into the MERESE's processes.			GIZ-MERESSE advisor	
14	2.1.3 Support the greening of microcredits to promote EbA and Climate Resilient Value Chains /Barrier 1	Failure to promote new and/or adjusted financial products meeting women's needs.	Support the design and/or adjustment of gender-responsive financial products to promote EbA and Climate Resilient Value Chains. The design will include a diagnostic identifying gender gaps and needs.	N° of gender-responsive financial products targeting woman and young producers.	0	At least 2 gender-responsive financial products designed and/or adjusted that include women needs.	Y: 3-5	Trainings GIZ- Microfinance advisor GIZ-Safeguards and gender advisor	GIZ- Microfinance advisor GIZ- Safeguards and Gender advisor	Included in the activity 2.1.3 1,295,103 GIZ- Microfinance advisor 228,000 GIZ- Safeguards and Gender advisor (see ID 3)

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15	Activity 3.1.1. Strengthen the capacities for territorial planning and governance processes integrating EbA and climate resilience/Barrier 3	Failure to promote planning processes with gender perspective due to the refusal of public officials acceptance, even though there is a fairly advanced regulatory framework that obliges all sectors of the State and all levels of government to mainstream a gender perspective in public policies	Ensure gender perspective during the planning processes training public officials.	N° state officials involved within the project target areas receiving trainings for the incorporation of EbA and Climate Resilient Value Chains measures with gender perspective in their planning processes.	0	X state officials (X men and X women) trained using existing state training programs.	Y: 2-6	GIZ-Safeguards and Gender advisor	GIZ- Team leader component 3 GIZ- Safeguards and Gender advisor	GIZ- Team leader component 376,920 GIZ- Safeguards and Gender advisor (see ID 3)

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16		Women have difficulties in participation in planning processes due to the long distances and family burden	Promote and support the participation of women in decision making planning processes.	% of female project beneficiaries take part in local and regional decision-making planning processes.	0	At least 5% of female project beneficiaries take part in local and regional decision-making planning processes.	Y: 3-6	Facilitation of women participation through provision of transport Family orientated information and training sessions	GIZ-Safeguards and Gender advisor GIZ-Local and regional advisors for component 3	Included within the budget of activity 3.1.1 2,975,216
17	Activity 3.1.2 Strengthen regulatory frameworks and M&E systems at national level/Barrier 3	Failure to collect, monitor and analyse disaggregated data at national level to be used afterwards for gender equal policy development.	Strengthening the M&E systems of MIDAGRIs programs ⁷ within the project to ensure the disaggregation of data.	N° of M&E systems with disaggregated (gender and EbA approach) information.	1	At least 3 M&E systems include disaggregated data (gender and EbA approach) information.	Y: 3-5	Trainings Technical Assistance	GIZ-Safeguards and Gender advisor GIZ-M&E Specialist MIDAGRI	Included within the budget of activity 3.1.2 1,928,804

⁷ AGROIDEAS, SSE, SENASA, AGORURAL, SIERRA AZUL, INIA, DSFFA, PSI and SERFOR

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